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Quantum mechanics for Many-particle Systems

From standard methods to quantum computing and machine learning

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1.1 Notations and conventions

Throughout these notes vectors, matrices and higher-order tensors are always set in boldface. Vectors are denoted by lower-case letters ($\boldsymbol{x}$, $\boldsymbol{v}$, ...) and matrices and tensors by upper-case letters ($\boldsymbol{A}$, $\boldsymbol{U}$, ...).

Unless stated otherwise the elements v_i of a vector $\boldsymbol{v}$ are assumed real. A real vector of length n satisfies $\boldsymbol{x} \in \mathbb{R}^n$; for complex vectors we write $\boldsymbol{x} \in \mathbb{C}^n$. A real $n \times n$ matrix satisfies $\boldsymbol{A} \in \mathbb{R}^{n\times n}$. Index counting starts at zero: matrix elements run from a_{00} to $a_{n-1,n-1}$.

We use the standard shorthand symbols: $\forall$ (for all), $\Rightarrow$ (implies), $\equiv$ (equivalent / defined as), with $\mathbb{R}$, $\mathbb{I}$, $\mathbb{C}$ for the real, integer and complex number fields.

1.2 Vectors

A column vector of length n and its transpose are

$$\boldsymbol{x} = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix}, \qquad \boldsymbol{x}^T = \begin{bmatrix} x_0 \ x_1 \ \cdots \ x_{n-1} \end{bmatrix}.$$

For complex vectors the *Hermitian conjugate* (adjoint) is

$$\boldsymbol{x}^\dagger = \begin{bmatrix} x_0^* \ x_1^* \ \cdots \ x_{n-1}^* \end{bmatrix}.$$

The *inner (dot) product* is the scalar

$$\boldsymbol{x}^T\boldsymbol{x} = \sum_{i=0}^{n-1} x_i x_i = x_0^2 + x_1^2 + \cdots + x_{n-1}^2,$$

and for complex vectors $\boldsymbol{x}^\dagger\boldsymbol{x} = \sum_{i=0}^{n-1} x_i^* x_i = \|\boldsymbol{x}\|^2$.

The *outer product* of two real vectors $\boldsymbol{y}$ (length n) and $\boldsymbol{z}$ (length m) is the $n \times m$ matrix

$$\boldsymbol{y}\boldsymbol{z}^T \implies (\boldsymbol{y}\boldsymbol{z}^T)_{ij} = y_i z_j.$$

For complex vectors one writes $\boldsymbol{y}\boldsymbol{z}^\dagger$ with $(\boldsymbol{y}\boldsymbol{z}^\dagger)_{ij} = y_i z_j^*$.

Important vector operations:

$$\begin{aligned}
\text{Addition / subtraction:}\quad & \boldsymbol{x} = \boldsymbol{y} \pm \boldsymbol{z} \;\Rightarrow\; x_i = y_i \pm z_i. \\
\text{Scalar multiplication:}\quad & \boldsymbol{x} = \gamma \boldsymbol{y} \;\Rightarrow\; x_i = \gamma y_i. \\
\text{Hadamard (element-wise) product:}\quad & \boldsymbol{x} = \boldsymbol{y} \circ \boldsymbol{z} \;\Rightarrow\; x_i = y_i z_i.
\end{aligned}$$

1.3 Matrices: definitions and algebraic operations

A general $n \times n$ matrix and its column-vector representation:

$$\boldsymbol{A} = \begin{bmatrix} a_{00} & a_{01} & \cdots & a_{0,n-1} \\ a_{10} & a_{11} & \cdots & a_{1,n-1} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n-1,0} & \cdots & \cdots & a_{n-1,n-1} \end{bmatrix} = \begin{bmatrix} \boldsymbol{a}_0 \; \boldsymbol{a}_1 \; \cdots \; \boldsymbol{a}_{n-1} \end{bmatrix}.$$

The fundamental algebraic operations are:

$$\begin{aligned}
\text{Scalar mult.:}\quad & \boldsymbol{A} = \gamma \boldsymbol{B} \;\Rightarrow\; a_{ij} = \gamma b_{ij}. \\
\text{Addition:}\quad & \boldsymbol{A} = \boldsymbol{B} \pm \boldsymbol{C} \;\Rightarrow\; a_{ij} = b_{ij} \pm c_{ij}. \\
\text{Matrix-vector product:}\quad & \boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} \;\Rightarrow\; y_i = \sum_{j=0}^{n-1} a_{ij} x_j. \\
\text{Matrix-matrix product:}\quad & \boldsymbol{A} = \boldsymbol{B}\boldsymbol{C} \;\Rightarrow\; a_{ij} = \sum_{k=0}^{n-1} b_{ik} c_{kj}. \\
\text{Transpose:}\quad & \boldsymbol{A} = \boldsymbol{B}^T \;\Rightarrow\; a_{ij} = b_{ji}.
\end{aligned}$$

Inverse.

If $\boldsymbol{A}$ is square and nonsingular, its inverse satisfies $\boldsymbol{A}^{-1}\boldsymbol{A} = \boldsymbol{I}$, where $\boldsymbol{I}$ is the identity matrix.

Hermitian conjugate.

The Hermitian conjugate $\boldsymbol{A}^\dagger$ is obtained by transposing $\boldsymbol{A}$ and complex-conjugating every element: $(A^\dagger)_{ij} = a_{ji}^*$.

Nonsingularity: equivalent characterisations.

For an $n \times n$ matrix $\boldsymbol{A}$ the following properties are all equivalent: (i) $\boldsymbol{A}^{-1}$ exists; (ii) $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$ implies $\boldsymbol{x} = \boldsymbol{0}$; (iii) the rows of $\boldsymbol{A}$ form a basis of $\mathbb{C}^n$; (iv) the columns of $\boldsymbol{A}$ form a basis of $\mathbb{C}^n$; (v) $\boldsymbol{A}$ is a product of elementary matrices; (vi) zero is not an eigenvalue of $\boldsymbol{A}$.

1.4 Special matrix types

Table 1.1 lists the matrix types that appear most frequently in quantum mechanics and many-body physics.
Structured (sparse) matrices important for numerical many-body work:

- **Diagonal**: $a_{ij} = 0$ for $i \neq j$.

Name	Defining relation	Element condition
Symmetric	$\boldsymbol{A} = \boldsymbol{A}^T$	$a_{ij} = a_{ji}$
Real orthogonal	$\boldsymbol{A} = (\boldsymbol{A}^T)^{-1}$	$\sum_k a_{ik} a_{jk} = \delta_{ij}$
Hermitian	$\boldsymbol{A} = \boldsymbol{A}^\dagger$	$a_{ij} = a_{ji}^*$
Unitary	$\boldsymbol{A} = (\boldsymbol{A}^\dagger)^{-1}$	$\sum_k a_{ik} a_{jk}^* = \delta_{ij}$
Real matrix	$\boldsymbol{A} = \boldsymbol{A}^*$	$a_{ij} = a_{ij}^*$

Table 1.1 Important special matrix types and their defining properties.

- **Upper/lower triangular**: $a_{ij} = 0$ for $i > j$ / $i < j$.
- **Upper/lower Hessenberg**: $a_{ij} = 0$ for $i > j+1$ / $i < j+1$.
- **Tridiagonal**: $a_{ij} = 0$ for $|i-j| > 1$.
- **Banded with bandwidth** p: $a_{ij} = 0$ for $|i-j| > p$.
- Block upper/lower triangular matrices, and more.

Many-body connection. Hamiltonian matrices in shell-model (full configuration-interaction, FCI) calculations are typically large, sparse, and often banded. Exploiting sparsity is essential to diagonalise them for systems with many single-particle states.

1.5 Dirac bra-ket notation

In quantum mechanics it is conventional to write state vectors using Dirac's bra-ket formalism. A ket $|x\rangle$ is a column vector

$$|x\rangle = \begin{bmatrix} x_0 \\ x_1 \\ \vdots \\ x_{n-1} \end{bmatrix},$$

and the corresponding bra $\langle x|$ is the row vector (dual)

$$\langle x| = \begin{bmatrix} x_0^* & x_1^* & \cdots & x_{n-1}^* \end{bmatrix},$$

so $|x\rangle^\dagger = \langle x|$. The *inner product* (a complex number) is

$$\langle y|x\rangle = \sum_{i=0}^{n-1} y_i^* x_i.$$

Note that $\langle x|x\rangle \geq 0$ is always real and non-negative, and $\langle y|x\rangle^* = \langle x|y\rangle$.

The *outer product* $|x\rangle\langle y|$ is an $n \times n$ matrix with elements $(|x\rangle\langle y|)_{ij} = x_i y_j^*$:

$$|x\rangle\langle y| = \begin{bmatrix} x_0 y_0^* & x_0 y_1^* & \cdots & x_0 y_{n-1}^* \\ x_1 y_0^* & x_1 y_1^* & \cdots & x_1 y_{n-1}^* \\ \vdots & \vdots & \ddots & \vdots \\ x_{n-1} y_0^* & \cdots & \cdots & x_{n-1} y_{n-1}^* \end{bmatrix}.$$

Worked example.

Let $|x\rangle = \begin{bmatrix} 1-i \\ 2+i \end{bmatrix}$. Then

$$\langle x|x\rangle = (1+i)(1-i) + (2-i)(2+i) = 2+5 = 7 \in \mathbb{R}.$$

For two different vectors $|x\rangle = \begin{bmatrix} -1 \\ 2i \\ 1 \end{bmatrix}$, $|y\rangle = \begin{bmatrix} 1 \\ 0 \\ i \end{bmatrix}$: $\langle x|y\rangle = -1+i \neq \langle y|x\rangle = -1-i$, confirming the rule $\langle x|y\rangle^* = \langle y|x\rangle$.

1.6 Computational basis states and superposition

A *two-level system* (qubit) is described by two orthonormal basis states, which we label

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad \langle 0|1\rangle = 0, \quad \langle 0|0\rangle = \langle 1|1\rangle = 1.$$

These are the *computational basis* states. A general state is a *superposition*

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

where $\alpha = \langle 0|\psi\rangle$ and $\beta = \langle 1|\psi\rangle$ are the probability amplitudes, satisfying $|\alpha|^2 + |\beta|^2 = 1$.

The identity operator decomposes as a sum of outer products:

$$\boldsymbol{I} = \sum_{i=0}^{1} |i\rangle\langle i| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Projection operators.

The outer products

$$\boldsymbol{P} = |0\rangle\langle 0| = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad \boldsymbol{Q} = |1\rangle\langle 1| = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$$

are *projection operators*. They are *idempotent* ($\boldsymbol{P}^2 = \boldsymbol{P}$, $\boldsymbol{Q}^2 = \boldsymbol{Q}$), commute with each other ($[\boldsymbol{P},\boldsymbol{Q}] = 0$), and satisfy $\boldsymbol{PQ} = 0$, $\boldsymbol{P}+\boldsymbol{Q} = \boldsymbol{I}$. Acting on a superposition state:

$$\boldsymbol{P}|\psi\rangle = \alpha|0\rangle, \qquad \boldsymbol{Q}|\psi\rangle = \beta|1\rangle.$$

The measurement postulate states that a measurement on $|\psi\rangle$ yields $|0\rangle$ with probability $|\alpha|^2$ and $|1\rangle$ with probability $|\beta|^2$.

Many-body connection. In many-body theory, projection operators $\hat{P}$ (onto occupied states) and $\hat{Q}$ (onto unoccupied / virtual states) define the particle-hole decomposition that is fundamental to perturbation theory, Hartree-Fock theory, and coupled-cluster methods.

Density matrix.

The density matrix (density operator) of a pure state $|\psi\rangle$ is the outer product

$$\boldsymbol{\rho} = |\psi\rangle\langle\psi| = \begin{bmatrix} \alpha\alpha^* & \alpha\beta^* \\ \beta\alpha^* & \beta\beta^* \end{bmatrix},$$

with $\operatorname{tr}\boldsymbol{\rho} = |\alpha|^2 + |\beta|^2 = 1$. Density matrices generalise to mixed states and subsystems; they are central to quantum information theory and to reduced-density-matrix many-body methods.

1.7 Pauli matrices

The three Pauli matrices are

$$\sigma_x = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \sigma_y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \quad \sigma_z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

Key properties (easy to verify directly):

- **Involutory**: $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = \boldsymbol{I}$.
- **Unitary**: $\sigma_\alpha^\dagger = \sigma_\alpha^{-1}$ (Hermitian conjugate equals inverse).
- **Hermitian**: $\sigma_\alpha^\dagger = \sigma_\alpha$.
- $\det(\sigma_\alpha) = -1$ for each α.
- **Commutation**: $[\sigma_x, \sigma_y] = 2i\sigma_z$ (and cyclic permutations).

The Pauli matrices act on the computational basis as

$$\sigma_x|0\rangle = |1\rangle, \quad \sigma_x|1\rangle = |0\rangle; \qquad \sigma_z|0\rangle = +|0\rangle, \quad \sigma_z|1\rangle = -|1\rangle.$$

Acting on a superposition $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$:

$$\sigma_x|\psi\rangle = \beta|0\rangle + \alpha|1\rangle, \qquad \sigma_y|\psi\rangle = i\beta|0\rangle - i\alpha|1\rangle.$$

Quantum computing connection. The Pauli matrices are the elementary single-qubit gates X, Y, Z. Together with the Hadamard gate $H = (\sigma_x + \sigma_z)/\sqrt{2}$ and the phase gate T they form a universal single-qubit gate set.

Many-body connection. Via the Jordan-Wigner transformation, the creation/annihilation operators of second quantisation are mapped directly onto products (strings) of Pauli matrices, enabling the representation of fermionic Hamiltonians on qubit registers.

1.8 Orthonormal bases and the completeness relation

A set of states $\{|\phi_i\rangle\}_{i=0}^{n-1}$ is an *orthonormal basis* (ONB) of an n-dimensional Hilbert space if

$$\langle\phi_j|\phi_i\rangle = \delta_{ij} \quad \text{and} \quad \sum_{i=0}^{n-1} |\phi_i\rangle\langle\phi_i| = \boldsymbol{I} \quad \text{(completeness / resolution of the identity).}$$

Any state can then be expanded as $|\psi\rangle = \sum_i \langle\phi_i|\psi\rangle|\phi_i\rangle$, with $\langle\phi_i|\psi\rangle$ the projection (overlap) of $|\psi\rangle$ onto $|\phi_i\rangle$.

Many-body connection. The natural ONB in many-body physics is the set of single-particle eigenstates of the one-body Hamiltonian:

$$\hat{h}_0|\phi_i\rangle = \varepsilon_i|\phi_i\rangle, \qquad \phi_i(\mathbf{r}) = \langle\mathbf{r}|\phi_i\rangle.$$

This single-particle basis $\{|\phi_i\rangle\}$ is the starting point for virtually all many-body methods: Hartree-Fock, MBPT, FCI, coupled-cluster, and quantum Monte Carlo.

1.9 Unitary transformations and preservation of orthogonality

An $n \times n$ matrix $\boldsymbol{U}$ is *unitary* if $\boldsymbol{U}^\dagger \boldsymbol{U} = \boldsymbol{U}\boldsymbol{U}^\dagger = \boldsymbol{I}$, which means $\boldsymbol{U}^{-1} = \boldsymbol{U}^\dagger$.

Orthogonality is preserved.

If $\{\boldsymbol{v}_i\}$ is an orthonormal set, $\boldsymbol{v}_j^T \boldsymbol{v}_i = \delta_{ij}$, then the transformed set $\boldsymbol{w}_i = \boldsymbol{U}\boldsymbol{v}_i$ is also orthonormal:

$$\boldsymbol{w}_j^T \boldsymbol{w}_i = (\boldsymbol{U}\boldsymbol{v}_j)^T \boldsymbol{U}\boldsymbol{v}_i = \boldsymbol{v}_j^T \underbrace{\boldsymbol{U}^T \boldsymbol{U}}_{=\boldsymbol{I}} \boldsymbol{v}_i = \boldsymbol{v}_j^T \boldsymbol{v}_i = \delta_{ij}.$$

In Dirac notation, if $|\psi_i\rangle = \sum_j u_{ij} |\phi_j\rangle$ with (u_{ij}) unitary, then $\langle \psi_j | \psi_i \rangle = \delta_{ij}$.

Why only unitary transformations?

For the inner product (and hence the norm) to be preserved under $|\phi\rangle = \boldsymbol{U}|\psi\rangle$ we need

$$\langle \phi | \phi \rangle = \langle \psi | \boldsymbol{U}^\dagger \boldsymbol{U} | \psi \rangle = \langle \psi | \psi \rangle = 1,$$

which requires $\boldsymbol{U}^\dagger \boldsymbol{U} = \boldsymbol{I}$. Since probabilities must sum to unity, this is a fundamental physical requirement. Unitary transformations are rotations (and reflections) in Hilbert space. The fact that $\boldsymbol{U}^{-1} = \boldsymbol{U}^\dagger$ also means that every unitary transformation is perfectly reversible.

Example: two-state rotation.

Given $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$ and a unitary $\boldsymbol{U}$, the rotated state $|\phi\rangle = \gamma|0\rangle + \delta|1\rangle = \boldsymbol{U}|\psi\rangle$ satisfies

$$\begin{bmatrix} \gamma \\ \delta \end{bmatrix} = \begin{bmatrix} u_{00} & u_{01} \\ u_{10} & u_{11} \end{bmatrix} \begin{bmatrix} \alpha \\ \beta \end{bmatrix}.$$

Since $|\alpha|^2 + |\beta|^2 = 1$ and $\boldsymbol{U}$ is unitary, $|\gamma|^2 + |\delta|^2 = 1$ automatically.

Many-body connection. In Hartree-Fock theory the self-consistent field (SCF) procedure generates a sequence of unitary rotations of the single-particle basis. The total HF energy is invariant under unitary rotations within the occupied or virtual subspace separately, which is the Thouless theorem.

Quantum computing connection. Every quantum gate is a unitary matrix. The evolution of a closed quantum system is described entirely by unitary operations; this is the quantum analogue of classical reversibility.

1.10 Tensor products

The *tensor (Kronecker) product* of two vectors $|x\rangle$ (length n) and $|y\rangle$ (length m) is the vector of length nm:

$$|x\rangle \otimes |y\rangle = |xy\rangle = \begin{bmatrix} x_0 y_0 \\ x_0 y_1 \\ x_1 y_0 \\ x_1 y_1 \end{bmatrix} \quad (\text{for } n = m = 2).$$

For the computational basis:

$$|0\rangle \otimes |0\rangle = |00\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad |0\rangle \otimes |1\rangle = |01\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad |1\rangle \otimes |0\rangle = |10\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad |1\rangle \otimes |1\rangle = |11\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

These four states form an ONB for the two-qubit Hilbert space $\mathbb{C}^4 = \mathbb{C}^2 \otimes \mathbb{C}^2$.

The most general state of N qubits is

$$|\Psi\rangle = \sum_{i_1 i_2 \cdots i_N} \Psi_{i_1 i_2 \cdots i_N} |i_1\rangle \otimes |i_2\rangle \otimes \cdots \otimes |i_N\rangle,$$

requiring 2^N complex amplitudes. For $N \approx 300$ this exceeds the number of atoms in the observable universe — this is the *exponential complexity* at the heart of quantum many-body physics.

Many-body connection. The full N-particle Hilbert space is a tensor product of N single-particle spaces. The FCI method works in this exponentially large space; all approximate methods (HF, MBPT, coupled cluster, DMRG, ...) are strategies to identify the relevant low-dimensional submanifold.

Quantum computing connection. A quantum computer with N qubits *inherently* represents a 2^N-dimensional state vector. This is both the source of its potential power and the reason that quantum simulation of many-body systems is a leading application.

1.11 Spectral decomposition of Hermitian operators

A Hermitian operator $\boldsymbol{A} = \boldsymbol{A}^\dagger$ has the following key properties:

1. All eigenvalues λ_i are *real*.
2. Eigenvectors belonging to distinct eigenvalues are *orthogonal*.
3. $\boldsymbol{A}$ is diagonalisable (it is a *normal* operator, $[\boldsymbol{A}, \boldsymbol{A}^\dagger] = 0$).

The eigenvalue equation $\boldsymbol{A}|\psi\rangle = \lambda|\psi\rangle$ has non-trivial solutions only when

$$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0 \quad (\text{characteristic polynomial}).$$

Spectral decomposition.

Let $\{|i\rangle\}$ be the eigenbasis of $\boldsymbol{A}$ with eigenvalues λ_i. Any state expands as $|\psi\rangle = \sum_i \alpha_i |i\rangle$. Using projection operators $\boldsymbol{P}_i = |i\rangle\langle i|$ we obtain:

$$\boldsymbol{A}|\psi\rangle = \sum_i \alpha_i \lambda_i |i\rangle = \sum_i \lambda_i \boldsymbol{P}_i |\psi\rangle,$$

from which, since this holds for any $|\psi\rangle$,

$$\boxed{\boldsymbol{A} = \sum_{i=0}^{n-1} \lambda_i |i\rangle\langle i| = \sum_{i=0}^{n-1} \lambda_i \boldsymbol{P}_i.}$$

This is the *spectral decomposition*: the operator is expressed as a weighted sum of its projection operators.

Example for a 2×2 operator.

If $\boldsymbol{A}$ has eigenstates $|\psi_a\rangle = \alpha_0|0\rangle + \alpha_1|1\rangle$ and $|\psi_b\rangle = \beta_0|0\rangle + \beta_1|1\rangle$ with eigenvalues λ_a, λ_b, then

$$\boldsymbol{A} = \lambda_a \begin{bmatrix} |\alpha_0|^2 & \alpha_0\alpha_1^* \\ \alpha_1\alpha_0^* & |\alpha_1|^2 \end{bmatrix} + \lambda_b \begin{bmatrix} |\beta_0|^2 & \beta_0\beta_1^* \\ \beta_1\beta_0^* & |\beta_1|^2 \end{bmatrix}.$$

Many-body connection. The spectral decomposition is used every time one diagonalises a Hamiltonian matrix. In FCI the full N-body Hamiltonian is diagonalised in the many-determinant basis; in Hartree-Fock the Fock matrix (a single-particle Hermitian operator) is diagonalised to find the optimal orbital basis.

Quantum computing connection. Any unitary gate $\boldsymbol{U}$ can be written as $\boldsymbol{U} = e^{i\boldsymbol{H}}$ for a Hermitian generator $\boldsymbol{H}$. The spectral decomposition of $\boldsymbol{H}$ gives $\boldsymbol{U} = \sum_i e^{i\lambda_i}|i\rangle\langle i|$. This exponential relationship connects the physical Hamiltonian to the time-evolution operator $\hat{U} = e^{-i\hat{H}t/\hbar}$ and is the basis of the variational quantum eigensolver (VQE) and unitary coupled-cluster (UCC) ansatz.

1.12 From algebra to computation

Everything said so far is exact algebra. The moment we hand a matrix to a computer, however, two new questions arise. The first is one of *accuracy*: floating-point numbers carry a finite number of digits, and we need a way of measuring how large the resulting error is. The second is one of *cost*: a many-body Hamiltonian matrix may have a dimension of 10^6 or more, and an algorithm whose work grows as n^3 is then simply not available to us.

The remainder of this chapter is devoted to these two questions. We first introduce the norms with which errors are measured, then develop the direct (elimination) methods for linear systems, the iterative alternatives, and finally the algorithms for the algebraic eigenvalue problem $\boldsymbol{A}|x\rangle = \lambda|x\rangle$, which is the computational problem at the heart of essentially every method in this book. All programs are written in Python; the complete, runnable versions are collected in the `BookMaterial/Programs` directory.

Throughout we keep the convention of Section 1.1 that indices run from 0 to $n-1$, so that the mathematical formulae and the Python code carry the same indices.

1.13 Vector and matrix norms

A class of vector norms is given by the so-called p-norms

$$\|\boldsymbol{x}\|_p = \left(|x_0|^p + |x_1|^p + \cdots + |x_{n-1}|^p\right)^{1/p}, \qquad p \geq 1. \tag{1.1}$$

The three cases we shall need are $p = 1$, $p = 2$ and the limit $p \to \infty$,

$$\|\boldsymbol{x}\|_1 = |x_0| + |x_1| + \cdots + |x_{n-1}|, \tag{1.2}$$

$$\|\boldsymbol{x}\|_2 = \left(|x_0|^2 + \cdots + |x_{n-1}|^2\right)^{1/2} = \left(\boldsymbol{x}^\dagger \boldsymbol{x}\right)^{1/2}, \tag{1.3}$$

$$\|\boldsymbol{x}\|_\infty = \max_{0\leq i\leq n-1} |x_i|. \tag{1.4}$$

The 2-norm is the one we have already been using implicitly: it is the square root of the inner product of Section 1.2, and in Dirac notation $\||x\rangle\|_2^2 = \langle x|x\rangle$. A quantum state is normalised precisely when its 2-norm equals unity.

Two inequalities follow from these definitions and will be used repeatedly. The first is the Cauchy-Schwarz inequality, valid for any two vectors in an inner product space,

$$\left|\boldsymbol{x}^\dagger \boldsymbol{y}\right| \leq \|\boldsymbol{x}\|_2 \,\|\boldsymbol{y}\|_2, \tag{1.5}$$

with equality only when $\boldsymbol{x}$ and $\boldsymbol{y}$ are linearly dependent. In Dirac notation this reads $|\langle x|y\rangle|^2 \leq \langle x|x\rangle\langle y|y\rangle$, and it is the inequality behind the variational principle: the overlap of a trial state with the exact ground state can never exceed the product of their norms. The second is the triangle inequality

$$\|\boldsymbol{x}+\boldsymbol{y}\|_2 \leq \|\boldsymbol{x}\|_2 + \|\boldsymbol{y}\|_2, \tag{1.6}$$

which follows from Eq. (1.5). Proofs may be found in Golub and Van Loan [1].

For matrices the most frequently used norms are the Frobenius norm

$$\|\boldsymbol{A}\|_F = \sqrt{\sum_{i=0}^{m-1}\sum_{j=0}^{n-1} |a_{ij}|^2}, \tag{1.7}$$

which is simply the 2-norm of the matrix read as a long vector, and the induced or operator p-norms

$$\|\boldsymbol{A}\|_p = \max_{\boldsymbol{x}\neq 0} \frac{\|\boldsymbol{A}\boldsymbol{x}\|_p}{\|\boldsymbol{x}\|_p}. \tag{1.8}$$

The induced norm measures the largest factor by which $\boldsymbol{A}$ can stretch a vector. For a Hermitian matrix $\|\boldsymbol{A}\|_2$ is the largest absolute eigenvalue, a fact which follows immediately from the spectral decomposition of Section 1.11. A unitary matrix has $\|\boldsymbol{U}\|_2 = 1$; it rotates but never stretches, which is Section 1.9 restated in the language of norms.

Relative error.

Let $fl(\boldsymbol{x})$ denote the machine representation of a vector $\boldsymbol{x} \neq 0$. The relative error is

$$\varepsilon = \frac{\|fl(\boldsymbol{x}) - \boldsymbol{x}\|}{\|\boldsymbol{x}\|}, \tag{1.9}$$

and if we use the ∞-norm the statement $\|fl(\boldsymbol{x}) - \boldsymbol{x}\|_\infty / \|\boldsymbol{x}\|_\infty \approx 10^{-l}$ says that the largest component of $fl(\boldsymbol{x})$ carries roughly l correct significant digits.

The condition number.

Suppose we solve $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ but, because of rounding, actually solve a slightly perturbed problem with right-hand side $\boldsymbol{b} + \delta\boldsymbol{b}$. The resulting error $\delta\boldsymbol{x}$ satisfies $\boldsymbol{A}\,\delta\boldsymbol{x} = \delta\boldsymbol{b}$, hence $\|\delta\boldsymbol{x}\| \leq \|\boldsymbol{A}^{-1}\|\,\|\delta\boldsymbol{b}\|$, while $\|\boldsymbol{b}\| \leq \|\boldsymbol{A}\|\,\|\boldsymbol{x}\|$. Combining the two gives

$$\frac{\|\delta\boldsymbol{x}\|}{\|\boldsymbol{x}\|} \leq \kappa(\boldsymbol{A})\,\frac{\|\delta\boldsymbol{b}\|}{\|\boldsymbol{b}\|}, \qquad \kappa(\boldsymbol{A}) = \|\boldsymbol{A}\|\,\|\boldsymbol{A}^{-1}\|. \tag{1.10}$$

The quantity $\kappa(\boldsymbol{A})$ is the *condition number*. It tells us how much an input error can be amplified, and it is a property of the problem, not of the algorithm: no amount of cleverness can rescue an ill-conditioned system. In the 2-norm and for a Hermitian matrix

$$\kappa_2(\boldsymbol{A}) = \frac{\max_i |\lambda_i|}{\min_i |\lambda_i|}, \tag{1.11}$$

so a matrix is ill-conditioned exactly when it has eigenvalues of very different magnitude, that is, when it is close to being singular. As a rule of thumb, if $\kappa \approx 10^k$ one loses about k significant digits.

Many-body connection. Ill-conditioning is not an academic worry in many-body physics. Overlap matrices of non-orthogonal single-particle bases, such as the Gaussian basis sets of quantum chemistry, become nearly singular as the basis approaches linear dependence, and their condition numbers can reach 10^{10} and beyond. The standard cure is to diagonalise the overlap matrix and discard the eigenvectors with the smallest eigenvalues, which is the canonical orthogonalisation procedure met again in Hartree-Fock theory.

1.14 Linear systems and Gaussian elimination

We now turn to the solution of

$$\boldsymbol{A}\boldsymbol{x} = \boldsymbol{w}, \tag{1.12}$$

with $\boldsymbol{A}$ non-singular. Before describing the algorithms it is worth seeing why such systems appear at all.

Where linear systems come from.

Consider the boundary value problem

$$-\frac{d^2u(x)}{dx^2} = f(x), \qquad x \in (a,b), \qquad u(a) = u(b) = 0. \tag{1.13}$$

We subdivide the interval into $n+1$ subintervals by setting $x_i = a + ih$ with step $h = (b-a)/(n+1)$, and replace the second derivative by the three-point formula

$$u''(x_i) \approx \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} + \mathscr{O}(h^2). \tag{1.14}$$

For the interior points $i = 0, 1, \ldots, n-1$ this turns Eq. (1.13) into

$$\frac{-u_{i+1} + 2u_i - u_{i-1}}{h^2} = f_i, \tag{1.15}$$

where the boundary values u_{-1} and u_n are known and equal to zero. Collecting the coefficients we obtain $\boldsymbol{A}\boldsymbol{u} = \boldsymbol{f}$ with the tridiagonal matrix

$$\boldsymbol{A} = \frac{1}{h^2}\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & \ddots & \\ & & \ddots & \ddots & -1 \\ & & & -1 & 2 \end{bmatrix}. \tag{1.16}$$

A differential equation has become a matrix equation. This is the single most important lesson of this chapter, and it will be repeated in a many-body setting in Section 1.27: discretising a continuous problem, or expanding a state in a finite basis, always produces a matrix problem.

The elimination.

The idea of Gaussian elimination is to use the first equation to eliminate x_0 from the remaining $n-1$ equations, then the new second equation to eliminate x_1 from the remaining $n-2$, and so on. After $n-1$ such steps we are left with an upper triangular system

$$\boldsymbol{U}\boldsymbol{x} = \boldsymbol{y}, \tag{1.17}$$

which is solved from the bottom upwards by *backward substitution*

$$x_m = \frac{1}{u_{mm}}\left(y_m - \sum_{k=m+1}^{n-1} u_{mk}x_k\right), \qquad m = n-1, n-2, \ldots, 0. \tag{1.18}$$

To eliminate x_0 we multiply the first equation by a_{j0}/a_{00} and subtract it from equation j, for $j = 1, \ldots, n-1$. The elements of the remaining $(n-1) \times (n-1)$ block are updated according to

$$a_{jk}^{(1)} = a_{jk} - \frac{a_{j0}\,a_{0k}}{a_{00}}, \qquad j,k = 1, \ldots, n-1, \tag{1.19}$$

and the same formula, applied to successively smaller blocks, defines the whole algorithm. Counting operations, the elimination requires $2n^3/3 + \mathscr{O}(n^2)$ floating point operations, while the backward substitution costs only $\mathscr{O}(n^2)$. The elimination dominates, and this cubic scaling is the reason why direct methods become unusable for the large matrices of Section 1.26.

Pivoting.

The derivation assumed $a_{00} \neq 0$, and more generally that no pivot element vanishes. Even a small pivot is dangerous, since dividing by it amplifies rounding errors. The cure is *partial pivoting*: before eliminating column k we search the column for the element of largest absolute value and interchange rows so that this element becomes the pivot. Row interchanges do not alter the solution, and they keep all multipliers bounded by unity. In practice one always pivots.

The following class implements the algorithm.

```
class GaussianElimination(DirectSolver):
    """Gaussian elimination with partial pivoting."""

    def solve(self, b):
        n = self.n
        M = self.A.copy()                 # the elimination destroys its input
        y = np.asarray(b, dtype=float).copy()

        for k in range(n - 1):            # forward elimination
            # partial pivoting: use the largest element in the column
            p = k + np.argmax(np.abs(M[k:, k]))
            if abs(M[p, k]) < 1.0e-14:
                raise np.linalg.LinAlgError("matrix is singular")
            if p != k:
                M[[k, p]] = M[[p, k]]
                y[k], y[p] = y[p], y[k]
            for i in range(k + 1, n):
                factor = M[i, k] / M[k, k]
                M[i, k:] -= factor * M[k, k:]
                y[i] -= factor * y[k]

        return self._back_substitute(M, y)

    @staticmethod
    def _back_substitute(U, y):
        """Solve U x = y for an upper triangular U."""
        n = U.shape[0]
        x = np.zeros(n)
        for m in range(n - 1, -1, -1):
            x[m] = (y[m] - U[m, m+1:] @ x[m+1:]) / U[m, m]
        return x
```

1.15 LU decomposition

Gaussian elimination as written above throws away the multipliers once it has used them. If we keep them instead, we obtain a factorisation of $\boldsymbol{A}$ itself, and this is what makes the method reusable. The LU decomposition writes

$$\boldsymbol{A} = \boldsymbol{L}\boldsymbol{U}, \tag{1.20}$$

with $\boldsymbol{L}$ unit lower triangular and $\boldsymbol{U}$ upper triangular; for a 4×4 matrix

$$\begin{bmatrix} a_{00} & a_{01} & a_{02} & a_{03} \\ a_{10} & a_{11} & a_{12} & a_{13} \\ a_{20} & a_{21} & a_{22} & a_{23} \\ a_{30} & a_{31} & a_{32} & a_{33} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ l_{10} & 1 & 0 & 0 \\ l_{20} & l_{21} & 1 & 0 \\ l_{30} & l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{00} & u_{01} & u_{02} & u_{03} \\ 0 & u_{11} & u_{12} & u_{13} \\ 0 & 0 & u_{22} & u_{23} \\ 0 & 0 & 0 & u_{33} \end{bmatrix}. \tag{1.21}$$

The factorisation exists whenever $\boldsymbol{A}$ is non-singular and, with the normalisation $l_{ii} = 1$, it is unique. Multiplying out the first column gives $a_{00} = u_{00}$ and $a_{j0} = l_{j0}u_{00}$, which determines u_{00} and the multipliers l_{j0}. The second column then gives u_{01}, u_{11} and l_{j1}, and so on: at every stage the unknowns are determined by quantities already computed. In practice $\boldsymbol{L}$ and $\boldsymbol{U}$ are stored in the same array as $\boldsymbol{A}$, since the unit diagonal of $\boldsymbol{L}$ need not be kept.

With partial pivoting the factorisation reads $\boldsymbol{PA} = \boldsymbol{LU}$ with $\boldsymbol{P}$ a permutation matrix, which in the program is recorded as a permutation array rather than as a matrix.

The determinant.

Because $\det(\boldsymbol{L}) = 1$ and the determinant of a triangular matrix is the product of its diagonal elements,

$$|\boldsymbol{A}| = \pm u_{00}u_{11}\cdots u_{n-1,n-1}, \tag{1.22}$$

where the sign is $(-1)^r$ with r the number of row interchanges. This is the only sensible way of computing a determinant numerically; expanding in minors would cost $\mathscr{O}(n!)$ operations.

Solving the system.

Writing $\boldsymbol{Ax} = \boldsymbol{LUx} = \boldsymbol{w}$ and introducing the intermediate vector $\boldsymbol{y} = \boldsymbol{Ux}$, the problem splits into two triangular systems,

$$\boldsymbol{Ly} = \boldsymbol{Pw} \quad \text{(forward substitution)}, \qquad \boldsymbol{Ux} = \boldsymbol{y} \quad \text{(backward substitution)}, \tag{1.23}$$

each costing only $\mathscr{O}(n^2)$ operations. This is the decisive advantage of LU over plain elimination: the expensive $\mathscr{O}(n^3)$ factorisation is performed once, after which any number of right-hand sides can be treated cheaply.

The inverse.

Column j of $\boldsymbol{A}^{-1}$ is the solution of $\boldsymbol{Ax} = \boldsymbol{e}_j$, where $\boldsymbol{e}_j$ is the jth unit vector. The inverse therefore costs one factorisation plus n substitutions. It is worth stressing that one should almost never compute an inverse: if the aim is to solve a linear system, solving it directly is both faster and more accurate than forming $\boldsymbol{A}^{-1}$ and multiplying.

```
class LUDecomposition(DirectSolver):
    """Doolittle LU factorisation with partial pivoting, P A = L U."""

    def _factorize(self):
```

```
        n = self.n
        LU = self.A.copy()
        perm = np.arange(n)
        sign = 1.0

        for k in range(n):
            p = k + np.argmax(np.abs(LU[k:, k]))
            if abs(LU[p, k]) < 1.0e-14:
                raise np.linalg.LinAlgError("matrix is singular")
            if p != k:
                LU[[k, p]] = LU[[p, k]]
                perm[[k, p]] = perm[[p, k]]
                sign = -sign
            # the multipliers l_ik are stored in place, below the diagonal
            LU[k+1:, k] /= LU[k, k]
            # rank-one update of the trailing submatrix
            LU[k+1:, k+1:] -= np.outer(LU[k+1:, k], LU[k, k+1:])

        self.LU, self.perm, self.sign = LU, perm, sign

    def solve(self, b):
        """Solve A x = b in two steps: L y = P b, then U x = y."""
        y = np.asarray(b, dtype=float)[self.perm].copy()
        n = self.n
        for i in range(1, n):                    # forward, L has unit diagonal
            y[i] -= self.LU[i, :i] @ y[:i]
        x = np.zeros(n)
        for i in range(n - 1, -1, -1):           # backward
            x[i] = (y[i] - self.LU[i, i+1:] @ x[i+1:]) / self.LU[i, i]
        return x

    def determinant(self):
        return self.sign * np.prod(np.diag(self.LU))
```

Cholesky's factorisation.

If $\boldsymbol{A}$ is real, symmetric and positive definite it admits the special factorisation

$$\boldsymbol{A} = \boldsymbol{L}\boldsymbol{L}^T, \tag{1.24}$$

with $\boldsymbol{L}$ lower triangular. The elements follow from

$$L_{ii} = \left(A_{ii} - \sum_{k=0}^{i-1} L_{ik}^2 \right)^{1/2}, \tag{1.25}$$

$$L_{ji} = \frac{1}{L_{ii}} \left(A_{ji} - \sum_{k=0}^{i-1} L_{ik} L_{jk} \right), \qquad j = i+1, \ldots, n-1. \tag{1.26}$$

The algorithm costs half of a general LU factorisation, and since the pivots L_{ii} are guaranteed positive there is no need for pivoting at all. The square root in Eq. (1.25) also provides the cheapest available test of positive definiteness: if the argument turns negative, the matrix is not positive definite.

Many-body connection. Positive definite matrices are everywhere in many-body theory. Overlap matrices of linearly independent basis functions are positive definite by construction, and the Cholesky factor of the overlap matrix is the standard tool for transforming a generalised eigenvalue problem $\boldsymbol{Hc} = E\boldsymbol{Sc}$ into an ordinary one. The Hessian of the variance in variational Monte Carlo is likewise positive definite, which is what makes the conjugate gradient method of Section 1.18 applicable to the optimisation of trial wave functions.

1.16 Tridiagonal systems

The matrix of Eq. (1.16) is tridiagonal, and applying general Gaussian elimination to it would be an extravagant waste: almost all the operations would be spent on elements that are already zero. Writing the system as

$$a_i u_{i-1} + b_i u_i + c_i u_{i+1} = f_i, \qquad i = 0, 1, \ldots, n-1, \tag{1.27}$$

with a_0 and c_{n-1} not referenced, the elimination reduces to a single forward sweep followed by a single backward sweep. The cost is $\mathscr{O}(n)$ operations rather than $2n^3/3$, and the storage is three vectors rather than a full matrix. Whenever a problem produces a tridiagonal, banded or otherwise sparse matrix, exploiting that structure is not an optimisation but a necessity.

```
class TridiagonalSolver:
    """Solve a tridiagonal system in O(n) operations."""

    def solve(self, f):
        n = self.n
        f = np.asarray(f, dtype=float)
        temp = np.zeros(n)
        u = np.zeros(n)

        btemp = self.b[0]                        # forward substitution
        u[0] = f[0] / btemp
        for i in range(1, n):
            temp[i] = self.c[i-1] / btemp
            btemp = self.b[i] - self.a[i] * temp[i]
            u[i] = (f[i] - self.a[i] * u[i-1]) / btemp

        for i in range(n - 2, -1, -1):           # backward substitution
            u[i] -= temp[i+1] * u[i+1]
        return u
```

Applied to $-u'' = 2$ on $(0,1)$ with $u(0) = u(1) = 0$, whose exact solution is $u(x) = x(1-x)$, the solver reproduces the analytic answer to 8.6×10^{-16} with $n = 100$ interior points. The three-point formula is exact for polynomials of degree two, which is why the discretisation error vanishes here; for a general right-hand side the error would be $\mathscr{O}(h^2)$, as we shall see in Section 1.27.

1.17 Iterative methods for linear systems

The methods of the previous sections are *direct*: up to rounding they produce the exact answer in a finite, known number of operations. Their cost, however, grows as n^3, and they require the matrix to be stored. Iterative methods take the opposite view. We start from a guess and improve it until the change falls below a tolerance. The matrix enters only through the product $\boldsymbol{Ax}$, so a matrix that is sparse, or that is never formed at all but only applied, costs nothing extra. This is the reason iterative methods, and not the direct ones, are what one uses for large-scale many-body calculations.

We split the matrix as

$$\boldsymbol{A} = \boldsymbol{D} + \boldsymbol{L} + \boldsymbol{U}, \tag{1.28}$$

with $\boldsymbol{D}$ diagonal, $\boldsymbol{L}$ strictly lower and $\boldsymbol{U}$ strictly upper triangular.

Jacobi's method.

Solving row i for x_i and evaluating the right-hand side with the previous iterate gives

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j \neq i} a_{ij} x_j^{(k)} \right), \tag{1.29}$$

or, in matrix form,

$$\boldsymbol{x}^{(k+1)} = \boldsymbol{D}^{-1} \left(\boldsymbol{b} - (\boldsymbol{L} + \boldsymbol{U}) \boldsymbol{x}^{(k)} \right). \tag{1.30}$$

Every component of the new iterate is computed from the old ones only, so all n updates are independent and the method parallelises trivially. It converges whenever $\boldsymbol{A}$ is strictly diagonally dominant or symmetric positive definite.

Gauss-Seidel.

Nothing forces us to wait: the components $x_0^{(k+1)}, \ldots, x_{i-1}^{(k+1)}$ are already available when we compute $x_i^{(k+1)}$. Using them immediately gives the Gauss-Seidel iteration

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(b_i - \sum_{j<i} a_{ij} x_j^{(k+1)} - \sum_{j>i} a_{ij} x_j^{(k)} \right). \tag{1.31}$$

This is forward substitution applied to the splitting, and it typically halves the number of iterations. The price is that the sweep is now inherently sequential.

Successive over-relaxation.

Gauss-Seidel usually moves in the right direction but not far enough. Over-relaxation exaggerates the step by a factor ω,

$$x_i^{(k+1)} = (1-\omega) x_i^{(k)} + \frac{\omega}{a_{ii}} \left(b_i - \sum_{j<i} a_{ij} x_j^{(k+1)} - \sum_{j>i} a_{ij} x_j^{(k)} \right), \tag{1.32}$$

which in matrix form reads

$$\boldsymbol{x}^{(k+1)} = (\boldsymbol{D} + \omega \boldsymbol{L})^{-1} \left(\omega \boldsymbol{b} - [\omega \boldsymbol{U} + (\omega - 1) \boldsymbol{D}] \, \boldsymbol{x}^{(k)} \right). \tag{1.33}$$

For symmetric positive definite matrices convergence is guaranteed for $0 < \omega < 2$, and $\omega = 1$ returns Gauss-Seidel. A well chosen ω can reduce the iteration count by an order of magnitude, but the optimal value depends on the spectrum of $\boldsymbol{A}$ and is rarely known in advance.

1.18 The conjugate gradient method

The methods above treat the linear system as a fixed-point problem. The conjugate gradient method instead treats it as a minimisation. For a symmetric positive definite $\boldsymbol{A}$ the solution of $\boldsymbol{Ax} = \boldsymbol{b}$ is the unique minimum of the quadratic form

$$P(\boldsymbol{x}) = \tfrac{1}{2} \boldsymbol{x}^T \boldsymbol{A} \boldsymbol{x} - \boldsymbol{x}^T \boldsymbol{b}, \tag{1.34}$$

since $\nabla P = \boldsymbol{Ax} - \boldsymbol{b}$, which vanishes precisely at the solution. The residual

$$\boldsymbol{r} = \boldsymbol{b} - \boldsymbol{Ax} \tag{1.35}$$

is therefore the negative gradient of P, and steepest descent would move along it. Steepest descent is however notoriously slow, because successive steps undo part of the progress of their predecessors.

The remedy is to search along directions that are *conjugate*, meaning orthogonal with respect to the inner product defined by $\boldsymbol{A}$,

$$\boldsymbol{p}_i^T \boldsymbol{A} \boldsymbol{p}_j = 0, \qquad i \neq j. \tag{1.36}$$

The eigenvectors of $\boldsymbol{A}$ are an example, since $\boldsymbol{v}_i^T \boldsymbol{A} \boldsymbol{v}_j = \lambda_j \boldsymbol{v}_i^T \boldsymbol{v}_j$ vanishes for $i \neq j$; but the whole point of the method is that we shall construct conjugate directions without knowing any eigenvectors. A set of n mutually conjugate vectors forms a basis of $\mathbb{R}^n$, so we may expand

$$\boldsymbol{x} = \sum_{i=0}^{n-1} \alpha_i \boldsymbol{p}_i. \tag{1.37}$$

Multiplying $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b}$ from the left by $\boldsymbol{p}_k^T$ and using Eq. (1.36) collapses the sum to a single term,

$$\alpha_k = \frac{\boldsymbol{p}_k^T \boldsymbol{b}}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k}, \tag{1.38}$$

so each direction can be treated independently. In exact arithmetic the method therefore terminates after at most n steps, which makes it a direct method in disguise; what makes it useful is that a good approximation is normally reached in far fewer.

The directions are generated on the fly. Starting from $\boldsymbol{x}_0 = \boldsymbol{0}$ and $\boldsymbol{p}_0 = \boldsymbol{r}_0 = \boldsymbol{b}$, the new direction is the residual with its component along the previous direction projected out,

$$\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} - \frac{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{r}_{k+1}}{\boldsymbol{p}_k^T \boldsymbol{A} \boldsymbol{p}_k} \boldsymbol{p}_k, \tag{1.39}$$

and the residual itself is updated recursively. From $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k$ we get

$$\boldsymbol{r}_{k+1} = \boldsymbol{b} - \boldsymbol{A}(\boldsymbol{x}_k + \alpha_k \boldsymbol{p}_k) = \boldsymbol{r}_k - \alpha_k \boldsymbol{A} \boldsymbol{p}_k, \tag{1.40}$$

which saves one matrix-vector product per iteration. Remarkably, the orthogonality relations that hold between the residuals allow Eq. (1.39) to be reduced to the two-term recurrence

$$\boldsymbol{p}_{k+1} = \boldsymbol{r}_{k+1} + \beta_k \boldsymbol{p}_k, \qquad \beta_k = \frac{\boldsymbol{r}_{k+1}^T \boldsymbol{r}_{k+1}}{\boldsymbol{r}_k^T \boldsymbol{r}_k}, \tag{1.41}$$

so that only the current residual and the current direction need to be stored. This is what the implementation below uses.

```
class ConjugateGradient(IterativeSolver):
    """Conjugate gradient for symmetric positive definite A.

    Only the matrix-vector product A p is required, so a `matvec` callable
    may be supplied instead of a dense matrix.
    """

    def solve(self, b, x0=None):
        b = np.asarray(b, dtype=float)
        x = self._initial_guess(x0)
        r = b - self.matvec(x)            # residual
        p = r.copy()                      # first search direction
        rs = r @ r

        for k in range(self.max_iter):
            Ap = self.matvec(p)
            alpha = rs / (p @ Ap)
            x += alpha * p                # x_{k+1} = x_k + alpha_k p_k
```

```
            r -= alpha * Ap                # r_{k+1} = r_k - alpha_k A p_k
            rs_new = r @ r
            if np.sqrt(rs_new) < self.tol:
                break
            p = r + (rs_new / rs) * p      # beta_k = |r_{k+1}|^2 / |r_k|^2
            rs = rs_new
        return x
```

Table 1.2 compares the four methods on the discretised second derivative of Eq. (1.16) with $n = 50$ interior points, for a residual tolerance of 10^{-8}. The ordering is the one to remember: Gauss-Seidel is roughly twice as fast as Jacobi, a well chosen relaxation parameter gains an order of magnitude, and the conjugate gradient method gains another.

Method	Iterations	Error
Jacobi	11049	1.0×10^{-9}
Gauss-Seidel	5526	1.0×10^{-9}
Successive over-relaxation ($\omega = 1.9$)	220	2.8×10^{-10}
Conjugate gradient	25	1.7×10^{-16}

Table 1.2 Iterations needed to solve $-u'' = 2$ on $(0,1)$ with $u(0) = u(1) = 0$, discretised on $n = 50$ interior points, to a residual tolerance of 10^{-8}. The error is measured against the analytic solution $u(x) = x(1-x)$.

Many-body connection. The conjugate gradient method appears in two distinct roles in this book. As a linear solver it is used wherever a large sparse positive definite system arises. As a minimiser it is the workhorse of variational calculations: in variational Monte Carlo the parameters of a trial wave function are optimised by conjugate gradient steps on the energy or the variance, and the Hessian of the variance plays the role of $\boldsymbol{A}$. The non-linear generalisations of the method, discussed in the chapter on gradient methods, all rest on the linear theory developed here.

1.19 Cubic spline interpolation

Tridiagonal systems appear in a second, quite different context, and the solver of Section 1.16 can be reused verbatim. Suppose we are given $n+1$ knots $x_0 < x_1 < \cdots < x_n$ and the corresponding function values $y_i = f(x_i)$, and we want a smooth interpolant. A spline function s of degree k is a piecewise polynomial: on every subinterval $[x_i, x_{i+1})$ it is a polynomial of degree at most k, and it has $k-1$ continuous derivatives on the whole interval. For $k = 1$ the spline is the broken line through the data points, with no continuous derivative at all. The useful choice is $k = 3$, the cubic spline, which is twice continuously differentiable and therefore looks smooth to the eye.

On each of the n subintervals we have a cubic

$$s_i(x) = a_{i0} + a_{i1}x + a_{i2}x^2 + a_{i3}x^3, \tag{1.42}$$

giving $4n$ unknown coefficients. Interpolation at both ends of every subinterval provides $2n$ conditions, continuity of s' at the interior knots $n-1$ more, and continuity of s'' another $n-1$, for a total of $4n-2$. Two degrees of freedom remain, and they are fixed by a condition at the two ends; setting the second derivative to zero there gives the *natural* cubic spline.

The elegant way to construct the spline is to take the second derivatives $f_i = s''(x_i)$ as the unknowns. Since s'' is piecewise linear,

$$s_i''(x) = \frac{f_i}{x_{i+1}-x_i}(x_{i+1}-x) + \frac{f_{i+1}}{x_{i+1}-x_i}(x-x_i), \tag{1.43}$$

and integrating twice, fixing the integration constants by the interpolation conditions, gives $s_i(x)$ in terms of f_i and f_{i+1}. Demanding that s' be continuous at the interior knots then produces $n-1$ equations for the $n-1$ unknown interior second derivatives, and those equations are tridiagonal and diagonally dominant. A cubic spline through n points therefore costs $\mathscr{O}(n)$ operations, which is why splines rather than global polynomials are used to interpolate tabulated potentials, form factors and matrix elements.

1.20 Applications: integral equations and matrix inversion

Two further examples show how naturally physical problems turn into the matrix problems of this chapter.

A linear integral equation.

The scattering problem in quantum mechanics leads to the Lippmann-Schwinger equation, an integral equation of the form

$$R(k,k') = V(k,k') + \frac{2}{\pi}\mathscr{P}\int_0^\infty dq\, q^2 \frac{V(k,q)R(q,k')}{(k_0^2-q^2)/m}, \tag{1.44}$$

where $\mathscr{P}$ denotes the Cauchy principal value. If we approximate the integral by a quadrature rule with N points q_j and weights w_j, the unknown function R is replaced by the $N+1$ numbers $R(q_i,k_0)$, and Eq. (1.44) becomes

$$\sum_j A_{ij}R_j = V_i, \qquad A_{ij} = \delta_{ij} - \frac{2}{\pi}u_j V(k_i,q_j), \tag{1.45}$$

with the weights u_j containing the quadrature weights and the propagator. The principal value is handled by the standard subtraction trick, which removes the singularity at $q=k_0$ before the quadrature is applied. What began as an integral equation is now a linear system of the type treated in Section 1.15: the physics is entirely in the construction of the matrix, and the solution is one call to an LU solver.

An ill-conditioned inverse.

Not every matrix inversion is benign. Fitting a polynomial of degree $n-1$ through n points $x_0,\ldots,x_{n-1}$ requires inverting the Vandermonde matrix

$$\boldsymbol{V} = \begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^{n-1} \\ 1 & x_1 & x_1^2 & \cdots & x_1^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n-1} & x_{n-1}^2 & \cdots & x_{n-1}^{n-1} \end{bmatrix}. \tag{1.46}$$

Its inverse is known in closed form, but its condition number grows exponentially with n: for equidistant points on $[0,1]$ and $n=20$ it exceeds 10^{20}, so that not a single significant digit survives in double precision. This is the numerical statement of a fact familiar from interpolation theory, namely that high-degree polynomial interpolation on a uniform grid is a bad idea, and it is the practical reason for preferring the piecewise construction of Section 1.19.

1.21 The algebraic eigenvalue problem

We return to the eigenvalue problem introduced in Section 1.11, this time asking how it is actually solved. The eigenvalues of a matrix $\boldsymbol{A}$ of dimension n are defined by

$$\boldsymbol{A}\boldsymbol{x}^{(\nu)} = \lambda^{(\nu)}\boldsymbol{x}^{(\nu)}, \tag{1.47}$$

where, unless stated otherwise, eigenvector means right eigenvector; the left eigenvectors satisfy $\boldsymbol{x}_L^{(\nu)}\boldsymbol{A} = \lambda^{(\nu)}\boldsymbol{x}_L^{(\nu)}$. Rewriting Eq. (1.47) as $(\boldsymbol{A} - \lambda^{(\nu)}\boldsymbol{I})\boldsymbol{x}^{(\nu)} = \boldsymbol{0}$, a non-trivial solution exists only if the matrix is singular, that is only if

$$P(\lambda) = \det(\lambda\boldsymbol{I} - \boldsymbol{A}) = 0. \tag{1.48}$$

The characteristic polynomial $P(\lambda)$ has degree n, and its roots

$$P(\lambda) = \prod_{i=0}^{n-1}(\lambda_i - \lambda) \tag{1.49}$$

form the *spectrum* $\lambda(\boldsymbol{A})$. Two useful identities follow at once from the factorisation,

$$|\boldsymbol{A}| = \lambda_0\lambda_1\cdots\lambda_{n-1}, \qquad \mathrm{Tr}(\boldsymbol{A}) = \lambda_0 + \lambda_1 + \cdots + \lambda_{n-1}. \tag{1.50}$$

It is tempting to conclude that one should find eigenvalues by computing the characteristic polynomial and looking for its roots. This is almost always a bad idea: the roots of a polynomial are extremely sensitive to its coefficients, so the detour through $P(\lambda)$ throws away accuracy that the original matrix still contained. The one case where the polynomial route is respectable is a tridiagonal matrix, for which the leading principal minors obey the recursion

$$P_k(\lambda) = (d_k - \lambda)P_{k-1}(\lambda) - e_{k-1}^2 P_{k-2}(\lambda), \tag{1.51}$$

with $P_0(\lambda) = d_0 - \lambda$ and $P_{-1} = 1$. The sequence $\{P_k(\lambda)\}$ is a Sturm sequence, and counting its sign changes tells us how many eigenvalues lie below λ, which is a robust way of isolating selected eigenvalues.

The standard approach is entirely different. We apply a sequence of similarity transformations that bring $\boldsymbol{A}$ either to diagonal form directly, or first to tridiagonal form and then to diagonal form. The second route is the one we follow, because diagonalising a tridiagonal matrix is vastly cheaper than diagonalising a full one.

1.22 Similarity transformations

Let $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ be real and symmetric. Then there exists a real orthogonal matrix $\boldsymbol{S}$ such that

$$\boldsymbol{S}^T\boldsymbol{A}\boldsymbol{S} = \mathrm{diag}(\lambda_0, \lambda_1, \ldots, \lambda_{n-1}) \equiv \boldsymbol{D}, \tag{1.52}$$

and the jth column of $\boldsymbol{S}$ is the eigenvector belonging to λ_j [1]. We say that $\boldsymbol{B}$ is a similarity transform of $\boldsymbol{A}$ if

$$\boldsymbol{B} = \boldsymbol{S}^T\boldsymbol{A}\boldsymbol{S}, \qquad \boldsymbol{S}^T\boldsymbol{S} = \boldsymbol{S}^{-1}\boldsymbol{S} = \boldsymbol{I}. \tag{1.53}$$

The importance of such a transformation is that it leaves the eigenvalues unchanged while altering the eigenvectors in a controlled way. The proof is one line. Multiply $\boldsymbol{A}\boldsymbol{x} = \lambda\boldsymbol{x}$ from the left by $\boldsymbol{S}^T$ and insert $\boldsymbol{S}\boldsymbol{S}^T = \boldsymbol{I}$ between $\boldsymbol{A}$ and $\boldsymbol{x}$,

$$(\boldsymbol{S}^T\boldsymbol{A}\boldsymbol{S})(\boldsymbol{S}^T\boldsymbol{x}) = \lambda\,(\boldsymbol{S}^T\boldsymbol{x}), \tag{1.54}$$

that is $\boldsymbol{B}(\boldsymbol{S}^T\boldsymbol{x}) = \lambda(\boldsymbol{S}^T\boldsymbol{x})$. So λ is an eigenvalue of $\boldsymbol{B}$ as well, with eigenvector $\boldsymbol{S}^T\boldsymbol{x}$.

The strategy is therefore either to apply transformations until

$$\boldsymbol{S}_N^T \cdots \boldsymbol{S}_1^T \boldsymbol{A} \boldsymbol{S}_1 \cdots \boldsymbol{S}_N = \boldsymbol{D}, \tag{1.55}$$

which is Jacobi's method of successive plane rotations, or to reduce $\boldsymbol{A}$ to tridiagonal form in a finite number of steps and then attack the tridiagonal matrix. We follow the second route, which is what all modern library routines do.

Many-body connection. Similarity transformations are not merely a numerical device. The whole of coupled-cluster theory rests on the similarity-transformed Hamiltonian $\bar{H} = e^{-T}\widehat{H}e^{T}$, which has the same eigenvalues as $\widehat{H}$ but is not Hermitian, since e^T is not unitary. The in-medium similarity renormalisation group instead uses a continuous sequence of *unitary* transformations to drive the Hamiltonian towards diagonal form, which is Eq. (1.55) carried out with infinitesimal steps.

1.23 The power method

The simplest of all eigenvalue algorithms is worth understanding, because everything that follows can be seen as a refinement of it. Assume $\boldsymbol{A}$ can be diagonalised, with eigenvalues $\lambda_0,\dots,\lambda_{n-1}$ and eigenvectors $\boldsymbol{v}_0,\dots,\boldsymbol{v}_{n-1}$, and assume there is a dominant eigenvalue, $|\lambda_0| > |\lambda_j|$ for $j>0$. An arbitrary starting vector expands as

$$\boldsymbol{b}_0 = c_0\boldsymbol{v}_0 + c_1\boldsymbol{v}_1 + \cdots + c_{n-1}\boldsymbol{v}_{n-1}, \tag{1.56}$$

and applying $\boldsymbol{A}$ repeatedly gives

$$\boldsymbol{A}^k\boldsymbol{b}_0 = c_0\lambda_0^k\left(\boldsymbol{v}_0 + \sum_{j>0}\frac{c_j}{c_0}\left(\frac{\lambda_j}{\lambda_0}\right)^k\boldsymbol{v}_j\right). \tag{1.57}$$

Every ratio $|\lambda_j/\lambda_0|$ is smaller than one, so the bracket tends to $\boldsymbol{v}_0$: repeated multiplication filters out everything except the dominant eigenvector. Normalising at every step, $\boldsymbol{b}_k = \boldsymbol{A}^k\boldsymbol{b}_0/\|\boldsymbol{A}^k\boldsymbol{b}_0\|$, and estimating the eigenvalue by the Rayleigh quotient

$$\mu_k = \frac{\boldsymbol{b}_k^\dagger\boldsymbol{A}\boldsymbol{b}_k}{\boldsymbol{b}_k^\dagger\boldsymbol{b}_k}, \tag{1.58}$$

we obtain a method that converges geometrically with ratio $|\lambda_1/\lambda_0|$, where λ_1 is the second largest eigenvalue in magnitude. When the two are close, convergence is correspondingly slow.

The power method has two obvious weaknesses: it gives only one eigenpair, and it discards all the information contained in the intermediate vectors $\boldsymbol{b}_0,\boldsymbol{A}\boldsymbol{b}_0,\dots,\boldsymbol{A}^{k-1}\boldsymbol{b}_0$. Removing the second weakness by working in the whole space spanned by those vectors, the *Krylov space*, leads directly to the Lanczos algorithm of Section 1.26. Applying the power method to $\boldsymbol{A}^{-1}$ instead gives inverse iteration, which converges to the eigenvalue of smallest magnitude and, with a shift, to any eigenvalue one can bracket.

1.24 Householder's method for tridiagonalisation

The first of the two steps is to construct an orthogonal $\boldsymbol{S}$ that brings $\boldsymbol{A}$ to tridiagonal form. It is built as a product of $n-2$ elementary transformations,

$$\boldsymbol{S} = \boldsymbol{S}_0\boldsymbol{S}_1\cdots\boldsymbol{S}_{n-3}, \tag{1.59}$$

each of which clears one column and one row. Only $n-2$ steps are needed, since the last two rows are already tridiagonal. After the first transformation

$$\boldsymbol{S}_0^T \boldsymbol{A} \boldsymbol{S}_0 = \begin{bmatrix} a_{00} & e_0 & 0 & \cdots & 0 \\ e_0 & a'_{11} & a'_{12} & \cdots & a'_{1,n-1} \\ 0 & a'_{21} & a'_{22} & \cdots & a'_{2,n-1} \\ \vdots & & & & \vdots \\ 0 & a'_{n-1,1} & \cdots & \cdots & a'_{n-1,n-1} \end{bmatrix}, \tag{1.60}$$

where the primed block is a matrix of dimension $n-1$ which the next transformation will treat in the same way. Note that the effective size of the problem shrinks at every step; in Jacobi's method, by contrast, each rotation acts on the full matrix and previously created zeros are destroyed. After all $n-2$ steps we are left with the diagonal elements $a_{00}, a'_{11}, a''_{22}, \ldots$ and the off-diagonal elements $e_0, e_1, \ldots, e_{n-2}$ of a tridiagonal matrix.

It remains to find the transformations. We take

$$\boldsymbol{S}_0 = \begin{bmatrix} 1 & \boldsymbol{0}^T \\ \boldsymbol{0} & \boldsymbol{P} \end{bmatrix}, \qquad \boldsymbol{P} = \boldsymbol{I} - 2\boldsymbol{u}\boldsymbol{u}^T, \qquad \boldsymbol{u}^T\boldsymbol{u} = 1, \tag{1.61}$$

where $\boldsymbol{u}$ is a column vector of length $n-1$. The matrix $\boldsymbol{P}$, whose elements are $P_{ij} = \delta_{ij} - 2u_i u_j$, is a *Householder reflector*: it is symmetric and satisfies $\boldsymbol{P}^2 = \boldsymbol{I}$, hence $\boldsymbol{P}^T = \boldsymbol{P}^{-1}$, so it is orthogonal. Geometrically it reflects a vector in the hyperplane perpendicular to $\boldsymbol{u}$.

Applying $\boldsymbol{S}_0$ gives

$$\boldsymbol{S}_0^T \boldsymbol{A} \boldsymbol{S}_0 = \begin{bmatrix} a_{00} & (\boldsymbol{P}\boldsymbol{v})^T \\ \boldsymbol{P}\boldsymbol{v} & \boldsymbol{A}' \end{bmatrix}, \qquad \boldsymbol{v}^T = \{a_{10}, a_{20}, \ldots, a_{n-1,0}\}, \tag{1.62}$$

so we must choose $\boldsymbol{u}$ such that $\boldsymbol{P}\boldsymbol{v}$ has only one non-zero component,

$$\boldsymbol{P}\boldsymbol{v} = \boldsymbol{v} - 2\boldsymbol{u}(\boldsymbol{u}^T\boldsymbol{v}) = \kappa\,\boldsymbol{e}, \qquad \boldsymbol{e}^T = \{1, 0, \ldots, 0\}. \tag{1.63}$$

The constant κ follows from taking the scalar product of Eq. (1.63) with itself and using $\boldsymbol{P}^2 = \boldsymbol{I}$,

$$(\boldsymbol{P}\boldsymbol{v})^T \boldsymbol{P}\boldsymbol{v} = \kappa^2 = \boldsymbol{v}^T\boldsymbol{v} = \sum_{i=1}^{n-1} a_{i0}^2, \tag{1.64}$$

so $\kappa = \pm\|\boldsymbol{v}\|_2$. Rewriting Eq. (1.63) as $\boldsymbol{v} - \kappa\boldsymbol{e} = 2\boldsymbol{u}(\boldsymbol{u}^T\boldsymbol{v})$ shows that $\boldsymbol{u}$ is proportional to $\boldsymbol{v} - \kappa\boldsymbol{e}$, and normalising gives

$$\boldsymbol{u} = \frac{\boldsymbol{v} - \kappa\boldsymbol{e}}{\|\boldsymbol{v} - \kappa\boldsymbol{e}\|_2}. \tag{1.65}$$

The sign of κ is free, and here numerical care is essential: if $v_0 = a_{10}$ and κ have the same sign, the leading component of $\boldsymbol{v} - \kappa\boldsymbol{e}$ is a difference of nearly equal numbers and precision is lost. We therefore always choose

$$\kappa = -\operatorname{sign}(v_0)\,\|\boldsymbol{v}\|_2, \tag{1.66}$$

which makes the subtraction an addition.

Finally we need the transformed block. Writing $\boldsymbol{p} = \boldsymbol{A}'\boldsymbol{u}$ and $K = \boldsymbol{u}^T\boldsymbol{p}$, and introducing $\boldsymbol{z} = \boldsymbol{p} - K\boldsymbol{u}$, a short calculation gives the symmetric rank-two update

$$\boldsymbol{P}\boldsymbol{A}'\boldsymbol{P} = \boldsymbol{A}' - 2\boldsymbol{z}\boldsymbol{u}^T - 2\boldsymbol{u}\boldsymbol{z}^T, \tag{1.67}$$

which costs $\mathscr{O}(n^2)$ operations rather than the $\mathscr{O}(n^3)$ of forming the products explicitly. Summed over the $n-2$ steps, the whole reduction costs $\frac{4}{3}n^3$ operations, or twice that if the transformation matrix is accumulated.

```
def tridiagonalize(self):
    """Return (d, e, S) with S^T A S tridiagonal."""
    n = self.n
    M = self.A.copy()
```

```
    S = np.eye(n)

    for k in range(n - 2):
        v = M[k+1:, k]                          # the column to be reduced
        norm_v = np.linalg.norm(v)
        if norm_v < self.tol:
            continue
        # the sign is chosen to avoid cancellation in v[0] - kappa
        kappa = -np.copysign(norm_v, v[0])
        w = v.copy()
        w[0] -= kappa
        u = w / np.linalg.norm(w)               # unit vector, u^T u = 1

        # similarity transformation of the trailing block,
        #   P A P = A - 2 z u^T - 2 u z^T,  z = A u - (u^T A u) u
        block = M[k+1:, k+1:]
        p = block @ u
        z = p - (u @ p) * u
        M[k+1:, k+1:] = block - 2.0*np.outer(z, u) - 2.0*np.outer(u, z)

        M[k+1:, k] = 0.0                        # the reduced column and row
        M[k+1, k] = kappa
        M[k, k+1:] = 0.0
        M[k, k+1] = kappa

        S[:, k+1:] -= 2.0 * np.outer(S[:, k+1:] @ u, u)    # S <- S P

    d = np.diag(M).copy()
    e = np.diag(M, -1).copy()
    return d, e, S
```

For a random symmetric 8×8 matrix the routine returns an $\boldsymbol{S}$ with $\|\boldsymbol{S}^T\boldsymbol{S}-\boldsymbol{I}\| = 1.1\times 10^{-15}$ and reproduces the original matrix, $\|\boldsymbol{S}\boldsymbol{T}\boldsymbol{S}^T-\boldsymbol{A}\| = 3.1\times 10^{-15}$. Orthogonality is preserved to machine precision, which is exactly why reflectors are preferred to the more obvious Gaussian-elimination-like schemes.

1.25 Diagonalising a tridiagonal matrix

The second step diagonalises the tridiagonal matrix

$$\boldsymbol{T} = \begin{bmatrix} d_0 & e_0 & & \\ e_0 & d_1 & e_1 & \\ & e_1 & d_2 & \ddots \\ & & \ddots & \ddots \end{bmatrix}. \tag{1.68}$$

The basic observation is that if any off-diagonal element e_i vanishes, the matrix splits into two independent blocks; in particular if $e_0 = 0$ then d_0 is already an eigenvalue. The algorithm therefore works by making the off-diagonal elements vanish one after another.

Francis' original scheme, and the QL and QR algorithms that grew out of it, apply a sequence of plane rotations

$$\boldsymbol{S}_i = \begin{bmatrix} \cos\theta & \cdots & \sin\theta \\ \vdots & & \vdots \\ -\sin\theta & \cdots & \cos\theta \end{bmatrix}, \tag{1.69}$$

choosing θ at each step so as to annihilate one off-diagonal element. Two refinements turn this idea into the algorithm actually used. The first is *shifting*: instead of working with $\boldsymbol{T}$ one works with $\boldsymbol{T}-\sigma\boldsymbol{I}$, where σ is an estimate of the eigenvalue being sought. The rate of convergence of the element e_{m-1} to zero is governed by the ratio of the distances of σ to the two nearest eigenvalues, so a good shift produces cubic convergence and typically two or three iterations per eigenvalue. The Wilkinson shift, obtained from the trailing 2×2 submatrix, is the standard choice. The second refinement is to apply the shift *implicitly*, by chasing an unwanted element down the subdiagonal with successive rotations, which avoids ever forming $\boldsymbol{T} - \sigma \boldsymbol{I}$ explicitly and so avoids the associated cancellation.

The resulting routine is traditionally called `tqli`. It takes the diagonal d and the off-diagonal e produced by the Householder step, together with the accumulated transformation matrix, and returns the eigenvalues in d and the eigenvectors as the columns of that matrix.

```
@staticmethod
def tqli(d, e, z=None, max_sweeps=50):
    """QL algorithm with implicit shifts for a symmetric tridiagonal matrix.

    Pass the S of the Householder step as z to obtain the eigenvectors of
    the original matrix A.
    """
    d = np.asarray(d, dtype=float).copy()
    n = len(d)
    e = np.concatenate([np.asarray(e, dtype=float), [0.0]])
    z = np.eye(n) if z is None else np.asarray(z, dtype=float).copy()
    eps = np.finfo(float).eps

    for l in range(n):
        for _ in range(max_sweeps):
            # look for a negligible off-diagonal element to split off
            m = n - 1
            for i in range(l, n - 1):
                if abs(e[i]) <= eps * (abs(d[i]) + abs(d[i+1])):
                    m = i
                    break
            if m == l:                          # d[l] has converged
                break

            # Wilkinson shift, computed without cancellation
            g = (d[l+1] - d[l]) / (2.0 * e[l])
            r = np.hypot(g, 1.0)
            g = d[m] - d[l] + e[l] / (g + np.copysign(r, g))

            s = c = 1.0
            p = 0.0
            for i in range(m - 1, l - 1, -1):
                f = s * e[i]
                b = c * e[i]
                r = np.hypot(f, g)
                e[i+1] = r
                s, c = f / r, g / r
                g = d[i+1] - p
                r = (d[i] - g) * s + 2.0 * c * b
                p = s * r
                d[i+1] = g + p
                g = c * r - b
                col = z[:, i+1].copy()        # accumulate the plane rotation
                z[:, i+1] = s * z[:, i] + c * col
                z[:, i] = c * z[:, i] - s * col
            d[l] -= p
            e[l] = g
            e[m] = 0.0
    return d, z
```

Together, the two steps constitute a complete eigenvalue solver. On a random symmetric 8×8 matrix it reproduces the eigenvalues returned by `numpy.linalg.eigvalsh` to 2.2×10^{-15}, with $\|\boldsymbol{AV} - \boldsymbol{VD}\| =$

4.9×10^{-15} and $\|\boldsymbol{V}^T\boldsymbol{V} - \boldsymbol{I}\| = 2.7 \times 10^{-15}$. The total cost is $\mathscr{O}(n^3)$, dominated by the Householder reduction.

Many-body connection. The Householder and QL combination is what lies behind every call to a library eigenvalue routine, and it is entirely adequate as long as the matrix fits in memory. For the Hartree-Fock and random-phase-approximation matrices of the following chapters, whose dimensions are set by the number of single-particle states, this is the method of choice. It becomes useless the moment the dimension is set by the number of Slater determinants, which is the situation of the next section.

1.26 Lanczos' algorithm

The methods of the previous two sections transform the matrix, and therefore need to have it. In large-scale many-body calculations we do not. A full configuration-interaction Hamiltonian for a realistic system may have a dimension of 10^9 or more; it is sparse, it is generated on the fly from the occupation numbers of the Slater determinants, and it is never stored. What we can always do is compute the product $\boldsymbol{A}\boldsymbol{q}$ for a given vector $\boldsymbol{q}$. And we rarely want the whole spectrum: the ground state and a handful of excited states are enough.

The Lanczos algorithm is built for exactly this situation. It applies to real symmetric matrices and has the following properties:

- it generates a sequence of real tridiagonal matrices $\boldsymbol{T}_k$ of dimension $k \times k$ with $k \le n$, whose extremal eigenvalues are progressively better estimates of the extremal eigenvalues of $\boldsymbol{A}$;
- the matrix enters only through the product $\boldsymbol{A}\boldsymbol{q}$;
- the underlying transformation is the similarity transformation $\boldsymbol{T} = \boldsymbol{Q}^T\boldsymbol{A}\boldsymbol{Q}$, with the first Lanczos vector $\boldsymbol{Q}\boldsymbol{e}_0 = \boldsymbol{q}_0$ chosen at will.

The three-term recurrence.

Write $\boldsymbol{Q}$ in terms of its column vectors, $\boldsymbol{Q} = [\boldsymbol{q}_0\,\boldsymbol{q}_1 \cdots \boldsymbol{q}_{n-1}]$, and let

$$\boldsymbol{T} = \begin{bmatrix} \alpha_0 & \beta_0 & & & \\ \beta_0 & \alpha_1 & \beta_1 & & \\ & \beta_1 & \alpha_2 & \ddots & \\ & & \ddots & \ddots & \beta_{n-2} \\ & & & \beta_{n-2} & \alpha_{n-1} \end{bmatrix}. \tag{1.70}$$

Using $\boldsymbol{Q}\boldsymbol{Q}^T = \boldsymbol{I}$ we may rewrite $\boldsymbol{T} = \boldsymbol{Q}^T\boldsymbol{A}\boldsymbol{Q}$ as $\boldsymbol{Q}\boldsymbol{T} = \boldsymbol{A}\boldsymbol{Q}$, and equating column k on both sides gives the three-term recurrence

$$\boldsymbol{A}\boldsymbol{q}_k = \beta_{k-1}\boldsymbol{q}_{k-1} + \alpha_k\boldsymbol{q}_k + \beta_k\boldsymbol{q}_{k+1}, \tag{1.71}$$

with $\beta_{-1}\boldsymbol{q}_{-1} = \boldsymbol{0}$. This single equation contains the whole algorithm. Since the Lanczos vectors are orthonormal, multiplying Eq. (1.71) from the left by $\boldsymbol{q}_k^T$ gives

$$\alpha_k = \boldsymbol{q}_k^T\boldsymbol{A}\boldsymbol{q}_k, \tag{1.72}$$

and the remainder

$$\boldsymbol{r}_k = (\boldsymbol{A} - \alpha_k\boldsymbol{I})\boldsymbol{q}_k - \beta_{k-1}\boldsymbol{q}_{k-1} \tag{1.73}$$

determines the next vector. If $\boldsymbol{r}_k \neq \boldsymbol{0}$ then

$$\boldsymbol{q}_{k+1} = \boldsymbol{r}_k/\beta_k, \qquad \beta_k = \pm\|\boldsymbol{r}_k\|_2. \tag{1.74}$$

If $\boldsymbol{r}_k$ does vanish we have found an invariant subspace, and the eigenvalues of $\boldsymbol{T}_k$ are exact eigenvalues of $\boldsymbol{A}$. The algorithm is therefore

$$
\begin{aligned}
&\boldsymbol{r}_0 = \boldsymbol{q}_0, \quad \beta_{-1} = 1, \quad \boldsymbol{q}_{-1} = \boldsymbol{0}, \quad k = 0 \\
&\textbf{while } \beta_{k-1} \neq 0 \\
&\quad \boldsymbol{q}_k = \boldsymbol{r}_{k-1}/\beta_{k-1} \\
&\quad \alpha_k = \boldsymbol{q}_k^T \boldsymbol{A} \boldsymbol{q}_k \\
&\quad \boldsymbol{r}_k = (\boldsymbol{A} - \alpha_k \boldsymbol{I})\boldsymbol{q}_k - \beta_{k-1}\boldsymbol{q}_{k-1} \\
&\quad \beta_k = \|\boldsymbol{r}_k\|_2, \quad k \leftarrow k+1 \\
&\textbf{end while}
\end{aligned}
\tag{1.75}
$$

Krylov spaces and Ritz values.

After m steps the vectors $\boldsymbol{q}_0, \dots, \boldsymbol{q}_{m-1}$ span the Krylov space

$$
\mathcal{K}_m(\boldsymbol{A}, \boldsymbol{q}_0) = \operatorname{span}\{\boldsymbol{q}_0, \boldsymbol{A}\boldsymbol{q}_0, \boldsymbol{A}^2\boldsymbol{q}_0, \dots, \boldsymbol{A}^{m-1}\boldsymbol{q}_0\}, \tag{1.76}
$$

which is precisely the sequence of vectors the power method of Section 1.23 generates and then throws away. Lanczos keeps them, orthonormalises them, and diagonalises $\boldsymbol{A}$ restricted to their span. The eigenvalues θ_i of the small matrix $\boldsymbol{T}_m$ are called *Ritz values* and the vectors $\boldsymbol{Q}\boldsymbol{y}_i$, with $\boldsymbol{y}_i$ the eigenvectors of $\boldsymbol{T}_m$, are the *Ritz vectors*. They are the best approximations to eigenpairs of $\boldsymbol{A}$ obtainable within the Krylov space, in the variational sense: the lowest Ritz value is the minimum of the Rayleigh quotient over $\mathcal{K}_m$, and is therefore an upper bound on the exact ground-state energy. This is the same variational principle that underlies Hartree-Fock theory and the variational Monte Carlo method.

Because $\boldsymbol{T}_m$ is tridiagonal and small, it is diagonalised with the QL algorithm of Section 1.25. The two algorithms are thus used together: Lanczos reduces the enormous problem to a tiny tridiagonal one, and `tqli` finishes the job.

```
def run(self, m, q0=None, tol=1.0e-12):
    """Perform m Lanczos steps, returning the tridiagonal matrix elements."""
    n = self.n
    q = np.random.default_rng(2026).normal(size=n) if q0 is None \
        else np.asarray(q0, dtype=float).copy()
    q /= np.linalg.norm(q)

    Q = np.zeros((n, m))
    alpha = np.zeros(m)
    beta = np.zeros(max(m - 1, 0))
    q_prev = np.zeros(n)
    b_prev = 0.0

    for k in range(m):
        Q[:, k] = q
        w = self.matvec(q)                     # the only use made of A
        alpha[k] = q @ w                       # alpha_k = q_k^T A q_k
        w = w - alpha[k]*q - b_prev*q_prev     # r_k

        if self.reorthogonalize:
            for _ in range(2):                 # modified Gram-Schmidt
                w -= Q[:, :k+1] @ (Q[:, :k+1].T @ w)

        if k == m - 1:
            break
        b = np.linalg.norm(w)
        if b < tol:                            # invariant subspace found
            alpha, beta, Q = alpha[:k+1], beta[:k], Q[:, :k+1]
            break
        beta[k] = b
```

```
        q_prev, b_prev = q, b
        q = w / b

    self.alpha, self.beta, self.Q = alpha, beta, Q
    return alpha, beta, Q
```

Loss of orthogonality.

In exact arithmetic the three-term recurrence produces orthogonal vectors automatically. In floating-point arithmetic it does not. As soon as one Ritz value converges, the corresponding direction starts to reappear in the remainder $\boldsymbol{r}_k$, orthogonality is lost, and the algorithm produces spurious extra copies of eigenvalues it has already found. These are the notorious *ghost* eigenvalues. The simplest cure, used in the code above, is full reorthogonalisation: at every step the new vector is explicitly orthogonalised against all previous ones. This requires storing all the Lanczos vectors and costs $\mathcal{O}(m^2 n)$ operations, which is acceptable when m is a few hundred. For very long runs one uses selective or periodic reorthogonalisation instead.

A many-body example.

Table 1.3 shows the algorithm at work on the pairing model that runs through this book: twelve doubly degenerate single-particle levels, six pairs, single-particle spacing $\delta = 1$ and pairing strength $g = 0.5$. In the seniority-zero space a basis state is a choice of which levels carry a pair, so the dimension is $\binom{12}{6} = 924$. The ground-state energy is converged to ten digits after thirty matrix-vector products and to machine precision after forty, in a space of dimension 924.

m	**Lowest Ritz value**	**Error**
5	32.865169933329	8.0×10^{0}
10	25.140140936891	3.0×10^{-1}
20	24.839267115133	9.4×10^{-5}
30	24.839172748689	2.4×10^{-10}
40	24.839172748451	5.7×10^{-14}
exact	24.839172748451	

Table 1.3 Convergence of the lowest Ritz value for the pairing model with twelve levels and six pairs, $\delta = 1$, $g = 0.5$, in a space of dimension 924. The exact value is obtained by full diagonalisation.

Run without reorthogonalisation for 120 steps on the same matrix, the Lanczos vectors lose orthogonality completely, $\|\boldsymbol{Q}^T\boldsymbol{Q} - \boldsymbol{I}\| = 4.9$, and three copies of the non-degenerate ground state appear among the Ritz values. With reorthogonalisation the deviation is 8.3×10^{-15} and the ground state appears exactly once.

Many-body connection. The Lanczos algorithm is the engine of large-scale shell-model and full configuration-interaction codes. Its combination of properties – only matrix-vector products are needed, the extremal eigenvalues converge first, and the lowest Ritz value is a variational upper bound – matches the structure of the many-body problem exactly: we want the ground state, the Hamiltonian is sparse, and the dimension is astronomical. The same Krylov idea reappears in quantum computing, where the quantum phase estimation and variational quantum eigensolver algorithms are, in different ways, attempts to extract the same information with quantum rather than classical resources.

1.27 Application: Schrödinger's equation through diagonalisation

We close the chapter by putting the machinery to work on a genuine quantum mechanical problem, and one that will return in a many-body guise later: solving Schrödinger's equation not as a differential equation, but as a matrix eigenvalue problem.

The radial equation for a particle in a central potential is

$$-\frac{\hbar^2}{2m}\left(\frac{1}{r^2}\frac{d}{dr}r^2\frac{d}{dr}-\frac{l(l+1)}{r^2}\right)R(r)+V(r)R(r)=ER(r). \tag{1.77}$$

Substituting $R(r)=u(r)/r$ removes the first-derivative term,

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2}+\left(V(r)+\frac{l(l+1)}{r^2}\frac{\hbar^2}{2m}\right)u(r)=Eu(r), \tag{1.78}$$

and introducing the dimensionless variable $\rho=r/\alpha$ with α a length makes the equation dimensionless. We specialise to one dimension, drop the centrifugal term, and take the harmonic oscillator potential $V(x)=\frac{1}{2}kx^2$. In units where $k=\hbar=m=\alpha=1$ the equation becomes

$$-\frac{d^2u}{dx^2}+x^2u(x)=2E\,u(x). \tag{1.79}$$

The factor two on the right is worth remembering: the program will return the values $2E_n=2n+1$, that is $1,3,5,7,\dots$, rather than $E_n=n+\frac{1}{2}$.

We now repeat the discretisation of Section 1.14. Choose $R_{\min}$ and $R_{\max}$, a number of mesh points N_{step}, and the step $h=(R_{\max}-R_{\min})/N_{\text{step}}$. With $x_i=R_{\min}+ih$ and the three-point formula (1.14), Eq. (1.79) becomes

$$-\frac{u_{i+1}-2u_i+u_{i-1}}{h^2}+V_iu_i=2E\,u_i, \tag{1.80}$$

that is the tridiagonal eigenvalue problem

$$\begin{bmatrix} \frac{2}{h^2}+V_1 & -\frac{1}{h^2} & & \\ -\frac{1}{h^2} & \frac{2}{h^2}+V_2 & -\frac{1}{h^2} & \\ & \ddots & \ddots & \ddots \\ & & -\frac{1}{h^2} & \frac{2}{h^2}+V_{N_{\text{step}}-1} \end{bmatrix}\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N_{\text{step}}-1}\end{bmatrix}=2E\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_{N_{\text{step}}-1}\end{bmatrix}. \tag{1.81}$$

The boundary conditions $u(R_{\min})=u(R_{\max})=0$ are imposed simply by leaving the two endpoints out of the matrix, whose dimension is therefore $N_{\text{step}}-1$. The eigenvectors are the wave functions on the grid, normalised with the trapezoidal rule,

$$\int_{R_{\min}}^{R_{\max}}|u(x)|^2dx \longrightarrow h\sum_i u_i^2=1. \tag{1.82}$$

Since the matrix is already tridiagonal, the Householder step is unnecessary and we go straight to `tqli`. Alternatively, and this is the point of the exercise, we may pretend that the matrix is far too large to store and use the Lanczos algorithm with a matrix-free operator.

```
class SchrodingerDiagonalization:
    """Discretise -u'' + V(x) u = 2E u and diagonalise the result."""

    def __init__(self, potential, rmin=-10.0, rmax=10.0, nsteps=400):
        self.potential = potential
        self.h = (rmax - rmin) / nsteps
        # interior mesh points only; u vanishes at the two endpoints
        self.x = rmin + self.h * np.arange(1, nsteps)
        self.n = len(self.x)
```

```
    @property
    def diagonal(self):
        return 2.0 / self.h**2 + self.potential(self.x)

    @property
    def offdiagonal(self):
        return np.full(self.n - 1, -1.0 / self.h**2)

    def matvec(self, v):
        """Apply the Hamiltonian without ever forming the matrix."""
        w = self.diagonal * v
        w[:-1] -= v[1:] / self.h**2
        w[1:] -= v[:-1] / self.h**2
        return w

    def solve(self, k=5):
        """The k lowest eigenvalues and normalised eigenfunctions."""
        values, vectors = SymmetricEigenSolver.tqli(self.diagonal,
                                                     self.offdiagonal)
        order = np.argsort(values)
        return (values[order][:k],
                self._normalise(vectors[:, order][:, :k]))

    def solve_lanczos(self, k=5, m=None):
        """The same eigenvalues from the Lanczos algorithm."""
        lan = Lanczos(matvec=self.matvec, n=self.n)
        values, vectors = lan.ritz(10*k if m is None else m, k=k)
        return values, self._normalise(vectors)
```

Table 1.4 shows the result on $x \in [-10, 10]$. The eigenvalues converge to the exact values $1, 3, 5, \dots$ as h^2: doubling N_{step} divides the error by four, exactly as the truncation error of the three-point formula demands. Running the same problem through the Lanczos algorithm with $m = 200$ steps reproduces the three lowest eigenvalues to 6×10^{-13}, and the computed wave functions agree with the analytic $u_0(x) = \pi^{-1/4} e^{-x^2/2}$ to 7×10^{-5}.

N_{step}	$2E_0$	$2E_1$	$2E_2$	**Max error**
100	0.9974937026	2.9874430342	4.9672772610	1.0×10^{-1}
200	0.9993746086	2.9968714733	4.9918612670	2.6×10^{-2}
400	0.9998437256	2.9992185301	4.9979678946	6.4×10^{-3}
800	0.9999609360	2.9998046738	4.9994921341	1.6×10^{-3}
exact	1	3	5	

Table 1.4 The one-dimensional harmonic oscillator obtained by diagonalising Eq. (1.81) on $x \in [-10, 10]$. The error decreases as h^2.

Many-body connection. This calculation is the one-particle ancestor of everything that follows. Expanding a state in a finite basis and diagonalising the resulting matrix is exactly what full configuration interaction does, the only difference being that the basis consists of Slater determinants rather than grid points, and that the dimension grows exponentially with the number of particles rather than linearly with the number of mesh points. The eigenvectors computed here are the natural single-particle basis $\{|\phi_i\rangle\}$ of Section 1.8 from which those Slater determinants are built. Chapter 2 takes the first step in that direction.

1.28 The singular value decomposition

Every algorithm so far has assumed a square matrix, and most of them a symmetric one. Many of the objects we meet in many-body physics are neither. The coefficients of a state written across a partition of the system into two parts form a rectangular array; a set of sampled wave functions collected as columns forms a rectangular array; the two-body matrix elements arranged in a composite pair index form a matrix whose row and column dimensions have nothing to do with the dimension of the Hilbert space. For all of these we need a decomposition that does not require squareness, and preferably one that does not require diagonalisability either.

Recall that a square matrix can be diagonalised if and only if it is *normal*, that is if $\boldsymbol{X}\boldsymbol{X}^T = \boldsymbol{X}^T\boldsymbol{X}$ in the real case or $\boldsymbol{X}\boldsymbol{X}^\dagger = \boldsymbol{X}^\dagger\boldsymbol{X}$ in the complex one. Not every matrix satisfies this. The simplest counterexample is

$$\boldsymbol{X} = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, \tag{1.83}$$

whose determinant vanishes and for which $\boldsymbol{X}\boldsymbol{X}^T \neq \boldsymbol{X}^T\boldsymbol{X}$. It is *defective*: it cannot be diagonalised, and it has no inverse.

The singular value decomposition has no such restriction.

The theorem.

Any $m \times n$ matrix $\boldsymbol{X}$, real or complex, square or rectangular, can be written

$$\boldsymbol{X} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T, \tag{1.84}$$

where $\boldsymbol{U}$ is an $m \times m$ orthogonal matrix, $\boldsymbol{U}\boldsymbol{U}^T = \boldsymbol{U}^T\boldsymbol{U} = \boldsymbol{I}_m$, where $\boldsymbol{V}$ is an $n \times n$ orthogonal matrix, $\boldsymbol{V}\boldsymbol{V}^T = \boldsymbol{V}^T\boldsymbol{V} = \boldsymbol{I}_n$, and where $\boldsymbol{\Sigma}$ is an $m \times n$ matrix which is zero except on the main diagonal, on which sit the *singular values*

$$\sigma_0 \geq \sigma_1 \geq \cdots \geq \sigma_{r-1} \geq 0, \qquad r = \min(m,n). \tag{1.85}$$

The columns of $\boldsymbol{U}$ are the *left* singular vectors and the columns of $\boldsymbol{V}$ the *right* singular vectors. The decomposition always exists. For the defective matrix of Eq. (1.83) it reads

$$\boldsymbol{X} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T, \tag{1.86}$$

with $\sigma_0 = 2$ and $\sigma_1 = 0$. A matrix that cannot be diagonalised at all is decomposed without difficulty, and the vanishing singular value states plainly what the determinant only hinted at.

Geometry.

Equation (1.84) says that every linear map is a rotation, followed by a scaling along the new axes, followed by another rotation. The singular values are the scale factors, and $\|\boldsymbol{X}\|_2 = \sigma_0$, which is the statement of Section 1.13 that the induced 2-norm measures the largest stretching a matrix can produce.

Relation to the eigenvalue problem.

The SVD is not an independent construction; it is the symmetric eigenvalue problem of Section 1.21 in disguise. Using Eq. (1.84) and the orthogonality of $\boldsymbol{U}$,

$$\boldsymbol{X}^T\boldsymbol{X} = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{U}^T\boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{V}\boldsymbol{\Sigma}^T\boldsymbol{\Sigma}\boldsymbol{V}^T \equiv \boldsymbol{V}\tilde{\boldsymbol{\Sigma}}^2\boldsymbol{V}^T, \tag{1.87}$$

where $\tilde{\boldsymbol{\Sigma}}$ is the $n \times n$ diagonal matrix of singular values. Multiplying from the right by $\boldsymbol{V}$ gives

$$\left(\boldsymbol{X}^T \boldsymbol{X}\right) \boldsymbol{v}_i = \sigma_i^2 \boldsymbol{v}_i, \tag{1.88}$$

so the right singular vectors are the eigenvectors of $\boldsymbol{X}^T \boldsymbol{X}$ and the squared singular values its eigenvalues. In exactly the same way

$$\boldsymbol{X}\boldsymbol{X}^T = \boldsymbol{U}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^T\boldsymbol{U}^T, \qquad \left(\boldsymbol{X}\boldsymbol{X}^T\right) \boldsymbol{u}_i = \sigma_i^2 \boldsymbol{u}_i, \tag{1.89}$$

with the remaining $m-r$ eigenvalues equal to zero. Every singular value is therefore the non-negative square root of an eigenvalue of $\boldsymbol{X}^T\boldsymbol{X}$, and if $\boldsymbol{X}$ is itself symmetric the singular values are the absolute values of its eigenvalues. These two relations will do a great deal of work in the sections that follow: $\boldsymbol{X}\boldsymbol{X}^T$ and $\boldsymbol{X}^T\boldsymbol{X}$ will turn out to be the two reduced density matrices of a bipartite quantum state.

The economy-size decomposition.

If $m > n$ the last $m-n$ columns of $\boldsymbol{U}$ are multiplied by zeros in $\boldsymbol{\Sigma}$ and can never contribute. Removing them, and the corresponding rows of zeros in $\boldsymbol{\Sigma}$, gives the economy-size decomposition, in which $\boldsymbol{U}$ is $m \times n$ and $\boldsymbol{\Sigma}$ is $n \times n$. If $n > m$ one keeps instead the first m columns of $\boldsymbol{V}$. Nothing is lost and both storage and work are reduced; this is the default in most library routines.

How not to compute it.

Equation (1.88) suggests an algorithm: form $\boldsymbol{X}^T\boldsymbol{X}$, diagonalise it with the Householder and QL machinery of Sections 1.24 and 1.25, take square roots. This is mathematically correct and numerically poor. From Eq. (1.11) the condition number of the product is the square of that of $\boldsymbol{X}$,

$$\kappa_2(\boldsymbol{X}^T\boldsymbol{X}) = \kappa_2(\boldsymbol{X})^2, \tag{1.90}$$

so half of the available significant digits are destroyed before the eigenvalue solver is even called. Table 1.5 shows the damage for a 60×12 matrix constructed with singular values spread logarithmically between 1 and 10^{-7}: the direct SVD returns all twelve to machine precision, the detour through $\boldsymbol{X}^T\boldsymbol{X}$ loses seven digits on the smallest.

Exact σ_i	**From the SVD**	**From $\boldsymbol{X}^T\boldsymbol{X}$**	**Relative error**
1.00000000×10^{0}	1.00000000×10^{0}	1.00000000×10^{0}	4.4×10^{-16}
$6.57933225 \times 10^{-4}$	$6.57933225 \times 10^{-4}$	$6.57933225 \times 10^{-4}$	2.0×10^{-11}
$1.87381742 \times 10^{-6}$	$1.87381742 \times 10^{-6}$	$1.87381889 \times 10^{-6}$	7.8×10^{-7}
$4.32876128 \times 10^{-7}$	$4.32876128 \times 10^{-7}$	$4.32893930 \times 10^{-7}$	4.1×10^{-5}
$1.00000000 \times 10^{-7}$	$1.00000000 \times 10^{-7}$	$9.99683051 \times 10^{-8}$	3.2×10^{-4}

Table 1.5 Singular values of a 60×12 matrix with $\kappa_2(\boldsymbol{X}) = 10^7$, computed directly and through the eigenvalues of $\boldsymbol{X}^T\boldsymbol{X}$, for which $\kappa_2 = 10^{14}$. Forming the product costs roughly half the significant digits.

The stable algorithm never forms the product. It reduces $\boldsymbol{X}$ to bidiagonal form by Householder reflectors applied alternately from the left and the right – the same reflectors as in Section 1.24 – and then runs an implicitly shifted QR sweep on the bidiagonal matrix. This is the Golub-Kahan algorithm, and it is what `numpy.linalg.svd` calls through LAPACK. We use the library routine in what follows: the interest here lies in what the decomposition is used for.

Many-body connection. Equations (1.88) and (1.89) are the reason the SVD keeps reappearing in many-body physics. Whenever a quantity is built as a product of a matrix with its own transpose – a density matrix, an overlap matrix, a two-body interaction written in a pair index – the SVD of the underlying rectangular object gives its spectrum directly, with better accuracy and often at lower cost than forming the product at all.

1.29 Rank, the pseudoinverse and ill-conditioned bases

The *rank* of a matrix is the number of linearly independent columns, and the SVD reads it off immediately: it is the number of non-zero singular values. In floating-point arithmetic one asks instead for the *numerical rank*, the number of singular values above a threshold, typically $\max(m,n)\,\varepsilon\,\sigma_0$ with ε the machine precision. A matrix is rank deficient when some columns are linear combinations of others, and near rank deficient – which is the practically dangerous case – when they nearly are.

The condition number of Section 1.13 is likewise a statement about singular values,

$$\kappa_2(\boldsymbol{X}) = \frac{\sigma_0}{\sigma_{r-1}}, \tag{1.91}$$

which is infinite exactly when $\boldsymbol{X}$ is singular. The SVD thus not only detects near-singularity, it quantifies it.

The Moore-Penrose pseudoinverse.

When $\boldsymbol{X}^{-1}$ does not exist, or exists but is useless because $\boldsymbol{X}$ is nearly singular, the SVD provides the natural substitute

$$\boldsymbol{X}^{+} = \boldsymbol{V}\boldsymbol{\Sigma}^{+}\boldsymbol{U}^{T}, \tag{1.92}$$

where $\boldsymbol{\Sigma}^{+}$ is obtained from $\boldsymbol{\Sigma}$ by transposing it and replacing every non-zero singular value by its reciprocal, *leaving the zeros as zeros*. In practice one treats as zero every singular value below some $\varepsilon_{\text{cut}}\sigma_0$. If $\boldsymbol{X}$ is invertible and well conditioned, $\boldsymbol{X}^{+} = \boldsymbol{X}^{-1}$; otherwise $\boldsymbol{X}^{+}\boldsymbol{b}$ is the least-squares solution of smallest norm. Inverting a singular value of 10^{-13} would amplify the rounding noise in the corresponding direction by 10^{13}; discarding it simply declares that the problem contains no information about that direction, which is the truth.

Near-linearly-dependent basis sets.

This is not an abstract concern. Many-body calculations are routinely carried out in non-orthogonal single-particle bases – Gaussians in quantum chemistry, harmonic-oscillator states in different oscillator lengths, generator coordinates in nuclear physics – and such bases become linearly dependent long before one would like them to. The eigenvalue problem is then the generalised one,

$$\boldsymbol{H}\boldsymbol{c} = E\,\boldsymbol{S}\boldsymbol{c}, \qquad S_{ij} = \langle \phi_i | \phi_j \rangle, \tag{1.93}$$

with $\boldsymbol{S}$ the overlap matrix. Diagonalising $\boldsymbol{S} = \boldsymbol{V}\boldsymbol{s}\boldsymbol{V}^{T}$ and forming

$$\boldsymbol{X} = \boldsymbol{V}\boldsymbol{s}^{-1/2} \tag{1.94}$$

turns Eq. (1.93) into the ordinary problem $(\boldsymbol{X}^{T}\boldsymbol{H}\boldsymbol{X})\boldsymbol{c}' = E\boldsymbol{c}'$. The transformation involves $s_i^{-1/2}$, so a tiny overlap eigenvalue is precisely where the procedure breaks down. *Canonical orthogonalisation* is the fix: drop the eigenvectors whose eigenvalues lie below a threshold before forming Eq. (1.94).

Table 1.6 shows this for twelve Gaussians of width 1.4 placed on a uniform grid, a deliberately overcomplete basis of the kind used in quantum chemistry. The overlap matrix has eigenvalues spanning thirteen orders of magnitude, from 6.9 down to 6.5×10^{-13}, and a condition number of 1.1×10^{13}: in double precision only three significant digits would survive. Discarding three basis directions removes six orders of magnitude of ill-conditioning at negligible cost in completeness.

Threshold	**Functions kept**	**Effective $\kappa_2(\boldsymbol{S})$**
none	12	1.1×10^{13}
10^{-12}	11	6.9×10^{10}
10^{-8}	9	2.1×10^{7}
10^{-5}	7	3.3×10^{4}

Table 1.6 Canonical orthogonalisation of a basis of twelve overlapping Gaussians. Eigenvectors of the overlap matrix with eigenvalue below the threshold, relative to the largest, are discarded.

```
class CanonicalOrthogonalization:
    """Orthogonalise a non-orthogonal basis with a near-singular overlap.

    The generalised problem H c = E S c is reduced to an ordinary one by
    X = V s^{-1/2}, keeping only the eigenvalues of S above a threshold.
    """

    def __init__(self, S, threshold=1.0e-8):
        self.S = np.asarray(S, dtype=float)
        w, V = np.linalg.eigh(self.S)
        order = np.argsort(w)[::-1]
        self.eigenvalues = w[order]
        self.eigenvectors = V[:, order]
        keep = self.eigenvalues > threshold * self.eigenvalues[0]
        self.kept = int(np.sum(keep))
        self.X = self.eigenvectors[:, keep] / np.sqrt(self.eigenvalues[keep])

    def solve_generalized(self, H):
        """Solve H c = E S c through the orthogonalised problem."""
        values, vectors = np.linalg.eigh(self.X.T @ H @ self.X)
        return values, self.X @ vectors
```

Many-body connection. The threshold is a physical choice, not a numerical detail. Setting it too low keeps directions that carry only rounding noise and poisons the whole calculation; setting it too high throws away genuine variational freedom and raises the computed energy. The same trade-off appears in the generator-coordinate method, where the overlap kernel of non-orthogonal mean-field states is routinely rank deficient, and in explicitly correlated Gaussian expansions, where it is the principal practical obstacle to enlarging the basis.

1.30 Low-rank approximation and the Schmidt decomposition

We now come to the property that makes the SVD indispensable in many-body physics. Truncating the sum

$$\boldsymbol{X} = \sum_{k=0}^{r-1} \sigma_k \boldsymbol{u}_k \boldsymbol{v}_k^T \tag{1.95}$$

after χ terms gives a matrix $\boldsymbol{X}_\chi$ of rank χ. The Eckart-Young theorem states that this is the *best* rank-χ approximation to $\boldsymbol{X}$, in both the 2-norm and the Frobenius norm, and that the error it makes is

$$\|\boldsymbol{X}-\boldsymbol{X}_\chi\|_2 = \sigma_\chi, \qquad \|\boldsymbol{X}-\boldsymbol{X}_\chi\|_F^2 = \sum_{k\geq\chi}\sigma_k^2. \tag{1.96}$$

No cleverer choice of χ vectors exists. If the singular values decay quickly, a matrix with mn entries is captured by $\chi(m+n+1)$ numbers.

The Schmidt decomposition.

Consider a quantum system divided into two parts, A and B, with orthonormal bases $\{|i\rangle_A\}$ and $\{|j\rangle_B\}$. Any state of the whole system can be written

$$|\Psi\rangle = \sum_{ij} C_{ij}\,|i\rangle_A \otimes |j\rangle_B, \tag{1.97}$$

and the coefficients form a matrix $\boldsymbol{C}$ which is rectangular whenever the two parts have different dimensions. Inserting its SVD $\boldsymbol{C}=\boldsymbol{U\Sigma V}^T$ and absorbing $\boldsymbol{U}$ and $\boldsymbol{V}$ into new basis states $|u_k\rangle_A = \sum_i U_{ik}|i\rangle_A$ and $|v_k\rangle_B = \sum_j V_{jk}|j\rangle_B$, which are orthonormal because $\boldsymbol{U}$ and $\boldsymbol{V}$ are orthogonal, gives

$$\boxed{|\Psi\rangle = \sum_{k=0}^{\chi-1}\sigma_k\,|u_k\rangle_A\otimes|v_k\rangle_B} \tag{1.98}$$

This is the *Schmidt decomposition*. A double sum over the product basis has collapsed to a single sum, and the SVD has done all the work. The singular values σ_k are the Schmidt coefficients, and normalisation $\langle\Psi|\Psi\rangle = 1$ gives $\sum_k \sigma_k^2 = 1$.

The reduced density matrices follow at once. Tracing out B,

$$\boldsymbol{\rho}_A = \mathrm{Tr}_B\,|\Psi\rangle\langle\Psi| = \boldsymbol{CC}^T = \boldsymbol{U\Sigma}^2\boldsymbol{U}^T, \qquad \boldsymbol{\rho}_B = \boldsymbol{C}^T\boldsymbol{C} = \boldsymbol{V\Sigma}^2\boldsymbol{V}^T, \tag{1.99}$$

which is Eqs. (1.88) and (1.89) read as physics. The two reduced density matrices have the *same* non-zero eigenvalues σ_k^2, however different the sizes of the two parts, and their eigenvectors are the Schmidt states. The number of non-zero σ_k is the *Schmidt rank*: it equals one precisely when $|\Psi\rangle$ is a product state, so a single singular value is the signature of an unentangled state. The *entanglement entropy*

$$S_A = -\mathrm{Tr}\,\boldsymbol{\rho}_A\ln\boldsymbol{\rho}_A = -\sum_k \sigma_k^2\ln\sigma_k^2 \tag{1.100}$$

measures how far from a product the state is, and it is manifestly symmetric between A and B.

Truncation.

Keeping only the χ largest Schmidt coefficients gives, by Eq. (1.96), the best possible approximation to $|\Psi\rangle$ of that rank, with discarded weight

$$\varepsilon_\chi = \sum_{k\geq\chi}\sigma_k^2. \tag{1.101}$$

This single quantity is the truncation error of the density-matrix renormalisation group, and χ is its bond dimension. Whether a many-body state can be compressed is therefore a question about the decay of the Schmidt spectrum, and that decay is what area laws for the entanglement entropy assert: for ground states of local, gapped Hamiltonians the entropy across a cut grows with the *boundary* of the cut rather than with the volume of the subsystem, so a modest χ suffices no matter how large the system. Matrix product states are the systematic version of this observation, obtained by applying Eq. (1.98) at every bond of a one-dimensional lattice in turn.

The pairing model across a cut.

Take the ground state of the pairing model of Section 1.26 and split the twelve single-particle levels into the lowest six and the highest six. A seniority-zero basis state is a set of occupied levels, which factorises into its part in A and its part in B, so the coefficients form a 64×64 matrix. Its singular values are the Schmidt coefficients across the cut. Table 1.7 gives the discarded weight as a function of χ: three Schmidt states out of 64 already capture all but 5×10^{-3} of the state, and ten capture all but 8×10^{-7}. The entanglement entropy is $S_A = 0.9238$, well below the maximum $\ln 64 = 4.16$ that an arbitrary state of this size could carry.

Schmidt states kept, χ	**Discarded weight ε_χ**
1	4.831×10^{-1}
2	7.461×10^{-2}
3	5.448×10^{-3}
5	1.854×10^{-4}
7	7.363×10^{-6}
10	7.530×10^{-7}
15	4.225×10^{-9}

Table 1.7 Schmidt spectrum of the pairing-model ground state (12 levels, 6 pairs, $\delta = 1$, $g = 0.5$) across the cut separating the lowest six levels from the highest six. The coefficient matrix is 64×64 and the entanglement entropy is 0.9238.

```
class SchmidtDecomposition:
    """The SVD of a bipartitioned state vector.

    For |Psi> = sum_ij C_ij |i>_A |j>_B the SVD C = U Sigma V^T gives
    |Psi> = sum_k sigma_k |u_k>_A |v_k>_B.  The eigenvalues of both reduced
    density matrices are sigma_k^2.
    """

    def __init__(self, C):
        self.C = np.asarray(C, dtype=float)
        self.U, self.sigma, self.Vt = np.linalg.svd(C, full_matrices=False)
        self.weights = self.sigma**2         # normalised: sum_k sigma_k^2 = 1

    def schmidt_rank(self, tol=1.0e-10):
        return int(np.sum(self.sigma > tol * self.sigma[0]))

    def entanglement_entropy(self, base=np.e):
        w = self.weights[self.weights > 1.0e-16]
        entropy = -np.sum(w * np.log(w)) / np.log(base)
        return float(entropy) if entropy > 0.0 else 0.0

    def truncation_error(self, k):
        """Weight discarded when only k Schmidt states are kept."""
        return float(np.sum(self.weights[k:]))

    def truncate(self, k):
        """The best rank-k approximation to the state, renormalised."""
        C = (self.U[:, :k] * self.sigma[:k]) @ self.Vt[:k]
```

```python
    return C / np.linalg.norm(C)
```

Many-body connection. The Schmidt decomposition is the bridge between linear algebra and the modern theory of many-body wave functions. The density-matrix renormalisation group truncates the Schmidt spectrum at every bond; matrix product states, projected entangled pair states and the whole family of tensor networks are built by iterating Eq. (1.98); and the entanglement entropy, computed from nothing more than a list of singular values, is what tells us in advance whether such a compression can work at all.

Quantum computing connection. The same numbers govern how hard a state is to simulate classically. A state of N qubits whose Schmidt rank across every cut is bounded by χ can be stored in $\mathscr{O}(N\chi^2)$ numbers and evolved classically; only states whose Schmidt rank grows exponentially require a quantum computer. Entanglement, quantified by Eq. (1.100), is thus simultaneously the resource that quantum algorithms exploit and the obstacle that classical ones must overcome.

1.31 Natural orbitals and occupation numbers

For two identical particles the Schmidt decomposition has a name of its own. Write the two-particle wave function in a single-particle basis $\{\varphi_p\}$,

$$\Psi(x_1,x_2) = \sum_{pq} C_{pq}\,\varphi_p(x_1)\varphi_q(x_2), \tag{1.102}$$

where symmetry under exchange makes $\boldsymbol{C}$ symmetric (or antisymmetric). The SVD of $\boldsymbol{C}$, which for a symmetric matrix is its eigen-decomposition up to signs, gives the *natural expansion*

$$\Psi(x_1,x_2) = \sum_{k} \sigma_k\,\chi_k(x_1)\chi_k(x_2), \tag{1.103}$$

a single sum in terms of the transformed orbitals $\chi_k = \sum_p U_{pk}\varphi_p$. These are Löwdin's *natural orbitals*, and they are the eigenvectors of the one-body density matrix

$$\rho_{pq} = \langle\Psi|a_p^\dagger a_q|\Psi\rangle, \qquad \boldsymbol{\rho} \propto \boldsymbol{C}\boldsymbol{C}^\dagger = \boldsymbol{U}\boldsymbol{\Sigma}^2\boldsymbol{U}^\dagger, \tag{1.104}$$

with eigenvalues $n_k \propto \sigma_k^2$: the *occupation numbers*. For a single Slater determinant exactly N occupation numbers equal one and the rest vanish; correlation depletes the occupied orbitals and populates the others, and the deviation of the largest n_k from unity is a direct measure of how far the state is from a mean-field one.

Table 1.8 illustrates this for two particles in a one-dimensional harmonic trap interacting through a softened Coulomb potential of strength g, solved on a grid of 40 points by full diagonalisation of the 1600-dimensional two-particle problem. At $g = 0$ the ground state is a product, there is a single natural orbital with $n_0 = 1$, and the entanglement entropy vanishes exactly. Switching on the interaction depletes n_0 and populates n_1, n_2 and so on, and the entropy rises.

Many-body connection. Natural orbitals are the optimal single-particle basis in a precise sense: for a given number of retained orbitals, no other choice gives a better representation of the state, which is Eq. (1.96) again. This is why natural-orbital truncation is the standard way of reducing a correlated basis in quantum chemistry and in nuclear structure, and why occupation numbers are reported as a diagnostic: a system whose largest occupation number is 0.99 is well described by a single determinant and Hartree-Fock theory of Chapter 2 will be a good starting point, while one whose occupation numbers are spread over several orbitals is strongly correlated and demands a multireference treatment.

g	E_0	S_A	n_0	n_1	χ for 10^{-8}
0	1.992536	0.000000	1.000000	0.000000	1
1	3.173866	0.040874	0.993528	0.006017	10
2	4.245117	0.139618	0.970732	0.027407	11
5	6.755062	0.507970	0.821806	0.168066	11

Table 1.8 Natural occupation numbers and entanglement entropy for two particles in a harmonic trap with a softened Coulomb repulsion of strength g, on a grid of 40 points. The last column gives the number of natural orbitals needed to recover all but 10^{-8} of the norm.

1.32 Low-rank factorisation of the two-body interaction

The last application concerns not the wave function but the Hamiltonian. The two-body matrix elements V_{pqrs} form a four-index array with n^4 entries, where n is the number of single-particle states, and for n of a few hundred this is already the dominant storage cost of a many-body calculation. Reading the pairs (pq) and (rs) as composite indices turns the array into an ordinary $n^2 \times n^2$ matrix,

$$V_{(pq),(rs)} = \int\int dxdy\, \varphi_p(x)\varphi_q(x)\, K(x,y)\, \varphi_r(y)\varphi_s(y), \tag{1.105}$$

and for a Coulomb-like kernel K this matrix is symmetric and positive semidefinite. Its eigen-decomposition, equivalently its Cholesky factorisation of Section 1.15, gives

$$V_{(pq),(rs)} = \sum_{\gamma=0}^{M-1} L^{\gamma}_{pq} L^{\gamma}_{rs}, \qquad L^{\gamma}_{pq} = \sqrt{\lambda_\gamma}\, U_{(pq),\gamma}. \tag{1.106}$$

The decisive fact is that the eigenvalues λ_γ decay rapidly, because the pair densities $\varphi_p\varphi_q$ are far from linearly independent as functions. Keeping only $M \ll n^2$ of them reduces the storage from n^4 to Mn^2 and turns the contraction of the interaction with a wave function into a sequence of matrix products.

Table 1.9 shows the decay for the twelve-Gaussian basis of Section 1.29, for which there are 144 pair indices. Nine vectors reproduce the interaction to one part in 10^6, and ten to one part in 10^8: a reduction of the rank by more than a factor of ten.

Target weight	**Vectors** M	$\|\boldsymbol{V} - \boldsymbol{L}^T\boldsymbol{L}\|/\|\boldsymbol{V}\|$
10^{-4}	7	2.1×10^{-5}
10^{-6}	9	1.7×10^{-7}
10^{-8}	10	1.2×10^{-8}

Table 1.9 Low-rank factorisation of the two-body matrix elements in a basis of twelve Gaussians, for which the pair index runs over 144 values. The leading eigenvalues of the pair matrix are 44.7, 15.6, 5.18, 1.38, 0.309, 0.055, …, a geometric decay.

```
class TwoBodyFactorization:
    """Low-rank factorisation of a two-body interaction.

    The pair-index matrix V_{(pq),(rs)} is positive semidefinite for a
    Coulomb-like interaction, so it factorises as a sum of outer products,
    V = sum_gamma L^gamma (L^gamma)^T, with few terms.
    """

    def __init__(self, V):
```

```python
        self.V = np.asarray(V, dtype=float)
        w, U = np.linalg.eigh(0.5 * (self.V + self.V.T))
        order = np.argsort(w)[::-1]
        self.eigenvalues = w[order]
        self.vectors = U[:, order]

    def rank_for(self, tol):
        """Smallest M such that the discarded weight is below tol."""
        w = np.maximum(self.eigenvalues, 0.0)
        tail = np.cumsum(w[::-1])[::-1]
        below = np.where(tail <= tol * np.sum(w))[0]
        return int(below[0]) if len(below) else len(w)

    def cholesky_vectors(self, M):
        """The M leading vectors L^gamma."""
        w = np.maximum(self.eigenvalues[:M], 0.0)
        return (self.vectors[:, :M] * np.sqrt(w)).T
```

Many-body connection. Equation (1.106) is known as the Cholesky decomposition of the two-electron integrals, or, when the auxiliary vectors are chosen from a fitted basis rather than computed, as density fitting or the resolution of the identity. It is what makes large-basis coupled-cluster and many-body perturbation theory calculations feasible, since the rate-determining contractions can be performed on the factors L^γ rather than on the full four-index array.

Quantum computing connection. Applying the same idea twice gives the *double factorisation*: each matrix L^γ_{pq} is itself diagonalised, $\boldsymbol{L}^\gamma = \boldsymbol{W}^\gamma \boldsymbol{d}^\gamma (\boldsymbol{W}^\gamma)^T$, so that the Hamiltonian becomes a short sum of squares of one-body operators. Since a one-body operator maps under the Jordan-Wigner transformation of Section 1.7 onto a small number of commuting Pauli strings, this reduces the number of terms that must be measured in a variational quantum eigensolver, and the cost of block encoding the Hamiltonian for qubitisation, from $\mathscr{O}(n^4)$ to $\mathscr{O}(n^3)$ or better. The same singular values control the truncation in both the classical and the quantum setting.

1.33 Summary and the programs

The chapter has moved from notation to algorithms. The linear algebra of Sections 1.1-1.11 supplies the language in which quantum mechanics is written: states are vectors, observables are Hermitian matrices, symmetries and basis changes are unitary transformations, and the physical content of an operator is its spectral decomposition. The numerical sections supply the means of extracting numbers from that language.

Three dividing lines are worth keeping in mind. The first separates direct from iterative methods. Direct methods – Gaussian elimination, LU, Cholesky, Householder with QL – deliver the complete answer in a fixed $\mathscr{O}(n^3)$ number of operations, and require the matrix to be stored. Iterative methods – Jacobi, Gauss-Seidel, over-relaxation, conjugate gradient, Lanczos – deliver an approximation that improves with each step, and require only the action of the matrix on a vector. The second line separates dense from structured problems: a tridiagonal system costs $\mathscr{O}(n)$ rather than $\mathscr{O}(n^3)$, and recognising structure is usually worth more than any amount of tuning.

The third line is the one drawn by the singular value decomposition, and it is of a different kind. It separates the part of a problem that carries information from the part that does not. The same list of singular values tells us the numerical rank of a matrix, how far a basis is from linear dependence, how much of a wave function survives a truncation, how entangled a state is across a cut, which single-particle orbitals are actually occupied, and how few auxiliary vectors are needed to represent a two-body interaction. Sections 1.28-1.32 are five readings of one decomposition, and the thread connecting them – Eq. (1.96), that truncating the singular values is the best approximation of its rank – is the mathematical justification for a large part of modern many-body theory, from natural-orbital bases to the density-matrix renormalisation group.

The complete programs are collected in `doc/BookManybody/BookMaterial/Programs`:

- `direct_solvers.py` – Gaussian elimination, LU, Cholesky and the tridiagonal solver, with determinants and inverses.
- `iterative_solvers.py` – Jacobi, Gauss-Seidel, over-relaxation and the conjugate gradient method, in both a matrix-based and a matrix-free form.
- `householder.py` – Householder tridiagonalisation, the QL algorithm with implicit shifts, and the power method.
- `lanczos.py` – the Lanczos algorithm with optional full reorthogonalisation, together with the pairing-model Hamiltonian used in Table 1.3.
- `schrodinger_diagonalization.py` – Schrödinger's equation by diagonalisation, solved both directly and by Lanczos.
- `svd.py` – the singular value decomposition: rank, the pseudoinverse, canonical orthogonalisation, the Schmidt decomposition with entanglement entropies and natural occupation numbers, and the low-rank factorisation of a two-body interaction. It also contains the two physical models used in Sections 1.30-1.32: two particles in a trap, and the pairing model bipartitioned across a cut.

Each file runs as a script and reproduces the numbers quoted in this chapter. An executable version of the same material, in which the reader can vary the parameters, is available as a Jupyter notebook in the accompanying Jupyter-book.

1.34 Exercises

Warm-up exercises

1. **Inner and outer products.** Let $\boldsymbol{u} = \begin{bmatrix}1-i \ 2+i\end{bmatrix}^T$ and $\boldsymbol{v} = \begin{bmatrix}1 \ i\end{bmatrix}^T$. (a) Compute $\boldsymbol{u}^\dagger\boldsymbol{v}$ and $\boldsymbol{v}^\dagger\boldsymbol{u}$. (b) Verify $(\boldsymbol{u}^\dagger\boldsymbol{v})^* = \boldsymbol{v}^\dagger\boldsymbol{u}$. (c) Compute the outer product $\boldsymbol{u}\boldsymbol{v}^\dagger$ and show $\mathrm{tr}(\boldsymbol{u}\boldsymbol{v}^\dagger) = \boldsymbol{v}^\dagger\boldsymbol{u}$.
2. **Special matrix types.** For each matrix below, identify which type(s) from Table 1.1 it belongs to, and verify your answer using the element conditions in the table: (a) σ_x; (b) σ_z; (c) the Hadamard gate $H = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix}$.
3. **Pauli algebra.** (a) Prove $\sigma_x^2 = \sigma_y^2 = \sigma_z^2 = \boldsymbol{I}$. (b) Compute all commutators $[\sigma_i, \sigma_j]$ for $i, j \in \{x,y,z\}$. (c) Which Pauli matrix has $|0\rangle$ and $|1\rangle$ as its eigenbasis? What are the eigenvalues?
4. **Projection operators.** (a) Verify that $\boldsymbol{P} = |0\rangle\langle 0|$ satisfies $\boldsymbol{P}^2 = \boldsymbol{P}$ and $\det(\boldsymbol{P}) = 0$. (b) Show $\boldsymbol{P} + \boldsymbol{Q} = \boldsymbol{I}$ and $\boldsymbol{P}\boldsymbol{Q} = 0$. (c) What does $\boldsymbol{P}|\psi\rangle$ equal for $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$?
5. **Commutator identities.** Prove the following for arbitrary operators $\hat{A}$, $\hat{B}$, $\hat{C}$: (a) $[\hat{A}+\hat{B},\hat{C}] = [\hat{A},\hat{C}] + [\hat{B},\hat{C}]$; (b) $[\hat{A},\hat{B}\hat{C}] = [\hat{A},\hat{B}]\hat{C} + \hat{B}[\hat{A},\hat{C}]$; (c) $[\hat{A},[\hat{B},\hat{C}]] + [\hat{B},[\hat{C},\hat{A}]] + [\hat{C},[\hat{A},\hat{B}]] = 0$ (Jacobi identity).
6. **Shared eigenstates.** Prove that if $[\hat{A},\hat{B}] = 0$ then $\hat{A}$ and $\hat{B}$ share a common eigenbasis.
7. **Tensor products.** (a) Write out all four two-qubit basis states $|00\rangle$, $|01\rangle$, $|10\rangle$, $|11\rangle$ as column vectors of length 4 and verify they are mutually orthogonal. (b) Compute $C_X|00\rangle$, $C_X|01\rangle$, $C_X|10\rangle$, $C_X|11\rangle$. Explain the action of the CNOT gate in words.
8. **Unitary transformation.** Let $\boldsymbol{U}$ be a 2×2 unitary matrix and $\{|v_0\rangle, |v_1\rangle\}$ an ONB. Prove that $\{\boldsymbol{U}|v_0\rangle, \boldsymbol{U}|v_1\rangle\}$ is also an ONB.
9. **Spectral decomposition.** (a) Find the eigenvalues and normalised eigenvectors of σ_x. (b) Write the spectral decomposition $\sigma_x = \sum_i \lambda_i |i\rangle\langle i|$ explicitly. (c) Verify that $\sigma_x|\psi_a\rangle = \lambda_a|\psi_a\rangle$ for each eigenvector.

Exercise session, week 34: unitary transformations and determinants

The exercises of this session are mainly meant as reminders of the specific linear algebra elements that we shall use throughout the course.

Exercise 1: unitary transformations and orthogonality.

In many-body theories it is common to expand a new basis in terms of a known one. Well-known examples are the analytical solutions of the quantum mechanical harmonic oscillator, or the hydrogen atom with a Coulomb interaction between the electron and the nucleus. Varying the expansion coefficients leads to a variational minimum which is expected to lie closer to the true ground state of a given Hamiltonian. This is the background for mean-field theories, full configuration interaction theory and many other many-body methods, and the expansion coefficients are normally the matrix elements of a unitary or orthogonal matrix.

Assume that the matrix $\boldsymbol{U}$ is unitary with dimension $n\times n$ and matrix elements $u_{p\lambda}$. Using Dirac notation we represent a new vector $|\psi_p\rangle$ of length n in terms of another vector $|\phi_\lambda\rangle$ of length n through

$$|\psi_p\rangle = \boldsymbol{U}|\phi_\lambda\rangle. \tag{1.107}$$

It is common to assume that the basis vectors $|\phi_\lambda\rangle$ are normalised and orthogonal, and that they are the eigenvectors of an eigenvalue problem

$$\boldsymbol{O}|\phi_\lambda\rangle = o_\lambda|\phi_\lambda\rangle, \tag{1.108}$$

where $\boldsymbol{O}$ is a Hermitian matrix of dimension $n\times n$ and o_λ one of its n eigenvalues. We have thus defined our new basis by performing a unitary (or orthogonal) transformation.

a) Write down the properties of unitary and orthogonal matrices.
b) Write down the properties of Hermitian matrices.
c) Assuming that the coefficients $u_{p\lambda}$ belong to a unitary or orthogonal transformation, show that the new basis is orthogonal and normalised as well.
d) Show that the eigenvalues obtained in the new basis are the same as those of the original basis.

Exercise 2: determinants.

a) Consider the determinant

$$\begin{vmatrix}\alpha_1 b_{11}+\alpha_2 b_{12} & a_{12}\\ \alpha_1 b_{21}+\alpha_2 b_{22} & a_{22}\end{vmatrix} = \alpha_1\begin{vmatrix}b_{11} & a_{12}\\ b_{21} & a_{22}\end{vmatrix} + \alpha_2\begin{vmatrix}b_{12} & a_{12}\\ b_{22} & a_{22}\end{vmatrix},$$

which generalises to the linearity in a column expressed by Eq. (2.12). Use it to show that, with every element a_{ij} given as a linear combination $a_{ij}=\sum_{k=1}^{n} b_{ik}c_{kj}$,

$$\begin{vmatrix}\sum_k b_{1k}c_{k1} & \cdots & \sum_k b_{1k}c_{kn}\\ \vdots & & \vdots\\ \sum_k b_{nk}c_{k1} & \cdots & \sum_k b_{nk}c_{kn}\end{vmatrix} = |\boldsymbol{C}|\,|\boldsymbol{B}|,$$

where $|\boldsymbol{B}|$ and $|\boldsymbol{C}|$ are the determinants of the $n\times n$ matrices with elements b_{ij} and c_{ij}. This is the product rule (2.13), and we shall use it repeatedly in the mean-field theories of the following chapters.

b) With the new basis of the previous exercise, now specialised to a basis of single-particle functions,

$$\psi_p(x) = \sum_\lambda C_{p\lambda}\phi_\lambda(x),$$

show that the determinant in the new basis $\psi_p(x)$ can be written as (omitting the factor $1/\sqrt{N!}$, with N the number of particles)

$$\begin{vmatrix} \psi_p(x_1) & \cdots & \psi_p(x_N) \\ \vdots & \ddots & \vdots \\ \psi_t(x_1) & \cdots & \psi_t(x_N) \end{vmatrix} = \begin{vmatrix} \sum_\lambda C_{p\lambda}\phi_\lambda(x_1) & \cdots & \sum_\lambda C_{p\lambda}\phi_\lambda(x_N) \\ \vdots & \ddots & \vdots \\ \sum_\lambda C_{t\lambda}\phi_\lambda(x_1) & \cdots & \sum_\lambda C_{t\lambda}\phi_\lambda(x_N) \end{vmatrix},$$

which is nothing but $|\boldsymbol{C}||\boldsymbol{\Phi}|$, with $|\boldsymbol{\Phi}|$ the determinant built from the basis functions $\phi_\lambda(x)$.

c) Show that the new determinant differs from $|\boldsymbol{\Phi}|$ by a complex phase only.

Answers, week 34

Exercise 1a.

A matrix $\boldsymbol{U} \in \mathbb{C}^{n\times n}$ is *unitary* if

$$\boldsymbol{U}^\dagger \boldsymbol{U} = \boldsymbol{U}\boldsymbol{U}^\dagger = \boldsymbol{I}, \qquad \text{equivalently} \qquad \boldsymbol{U}^{-1} = \boldsymbol{U}^\dagger. \tag{1.109}$$

In terms of matrix elements this reads $\sum_k u_{ik}u^*_{jk} = \sum_k u^*_{ki}u_{kj} = \delta_{ij}$: the rows form an orthonormal set, and so do the columns. Three consequences follow at once and were used throughout Section 1.9. The inner product is preserved, $(\boldsymbol{U}\boldsymbol{x})^\dagger(\boldsymbol{U}\boldsymbol{y}) = \boldsymbol{x}^\dagger\boldsymbol{y}$, hence so is the norm; the eigenvalues all have modulus one, since $\boldsymbol{U}\boldsymbol{x} = \lambda\boldsymbol{x}$ with $||\boldsymbol{x}|| = 1$ gives $|\lambda|^2 = \boldsymbol{x}^\dagger\boldsymbol{U}^\dagger\boldsymbol{U}\boldsymbol{x} = 1$; and $||\boldsymbol{U}|| = 1$, because $|\boldsymbol{U}|^*|\boldsymbol{U}| = |\boldsymbol{U}^\dagger\boldsymbol{U}| = 1$. A unitary matrix is thus a rotation, and it is perfectly reversible.

A *real* unitary matrix is called *orthogonal*: $\boldsymbol{O}^T\boldsymbol{O} = \boldsymbol{O}\boldsymbol{O}^T = \boldsymbol{I}$ and $\boldsymbol{O}^{-1} = \boldsymbol{O}^T$, with $|\boldsymbol{O}| = \pm 1$. These are the matrix types listed in Table 1.1.

Exercise 1b.

A matrix is *Hermitian* if $\boldsymbol{A} = \boldsymbol{A}^\dagger$, that is $a_{ij} = a^*_{ji}$; a real Hermitian matrix is symmetric. As shown in Section 1.11, the eigenvalues of a Hermitian matrix are real, eigenvectors belonging to distinct eigenvalues are orthogonal, and the matrix is diagonalisable by a unitary transformation, $\boldsymbol{U}^\dagger\boldsymbol{A}\boldsymbol{U} = \mathrm{diag}(\lambda_0,\ldots,\lambda_{n-1})$. It follows that expectation values $\langle\psi|\boldsymbol{A}|\psi\rangle$ are real, which is why observables are represented by Hermitian operators.

Exercise 1c.

Write the transformation in components, $|\psi_p\rangle = \sum_\lambda u_{p\lambda}|\phi_\lambda\rangle$, and use $\langle\phi_\lambda|\phi_\mu\rangle = \delta_{\lambda\mu}$:

$$\langle\psi_p|\psi_q\rangle = \sum_{\lambda\mu} u^*_{p\lambda}u_{q\mu}\langle\phi_\lambda|\phi_\mu\rangle = \sum_\lambda u^*_{p\lambda}u_{q\lambda} = \left(\boldsymbol{U}\boldsymbol{U}^\dagger\right)_{qp} = \delta_{pq}. \tag{1.110}$$

Orthonormality of the new basis is therefore *equivalent* to unitarity of the coefficient matrix; nothing else about $\boldsymbol{U}$ is needed, and nothing less would do. This is the statement proved geometrically in Eq. (2.19).

Exercise 1d.

In the new basis the operator $\boldsymbol{O}$ is represented by $\boldsymbol{O}' = \boldsymbol{U}\boldsymbol{O}\boldsymbol{U}^\dagger$. This is a similarity transformation of the kind studied in Section 1.22, and by Eq. (1.54) it leaves the spectrum unchanged: if $\boldsymbol{O}|\phi_\lambda\rangle = o_\lambda|\phi_\lambda\rangle$ then

$$\boldsymbol{O}'\left(\boldsymbol{U}|\phi_\lambda\rangle\right) = \boldsymbol{U}\boldsymbol{O}\boldsymbol{U}^\dagger\boldsymbol{U}|\phi_\lambda\rangle = o_\lambda\left(\boldsymbol{U}|\phi_\lambda\rangle\right). \tag{1.111}$$

The eigenvalues are the same and the eigenvectors are the rotated ones. Physically: the spectrum is a property of the operator, not of the basis we happen to write it in.

Exercise 2a.

Let $\boldsymbol{A} = \boldsymbol{B}\boldsymbol{C}$, so that column j of $\boldsymbol{A}$ is $\boldsymbol{a}_j = \sum_k c_{kj}\boldsymbol{b}_k$, a linear combination of the columns of $\boldsymbol{B}$. Applying the linearity (2.12) to the first column, then to the second, and so on through all n columns, gives

$$|\boldsymbol{A}| = \sum_{k_1=1}^{n}\sum_{k_2=1}^{n}\cdots\sum_{k_n=1}^{n} c_{k_1 1}c_{k_2 2}\cdots c_{k_n n}\left|\boldsymbol{b}_{k_1}\boldsymbol{b}_{k_2}\cdots\boldsymbol{b}_{k_n}\right|. \tag{1.112}$$

There are n^n terms, but a determinant with two equal columns vanishes, so every term in which two indices coincide drops out. Only the $n!$ terms in which $(k_1,\dots,k_n)$ is a permutation σ of $(1,\dots,n)$ survive, and reordering the columns of such a term into their natural order produces the sign $(-1)^\sigma$ together with $|\boldsymbol{B}|$. Hence

$$|\boldsymbol{A}| = \left(\sum_\sigma (-1)^\sigma c_{\sigma(1)1}c_{\sigma(2)2}\cdots c_{\sigma(n)n}\right)|\boldsymbol{B}| = |\boldsymbol{C}|\,|\boldsymbol{B}|, \tag{1.113}$$

where the bracket is precisely the definition (2.10) of $|\boldsymbol{C}|$.

Two remarks are worth making. First, the derivation is combinatorial and says nothing about how to *compute* either determinant; for that one uses the LU factorisation of Section 1.15 and Eq. (1.22), at a cost of $\mathscr{O}(n^3)$ rather than the $\mathscr{O}(n!)$ of the sum above. Second, the result immediately gives $|\boldsymbol{A}^{-1}| = 1/|\boldsymbol{A}|$ and, for a unitary matrix, $||\boldsymbol{U}|| = 1$.

Exercise 2b.

The matrix whose determinant we want has elements

$$M_{pj} = \psi_p(x_j) = \sum_\lambda C_{p\lambda}\phi_\lambda(x_j) = (\boldsymbol{C}\boldsymbol{\Phi})_{pj}, \qquad \Phi_{\lambda j} = \phi_\lambda(x_j), \tag{1.114}$$

so $\boldsymbol{M} = \boldsymbol{C}\boldsymbol{\Phi}$ and part a) gives directly

$$|\boldsymbol{M}| = |\boldsymbol{C}|\,|\boldsymbol{\Phi}|. \tag{1.115}$$

The rows of $\boldsymbol{M}$ are labelled by the new single-particle states and its columns by the particle coordinates, exactly as in Eq. (2.47).

Exercise 2c.

If $\boldsymbol{C}$ is unitary then $|\boldsymbol{C}|^*|\boldsymbol{C}| = |\boldsymbol{C}^\dagger\boldsymbol{C}| = |\boldsymbol{I}| = 1$, so $|\boldsymbol{C}| = e^{i\theta}$ for some real θ and

$$|\boldsymbol{M}| = e^{i\theta}\,|\boldsymbol{\Phi}|. \tag{1.116}$$

The two determinants differ by a global phase. Since all physical predictions depend on $|\Psi|^2$ or on expectation values $\langle\Psi|\widehat{O}|\Psi\rangle$, in which the phase cancels, the two describe the *same* state. This is the result exploited in Section 2.5: one may rotate the occupied orbitals among themselves at will without changing any physics, which is the freedom underlying Hartree-Fock theory and Thouless' theorem.

Chapter 2
From Linear Algebra to Many-Body Physics

Chapter 1 developed a language and a set of algorithms. This chapter puts them to work. We shall see that the central object of fermionic many-body theory, the Slater determinant, is a determinant in exactly the sense of Section 2.3; that changing the single-particle basis is a unitary transformation of the kind studied in Section 1.9; that the Hartree-Fock energy is minimised by solving a symmetric eigenvalue problem with the machinery of Sections 1.24 and 1.25; and that the obstacle which makes the whole subject difficult – the exponential growth of the Hilbert space – is the tensor product of Section 1.10.

Nothing in this chapter is new mathematics. What is new is the physics that the mathematics is asked to carry.

2.1 The many-body Hamiltonian

Throughout these lectures we assume that the interacting part of the Hamiltonian can be approximated by a two-body interaction, so the full N-particle Hamiltonian takes the form

$$\hat{H} = \hat{H}_0 + \hat{H}_I = \sum_{i=1}^{N} \hat{h}_0(x_i) + \sum_{i<j}^{N} \hat{v}(r_{ij}), \tag{2.1}$$

where x_i denotes the coordinates – position and spin – of particle i. The one-body operator splits into a kinetic and a potential part,

$$\hat{h}_0(x_i) = \hat{t}(x_i) + \hat{u}_{\text{ext}}(x_i), \tag{2.2}$$

where $\hat{u}_{\text{ext}}$ is typically a harmonic oscillator potential, the Coulomb attraction of a nucleus, or the self-consistent Hartree-Fock potential derived later in this chapter.

We shall assume that the single-particle functions ψ_α are eigenfunctions of $\hat{h}_0$,

$$\hat{h}_0(x)\,\psi_\alpha(x) = (\hat{t}(x) + \hat{u}_{\text{ext}}(x))\,\psi_\alpha(x) = \varepsilon_\alpha\,\psi_\alpha(x), \tag{2.3}$$

with ε_α the *unperturbed* or non-interacting single-particle energies. In the absence of $\hat{H}_I$ the total energy would simply be the sum of the occupied ε_α. Everything that makes many-body physics hard is contained in the correction to that statement.

Equation (2.3) is an eigenvalue problem, and in practice we solve it exactly as in Section 1.27: discretise on a grid, or expand in a finite basis, and diagonalise the resulting matrix with the algorithms of Sections 1.24 and 1.25. The harmonic oscillator eigenfunctions computed there are the single-particle basis used in the numerical examples of this chapter.

Finally, the Hamiltonian (2.1) is *symmetric* under the interchange of any two particles. Writing $\hat{P}_{ij}$ for the operator that permutes particles i and j,

$$[\hat{H}, \hat{P}_{ij}] = 0, \tag{2.4}$$

so the eigenfunctions of $\hat{H}$ can be chosen to be eigenfunctions of $\hat{P}_{ij}$ as well.

2.2 The Pauli exclusion principle and antisymmetry

Let Ψ_λ be such a simultaneous eigenfunction. Then

$$\hat{P}_{ij}\Psi_\lambda(x_1,\ldots,x_i,\ldots,x_j,\ldots,x_N) = \beta\,\Psi_\lambda(x_1,\ldots,x_i,\ldots,x_j,\ldots,x_N), \tag{2.5}$$

and since applying the permutation twice returns the original function, $\beta^2 = 1$ and $\beta = \pm 1$. For bosons $\beta = +1$; for fermions, which are our concern here, the Pauli principle requires

$$\beta = -1, \tag{2.6}$$

so the wave function changes sign under every transposition.

An arbitrary product of single-particle functions has no such symmetry, and must be projected onto the antisymmetric subspace. The projector is the *antisymmetrisation operator*

$$\hat{A} = \frac{1}{N!}\sum_p (-)^p \hat{P}, \tag{2.7}$$

where the sum runs over all $N!$ permutations $\hat{P}$ and p is the parity of each, that is the number of transpositions required to build it. Two properties of $\hat{A}$ will do all the work in Section 2.8. First, since $\hat{H}_0$ and $\hat{H}_I$ are invariant under every permutation,

$$[\hat{H}_0,\hat{A}] = [\hat{H}_I,\hat{A}] = 0. \tag{2.8}$$

Second, $\hat{A}$ is idempotent,

$$\hat{A}^2 = \hat{A}, \tag{2.9}$$

because permuting an already antisymmetrised function reproduces it up to the sign that the prefactor $(-)^p$ removes. A projector, in the sense of Section 1.6: $\hat{A}$ projects the full N-particle space onto its antisymmetric subspace, exactly as $\boldsymbol{P} = |0\rangle\langle 0|$ projects a qubit onto a computational basis state.

2.3 Determinants, and how they are actually computed

For an $n \times n$ matrix $\boldsymbol{A}$, the determinant is defined by summing over all $n!$ permutations of the column indices,

$$|\boldsymbol{A}| = |\boldsymbol{A}| = \sum_{i=1}^{n!} (-1)^{p_i} \hat{P}_i\, a_{11}a_{22}\cdots a_{nn}, \tag{2.10}$$

where $\hat{P}_i$ permutes the column indices and p_i is the number of transpositions needed to restore the natural ordering $a_{11}a_{22}\cdots a_{nn}$. For $n = 2$ the sum has two terms,

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = (-1)^0 a_{11}a_{22} + (-1)^1 a_{12}a_{21}, \tag{2.11}$$

the second obtained by interchanging the column indices 1 and 2.

The resemblance between Eq. (2.10) and the antisymmetriser (2.7) is not an accident, and it is the reason determinants appear in fermionic physics at all: both are the signed sum over the symmetric group, and the determinant is simply what the antisymmetriser produces when applied to a product of single-particle functions.

Two properties we shall need.

The determinant is linear in each column separately,

$$\begin{vmatrix} a_{11} & \cdots & \sum_{k=1}^{n} c_k b_{1k} & \cdots \\ a_{21} & \cdots & \sum_{k=1}^{n} c_k b_{2k} & \cdots \\ \vdots & & \vdots & \\ a_{n1} & \cdots & \sum_{k=1}^{n} c_k b_{nk} & \cdots \end{vmatrix} = \sum_{k=1}^{n} c_k \begin{vmatrix} a_{11} & \cdots & b_{1k} & \cdots \\ a_{21} & \cdots & b_{2k} & \cdots \\ \vdots & & \vdots & \\ a_{n1} & \cdots & b_{nk} & \cdots \end{vmatrix}, \tag{2.12}$$

and if every column of $\boldsymbol{A}$ is a linear combination of the columns of $\boldsymbol{B}$ with coefficient matrix $\boldsymbol{C}$, that is $a_{ij} = \sum_k b_{ik} c_{kj}$ or $\boldsymbol{A} = \boldsymbol{BC}$, then

$$|\boldsymbol{A}| = |\boldsymbol{B}| \cdot |\boldsymbol{C}|. \tag{2.13}$$

Equation (2.13) follows by applying Eq. (2.12) to each column in turn; the reader is invited to verify it for $n = 2$. It is the single most important algebraic fact in this chapter, and Section 2.5 is nothing but its physical interpretation.

A warning about Eq. (2.10).

The definition is a definition, not an algorithm. Evaluating a determinant by summing over permutations costs $\mathscr{O}(n!)$ operations, which for $n = 20$ is 2×10^{18} and for $n = 25$ exceeds the age of the universe on any machine. The LU decomposition of Section 1.15 computes the same number in $\mathscr{O}(n^3)$ operations as the product of the diagonal elements of $\boldsymbol{U}$, Eq. (1.22). Table 2.1 makes the comparison. Every Slater determinant evaluated in a Monte Carlo or configuration-interaction code is computed this way, never from Eq. (2.10).

n	**From Eq. (2.10),** $n!$	**From LU,** $\sim \frac{2}{3}n^3$
5	1.2×10^2	8.3×10^1
10	3.6×10^6	6.7×10^2
20	2.4×10^{18}	5.3×10^3
50	3.0×10^{64}	8.3×10^4

Table 2.1 Operation counts for evaluating an $n \times n$ determinant from the definition and from the LU decomposition of Section 1.15.

2.4 The Slater determinant

We now have everything needed to write down an antisymmetric N-particle wave function. Given N single-particle orbitals $\psi_\alpha, \psi_\beta, \ldots, \psi_\sigma$, the *Slater determinant* is

$$\Phi(x_1,\ldots,x_N;\alpha,\beta,\ldots,\sigma) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \psi_\alpha(x_1) & \psi_\alpha(x_2) & \cdots & \psi_\alpha(x_N) \\ \psi_\beta(x_1) & \psi_\beta(x_2) & \cdots & \psi_\beta(x_N) \\ \vdots & \vdots & \ddots & \vdots \\ \psi_\sigma(x_1) & \psi_\sigma(x_2) & \cdots & \psi_\sigma(x_N) \end{vmatrix}, \tag{2.14}$$

where $\alpha, \beta, \ldots, \sigma$ are the quantum numbers labelling the occupied orbitals. The two defining physical properties follow from two elementary properties of determinants:

- interchanging two particles interchanges two *columns*, which reverses the sign – the wave function is antisymmetric, as Eq. (2.6) demands;
- placing two particles in the same orbital gives two identical *rows*, and the determinant vanishes – no two fermions may occupy the same single-particle state.

The Pauli principle is thus not an extra postulate bolted onto the formalism; it is a property of determinants.

An equivalent form uses the antisymmetriser directly. Writing the unsymmetrised product, the *Hartree function*,

$$\Phi_H(x_1, \ldots, x_N; \alpha, \ldots, \nu) = \psi_\alpha(x_1)\psi_\beta(x_2)\cdots\psi_\nu(x_N), \tag{2.15}$$

we have

$$\Phi = \frac{1}{\sqrt{N!}}\sum_p (-)^p \hat{P}\,\Phi_H = \sqrt{N!}\hat{A}\,\Phi_H. \tag{2.16}$$

This is the form used in the derivations of Section 2.8.

Link to Chapter 1. The Hartree function (2.15) is precisely a tensor product, $|\alpha\rangle \otimes |\beta\rangle \otimes \cdots \otimes |\nu\rangle$, of the kind introduced in Section 1.10. The Slater determinant is its antisymmetric projection. A tensor product of N single-particle states is the least entangled state there is, and its Schmidt rank across any cut is one; antisymmetrisation introduces exactly the correlation demanded by statistics and no more. This is the precise sense in which a Slater determinant is an *uncorrelated* state, and it is made quantitative by the occupation numbers of Section 1.31: for a single Slater determinant they are exactly 1 for the N occupied orbitals and exactly 0 for all the rest. Any deviation from those values is a measure of genuine, dynamical correlation.

The numerical companion `slaterdeterminant.py` verifies both statements. Evaluating Eq. (2.14) for three particles at (x_1, x_2, x_3) and at (x_2, x_1, x_3) gives values that cancel to machine zero, and putting two particles in the same orbital gives $\Phi = -6 \times 10^{-18}$. The eigenvalues of the density matrix $\rho_{\lambda\mu} = \sum_i C^*_{i\lambda} C_{i\mu}$ come out as $(1,1,1,0,0,0,0,0)$ to machine precision.

2.5 Single-particle bases and unitary transformations

In practice one rarely knows the right orbitals in advance. The standard strategy is to expand them in a fixed, known basis $\{\phi_\lambda\}$ – harmonic oscillator functions, hydrogen-like functions, plane waves, Gaussians – and to vary the expansion coefficients,

$$\psi_p(x) = \sum_\lambda C_{p\lambda}\,\phi_\lambda(x). \tag{2.17}$$

The basis $\{\phi_\lambda\}$ is normally chosen as the eigenfunctions of the one-body Hamiltonian, Eq. (2.3), and is orthonormal. In Dirac notation Eq. (2.17) reads

$$|p\rangle = \sum_\lambda C_{p\lambda}|\lambda\rangle, \qquad \phi_\lambda(\boldsymbol{r}) = \langle \boldsymbol{r}|\lambda\rangle. \tag{2.18}$$

Two questions arise immediately. Does the new basis remain orthonormal? And what happens to the Slater determinant? Chapter 1 has already answered both.

Orthonormality is preserved if and only if $\boldsymbol{C}$ is unitary.

This was proved in Section 1.9: for an orthonormal set $\boldsymbol{v}_j^T\boldsymbol{v}_i = \delta_{ij}$ and a transformation $\boldsymbol{w}_i = \boldsymbol{U}\boldsymbol{v}_i$,

$$\boldsymbol{w}_j^T \boldsymbol{w}_i = \boldsymbol{v}_j^T \underbrace{\boldsymbol{U}^T \boldsymbol{U}}_{=\boldsymbol{I}} \boldsymbol{v}_i = \delta_{ij}, \tag{2.19}$$

so $\langle p|q\rangle = \delta_{pq}$ follows whenever $(C_{p\lambda})$ is unitary. We shall use this constantly: it is what allows us to search over single-particle bases without ever having to re-orthogonalise.

The Slater determinant picks up a phase.

Substituting Eq. (2.17) into Eq. (2.14), every entry of the determinant becomes a sum $\sum_\lambda C_{p\lambda}\phi_\lambda(x_i)$, that is, the matrix of orbitals is the product of the coefficient matrix $\boldsymbol{C}$ with the matrix $\boldsymbol{\Phi}$ of basis functions. By the product rule (2.13),

$$\frac{1}{\sqrt{N!}} \begin{vmatrix} \sum_\lambda C_{p\lambda}\phi_\lambda(x_1) & \cdots & \sum_\lambda C_{p\lambda}\phi_\lambda(x_N) \\ \vdots & \ddots & \vdots \\ \sum_\lambda C_{t\lambda}\phi_\lambda(x_1) & \cdots & \sum_\lambda C_{t\lambda}\phi_\lambda(x_N) \end{vmatrix} = |\boldsymbol{C}| \cdot |\boldsymbol{\Phi}|. \tag{2.20}$$

If $\boldsymbol{C}$ is unitary then $||\boldsymbol{C}|| = 1$, so the rotated Slater determinant differs from the original only by an overall phase. It describes the *same* physical state.

Many-body connection. Equation (2.20) is the algebraic foundation of a fact used everywhere in this book: one may perform any unitary rotation among the occupied orbitals without changing the many-body state, the density, or the total energy. In Hartree-Fock theory this is the freedom that makes canonical and localised orbitals equally valid descriptions of the same determinant, and it is the content of Thouless' theorem. In configuration interaction and coupled-cluster theory it is why the choice of reference orbitals is a matter of convenience rather than principle. The numerical check in `slaterdeterminant.py` makes it concrete: a random orthogonal rotation of three occupied orbitals leaves the energy unchanged to 2.7×10^{-15}, while a rotation that mixes occupied and unoccupied orbitals changes it by more than a factor of two.

Truncation, and what happens when the basis is not orthogonal.

A complete basis $\{|\lambda\rangle\}$ contains infinitely many states, and any calculation must truncate the sum in Eq. (2.17). This is the first and often the largest approximation in a many-body calculation, and its convergence has to be studied explicitly.

Furthermore, some of the most useful bases are not orthogonal at all. Gaussian basis sets in quantum chemistry, oscillator states of different oscillator length, and generator-coordinate states all have non-trivial overlap matrices $S_{ij} = \langle\phi_i|\phi_j\rangle$, and Eq. (2.19) no longer applies. The eigenvalue problem then becomes the generalised one, $\boldsymbol{Hc} = E\boldsymbol{Sc}$, treated in Section 1.29: one diagonalises $\boldsymbol{S}$, discards the directions whose eigenvalues lie below a threshold, and transforms with $\boldsymbol{X} = \boldsymbol{V}\boldsymbol{s}^{-1/2}$. The threshold is a physical choice, and Table 1.6 shows how much ill-conditioning it removes.

2.6 Particle-hole notation and the Fermi level

Before deriving the energy it is worth fixing the index conventions that will be used for the rest of the book. Given a reference Slater determinant – usually the one obtained by filling the N lowest single-particle levels – the single-particle states divide into those that are occupied in the reference and those that are not. The dividing line is the *Fermi level*, denoted F. We write

$$\begin{array}{ll} i,j,k,l,\dots \le F & \text{occupied states, or } \textit{hole} \text{ states,} \\ a,b,c,d,\dots > F & \text{unoccupied states, or } \textit{particle} \text{ states,} \\ p,q,r,s,\dots & \text{general single-particle states.} \end{array} \tag{2.21}$$

The names come from the picture in which the reference determinant is treated as a new vacuum: exciting a particle from an occupied state i to an unoccupied state a creates a particle in a and leaves a hole in i. Chapter 3 develops this into the particle-hole formalism of second quantisation, where it becomes the natural language for perturbation theory, coupled-cluster theory and the shell model. For now it is simply a convention, but it is one worth adopting at once: every equation from here on is easier to read with it.

2.7 The variational principle

Let E_0 be the exact ground-state energy of $\hat{H}$. For any normalised trial function Φ,

$$E_0 \le E[\Phi] = \int \Phi^* \hat{H}\, \Phi \, d\boldsymbol{\tau}, \qquad \int \Phi^* \Phi \, d\boldsymbol{\tau} = 1, \tag{2.22}$$

with the shorthand $d\boldsymbol{\tau} = dx_1 dx_2 \cdots dx_N$. The proof is the spectral decomposition of Section 1.11 and nothing more. Expanding Φ in the exact eigenstates, $|\Phi\rangle = \sum_k c_k |\Psi_k\rangle$ with $\hat{H}|\Psi_k\rangle = E_k|\Psi_k\rangle$ and $\sum_k |c_k|^2 = 1$, gives

$$E[\Phi] = \sum_k |c_k|^2 E_k \ \ge\ E_0 \sum_k |c_k|^2 = E_0, \tag{2.23}$$

with equality only when Φ is the ground state itself.

This one inequality is the engine of most of what follows. Restricting the trial function to a single Slater determinant and minimising gives Hartree-Fock theory; allowing a linear combination of determinants gives configuration interaction; parametrising it by a quantum circuit gives the variational quantum eigensolver of Section 2.16. It is also the reason the Lanczos algorithm of Section 1.26 is so well suited to many-body problems: the lowest Ritz value is the minimum of the Rayleigh quotient over the Krylov space, and therefore an upper bound on E_0 that can only improve as the space grows.

2.8 Expectation values with a Slater determinant

We now compute $E[\Phi]$ for the Slater determinant (2.14). The calculation uses only Eqs. (2.8), (2.9) and the orthonormality of the single-particle functions, and it is worth following in full: the same manipulations recur throughout many-body theory, and second quantisation exists largely to make them automatic.

The one-body contribution.

Using the form (2.16) of the Slater determinant,

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = N! \int \Phi_H^* \hat{A} \hat{H}_0 \hat{A}\, \Phi_H \, d\boldsymbol{\tau}. \tag{2.24}$$

Because $\hat{H}_0$ commutes with $\hat{A}$ we may move the left-hand antisymmetriser through it, and because $\hat{A}^2 = \hat{A}$ the two then collapse into one,

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = N! \int \Phi_H^* \hat{H}_0 \hat{A}\, \Phi_H \, d\boldsymbol{\tau}. \tag{2.25}$$

Replacing $\hat{A}$ by its definition (2.7), which cancels the factor $N!$, and $\hat{H}_0$ by the sum of one-body operators,

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = \sum_{i=1}^{N} \sum_{p} (-)^p \int \Phi_H^* \hat{h}_0(x_i) \hat{P} \Phi_H \, d\boldsymbol{\tau}. \tag{2.26}$$

Now the key step. The integrand factorises into single-particle integrals, and $\hat{h}_0(x_i)$ touches only one coordinate. If $\hat{P}$ permutes any two particles, at least one of the remaining factors is an overlap $\langle \mu | \nu \rangle$ between two *different* orbitals, which vanishes. Only the identity permutation survives,

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = \sum_{i=1}^{N} \int \Phi_H^* \hat{h}_0(x_i) \Phi_H \, d\boldsymbol{\tau}, \tag{2.27}$$

and integrating out the $N-1$ spectator coordinates, each of which contributes unity, leaves

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = \sum_{\mu=1}^{N} \int \psi_\mu^*(x) \, \hat{h}_0 \, \psi_\mu(x) dx. \tag{2.28}$$

Introducing the shorthand

$$\langle \mu | \hat{h}_0 | \mu \rangle = \int \psi_\mu^*(x) \hat{h}_0 \psi_\mu(x) dx, \tag{2.29}$$

this is simply

$$\int \Phi^* \hat{H}_0 \Phi \, d\boldsymbol{\tau} = \sum_{\mu=1}^{N} \langle \mu | \hat{h}_0 | \mu \rangle. \tag{2.30}$$

The one-body energy is the sum of the diagonal matrix elements over the occupied orbitals – the result one would have guessed, now derived.

The two-body contribution.

The same three steps apply,

$$\int \Phi^* \hat{H}_I \Phi \, d\boldsymbol{\tau} = N! \int \Phi_H^* \hat{A} \hat{H}_I \hat{A} \, \Phi_H \, d\boldsymbol{\tau} = \sum_{i<j}^{N} \sum_{p} (-)^p \int \Phi_H^* \hat{v}(r_{ij}) \hat{P} \Phi_H \, d\boldsymbol{\tau}, \tag{2.31}$$

but the conclusion is different. Because $\hat{v}(r_{ij})$ depends on *two* coordinates, the permutation that exchanges precisely those two particles no longer produces a vanishing overlap. Two permutations survive: the identity, and the transposition $\hat{P}_{ij}$. Hence

$$\int \Phi^* \hat{H}_I \Phi \, d\boldsymbol{\tau} = \sum_{i<j}^{N} \int \Phi_H^* \, \hat{v}(r_{ij}) \left(1 - \hat{P}_{ij}\right) \Phi_H \, d\boldsymbol{\tau}. \tag{2.32}$$

Writing out the two terms and integrating over the spectator coordinates,

$$\begin{aligned}\int \Phi^* \hat{H}_I \Phi \, d\boldsymbol{\tau} = \frac{1}{2} \sum_{\mu=1}^{N} \sum_{\nu=1}^{N} \Big[&\int \psi_\mu^*(x_i) \psi_\nu^*(x_j) \, \hat{v}(r_{ij}) \, \psi_\mu(x_i) \psi_\nu(x_j) \, dx_i dx_j \\ &- \int \psi_\mu^*(x_i) \psi_\nu^*(x_j) \, \hat{v}(r_{ij}) \, \psi_\nu(x_i) \psi_\mu(x_j) \, dx_i dx_j \Big],\end{aligned} \tag{2.33}$$

where the factor $\frac{1}{2}$ compensates for the unrestricted double sum, which counts every pair twice. The first term is the *direct* or *Hartree* term: it is the classical electrostatic energy of one charge distribution in the field of another. The second is the *exchange* or *Fock* term, and it has no classical counterpart whatsoever; it exists solely because the wave function is antisymmetric, and it is the sole trace left in the energy by the transposition $\hat{P}_{ij}$ in Eq. (2.32).

Notation for the matrix elements.

Writing the single-particle functions as overlaps, $\psi_\alpha(x) = \langle x|\alpha\rangle$, we abbreviate the two integrals as

$$\langle \mu\nu|\hat{v}|\mu\nu\rangle = \int \psi_\mu^*(x_i)\psi_\nu^*(x_j)\hat{v}(r_{ij})\psi_\mu(x_i)\psi_\nu(x_j)\,dx_i dx_j, \tag{2.34}$$

$$\langle \mu\nu|\hat{v}|\nu\mu\rangle = \int \psi_\mu^*(x_i)\psi_\nu^*(x_j)\hat{v}(r_{ij})\psi_\nu(x_i)\psi_\mu(x_j)\,dx_i dx_j, \tag{2.35}$$

and define the *antisymmetrised* matrix element

$$\langle \mu\nu|\hat{v}|\mu\nu\rangle_{\mathrm{AS}} = \langle \mu\nu|\hat{v}|\mu\nu\rangle - \langle \mu\nu|\hat{v}|\nu\mu\rangle. \tag{2.36}$$

Note that $\langle \mu\mu|\hat{v}|\mu\mu\rangle_{\mathrm{AS}} = 0$: a particle does not interact with itself. The self-interaction that the Hartree term would otherwise contain is cancelled exactly by the exchange term, which is one of the reasons Hartree-Fock theory is better behaved than the Hartree theory that preceded it.

2.9 The energy as a functional of the coefficients

Collecting Eqs. (2.30) and (2.33), the expectation value of the Hamiltonian in a Slater determinant is

$$E[\Phi] = \sum_{\mu=1}^{N}\langle \mu|\hat{h}_0|\mu\rangle + \frac{1}{2}\sum_{\mu=1}^{N}\sum_{\nu=1}^{N}\langle \mu\nu|\hat{v}|\mu\nu\rangle_{\mathrm{AS}}, \tag{2.37}$$

or, in the particle-hole notation of Section 2.6,

$$E[\Phi] = \sum_{i=1}^{N}\langle i|\hat{h}_0|i\rangle + \frac{1}{2}\sum_{i,j=1}^{N}\langle ij|\hat{v}|ij\rangle_{\mathrm{AS}}. \tag{2.38}$$

This is a functional of the occupied orbitals. To turn it into something we can minimise we insert the expansion (2.17) and let the coefficients carry the variation,

$$E[\Psi] = \sum_{i=1}^{N}\sum_{\alpha\beta} C_{i\alpha}^* C_{i\beta}\langle \alpha|\hat{h}_0|\beta\rangle + \frac{1}{2}\sum_{i,j=1}^{N}\sum_{\alpha\beta\gamma\delta} C_{i\alpha}^* C_{j\beta}^* C_{i\gamma} C_{j\delta}\langle \alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}}. \tag{2.39}$$

The Greek indices run over the full, fixed basis; the Latin indices run over the N occupied orbitals only. Everything on the right-hand side except the C's is computed once and stored.

Minimising Eq. (2.39) with respect to $C_{i\alpha}^*$ subject to the orthonormality constraint yields the Hartree-Fock equations. We defer the derivation, but the structure of the answer can be read off immediately and is worth stating now. Define the *density matrix* and the *Fock matrix*

$$\rho_{\gamma\delta} = \sum_{i=1}^{N} C_{i\gamma}^* C_{i\delta}, \qquad f_{\alpha\beta} = \langle \alpha|\hat{h}_0|\beta\rangle + \sum_{\gamma\delta}\rho_{\gamma\delta}\langle \alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}. \tag{2.40}$$

The condition for a stationary point is then

$$\sum_{\beta} f_{\alpha\beta}\, C_{i\beta} = \varepsilon_i C_{i\alpha}, \tag{2.41}$$

a Hermitian eigenvalue problem of exactly the kind solved in Sections 1.24 and 1.25 – with the single complication that the matrix $\boldsymbol{f}$ depends, through $\boldsymbol{\rho}$, on the eigenvectors we are trying to find. One therefore guesses, builds $\boldsymbol{f}$, diagonalises, rebuilds and repeats until nothing changes. This is the *self-consistent field* procedure, and it is an iteration in the sense of Section 1.17, with a matrix diagonalisation in place of a single sweep.

Summary: Hartree-Fock is Chapter 1 applied to fermions. Expand the states in an orthonormal basis (Section 1.8); change basis with a unitary transformation (Section 1.9), which leaves the determinant invariant up to a phase by the product rule (Section 2.3); minimise a quadratic form subject to a constraint, which produces a Hermitian eigenvalue problem (Section 1.11); solve it by Householder reduction followed by the QL algorithm (Sections 1.24 and 1.25); and iterate to self-consistency. Every step has already been developed.

The minimal implementation in `slaterdeterminant.py` does exactly this, using the eigenvalue solver written in Chapter 1. For spinless fermions in a one-dimensional trap with a softened Coulomb repulsion, and a basis of eight harmonic oscillator orbitals, it converges in nine iterations and lowers the energy below that of the naive determinant built from the N lowest oscillator states, as shown in Table 2.2.

N	E (lowest N orbitals)	E (Hartree-Fock)	**Gain**
2	2.69018342	2.66308878	-2.7×10^{-2}
3	6.46801021	6.39016133	-7.8×10^{-2}
4	11.77084712	11.62305385	-1.5×10^{-1}

Table 2.2 Self-consistent Hartree-Fock energies for N spinless fermions in a one-dimensional harmonic trap with a softened Coulomb interaction, in a basis of eight oscillator orbitals. The middle column is the energy of the determinant built from the N lowest oscillator states; the Hartree-Fock solution optimises the orbitals and always lies below it.

For $N=2$ the exact ground state in the same truncated basis can be obtained by diagonalising the Hamiltonian in the space of antisymmetric pairs, and the difference

$$E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}} \tag{2.42}$$

defines the *correlation energy*. Here it is -3.86×10^{-3}: the system is weakly correlated, and the largest amplitude in the exact state, 0.9930, sits on the Hartree-Fock configuration. Recovering the remaining 0.7% of the amplitude is what every method in the rest of this book is built to do.

2.10 The price of the two-body matrix elements

Equation (2.39) contains a sum over four basis indices, so a calculation in a basis of n single-particle states requires n^4 two-body matrix elements. For $n=100$ that is 10^8 numbers, for $n=1000$ it is 10^{12}, and long before the many-body problem becomes hard the mere storage of the interaction does.

Symmetries help. For a real, local interaction the matrix elements obey

$$\langle\alpha\beta|\hat{v}|\gamma\delta\rangle = \langle\beta\alpha|\hat{v}|\delta\gamma\rangle = \langle\gamma\delta|\hat{v}|\alpha\beta\rangle, \tag{2.43}$$

which reduces the count by roughly a factor of eight, and rotational or translational invariance can reduce it much further by making most elements vanish. But the scaling remains $\mathscr{O}(n^4)$.

The structural remedy is the one developed in Section 1.32. Reading the pairs $(\alpha\gamma)$ and $(\beta\delta)$ as composite indices turns the four-index array into an ordinary matrix, which for a Coulomb-like interaction is positive semidefinite with rapidly decaying eigenvalues, so that

$$\langle\alpha\beta|\hat{v}|\gamma\delta\rangle = \sum_{\eta=0}^{M-1} L^{\eta}_{\alpha\gamma} L^{\eta}_{\beta\delta}, \qquad M \ll n^2. \tag{2.44}$$

The storage drops from n^4 to Mn^2 and the contractions in Eq. (2.39) can be performed on the factors rather than on the full array. For the eight-orbital basis used above, the 64×64 pair matrix has numerical rank 15, not 64: less than a quarter of the nominal size. This is the same effect that Table 1.9 quantified, and it is what makes large-basis coupled-cluster and perturbation-theory calculations possible at all.

2.11 Evaluating Slater determinants in Monte Carlo simulations

The energy functional of Section 2.9 needs the Slater determinant once. A Monte Carlo calculation needs it several million times, and needs its gradient and its Laplacian as well. The variational and diffusion Monte Carlo methods developed later in this book propose a new configuration, accept or reject it, and repeat; the local energy requires

$$\frac{\nabla_i |\boldsymbol{D}|}{|\boldsymbol{D}|} \qquad \text{and} \qquad \frac{\nabla_i^2 |\boldsymbol{D}|}{|\boldsymbol{D}|} \tag{2.45}$$

for every particle i at every step. Since the cost of a single determinant is $\mathscr{O}(N^3)$ by Section 2.3, and there are $N \cdot d$ coordinates to differentiate, brute-force evaluation costs

$$Nd \cdot \mathscr{O}(N^3) \sim \mathscr{O}(d \cdot N^4) \tag{2.46}$$

per sweep. This is prohibitive, and it is entirely avoidable. The technical material of this section and the next reduces it to $\mathscr{O}(d \cdot N^2)$. It is placed here because it is nothing but linear algebra applied to the object introduced in Section 2.4; the physics that makes it necessary comes later.

The Slater matrix.

Define the $N \times N$ matrix $\boldsymbol{D}$ with elements

$$d_{ij} = \phi_j(\boldsymbol{r}_i), \tag{2.47}$$

where ϕ_j is a single-particle wave function. Rows are labelled by particles and columns by the single-particle states. The Slater determinant of Eq. (2.14) is $|\boldsymbol{D}|/\sqrt{N!}$.

The observation that makes everything work is this: *if we move only one particle at a time, only one row of $\boldsymbol{D}$ changes*. Recomputing the whole determinant then throws away almost all of the work already done. The solution is to keep track of the *inverse* of the Slater matrix.

The ratio of two determinants.

Let $\boldsymbol{r}^{\text{old}}$ be the current configuration and $\boldsymbol{r}^{\text{new}}$ the proposed one, differing only in the position of particle i. The Metropolis test needs the ratio

$$R \equiv \frac{|\boldsymbol{D}(\boldsymbol{r}^{\text{new}})|}{|\boldsymbol{D}(\boldsymbol{r}^{\text{old}})|}. \tag{2.48}$$

Expanding each determinant along row i in terms of the cofactors C_{ij},

$$R = \frac{\sum_{j=1}^{N} d_{ij}(\boldsymbol{r}^{\text{new}}) C_{ij}(\boldsymbol{r}^{\text{new}})}{\sum_{j=1}^{N} d_{ij}(\boldsymbol{r}^{\text{old}}) C_{ij}(\boldsymbol{r}^{\text{old}})}. \tag{2.49}$$

Now recall that the cofactor C_{ij} is the signed minor obtained by *deleting* row i and column j, so the ith row of the cofactor matrix does not depend on the ith row of $\boldsymbol{D}$ at all. Since the two configurations differ only in that row,

$$C_{ij}(\boldsymbol{r}^{\text{new}}) = C_{ij}(\boldsymbol{r}^{\text{old}}) \qquad \forall\, j \in \{1, \ldots, N\}. \tag{2.50}$$

Next, the inverse of a matrix is the transposed cofactor matrix divided by the determinant,

$$d_{ij}^{-1} = \frac{C_{ji}}{|\boldsymbol{D}|}, \tag{2.51}$$

and by definition $\sum_k d_{ik} d_{kj}^{-1} = \delta_{ij}$. Substituting Eq. (2.50) into Eq. (2.49) and replacing the cofactors using Eq. (2.51), the denominator becomes exactly unity, and we are left with

$$\boxed{R = \sum_{j=1}^{N} d_{ij}(\boldsymbol{r}^{\text{new}})\, d_{ji}^{-1}(\boldsymbol{r}^{\text{old}}) = \sum_{j=1}^{N} \phi_j(\boldsymbol{r}_i^{\text{new}})\, d_{ji}^{-1}(\boldsymbol{r}^{\text{old}})} \tag{2.52}$$

The ratio of two $N \times N$ determinants is a single dot product between the vector of single-particle functions evaluated at the new position and the ith column of the old inverse. The cost is $\mathscr{O}(N)$ instead of $\mathscr{O}(N^3)$, and no determinant is computed at all.

Updating the inverse.

The price is that we must maintain $\boldsymbol{D}^{-1}$. If the move is accepted, the inverse is repaired in $\mathscr{O}(N^2)$ operations rather than recomputed in $\mathscr{O}(N^3)$. For every column $j \neq i$ form

$$S_j = \Big(\boldsymbol{D}(\boldsymbol{r}^{\text{new}})\,\boldsymbol{D}^{-1}(\boldsymbol{r}^{\text{old}})\Big)_{ij} = \sum_{l=1}^{N} d_{il}(\boldsymbol{r}^{\text{new}})\, d_{lj}^{-1}(\boldsymbol{r}^{\text{old}}), \tag{2.53}$$

and update

$$d_{kj}^{-1}(\boldsymbol{r}^{\text{new}}) = d_{kj}^{-1}(\boldsymbol{r}^{\text{old}}) - \frac{S_j}{R}\, d_{ki}^{-1}(\boldsymbol{r}^{\text{old}}), \qquad k = 1, \ldots, N, \quad j \neq i. \tag{2.54}$$

The ith column is simply rescaled,

$$d_{ki}^{-1}(\boldsymbol{r}^{\text{new}}) = \frac{1}{R}\, d_{ki}^{-1}(\boldsymbol{r}^{\text{old}}), \qquad k = 1, \ldots, N. \tag{2.55}$$

Equations (2.54) and (2.55) are the Sherman-Morrison formula specialised to a rank-one change of a single row. Note that both contain a division by R: the algorithm degrades precisely when the proposed determinant is small, a point we return to in Section 2.14.

```python
def ratio(self, i, r_new):
    """R = |D(r_new)| / |D(r_old)|, in O(N) operations."""
    row = np.array([self.s.basis.value(j, r_new)[0]
                    for j in range(self.s.N)])
    return float(row @ self.s.Dinv[:, i]), row


def accept(self, i, r_new, R=None, row=None):
    """Update D and D^{-1} after the move of particle i is accepted."""
    if row is None:
        R, row = self.ratio(i, r_new)
    Dinv = self.s.Dinv

    # S_j = (D(new) D^{-1}(old))_{ij} for every column j
    S = row @ Dinv                              # length N, S[i] == R
    new_inv = Dinv.copy()
    for j in range(self.s.N):
        if j != i:
            new_inv[:, j] = Dinv[:, j] - (S[j] / R) * Dinv[:, i]
    new_inv[:, i] = Dinv[:, i] / R

    self.s.Dinv = new_inv
    self.s.D[i, :] = row
    self.s.positions[i] = r_new
```

For six particles in a two-dimensional oscillator basis the ratio computed from Eq. (2.52) agrees with the brute-force quotient of two determinants to 2×10^{-16}, as `slater_update.py` confirms.

2.12 Gradients, the Laplacian and the quantum force

The same argument applies to derivatives, and this is where the real saving lies. Differentiating the Slater determinant with respect to the coordinates of particle i changes only the ith row of $\boldsymbol{D}$, exactly as moving particle i did. The derivation leading to Eq. (2.52) therefore goes through unchanged with d_{ij} replaced by its derivative,

$$\frac{\nabla_i|\boldsymbol{D}(\boldsymbol{r})|}{|\boldsymbol{D}(\boldsymbol{r})|}=\sum_{j=1}^{N}\nabla_i d_{ij}(\boldsymbol{r})\,d_{ji}^{-1}(\boldsymbol{r})=\sum_{j=1}^{N}\nabla_i\phi_j(\boldsymbol{r}_i)\,d_{ji}^{-1}(\boldsymbol{r}), \tag{2.56}$$

$$\frac{\nabla_i^2|\boldsymbol{D}(\boldsymbol{r})|}{|\boldsymbol{D}(\boldsymbol{r})|}=\sum_{j=1}^{N}\nabla_i^2 d_{ij}(\boldsymbol{r})\,d_{ji}^{-1}(\boldsymbol{r})=\sum_{j=1}^{N}\nabla_i^2\phi_j(\boldsymbol{r}_i)\,d_{ji}^{-1}(\boldsymbol{r}). \tag{2.57}$$

To obtain every derivative of the Slater determinant we therefore need only the derivatives of the *single-particle* functions and the elements of the inverse Slater matrix. Each derivative is an $\mathscr{O}(N)$ dot product, and there are $d\cdot N$ of them, so the total evaluation costs $\mathscr{O}(d\cdot N^2)$. With the $\mathscr{O}(N^2)$ update of the inverse the overall scaling is no worse, which should be compared with the $\mathscr{O}(d\cdot N^4)$ of Eq. (2.46). Table 2.3 collects the counts.

Quantity	**Brute force**	**With the inverse**
Ratio R for one move	$\mathscr{O}(N^3)$	$\mathscr{O}(N)$
Update of $\boldsymbol{D}^{-1}$	$\mathscr{O}(N^3)$	$\mathscr{O}(N^2)$
All gradients and Laplacians	$\mathscr{O}(dN^4)$	$\mathscr{O}(dN^2)$

Table 2.3 Operation counts for the quantities a Monte Carlo step requires, with and without maintaining the inverse Slater matrix.

Equation (2.56) is the *quantum force* (up to a factor of two) that drives the importance-sampled random walk of diffusion Monte Carlo, and Eq. (2.57) supplies the kinetic energy in the local energy. Both are therefore evaluated at every single step of the simulation.

A practical remark.

In almost all cases one has closed-form expressions for the single-particle wave functions. It is then worth differentiating them analytically and writing separate functions for $\nabla_i\phi_j$ and $\nabla_i^2\phi_j$. Finite differences would cost both accuracy and speed, and the derivatives are needed far too often to tolerate either. The check in `slater_update.py` compares the analytic Eqs. (2.56) and (2.57) against numerical differentiation of $\ln|\boldsymbol{D}|$ and finds agreement to 5×10^{-10} and 9×10^{-6} respectively – the residual being the error of the finite differences, not of the formulae.

2.13 Spin factorisation of the Slater determinant

A further and quite substantial saving follows from a property of the spin degree of freedom. Consider beryllium, four electrons in the 1s and 2s orbitals. The Slater determinant reads

$$\Phi(\boldsymbol{r}_1,\boldsymbol{r}_2,\boldsymbol{r}_3,\boldsymbol{r}_4) = \frac{1}{\sqrt{4!}} \begin{vmatrix} \psi_{100\uparrow}(\boldsymbol{r}_1) & \psi_{100\uparrow}(\boldsymbol{r}_2) & \psi_{100\uparrow}(\boldsymbol{r}_3) & \psi_{100\uparrow}(\boldsymbol{r}_4) \\ \psi_{100\downarrow}(\boldsymbol{r}_1) & \psi_{100\downarrow}(\boldsymbol{r}_2) & \psi_{100\downarrow}(\boldsymbol{r}_3) & \psi_{100\downarrow}(\boldsymbol{r}_4) \\ \psi_{200\uparrow}(\boldsymbol{r}_1) & \psi_{200\uparrow}(\boldsymbol{r}_2) & \psi_{200\uparrow}(\boldsymbol{r}_3) & \psi_{200\uparrow}(\boldsymbol{r}_4) \\ \psi_{200\downarrow}(\boldsymbol{r}_1) & \psi_{200\downarrow}(\boldsymbol{r}_2) & \psi_{200\downarrow}(\boldsymbol{r}_3) & \psi_{200\downarrow}(\boldsymbol{r}_4) \end{vmatrix}. \tag{2.58}$$

As written this determinant *vanishes*, because the spatial parts of the spin-up and spin-down rows are identical: rows one and two are the same function of the spatial coordinates, and so are rows three and four. The spin labels carry the distinction, and a determinant of spatial functions alone cannot see them.

Expanding Eq. (2.58) in terms of 2×2 determinants built separately from the spin-up and spin-down orbitals gives

$$\begin{aligned} \Phi \propto\ & \det\!\uparrow(1,2)\det\!\downarrow(3,4) - \det\!\uparrow(1,3)\det\!\downarrow(2,4) \\ & - \det\!\uparrow(1,4)\det\!\downarrow(3,2) + \det\!\uparrow(2,3)\det\!\downarrow(1,4) \\ & - \det\!\uparrow(2,4)\det\!\downarrow(1,3) + \det\!\uparrow(3,4)\det\!\downarrow(1,2), \end{aligned} \tag{2.59}$$

with

$$\det\!\uparrow(1,2) = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_{100\uparrow}(\boldsymbol{r}_1) & \psi_{100\uparrow}(\boldsymbol{r}_2) \\ \psi_{200\uparrow}(\boldsymbol{r}_1) & \psi_{200\uparrow}(\boldsymbol{r}_2) \end{vmatrix}, \qquad \det\!\downarrow(3,4) = \frac{1}{\sqrt{2}} \begin{vmatrix} \psi_{100\downarrow}(\boldsymbol{r}_3) & \psi_{100\downarrow}(\boldsymbol{r}_4) \\ \psi_{200\downarrow}(\boldsymbol{r}_3) & \psi_{200\downarrow}(\boldsymbol{r}_4) \end{vmatrix}. \tag{2.60}$$

The total is of course still zero; the expansion has merely reorganised the sum over spin assignments.

The Moskowitz-Kalos ansatz.

What we would like is to avoid summing over spin variables altogether, particularly when the interaction does not depend on spin. It can be shown, see Moskowitz and Kalos [2], that for the variational energy one may replace the full determinant by the single product

$$\Phi(\boldsymbol{r}_1,\ldots,\boldsymbol{r}_N) \propto \det\!\uparrow\cdot\det\!\downarrow, \tag{2.61}$$

in which $\det\!\uparrow$ involves only the electrons with spin up and $\det\!\downarrow$ only those with spin down. This ansatz is *not* antisymmetric under the exchange of two electrons of opposite spin, but as long as the Hamiltonian is spin independent it yields the same expectation value for the energy as the full determinant. The reason is visible in Section 2.8: the exchange term of Eq. (2.33) connects only orbitals with the same spin, so the missing antisymmetry never enters the energy.

The saving.

The full determinant factorises,

$$|\boldsymbol{D}| = |\boldsymbol{D}|_{\uparrow}\cdot|\boldsymbol{D}|_{\downarrow}, \tag{2.62}$$

and the combined dimensionality of the two smaller determinants equals that of the original. The ratio then factorises too,

$$\frac{|\boldsymbol{D}|^{\mathrm{new}}}{|\boldsymbol{D}|^{\mathrm{old}}} = \frac{|\boldsymbol{D}|^{\mathrm{new}}_{\uparrow}}{|\boldsymbol{D}|^{\mathrm{old}}_{\uparrow}}\cdot\frac{|\boldsymbol{D}|^{\mathrm{new}}_{\downarrow}}{|\boldsymbol{D}|^{\mathrm{old}}_{\downarrow}}, \tag{2.63}$$

and when a single particle moves, one of the two factors is unchanged and cancels. Only a determinant of size $N/2$ is involved in each ratio and each inverse update, so the cost of one move drops from

$$\mathscr{O}_R(N) + \mathscr{O}_{\text{inverse}}(N^2) \qquad \text{to} \qquad \mathscr{O}_R(N/2) + \mathscr{O}_{\text{inverse}}(N^2/4), \tag{2.64}$$

and the cost of a full sweep over all N particles from $\mathscr{O}_R(N^2) + \mathscr{O}_{\text{inverse}}(N^3)$ to

$$\mathscr{O}_R(N^2/2) + \mathscr{O}_{\text{inverse}}(N^3/4). \tag{2.65}$$

The reduction is a constant factor, but a large one, and it is greatest when the numbers of spin-up and spin-down particles are equal so that the two factorised determinants are exactly half the size of the original.

2.14 Determinants, inverses and numerical stability

The updating scheme of Section 2.11 still needs the inverse and the determinant *once*, at the start of the simulation and whenever the accumulated error has to be reset. This is where the machinery of Chapter 1 enters directly.

The determinant from LU.

Never use the definition (2.10). The LU factorisation of Section 1.15 gives $\boldsymbol{PD} = \boldsymbol{LU}$ in $\mathscr{O}(N^3)$ operations, and the determinant follows from Eq. (1.22) as the signed product of the diagonal elements of $\boldsymbol{U}$,

$$|\boldsymbol{D}| = \pm u_{00} u_{11} \cdots u_{N-1,N-1}. \tag{2.66}$$

The same factorisation delivers the inverse for N additional back substitutions, Eq. (1.23). One factorisation therefore supplies both objects the Monte Carlo run needs.

In practice one should not store $|\boldsymbol{D}|$ at all but its logarithm. A product of N orbital values underflows or overflows long before N becomes physically interesting, whereas

$$\ln ||\boldsymbol{D}|| = \sum_{k=0}^{N-1} \ln |u_{kk}| \tag{2.67}$$

is perfectly well behaved, and the sign is tracked separately as the parity of the row interchanges. Since the Metropolis test uses only ratios, working with $\ln |\boldsymbol{\Phi}|$ throughout costs nothing.

The determinant from the SVD.

The singular value decomposition of Section 1.28 gives the same number,

$$||\boldsymbol{D}|| = \prod_{k=0}^{N-1} \sigma_k, \tag{2.68}$$

since $\boldsymbol{D} = \boldsymbol{U\Sigma V}^T$ with $||\boldsymbol{U}|| = ||\boldsymbol{V}|| = 1$. It costs several times more than LU and is never the right way to compute a determinant on its own. It is, however, exactly the right tool when one wants to know *how close to singular* the Slater matrix is, because it returns the condition number of Eq. (1.91) in the same breath,

$$\kappa_2(\boldsymbol{D}) = \frac{\sigma_0}{\sigma_{N-1}}. \tag{2.69}$$

The nodal surface.

This matters because the Slater determinant vanishes on the *nodal surface*, the set of configurations at which $|\boldsymbol{D}| = 0$. As a configuration approaches it the matrix becomes singular, the smallest singular value goes to zero, and the condition number diverges. Table 2.4 shows what happens as two of six particles are brought together: the determinant and the smallest singular value fall in step, the condition number rises by the same factor, and the accuracy of the computed inverse degrades exactly as Eq. (1.10) predicts.

Separation	$\|\|\boldsymbol{D}\|\|$	$\kappa_2(\boldsymbol{D})$	$\|\boldsymbol{D}\boldsymbol{D}^{-1} - \boldsymbol{I}\|$
10^0	2.81×10^0	2.58×10^1	8.7×10^{-16}
10^{-2}	3.21×10^{-2}	1.51×10^3	5.5×10^{-14}
10^{-4}	3.20×10^{-4}	1.54×10^5	4.3×10^{-12}
10^{-6}	3.20×10^{-6}	1.54×10^7	4.7×10^{-10}

Table 2.4 Six particles in a two-dimensional oscillator basis, with two of them brought to the separation shown. Each factor of 100 in the separation costs two digits in the inverse.

When to rebuild.

Equations (2.54) and (2.55) divide by R, so every near-nodal move multiplies the error in $\boldsymbol{D}^{-1}$ by roughly $1/R$, and the damage is permanent: subsequent updates propagate it. Under Metropolis sampling of $|\Phi|^2$ this is rare, since the sampling density itself vanishes on the node – in a run of 5000 moves the smallest accepted $|R|$ was 0.143 and the error in the inverse never exceeded 5×10^{-14}. But rare is not never. Forcing a sequence of near-nodal moves shows the mechanism plainly:

R **of the move**	$\|\boldsymbol{D}\boldsymbol{D}^{-1} - \boldsymbol{I}\|$ **after**
(start)	2.2×10^{-15}
-7.0×10^{-3}	3.4×10^{-14}
1.2×10^{-4}	3.5×10^{-12}
1.2×10^{-6}	2.0×10^{-10}
1.2×10^{-8}	3.2×10^{-8}
after an LU refresh	1.7×10^{-15}

Table 2.5 Error accumulated in the updated inverse by a sequence of moves that pass close to the nodal surface, and its removal by one LU refactorisation.

The remedy is simple and universal: rebuild $\boldsymbol{D}^{-1}$ from an LU factorisation every few hundred accepted moves. The $\mathcal{O}(N^3)$ cost, amortised over that many $\mathcal{O}(N^2)$ updates, is negligible, and the accumulated error is reset to machine precision. A production Monte Carlo code should also monitor $\|\boldsymbol{D}\boldsymbol{D}^{-1} - \boldsymbol{I}\|$, or the condition number of Eq. (2.69), and refresh whenever it exceeds a threshold.

Link to Chapter 1. Three separate results from the first chapter meet here. The LU decomposition of Section 1.15 supplies the determinant and the inverse at the start of the run; the condition number of Section 1.13, computed from the singular values of Section 1.28, tells us how many digits the inverse has lost; and the observation that the accuracy of a computed inverse is governed by κ_2 rather than by the algorithm, Eq. (1.10), explains why no cleverer updating formula could avoid the problem. The nodal surface of a fermionic wave function is, from the point of view of linear algebra, simply the locus where the Slater matrix becomes singular.

2.15 Quantum computing: the gate model and its linear-algebra foundation

A quantum computer is a well-controlled quantum many-body system, and the abstract *gate-based* model is defined entirely in the language of Chapter 1. Starting from N qubits initialised in $|\Psi\rangle^{(0)} = |00\cdots 0\rangle$, the state evolves by successive applications of unitary gates,

$$|\Psi\rangle^{(n+1)} = \hat{U}^{(n)}|\Psi\rangle^{(n)}, \qquad \hat{U}^{(n)} = T\exp\left(-i\int_{t_n}^{t_{n+1}} \hat{H}(t)\,dt\right).$$

A *single-qubit gate* acts on qubit a via a 2×2 unitary:

$$\Psi^{(n+1)}_{i_1\cdots i_N} = \sum_{i'_a} U^{(n)}_{i_a i'_a}\,\Psi^{(n)}_{i_1\cdots i_{a-1} i'_a i_{a+1}\cdots i_N}.$$

A *two-qubit gate* acts on qubits a, b via a 4×4 unitary. Standard examples are the Pauli gates X,Y,Z of Section 1.7, the Hadamard gate

$$H = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

the phase gate $T = \begin{pmatrix} e^{i\pi/8} & 0 \\ 0 & e^{-i\pi/8} \end{pmatrix}$, and the two-qubit CNOT (controlled-NOT):

$$C_X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

When qubit a is *measured*, the result $\alpha \in \{0,1\}$ occurs with probability

$$P_\alpha = \sum_{i_1\cdots\hat{i}_a\cdots i_N} \left|\Psi^{(n)}_{i_1\cdots\alpha\cdots i_N}\right|^2,$$

after which the state collapses to the corresponding projected state – the projection operators of Section 1.6 acting for real.

In a nutshell. Quantum computing is a restricted form of linear algebra: matrix-vector multiplications with special unitary matrices (gates) on exponentially large vectors (wave functions). The challenge – shared with classical many-body physics – is that only a tiny fraction of the 2^N-dimensional space is physically relevant; identifying that subspace is the core problem in both fields.

2.16 The exponential-dimension problem and why it matters

The state of N spin-$\frac{1}{2}$ particles requires 2^N complex numbers to describe, as the tensor-product construction of Section 1.10 already made clear. For $N \approx 300$, $2^{300} \gg 10^{82}$, the estimated number of atoms in the observable universe. The same counting applies to the fermionic problem: the number of Slater determinants that can be formed by distributing N particles over n single-particle states is $\binom{n}{N}$, which for $n = 40$ and $N = 20$ is already 1.4×10^{11}.

How do physicists actually compute?

1. **Clever approximations**: Bethe ansatz, perturbation theory, mean-field approximations such as the Hartree-Fock theory derived above.
2. **Computer simulations**: DFT, quantum Monte Carlo, FCI, coupled cluster, DMRG.
3. **Restricting to tractable subsystems**: systems where a single-particle basis is a good approximation.

Chapter 1 supplied two of the reasons this is not hopeless. The first is that we rarely want the whole spectrum: the Lanczos algorithm of Section 1.26 extracts the lowest eigenvalues of a matrix of dimension 10^9 using only matrix-vector products, and the pairing-model example there converged to machine precision in forty of them. The second is that physically relevant states are compressible: the Schmidt spectrum of a ground state decays rapidly (Section 1.30), and the Eckart-Young theorem, Eq. (1.96), guarantees that truncating it is the best approximation of its rank. Density-matrix renormalisation group and tensor-network methods are built on precisely that observation.

For a quantum circuit of depth D acting on N qubits, the accessible subspace has dimension $\mathscr{O}(DN)$, vastly smaller than 2^N. For $N = 100$ and $D = 10^6$, this subspace still has 10^{20} times fewer degrees of freedom than the full Hilbert space. The question is whether the *physically relevant* states lie in this reachable subspace – a central open problem in quantum many-body theory and quantum computing alike (see, e.g., LaRocca et al.).

The variational quantum eigensolver (VQE) tackles the problem by using a parametrised quantum circuit as an ansatz for the ground state and minimising $\langle \Psi(\boldsymbol{\theta})|\hat{H}|\Psi(\boldsymbol{\theta})\rangle$ classically over the circuit parameters. It is the quantum-computing form of the variational principle (2.22) that underlies Hartree-Fock and coupled-cluster theory: the bound $E(\boldsymbol{\theta}) \geq E_0$ holds for every choice of parameters, and the quality of the result is decided entirely by whether the ansatz can reach the true ground state.

2.17 Exercises

Warm-up exercises

1. **Determinant product rule.** For 2×2 matrices $\boldsymbol{B}$ and $\boldsymbol{C}$, define $a_{ij} = \sum_k b_{ik} c_{kj}$. Show by direct calculation that $\det(\boldsymbol{A}) = \det(\boldsymbol{B})\det(\boldsymbol{C})$.
2. **Slater determinant:** $N = 2$**.** Write the Slater determinant for two electrons in orbitals ψ_α and ψ_β. (a) Show it is antisymmetric under particle exchange. (b) Show it vanishes if $\psi_\alpha = \psi_\beta$.
3. **Antisymmetriser.** Verify for $N = 2$ that $\hat{A}^2 = \hat{A}$ and that $[\hat{H}_0, \hat{A}] = 0$ when $\hat{H}_0 = \hat{h}_0(x_1) + \hat{h}_0(x_2)$.
4. **Basis change of the Slater determinant.** With the expansion $\psi_p = \sum_\lambda C_{p\lambda} \phi_\lambda$ and a 2×2 unitary $\boldsymbol{C}$, show that the two-particle Slater determinant in the new basis equals $\det(\boldsymbol{C})\det(\boldsymbol{\Phi})$ and that $|\det(\boldsymbol{C})| = 1$ leaves the physics unchanged.
5. **Hartree-Fock energy.** For a two-electron system with orbitals ψ_α and ψ_β, write out the HF energy functional (2.37) explicitly, identifying the kinetic, direct (Hartree), and exchange (Fock) contributions.
6. **Variational principle.** Show that $E[\Phi] \geq E_0$ for any normalised trial state Φ, where E_0 is the exact ground state energy. Use the spectral decomposition of $\hat{H}$ in its eigenbasis.
7. **Exponential dimension.** How many complex numbers are needed to store the full N-particle wave function for $N = 10$, 20, 50? A modern GPU can perform roughly 10^{14} floating-point operations per second. Estimate how long a single matrix-vector multiplication in the $N = 50$ space would take.
8. **One-qubit gates numerically.** Write a short Python function that represents $|0\rangle$ and $|1\rangle$ as NumPy arrays and applies each Pauli matrix, the Hadamard gate H, and the phase gate T in turn. Verify the action of X (spin flip), Z (phase flip) and $H|0\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$ (creation of a superposition state).
9. **Pauli gate as a quantum NOT.** Explain why σ_x is the quantum analogue of the classical NOT gate. What is the difference between the classical and quantum cases when the input is a superposition $\alpha|0\rangle + \beta|1\rangle$?

10. **VQE and the variational principle.** The variational quantum eigensolver minimises $E(\boldsymbol{\theta}) = \langle \Psi(\boldsymbol{\theta})|\hat{H}|\Psi(\boldsymbol{\theta})\rangle$ over circuit parameters $\boldsymbol{\theta}$. (a) Explain why $E(\boldsymbol{\theta}) \geq E_0$ for any $\boldsymbol{\theta}$. (b) Describe the connection to the Hartree-Fock variational principle. (c) What determines the quality of the approximation in VQE compared to FCI?
11. **The antisymmetriser as a projector.** Using Eq. (2.7), verify for $N = 2$ that $\hat{A}^2 = \hat{A}$ and that $[\hat{H}_0, \hat{A}] = 0$ when $\hat{H}_0 = \hat{h}_0(x_1) + \hat{h}_0(x_2)$. Explain why $\hat{A}$ is a projection operator in the sense of Section 1.6, and identify the subspace it projects onto.
12. **Why only two permutations survive.** In the derivation of Eq. (2.32) it was argued that only the identity and the transposition $\hat{P}_{ij}$ contribute to the two-body expectation value. Write out the case $N = 3$ explicitly, listing all six permutations, and show that four of them give a vanishing integral. Then explain why the corresponding argument for the one-body operator leaves only one surviving permutation.
13. **Self-interaction.** Show from Eq. (2.36) that $\langle \mu\mu|\hat{v}|\mu\mu\rangle_{\mathrm{AS}} = 0$, and explain in words what would go wrong in Eq. (2.37) if the exchange term were omitted.
14. **Occupation numbers of a Slater determinant.** For the determinant (2.14) the density matrix is $\rho_{\lambda\mu} = \sum_i C_{i\lambda}^* C_{i\mu}$. (a) Show that $\boldsymbol{\rho}^2 = \boldsymbol{\rho}$ when the coefficients are orthonormal. (b) Conclude that the eigenvalues – the natural occupation numbers of Section 1.31 – can only be 0 or 1. (c) What does it mean physically when a computed occupation number comes out as 0.93?
15. **Cost of a determinant.** Reproduce Table 2.1. For $n = 20$, estimate how long the evaluation of a single determinant would take from Eq. (2.10) on a machine performing 10^{12} operations per second, and compare with the LU route of Section 1.15.
16. **Two-body storage.** How many two-body matrix elements $\langle \alpha\beta|\hat{v}|\gamma\delta\rangle$ are there for $n = 50$, $n = 200$ and $n = 1000$ single-particle states? How much memory does each case require in double precision? Using Eq. (2.44) with $M = 5n$, how much is saved?
17. **Hartree-Fock numerically.** Using `slaterdeterminant.py`, reproduce Table 2.2. Then vary the interaction strength and plot the correlation energy (2.42) and the largest natural occupation number against it. At what point would you no longer trust a single-determinant description?
18. **The ratio R by hand.** Take $N = 2$ and a Slater matrix $\boldsymbol{D}$ with elements $d_{ij} = \phi_j(\boldsymbol{r}_i)$. Move particle 1 and compute the ratio R both from the definition (2.48), by evaluating two 2×2 determinants, and from Eq. (2.52). Verify that they agree, and count the floating point operations each route requires.
19. **Why the cofactor row does not change.** Explain in your own words why $C_{ij}(\boldsymbol{r}^{\mathrm{new}}) = C_{ij}(\boldsymbol{r}^{\mathrm{old}})$ when only particle i has moved, Eq. (2.50). Where in the derivation of Eq. (2.52) would the argument fail if two particles were moved at once?
20. **Updating the inverse.** Implement Eqs. (2.53)–(2.55) for a 3×3 Slater matrix and check the result against `numpy.linalg.inv`. Confirm that the update costs $\mathscr{O}(N^2)$ operations by counting the loops.
21. **Derivatives of the determinant.** Using `slater_update.py`, compare the analytic gradient and Laplacian ratios of Eqs. (2.56) and (2.57) with numerical differentiation of $\ln|\boldsymbol{D}|$. Why does the Laplacian agree to fewer digits than the gradient?
22. **The nodal surface as an ill-conditioned matrix.** Reproduce Table 2.4. Plot $\ln \kappa_2(\boldsymbol{D})$ against $\ln(\mathrm{separation})$ and explain the slope. Using Eq. (1.10), predict how many digits of $\boldsymbol{D}^{-1}$ survive at a separation of 10^{-8}, and check your prediction.
23. **When to refresh.** Design a rule for deciding when to rebuild $\boldsymbol{D}^{-1}$ from an LU factorisation. Estimate the fraction of the total run time that the refreshes consume for $N = 50$ if you rebuild every 500 accepted moves, and compare with the cost of doing so at every move.
24. **Spin factorisation.** For $N = 4$ with two spin-up and two spin-down particles, write out Eq. (2.59) and verify that the full determinant of Eq. (2.58) vanishes. Then show, using the structure of Eq. (2.33), that the factorised ansatz (2.61) gives the same energy for a spin-independent Hamiltonian.

Exercise session, week 35: Slater determinants and a first FCI model

Exercise 1: expectation values with a Slater determinant.

Consider the fermion Slater determinant used as an ansatz for a quantum-mechanical state,

$$\Phi_{\lambda}^{AS}(x_1x_2\dots x_N;\alpha_1\alpha_2\dots\alpha_N) = \frac{1}{\sqrt{N!}}\sum_p(-)^p\hat{P}\prod_{i=1}^{N}\psi_{\alpha_i}(x_i), \tag{2.70}$$

where $\hat{P}$ permutes the particles and the sum runs over all $N!$ permutations. The number of particles equals the number of available single-particle states $\alpha_1\alpha_2\dots\alpha_N$, and the single-particle basis ψ_{α_i} is orthonormal.

a) Write out Φ^{AS} for $N=3$.
b) Show that

$$\int dx_1dx_2\cdots dx_N\left|\Phi_{\lambda}^{AS}(x_1x_2\dots x_N;\alpha_1\alpha_2\dots\alpha_N)\right|^2 = 1.$$

c) Define a general one-body operator $\hat{F}=\sum_i^N\hat{f}(x_i)$ and a general two-body operator $\hat{G}=\sum_{i>j}^N\hat{g}(x_i,x_j)$, with $\hat{g}$ invariant under the interchange of the coordinates of particles i and j. Calculate the matrix elements for a two-particle Slater determinant,

$$\langle\Phi_{\alpha_1\alpha_2}^{AS}|\hat{F}|\Phi_{\alpha_1\alpha_2}^{AS}\rangle \qquad \text{and} \qquad \langle\Phi_{\alpha_1\alpha_2}^{AS}|\hat{G}|\Phi_{\alpha_1\alpha_2}^{AS}\rangle.$$

Which properties do you expect these operators to have, in addition to permutation symmetry?
d) Repeat the calculation for a general N-particle Slater determinant.

Exercise 2: a three-level model.

We now consider a simple three-level problem, our first and very simple model of a many-fermion system and of what we shall later call full configuration interaction theory. The particles are fermions. The single-particle states are labelled by the quantum number p and each can accommodate up to two particles, so every single-particle state is doubly degenerate; one may think of the two as spin up and spin down. The spacing between the doubly degenerate states is constant with value d, the first state has energy d, and there are three available states $p=1,2,3$, as in Figure 2.1.

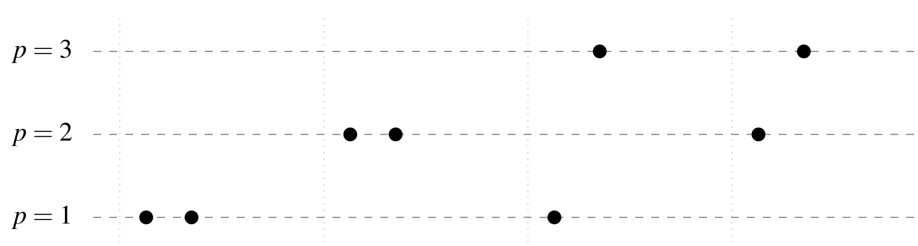

Fig. 2.1 Schematic plot of the possible single-particle levels with double degeneracy. The filled circles indicate occupied states and the spacing between the levels is constant. Four possible two-particle states are shown; the two leftmost are the configurations retained in parts b) and c), in which both particles occupy the same level.

a) How many two-particle Slater determinants can we construct in this space?
b) Limit the system to the two lowest orbits, $p=1$ and $p=2$, and two particles. Write the Hamiltonian as $\hat{H}=\hat{H}_0+\hat{H}_I$, with the one-body part satisfying

$$\hat{h}_0\psi_{p\sigma} = p\,d\,\psi_{p\sigma},$$

where σ is the spin quantum number. Assume that the only two-particle states that can exist are those in which both particles occupy the same level p, and that all two-particle matrix elements of $\hat{H}_I$ have the constant value $-g$. Show that the Hamiltonian matrix can be written as

$$\begin{pmatrix} 2d-g & -g \\ -g & 4d-g \end{pmatrix},$$

and find its eigenvalues and eigenvectors. What is the admixture of the state with two particles in $p=2$ to the wave function with two particles in $p=1$? Discuss the result in terms of a linear combination of Slater determinants.

c) Add the possibility that the two particles occupy $p=3$, and find the Hamiltonian matrix, the eigenvalues and the eigenvectors. We still insist that only two-particle states with both particles in the same level are allowed. The 3×3 matrix may be diagonalised numerically.

This small model catches several birds with one stone. It shows how linear combinations of Slater determinants are built and how the coefficients are interpreted as admixtures to a given state. The two-particle states above $p=1$ are excitations of the ground-state configuration, and the reliability of the single-determinant ansatz depends on the ratio of the interaction strength g to the level spacing d. Finally, the model is a schematic ansatz for pairing correlations, and hence for superfluidity and superconductivity in fermionic systems; it is the two-level ancestor of the pairing model used in Section 1.26.

Answers, week 35

Exercise 1a.

For $N=3$ the six permutations give

$$\begin{aligned}\Phi_\lambda^{AS} = \frac{1}{\sqrt{6}}\Big(&\psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\psi_{\alpha_3}(x_3) - \psi_{\alpha_1}(x_2)\psi_{\alpha_2}(x_1)\psi_{\alpha_3}(x_3)\\ &+ \psi_{\alpha_1}(x_2)\psi_{\alpha_2}(x_3)\psi_{\alpha_3}(x_1) - \psi_{\alpha_1}(x_3)\psi_{\alpha_2}(x_2)\psi_{\alpha_3}(x_1)\\ &+ \psi_{\alpha_1}(x_3)\psi_{\alpha_2}(x_1)\psi_{\alpha_3}(x_2) - \psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_3)\psi_{\alpha_3}(x_2)\Big),\end{aligned} \tag{2.71}$$

which is the expansion of the 3×3 determinant of Eq. (2.14) along its rows.

Exercise 1b.

Introduce the antisymmetriser (2.7) and the Hartree function (2.15), so that $\Phi_\lambda^{AS} = \sqrt{N!}\hat{A}\Phi_H$. Then

$$\langle \Phi_\lambda^{AS} | \Phi_\lambda^{AS} \rangle = N! \int \Phi_H^* \hat{A}^\dagger \hat{A} \Phi_H \, d\boldsymbol{\tau} = N! \int \Phi_H^* \hat{A} \Phi_H \, d\boldsymbol{\tau}, \tag{2.72}$$

where we used that $\hat{A}$ is Hermitian and idempotent, Eq. (2.9). Writing out the antisymmetriser cancels the factor $N!$,

$$\langle \Phi_\lambda^{AS} | \Phi_\lambda^{AS} \rangle = \sum_p (-)^p \int \Phi_H^* \hat{P} \Phi_H \, d\boldsymbol{\tau}. \tag{2.73}$$

The permutation operator acts on only one of the two Hartree functions, so any $\hat{P}$ other than the identity produces at least one overlap $\langle \alpha_i | \alpha_j \rangle$ with $i \neq j$, which vanishes by orthonormality. Only the identity survives and $\langle \Phi_\lambda^{AS} | \Phi_\lambda^{AS} \rangle = \int \Phi_H^* \Phi_H d\boldsymbol{\tau} = 1$.

Exercise 1c.

For two particles the same manipulation gives

$$\langle \Phi^{AS}_{\alpha_1\alpha_2}|\hat{F}|\Phi^{AS}_{\alpha_1\alpha_2}\rangle = \sum_p (-)^p \int \psi^*_{\alpha_1}(x_1)\psi^*_{\alpha_2}(x_2)\left[\hat{f}(x_1)+\hat{f}(x_2)\right]\hat{P}\,\psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\,d\boldsymbol{\tau}. \tag{2.74}$$

As in b), a permuted Hartree function makes the integral vanish, so only the identity contributes. In each remaining term the one-body operator acts on a single coordinate; integrating out the other gives unity, and

$$\langle \Phi^{AS}_{\alpha_1\alpha_2}|\hat{F}|\Phi^{AS}_{\alpha_1\alpha_2}\rangle = \langle\alpha_1|\hat{f}|\alpha_1\rangle + \langle\alpha_2|\hat{f}|\alpha_2\rangle, \tag{2.75}$$

with the shorthand of Eq. (2.29).

For the two-body operator there is only one term, $\hat{G} = \hat{g}(x_1,x_2)$, and now the transposition survives as well, since $\hat{g}$ depends on both coordinates. The two permutations give

$$\begin{aligned}\langle \Phi^{AS}_{\alpha_1\alpha_2}|\hat{G}|\Phi^{AS}_{\alpha_1\alpha_2}\rangle &= \int \psi^*_{\alpha_1}(x_1)\psi^*_{\alpha_2}(x_2)\hat{g}(x_1,x_2)\psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\,d\boldsymbol{\tau}\\ &\quad - \int \psi^*_{\alpha_1}(x_1)\psi^*_{\alpha_2}(x_2)\hat{g}(x_1,x_2)\psi_{\alpha_1}(x_2)\psi_{\alpha_2}(x_1)\,d\boldsymbol{\tau}\\ &= \langle\alpha_1\alpha_2|\hat{g}|\alpha_1\alpha_2\rangle_{\mathrm{AS}}.\end{aligned} \tag{2.76}$$

The first term is the direct or Hartree term, the second the exchange or Fock term, exactly as in Eq. (2.33).

Beyond permutation symmetry we expect both operators to be *Hermitian*, since they represent physical observables and must have real eigenvalues. Permutation symmetry is forced on us because the particles are indistinguishable, and it implies $[\hat{A},\hat{F}] = [\hat{A},\hat{G}] = 0$, which is what allowed the manipulations above.

Exercise 1d.

For general N the steps are the same. For the one-body operator,

$$\langle \Phi^{AS}|\hat{F}|\Phi^{AS}\rangle = \sum_{i=1}^{N}\sum_p (-)^p \int \Phi^*_H \hat{f}(x_i)\hat{P}\Phi_H\,d\boldsymbol{\tau} = \sum_{i=1}^{N}\langle\alpha_i|\hat{f}|\alpha_i\rangle. \tag{2.77}$$

For the two-body operator, each pair (x_i,x_j) receives two contributing permutations with opposite signs, giving the factor $(1-\hat{P}_{ij})$ of Eq. (2.32). Replacing the restricted sum $i<j$ by an unrestricted double sum with a compensating $\frac{1}{2}$, and noting that the $i=j$ terms cancel identically, leaves

$$\langle \Phi^{AS}|\hat{G}|\Phi^{AS}\rangle = \frac{1}{2}\sum_{i=1}^{N}\sum_{j=1}^{N}\langle\alpha_i\alpha_j|\hat{g}|\alpha_i\alpha_j\rangle_{\mathrm{AS}}. \tag{2.78}$$

Equations (2.77) and (2.78) together are Eq. (2.37), the Hartree-Fock energy functional.

Exercise 2a.

With three levels, each doubly degenerate, there are six single-particle states and hence $\binom{6}{2} = 15$ two-particle Slater determinants in general. If we count only the level labels and ignore spin, three states with two distinct levels and three with both particles in the same level give six configurations. Under the restriction of parts b) and c), that both particles occupy the *same* level, exactly three Slater determinants remain, one for each p.

Exercise 2b.

Denote by $|\Phi_p\rangle$ the state with both particles in level p. Since $\psi_{p\sigma}$ is an eigenstate of $\hat{h}_0$ with eigenvalue pd, the one-body part contributes $\langle\Phi_p|\hat{H}_0|\Phi_p\rangle = 2\langle p|\hat{h}_0|p\rangle = 2pd$, using Eq. (2.77) with $N = 2$. Every matrix element of $\hat{H}_I$ equals $-g$. With $|\Phi_1\rangle$ and $|\Phi_2\rangle$ as basis,

$$\boldsymbol{H} = \begin{pmatrix} 2d-g & -g \\ -g & 4d-g \end{pmatrix}, \tag{2.79}$$

which is the requested matrix. This is the matrix representation of an operator in a chosen basis, exactly as in Section 1.21: the Slater determinants are not eigenstates of $\hat{H}$, but they form a basis in which the eigenstates can be expanded, $|\Psi_i\rangle = c_{i0}|\Phi_1\rangle + c_{i1}|\Phi_2\rangle$.

The characteristic polynomial follows from the trace $6d-2g$ and the determinant $(2d-g)(4d-g) - g^2 = 8d^2 - 6dg$,

$$\lambda^2 + (2g-6d)\lambda + 8d^2 - 6dg = 0, \tag{2.80}$$

and solving it gives

$$\lambda_\pm = 3d - g \pm \sqrt{d^2+g^2}. \tag{2.81}$$

Substituting back, the unnormalised eigenvectors are

$$|\Psi_\mp\rangle = \frac{d \pm \sqrt{d^2+g^2}}{g}|\Phi_1\rangle + |\Phi_2\rangle, \tag{2.82}$$

or, with the dimensionless ratio $\gamma \equiv d/g$,

$$|\Psi_\mp\rangle = \left(\gamma \pm \sqrt{1+\gamma^2}\right)|\Phi_1\rangle + |\Phi_2\rangle. \tag{2.83}$$

The single parameter γ measures the strength of the interaction against the level spacing. For $g \to 0$ ($\gamma \to \infty$) the ground state becomes pure $|\Phi_1\rangle$: the single Slater determinant is exact. As g grows the admixture of $|\Phi_2\rangle$ grows with it, and at $\gamma = 0$ the two determinants enter with equal weight. This is correlation, in its simplest possible form, and it is the same story told quantitatively by the occupation numbers of Section 1.31.

Exercise 2c.

Adding $p = 3$ changes nothing structurally: the diagonal element is $2pd - g = 6d - g$ and the off-diagonal elements remain $-g$, so

$$\boldsymbol{H} = \begin{pmatrix} 2d-g & -g & -g \\ -g & 4d-g & -g \\ -g & -g & 6d-g \end{pmatrix} = g\begin{pmatrix} 2\gamma-1 & -1 & -1 \\ -1 & 4\gamma-1 & -1 \\ -1 & -1 & 6\gamma-1 \end{pmatrix}. \tag{2.84}$$

The eigenvalues may be found symbolically, or numerically with the Householder and QL solver of Sections 1.24 and 1.25 – three lines of Python. For $\gamma \gg 1$ the ground state is again dominated by $|\Phi_1\rangle$ with admixtures of order $1/\gamma$ from the two excited configurations. Continuing this construction to L levels and n pairs gives exactly the pairing model of Table 1.3, whose dimension $\binom{L}{n}$ grows fast enough that the Lanczos algorithm becomes the only practical route.

Exercise session, week 36: Condon-Slater rules and second quantisation

Exercise 1: Condon-Slater rules for expectation values.

Consider three N-particle Slater determinants $|SD\rangle$, $|SD_i^j\rangle$ and $|SD_{ij}^{kl}\rangle$, where $|SD_i^j\rangle$ differs from $|SD\rangle$ by a single single-particle state – the state ψ_i has been replaced by ψ_j – and $|SD_{ij}^{kl}\rangle$ differs by two. Define, as in week 35, a general one-body operator $\hat{F} = \sum_i^N \hat{f}(x_i)$ and a general two-body operator $\hat{G} = \sum_{i>j}^N \hat{g}(x_i, x_j)$ with $\hat{g}$ invariant under interchange of two coordinates. The single-particle states ψ_i are not necessarily eigenstates of $\hat{f}$.

a) Find $\langle SD|\hat{F}|SD\rangle$ and $\langle SD|\hat{G}|SD\rangle$.
b) Find $\langle SD|\hat{F}|SD_i^j\rangle$ and $\langle SD|\hat{G}|SD_i^j\rangle$.
c) Find $\langle SD|\hat{F}|SD_{ij}^{kl}\rangle$ and $\langle SD|\hat{G}|SD_{ij}^{kl}\rangle$. What happens to the two-body operator for a transition $\langle SD|\hat{G}|SD_{ijk}^{lmn}\rangle$, in which the determinant on the right differs by more than two single-particle states?

Exercise 2: a first encounter with second quantisation.

a) Show that the density of particles at $\boldsymbol{x}$,

$$n(\boldsymbol{x}) = N \int d\boldsymbol{x}_2 \cdots d\boldsymbol{x}_N \, |\Psi_{AS}(\boldsymbol{x}, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_N)|^2 ,$$

can be written in terms of the single-particle states as

$$n(\boldsymbol{x}) = \sum_k |\psi_k(\boldsymbol{x})|^2 .$$

b) Calculate the matrix elements $\langle \alpha_1 \alpha_2|\hat{F}|\alpha_1 \alpha_2\rangle$ and $\langle \alpha_1 \alpha_2|\hat{G}|\alpha_1 \alpha_2\rangle$ with

$$\begin{aligned}
|\alpha_1 \alpha_2\rangle &= a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger |0\rangle, \\
\hat{F} &= \sum_{\alpha\beta} \langle \alpha|f|\beta\rangle a_\alpha^\dagger a_\beta, \\
\hat{G} &= \frac{1}{2} \sum_{\alpha\beta\gamma\delta} \langle \alpha\beta|g|\gamma\delta\rangle a_\alpha^\dagger a_\beta^\dagger a_\delta a_\gamma,
\end{aligned}$$

where the matrix elements are the integrals of Eqs. (2.29), (2.34) and (2.35). Compare with exercise 1c) of week 35.

Answers, week 36

Throughout we assume the two determinants have been brought to *maximum coincidence*: the common single-particle states appear in the same order in both, so that no additional sign from reordering rows is left over. If they have not, every result below acquires the factor $(-1)^\sigma$ of the permutation needed to bring them into that form.

Exercise 1a: the diagonal elements.

These are the results already derived in Section 2.8 and in exercise 1d) of week 35,

$$\langle SD|\hat{F}|SD\rangle = \sum_{i=1}^{N}\langle i|\hat{f}|i\rangle, \qquad \langle SD|\hat{G}|SD\rangle = \frac{1}{2}\sum_{i,j=1}^{N}\langle ij|\hat{g}|ij\rangle_{\mathrm{AS}}. \tag{2.85}$$

The one-body operator sees each occupied orbital once; the two-body operator sees each occupied pair once, with the exchange term subtracted.

Exercise 1b: one differing state.

Write $|SD_i^j\rangle$ for the determinant in which ψ_i has been replaced by ψ_j, with j not among the states occupied in $|SD\rangle$. Using $\Phi = \sqrt{N!}\hat{A}\Phi_H$ and the same argument as in Section 2.8, a term survives only if every orbital not acted on by the operator overlaps with an *identical* orbital on the other side. Since exactly one orbital differs, the one-body operator must be the one that connects ψ_i to ψ_j, and every other factor contributes unity:

$$\langle SD|\hat{F}|SD_i^j\rangle = \langle i|\hat{f}|j\rangle. \tag{2.86}$$

A single term, not a sum: the one-body operator can repair exactly one mismatch.

For the two-body operator, one of the two coordinates must carry the $i \to j$ replacement while the other runs over all the orbitals common to both determinants. Both the direct and the exchange contraction survive, so

$$\langle SD|\hat{G}|SD_i^j\rangle = \sum_{k=1}^{N}\langle ik|\hat{g}|jk\rangle_{\mathrm{AS}} = \sum_{k=1}^{N}\left[\langle ik|\hat{g}|jk\rangle - \langle ik|\hat{g}|kj\rangle\right], \tag{2.87}$$

where the sum runs over the states occupied in $|SD\rangle$. The term $k=i$ contributes nothing, since $\langle ii|\hat{g}|ji\rangle_{\mathrm{AS}} = 0$ by the antisymmetry of Eq. (2.36).

Equation (2.87) is worth pausing on. It says that a two-body interaction, evaluated between determinants differing by one orbital, behaves like an *effective one-body* operator obtained by summing over the occupied states. That sum is precisely the interaction part of the Fock matrix of Eq. (2.40); the Hartree-Fock condition $\langle SD|\hat{H}|SD_i^j\rangle = 0$, that no single excitation couples to the reference, is Brillouin's theorem.

Exercise 1c: two differing states, and beyond.

The one-body operator can repair only one mismatch at a time, so with two differing orbitals at least one overlap $\langle i|k\rangle$ between different states is always left over and

$$\langle SD|\hat{F}|SD_{ij}^{kl}\rangle = 0. \tag{2.88}$$

The two-body operator can repair exactly two, and both coordinates are used up doing so. Nothing is left to sum over, and

$$\langle SD|\hat{G}|SD_{ij}^{kl}\rangle = \langle ij|\hat{g}|kl\rangle_{\mathrm{AS}} = \langle ij|\hat{g}|kl\rangle - \langle ij|\hat{g}|lk\rangle. \tag{2.89}$$

Finally, if the determinants differ by three or more single-particle states,

$$\langle SD|\hat{G}|SD_{ijk}^{lmn}\rangle = 0, \tag{2.90}$$

because a two-body operator acts on at most two coordinates and therefore cannot bridge three mismatches; at least one vanishing overlap always survives.

Equations (2.85)–(2.90) are the *Condon-Slater rules*, and the pattern is simple enough to memorise: an n-body operator connects Slater determinants differing by at most n single-particle states. Their consequence is the reason large-scale diagonalisation is feasible at all. A configuration-interaction Hamiltonian matrix contains only one- and two-body operators, so all matrix elements between determinants differing by three or more orbitals vanish identically. The matrix is therefore extremely *sparse*: of the $\binom{n}{N}^2$ possible elements only a tiny fraction are non-zero, which is exactly the structure the Lanczos algorithm of Section 1.26 was designed to exploit, and the reason the Hamiltonian need never be stored.

Exercise 2a: the density.

Expand the Slater determinant along the row belonging to the coordinate $\boldsymbol{x}$,

$$\Psi_{AS}(\boldsymbol{x},\boldsymbol{x}_2,\ldots,\boldsymbol{x}_N) = \frac{1}{\sqrt{N}}\sum_{k=1}^{N}(-)^{k+1}\psi_k(\boldsymbol{x})\,\Psi_{AS}^{(k)}(\boldsymbol{x}_2,\ldots,\boldsymbol{x}_N), \tag{2.91}$$

where $\Psi_{AS}^{(k)}$ is the normalised $(N-1)$-particle Slater determinant built from all the occupied orbitals *except* ψ_k. Because two such determinants with $k \neq k'$ differ by one orbital, they are orthogonal,

$$\int d\boldsymbol{x}_2\cdots d\boldsymbol{x}_N\,\Psi_{AS}^{(k)*}\Psi_{AS}^{(k')} = \delta_{kk'}. \tag{2.92}$$

Squaring Eq. (2.91) and integrating therefore kills every cross term and leaves

$$\int d\boldsymbol{x}_2\cdots d\boldsymbol{x}_N\,|\Psi_{AS}|^2 = \frac{1}{N}\sum_{k=1}^{N}|\psi_k(\boldsymbol{x})|^2, \tag{2.93}$$

so that

$$n(\boldsymbol{x}) = N\cdot\frac{1}{N}\sum_{k=1}^{N}|\psi_k(\boldsymbol{x})|^2 = \sum_{k=1}^{N}|\psi_k(\boldsymbol{x})|^2. \tag{2.94}$$

The density of a Slater determinant is the sum of the densities of its occupied orbitals, with no interference between them. Integrating gives $\int n(\boldsymbol{x})d\boldsymbol{x} = N$, as it must. Note that Eq. (2.94) is the diagonal of the one-body density matrix of Eq. (1.104) in coordinate space, whose eigenvalues we already know to be 0 and 1 for a single determinant.

Exercise 2b: the same results from second quantisation.

Start from the one-body operator. Acting with a_β on the two-particle state and using $\{a_\alpha, a_\beta^\dagger\} = \delta_{\alpha\beta}$,

$$a_\beta\, a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger|0\rangle = \delta_{\beta\alpha_1}a_{\alpha_2}^\dagger|0\rangle - \delta_{\beta\alpha_2}a_{\alpha_1}^\dagger|0\rangle, \tag{2.95}$$

so that

$$a_\alpha^\dagger a_\beta|\alpha_1\alpha_2\rangle = \delta_{\beta\alpha_1}a_\alpha^\dagger a_{\alpha_2}^\dagger|0\rangle - \delta_{\beta\alpha_2}a_\alpha^\dagger a_{\alpha_1}^\dagger|0\rangle. \tag{2.96}$$

Projecting on $\langle\alpha_1\alpha_2| = \langle 0|a_{\alpha_2}a_{\alpha_1}$ and using $\langle\alpha_1\alpha_2|\alpha\alpha_2\rangle = \delta_{\alpha\alpha_1}$ and $\langle\alpha_1\alpha_2|\alpha\alpha_1\rangle = -\delta_{\alpha\alpha_2}$ gives

$$\langle\alpha_1\alpha_2|a_\alpha^\dagger a_\beta|\alpha_1\alpha_2\rangle = \delta_{\alpha\alpha_1}\delta_{\beta\alpha_1} + \delta_{\alpha\alpha_2}\delta_{\beta\alpha_2}, \tag{2.97}$$

and therefore

$$\langle\alpha_1\alpha_2|\hat{F}|\alpha_1\alpha_2\rangle = \langle\alpha_1|f|\alpha_1\rangle + \langle\alpha_2|f|\alpha_2\rangle. \tag{2.98}$$

For the two-body operator, $a_\delta a_\gamma$ annihilates the state unless $\{\gamma,\delta\} = \{\alpha_1,\alpha_2\}$, and likewise $a_\alpha^\dagger a_\beta^\dagger$ must recreate the same pair. The four surviving index assignments give, after collecting the signs from the anti-commutators,

$$\langle\alpha_1\alpha_2|\hat{G}|\alpha_1\alpha_2\rangle = \langle\alpha_1\alpha_2|g|\alpha_1\alpha_2\rangle - \langle\alpha_1\alpha_2|g|\alpha_2\alpha_1\rangle = \langle\alpha_1\alpha_2|g|\alpha_1\alpha_2\rangle_{\mathrm{AS}}. \tag{2.99}$$

The factor $\frac{1}{2}$ in the definition of $\hat{G}$ is cancelled by the two equivalent assignments of the pair.

Comparison.

Equations (2.98) and (2.99) are *identical* to the results (2.75) and (2.76) obtained in week 35 by integrating over permutations of the Hartree function. This is the point of the exercise. The first-quantised route required us to write out the antisymmetriser, sort through $N!$ permutations, and argue term by term about which overlaps vanish; the second-quantised route required only the anticommutation relations $\{a_\alpha, a_\beta^\dagger\} = \delta_{\alpha\beta}$. Antisymmetry, which was an explicit and increasingly painful bookkeeping task in Section 2.8, has been built into the algebra of the operators themselves.

This is why the next chapter is devoted to second quantisation. The Slater determinant of Eq. (2.14) becomes a string of creation operators acting on the vacuum, the Condon-Slater rules of exercise 1 become elementary consequences of Wick's theorem, and calculations that are barely feasible by hand for $N = 3$ become mechanical for arbitrary N.

Chapter 3
Second quantization

Chapter 2 ended with a warning. Every expectation value there was obtained by writing out the antisymmetriser, sorting through $N!$ permutations, and arguing term by term about which overlaps vanish. For two particles this is an exercise; for the eight-operator strings of Section 2.8 it is already unpleasant; for the excited determinants that perturbation theory and coupled-cluster theory require it is impossible. Exercise 2 of the week-36 session made the point sharply: the same matrix elements fall out of the anticommutation relations in three lines.

This chapter develops that observation into a formalism. Antisymmetry, which was an explicit and increasingly costly bookkeeping task in Chapter 2, is built once and for all into the algebra of the creation and annihilation operators. The Slater determinant of Eq. (2.14) becomes a string of creation operators acting on the vacuum, the Condon-Slater rules of Section 2.6 become consequences of Wick's theorem, and calculations that were barely feasible by hand for $N = 3$ become mechanical for arbitrary N.

We introduce the time-independent operators $a_\alpha^\dagger$ and a_α which create and annihilate, respectively, a particle in the single-particle state φ_α. We define the fermion creation operator $a_\alpha^\dagger$

$$a_\alpha^\dagger|0\rangle \equiv |\alpha\rangle, \tag{3.1}$$

and

$$a_\alpha^\dagger|\alpha_1 \dots \alpha_n\rangle_{\mathrm{AS}} \equiv |\alpha\alpha_1 \dots \alpha_n\rangle_{\mathrm{AS}} \tag{3.2}$$

In Eq. (3.1) the operator $a_\alpha^\dagger$ acts on the vacuum state $|0\rangle$, which does not contain any particles. Alternatively, we could define a closed-shell nucleus or atom as our new vacuum, but then we need to introduce the particle-hole formalism, see the discussion to come.

In Eq. (3.2) $a_\alpha^\dagger$ acts on an antisymmetric n-particle state and creates an antisymmetric $(n+1)$-particle state, where the one-body state φ_α is occupied, under the condition that $\alpha \neq \alpha_1, \alpha_2, \dots, \alpha_n$. It follows that we can express an antisymmetric state as the product of the creation operators acting on the vacuum state.

$$|\alpha_1 \dots \alpha_n\rangle_{\mathrm{AS}} = a_{\alpha_1}^\dagger a_{\alpha_2}^\dagger \dots a_{\alpha_n}^\dagger|0\rangle \tag{3.3}$$

It is easy to derive the commutation and anticommutation rules for the fermionic creation operators $a_\alpha^\dagger$. Using the antisymmetry of the states (3.3)

$$|\alpha_1 \dots \alpha_i \dots \alpha_k \dots \alpha_n\rangle_{\mathrm{AS}} = -|\alpha_1 \dots \alpha_k \dots \alpha_i \dots \alpha_n\rangle_{\mathrm{AS}} \tag{3.4}$$

we obtain

$$a_{\alpha_i}^\dagger a_{\alpha_k}^\dagger = -a_{\alpha_k}^\dagger a_{\alpha_i}^\dagger \tag{3.5}$$

Using the Pauli principle

$$|\alpha_1 \dots \alpha_i \dots \alpha_i \dots \alpha_n\rangle_{\mathrm{AS}} = 0 \tag{3.6}$$

it follows that

$$a_{\alpha_i}^\dagger a_{\alpha_i}^\dagger = 0. \tag{3.7}$$

If we combine Eqs. (3.5) and (3.7), we obtain the well-known anti-commutation rule

$$a_\alpha^\dagger a_\beta^\dagger + a_\beta^\dagger a_\alpha^\dagger \equiv \{a_\alpha^\dagger, a_\beta^\dagger\} = 0 \tag{3.8}$$

The hermitian conjugate of $a_\alpha^\dagger$ is

$$a_\alpha = (a_\alpha^\dagger)^\dagger \tag{3.9}$$

If we take the hermitian conjugate of Eq. (3.8), we arrive at

$$\{a_\alpha, a_\beta\} = 0 \tag{3.10}$$

What is the physical interpretation of the operator a_α and what is the effect of a_α on a given state $|\alpha_1\alpha_2\dots\alpha_n\rangle_{\mathrm{AS}}$? Consider the following matrix element

$$\langle\alpha_1\alpha_2\dots\alpha_n|a_\alpha|\alpha_1'\alpha_2'\dots\alpha_m'\rangle \tag{3.11}$$

where both sides are antisymmetric. We distinguish between two cases. The first (1) is when $\alpha \in \{\alpha_i\}$. Using the Pauli principle of Eq. (3.6) it follows

$$\langle\alpha_1\alpha_2\dots\alpha_n|a_\alpha = 0 \tag{3.12}$$

The second (2) case is when $\alpha \notin \{\alpha_i\}$. It follows that an hermitian conjugation

$$\langle\alpha_1\alpha_2\dots\alpha_n|a_\alpha = \langle\alpha\alpha_1\alpha_2\dots\alpha_n| \tag{3.13}$$

Eq. (3.13) holds for case (1) since the lefthand side is zero due to the Pauli principle. We write Eq. (3.11) as

$$\langle\alpha_1\alpha_2\dots\alpha_n|a_\alpha|\alpha_1'\alpha_2'\dots\alpha_m'\rangle = \langle\alpha_1\alpha_2\dots\alpha_n|\alpha\alpha_1'\alpha_2'\dots\alpha_m'\rangle \tag{3.14}$$

Here we must have $m = n+1$ if Eq. (3.14) has to be trivially different from zero.

For the last case, the minus and plus signs apply when the sequence $\alpha, \alpha_1, \alpha_2, \dots, \alpha_n$ and $\alpha_1', \alpha_2', \dots, \alpha_{n+1}'$ are related to each other via even and odd permutations. If we assume that $\alpha \notin \{\alpha_i\}$ we obtain

$$\langle\alpha_1\alpha_2\dots\alpha_n|a_\alpha|\alpha_1'\alpha_2'\dots\alpha_{n+1}'\rangle = 0 \tag{3.15}$$

when $\alpha \in \{\alpha_i'\}$. If $\alpha \notin \{\alpha_i'\}$, we obtain

$$a_\alpha \underbrace{|\alpha_1'\alpha_2'\dots\alpha_{n+1}'\rangle}_{\neq\alpha} = 0 \tag{3.16}$$

and in particular

$$a_\alpha|0\rangle = 0 \tag{3.17}$$

If $\{\alpha\alpha_i\} = \{\alpha_i'\}$, performing the right permutations, the sequence $\alpha, \alpha_1, \alpha_2, \dots, \alpha_n$ is identical with the sequence $\alpha_1', \alpha_2', \dots, \alpha_{n+1}'$. This results in

$$\langle\alpha_1\alpha_2\dots\alpha_n|a_\alpha|\alpha\alpha_1\alpha_2\dots\alpha_n\rangle = 1 \tag{3.18}$$

and thus

$$a_\alpha|\alpha\alpha_1\alpha_2\dots\alpha_n\rangle = |\alpha_1\alpha_2\dots\alpha_n\rangle \tag{3.19}$$

The action of the operator a_α from the left on a state vector is to to remove one particle in the state α. If the state vector does not contain the single-particle state α, the outcome of the operation is zero. The operator a_α is normally called for a destruction or annihilation operator.

The next step is to establish the commutator algebra of $a_\alpha^\dagger$ and a_β.

The action of the anti-commutator $\{a_\alpha^\dagger, a_\alpha\}$ on a given n-particle state is

$$
\begin{aligned}
a_\alpha^\dagger a_\alpha \underbrace{|\alpha_1\alpha_2\dots\alpha_n\rangle}_{\neq\alpha} &= 0 \\
a_\alpha a_\alpha^\dagger \underbrace{|\alpha_1\alpha_2\dots\alpha_n\rangle}_{\neq\alpha} &= a_\alpha \underbrace{|\alpha\alpha_1\alpha_2\dots\alpha_n\rangle}_{\neq\alpha} = \underbrace{|\alpha_1\alpha_2\dots\alpha_n\rangle}_{\neq\alpha}
\end{aligned} \tag{3.20}
$$

if the single-particle state α is not contained in the state.

If it is present we arrive at

$$
\begin{aligned}
a_\alpha^\dagger a_\alpha |\alpha_1\alpha_2\dots\alpha_k\alpha\alpha_{k+1}\dots\alpha_{n-1}\rangle &= a_\alpha^\dagger a_\alpha (-1)^k |\alpha\alpha_1\alpha_2\dots\alpha_{n-1}\rangle \\
&= (-1)^k |\alpha\alpha_1\alpha_2\dots\alpha_{n-1}\rangle = |\alpha_1\alpha_2\dots\alpha_k\alpha\alpha_{k+1}\dots\alpha_{n-1}\rangle \\
a_\alpha a_\alpha^\dagger |\alpha_1\alpha_2\dots\alpha_k\alpha\alpha_{k+1}\dots\alpha_{n-1}\rangle &= 0
\end{aligned} \tag{3.21}
$$

From Eqs. (3.20) and (3.21) we arrive at

$$
\{a_\alpha^\dagger, a_\alpha\} = a_\alpha^\dagger a_\alpha + a_\alpha a_\alpha^\dagger = 1 \tag{3.22}
$$

The action of $\left\{a_\alpha^\dagger, a_\beta\right\}$, with $\alpha \neq \beta$ on a given state yields three possibilities. The first case is a state vector which contains both α and β, then either α or β and finally none of them.

The first case results in

$$
\begin{aligned}
a_\alpha^\dagger a_\beta |\alpha\beta\alpha_1\alpha_2\dots\alpha_{n-2}\rangle &= 0 \\
a_\beta a_\alpha^\dagger |\alpha\beta\alpha_1\alpha_2\dots\alpha_{n-2}\rangle &= 0
\end{aligned} \tag{3.23}
$$

while the second case gives

$$
\begin{aligned}
a_\alpha^\dagger a_\beta |\beta \underbrace{\alpha_1\alpha_2\dots\alpha_{n-1}}_{\neq\alpha}\rangle &= |\alpha \underbrace{\alpha_1\alpha_2\dots\alpha_{n-1}}_{\neq\alpha}\rangle \\
a_\beta a_\alpha^\dagger |\beta \underbrace{\alpha_1\alpha_2\dots\alpha_{n-1}}_{\neq\alpha}\rangle &= a_\beta |\alpha\beta \underbrace{\beta\alpha_1\alpha_2\dots\alpha_{n-1}}_{\neq\alpha}\rangle \\
&= -|\alpha \underbrace{\alpha_1\alpha_2\dots\alpha_{n-1}}_{\neq\alpha}\rangle
\end{aligned} \tag{3.24}
$$

Finally if the state vector does not contain α and β

$$
\begin{aligned}
a_\alpha^\dagger a_\beta |\underbrace{\alpha_1\alpha_2\dots\alpha_n}_{\neq\alpha,\beta}\rangle &= \qquad\qquad 0 \\
a_\beta a_\alpha^\dagger |\underbrace{\alpha_1\alpha_2\dots\alpha_n}_{\neq\alpha,\beta}\rangle &= a_\beta |\alpha \underbrace{\alpha_1\alpha_2\dots\alpha_n}_{\neq\alpha,\beta}\rangle = 0
\end{aligned} \tag{3.25}
$$

For all three cases we have

$$
\{a_\alpha^\dagger, a_\beta\} = a_\alpha^\dagger a_\beta + a_\beta a_\alpha^\dagger = 0, \quad \alpha \neq \beta \tag{3.26}
$$

We can summarize our findings in Eqs. (3.22) and (3.26) as

$$
\{a_\alpha^\dagger, a_\beta\} = \delta_{\alpha\beta} \tag{3.27}
$$

with $\delta_{\alpha\beta}$ is the Kroenecker δ-symbol.

The properties of the creation and annihilation operators can be summarized as (for fermions)

$$
a_\alpha^\dagger |0\rangle \equiv |\alpha\rangle,
$$

and

$$a^\dagger_\alpha|\alpha_1\dots\alpha_n\rangle_{\mathrm{AS}} \equiv |\alpha\alpha_1\dots\alpha_n\rangle_{\mathrm{AS}}.$$

from which follows

$$|\alpha_1\dots\alpha_n\rangle_{\mathrm{AS}} = a^\dagger_{\alpha_1}a^\dagger_{\alpha_2}\dots a^\dagger_{\alpha_n}|0\rangle.$$

The hermitian conjugate has the folowing properties

$$a_\alpha = (a^\dagger_\alpha)^\dagger.$$

Finally we found

$$a_\alpha \underbrace{|\alpha'_1\alpha'_2\dots\alpha'_{n+1}}\rangle_{\neq\alpha} = 0, \quad \text{in particular } a_\alpha|0\rangle = 0,$$

and

$$a_\alpha|\alpha\alpha_1\alpha_2\dots\alpha_n\rangle = |\alpha_1\alpha_2\dots\alpha_n\rangle,$$

and the corresponding commutator algebra

$$\{a^\dagger_\alpha, a^\dagger_\beta\} = \{a_\alpha, a_\beta\} = 0 \qquad \{a^\dagger_\alpha, a_\beta\} = \delta_{\alpha\beta}.$$

3.1 One-body operators in second quantization

The result we are about to derive was obtained once already, in Section 2.8, by integrating over permutations of the Hartree function. It is worth comparing the two routes as we go: what took a page of careful argument there follows here from the anticommutation relations alone.

A very useful operator is the so-called number-operator. Most physics cases we will study in this text conserve the total number of particles. The number operator is therefore a useful quantity which allows us to test that our many-body formalism conserves the number of particles. In for example (d,p) or (p,d) reactions it is important to be able to describe quantum mechanical states where particles get added or removed. A creation operator $a^\dagger_\alpha$ adds one particle to the single-particle state α of a give many-body state vector, while an annihilation operator a_α removes a particle from a single-particle state α.

Let us consider an operator proportional with $a^\dagger_\alpha a_\beta$ and $\alpha = \beta$. It acts on an n-particle state resulting in

$$a^\dagger_\alpha a_\alpha|\alpha_1\alpha_2\dots\alpha_n\rangle = \begin{cases} 0 & \alpha \notin \{\alpha_i\} \\ \\ |\alpha_1\alpha_2\dots\alpha_n\rangle & \alpha \in \{\alpha_i\} \end{cases} \tag{3.28}$$

Summing over all possible one-particle states we arrive at

$$\left(\sum_\alpha a^\dagger_\alpha a_\alpha\right)|\alpha_1\alpha_2\dots\alpha_n\rangle = n|\alpha_1\alpha_2\dots\alpha_n\rangle \tag{3.29}$$

The operator

$$\hat{N} = \sum_\alpha a^\dagger_\alpha a_\alpha \tag{3.30}$$

is called the number operator since it counts the number of particles in a give state vector when it acts on the different single-particle states. It acts on one single-particle state at the time and falls therefore under category one-body operators. Next we look at another important one-body operator, namely $\hat{H}_0$ and study its operator form in the occupation number representation.

We want to obtain an expression for a one-body operator which conserves the number of particles. Here we study the one-body operator for the kinetic energy plus an eventual external one-body potential. The action of this operator on a particular n-body state with its pertinent expectation value has already been studied in coordinate space. In coordinate space the operator reads

$$\hat{H}_0 = \sum_i \hat{h}_0(x_i) \tag{3.31}$$

and the anti-symmetric n-particle Slater determinant is defined as

$$\Phi(x_1, x_2, \dots, x_n, \alpha_1, \alpha_2, \dots, \alpha_n) = \frac{1}{\sqrt{n!}} \sum_p (-1)^p \hat{P} \psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\dots\psi_{\alpha_n}(x_n).$$

Defining

$$\hat{h}_0(x_i)\psi_{\alpha_i}(x_i) = \sum_{\alpha_k'} \psi_{\alpha_k'}(x_i)\langle \alpha_k'|\hat{h}_0|\alpha_k\rangle \tag{3.32}$$

we can easily evaluate the action of $\hat{H}_0$ on each product of one-particle functions in Slater determinant. From Eq. (3.32) we obtain the following result without permuting any particle pair

$$\begin{aligned}
& \left(\sum_i \hat{h}_0(x_i)\right) \psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\dots\psi_{\alpha_n}(x_n) \\
= & \sum_{\alpha_1'} \langle \alpha_1'|\hat{h}_0|\alpha_1\rangle \psi_{\alpha_1'}(x_1)\psi_{\alpha_2}(x_2)\dots\psi_{\alpha_n}(x_n) \\
+ & \sum_{\alpha_2'} \langle \alpha_2'|\hat{h}_0|\alpha_2\rangle \psi_{\alpha_1}(x_1)\psi_{\alpha_2'}(x_2)\dots\psi_{\alpha_n}(x_n) \\
+ & \dots \\
+ & \sum_{\alpha_n'} \langle \alpha_n'|\hat{h}_0|\alpha_n\rangle \psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\dots\psi_{\alpha_n'}(x_n)
\end{aligned} \tag{3.33}$$

If we interchange particles 1 and 2 we obtain

$$\begin{aligned}
& \left(\sum_i \hat{h}_0(x_i)\right) \psi_{\alpha_1}(x_2)\psi_{\alpha_1}(x_2)\dots\psi_{\alpha_n}(x_n) \\
= & \sum_{\alpha_2'} \langle \alpha_2'|\hat{h}_0|\alpha_2\rangle \psi_{\alpha_1}(x_2)\psi_{\alpha_2'}(x_1)\dots\psi_{\alpha_n}(x_n) \\
+ & \sum_{\alpha_1'} \langle \alpha_1'|\hat{h}_0|\alpha_1\rangle \psi_{\alpha_1'}(x_2)\psi_{\alpha_2}(x_1)\dots\psi_{\alpha_n}(x_n) \\
+ & \dots \\
+ & \sum_{\alpha_n'} \langle \alpha_n'|\hat{h}_0|\alpha_n\rangle \psi_{\alpha_1}(x_2)\psi_{\alpha_1}(x_2)\dots\psi_{\alpha_n'}(x_n)
\end{aligned} \tag{3.34}$$

We can continue by computing all possible permutations. We rewrite also our Slater determinant in its second quantized form and skip the dependence on the quantum numbers x_i. Summing up all contributions and taking care of all phases $(-1)^p$ we arrive at

$$\begin{aligned}
\hat{H}_0|\alpha_1, \alpha_2, \dots, \alpha_n\rangle = & \sum_{\alpha_1'} \langle \alpha_1'|\hat{h}_0|\alpha_1\rangle |\alpha_1'\alpha_2\dots\alpha_n\rangle \\
+ & \sum_{\alpha_2'} \langle \alpha_2'|\hat{h}_0|\alpha_2\rangle |\alpha_1\alpha_2'\dots\alpha_n\rangle \\
+ & \dots \\
+ & \sum_{\alpha_n'} \langle \alpha_n'|\hat{h}_0|\alpha_n\rangle |\alpha_1\alpha_2\dots\alpha_n'\rangle
\end{aligned} \tag{3.35}$$

In Eq. (3.35) we have expressed the action of the one-body operator of Eq. (3.31) on the n-body state in its second quantized form. This equation can be further manipulated if we use the properties of the creation and annihilation operator on each primed quantum number, that is

$$|\alpha_1\alpha_2\dots\alpha'_k\dots\alpha_n\rangle = a^\dagger_{\alpha'_k}a_{\alpha_k}|\alpha_1\alpha_2\dots\alpha_k\dots\alpha_n\rangle \tag{3.36}$$

Inserting this in the right-hand side of Eq. (3.35) results in

$$\begin{aligned}
\hat{H}_0|\alpha_1\alpha_2\dots\alpha_n\rangle = & \sum_{\alpha'_1}\langle\alpha'_1|\hat{h}_0|\alpha_1\rangle a^\dagger_{\alpha'_1}a_{\alpha_1}|\alpha_1\alpha_2\dots\alpha_n\rangle \\
+ & \sum_{\alpha'_2}\langle\alpha'_2|\hat{h}_0|\alpha_2\rangle a^\dagger_{\alpha'_2}a_{\alpha_2}|\alpha_1\alpha_2\dots\alpha_n\rangle \\
+ & \dots \\
+ & \sum_{\alpha'_n}\langle\alpha'_n|\hat{h}_0|\alpha_n\rangle a^\dagger_{\alpha'_n}a_{\alpha_n}|\alpha_1\alpha_2\dots\alpha_n\rangle \\
= & \sum_{\alpha,\beta}\langle\alpha|\hat{h}_0|\beta\rangle a^\dagger_\alpha a_\beta|\alpha_1\alpha_2\dots\alpha_n\rangle
\end{aligned} \tag{3.37}$$

In the number occupation representation or second quantization we get the following expression for a one-body operator which conserves the number of particles

$$\hat{H}_0 = \sum_{\alpha\beta}\langle\alpha|\hat{h}_0|\beta\rangle a^\dagger_\alpha a_\beta \tag{3.38}$$

Obviously, $\hat{H}_0$ can be replaced by any other one-body operator which preserved the number of particles. The stucture of the operator is therefore not limited to say the kinetic or single-particle energy only.

The opearator $\hat{H}_0$ takes a particle from the single-particle state β to the single-particle state α with a probability for the transition given by the expectation value $\langle\alpha|\hat{h}_0|\beta\rangle$.

It is instructive to verify Eq. (3.38) by computing the expectation value of $\hat{H}_0$ between two single-particle states

$$\langle\alpha_1|\hat{h}_0|\alpha_2\rangle = \sum_{\alpha\beta}\langle\alpha|\hat{h}_0|\beta\rangle\langle 0|a_{\alpha_1}a^\dagger_\alpha a_\beta a^\dagger_{\alpha_2}|0\rangle \tag{3.39}$$

Using the commutation relations for the creation and annihilation operators we have

$$a_{\alpha_1}a^\dagger_\alpha a_\beta a^\dagger_{\alpha_2} = (\delta_{\alpha\alpha_1} - a^\dagger_\alpha a_{\alpha_1})(\delta_{\beta\alpha_2} - a^\dagger_{\alpha_2}a_\beta), \tag{3.40}$$

which results in

$$\langle 0|a_{\alpha_1}a^\dagger_\alpha a_\beta a^\dagger_{\alpha_2}|0\rangle = \delta_{\alpha\alpha_1}\delta_{\beta\alpha_2} \tag{3.41}$$

and

$$\langle\alpha_1|\hat{h}_0|\alpha_2\rangle = \sum_{\alpha\beta}\langle\alpha|\hat{h}_0|\beta\rangle\delta_{\alpha\alpha_1}\delta_{\beta\alpha_2} = \langle\alpha_1|\hat{h}_0|\alpha_2\rangle \tag{3.42}$$

3.2 Two-body operators in second quantization

The antisymmetrised matrix elements $\langle pq|\hat{v}|rs\rangle_{AS}$ that appear below are exactly those of Eq. (2.36), and the n^4 of them are the storage problem discussed in Section 2.10. Nothing in this chapter changes that count; what changes is the ease with which they are contracted.

Let us now derive the expression for our two-body interaction part, which also conserves the number of particles. We can proceed in exactly the same way as for the one-body operator. In the coordinate representation our two-body interaction part takes the following expression

$$\hat{H}_I = \sum_{i<j}V(x_i,x_j) \tag{3.43}$$

where the summation runs over distinct pairs. The term V can be an interaction model for the nucleon-nucleon interaction or the interaction between two electrons. It can also include additional two-body interaction terms.

The action of this operator on a product of two single-particle functions is defined as

$$V(x_i,x_j)\psi_{\alpha_k}(x_i)\psi_{\alpha_l}(x_j) = \sum_{\alpha'_k\alpha'_l}\psi'_{\alpha_k}(x_i)\psi'_{\alpha_l}(x_j)\langle\alpha'_k\alpha'_l|\hat{v}|\alpha_k\alpha_l\rangle \tag{3.44}$$

We can now let $\hat{H}_I$ act on all terms in the linear combination for $|\alpha_1\alpha_2\ldots\alpha_n\rangle$. Without any permutations we have

$$\begin{aligned}
&\left(\sum_{i<j}V(x_i,x_j)\right)\psi_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\ldots\psi_{\alpha_n}(x_n)\\
=&\sum_{\alpha'_1\alpha'_2}\langle\alpha'_1\alpha'_2|\hat{v}|\alpha_1\alpha_2\rangle\psi'_{\alpha_1}(x_1)\psi'_{\alpha_2}(x_2)\ldots\psi_{\alpha_n}(x_n)\\
+&\ldots\\
+&\sum_{\alpha'_1\alpha'_n}\langle\alpha'_1\alpha'_n|\hat{v}|\alpha_1\alpha_n\rangle\psi'_{\alpha_1}(x_1)\psi_{\alpha_2}(x_2)\ldots\psi'_{\alpha_n}(x_n)\\
+&\ldots\\
+&\sum_{\alpha'_2\alpha'_n}\langle\alpha'_2\alpha'_n|\hat{v}|\alpha_2\alpha_n\rangle\psi_{\alpha_1}(x_1)\psi'_{\alpha_2}(x_2)\ldots\psi'_{\alpha_n}(x_n)\\
+&\ldots
\end{aligned} \tag{3.45}$$

where on the rhs we have a term for each distinct pairs.

For the other terms on the rhs we obtain similar expressions and summing over all terms we obtain

$$\begin{aligned}
H_I|\alpha_1\alpha_2\ldots\alpha_n\rangle =& \sum_{\alpha'_1,\alpha'_2}\langle\alpha'_1\alpha'_2|\hat{v}|\alpha_1\alpha_2\rangle|\alpha'_1\alpha'_2\ldots\alpha_n\rangle\\
+&\ldots\\
+&\sum_{\alpha'_1,\alpha'_n}\langle\alpha'_1\alpha'_n|\hat{v}|\alpha_1\alpha_n\rangle|\alpha'_1\alpha_2\ldots\alpha'_n\rangle\\
+&\ldots\\
+&\sum_{\alpha'_2,\alpha'_n}\langle\alpha'_2\alpha'_n|\hat{v}|\alpha_2\alpha_n\rangle|\alpha_1\alpha'_2\ldots\alpha'_n\rangle\\
+&\ldots
\end{aligned} \tag{3.46}$$

We introduce second quantization via the relation

$$\begin{aligned}
&a^\dagger_{\alpha'_k}a^\dagger_{\alpha'_l}a_{\alpha_l}a_{\alpha_k}|\alpha_1\alpha_2\ldots\alpha_k\ldots\alpha_l\ldots\alpha_n\rangle\\
=&(-1)^{k-1}(-1)^{l-2}a^\dagger_{\alpha'_k}a^\dagger_{\alpha'_l}a_{\alpha_l}a_{\alpha_k}|\alpha_k\alpha_l\underbrace{\alpha_1\alpha_2\ldots\alpha_n}_{\neq\alpha_k,\alpha_l}\rangle\\
=&(-1)^{k-1}(-1)^{l-2}|\alpha'_k\alpha'_l\underbrace{\alpha_1\alpha_2\ldots\alpha_n}_{\neq\alpha'_k,\alpha'_l}\rangle\\
=&|\alpha_1\alpha_2\ldots\alpha'_k\ldots\alpha'_l\ldots\alpha_n\rangle
\end{aligned} \tag{3.47}$$

Inserting this in (3.46) gives

$$
\begin{aligned}
H_I|\alpha_1\alpha_2\dots\alpha_n\rangle = & \sum_{\alpha_1',\alpha_2'} \langle \alpha_1'\alpha_2'|\hat{v}|\alpha_1\alpha_2\rangle a^\dagger_{\alpha_1'} a^\dagger_{\alpha_2'} a_{\alpha_2} a_{\alpha_1} |\alpha_1\alpha_2\dots\alpha_n\rangle \\
+ & \dots \\
= & \sum_{\alpha_1',\alpha_n'} \langle \alpha_1'\alpha_n'|\hat{v}|\alpha_1\alpha_n\rangle a^\dagger_{\alpha_1'} a^\dagger_{\alpha_n'} a_{\alpha_n} a_{\alpha_1} |\alpha_1\alpha_2\dots\alpha_n\rangle \\
+ & \dots \\
= & \sum_{\alpha_2',\alpha_n'} \langle \alpha_2'\alpha_n'|\hat{v}|\alpha_2\alpha_n\rangle a^\dagger_{\alpha_2'} a^\dagger_{\alpha_n'} a_{\alpha_n} a_{\alpha_2} |\alpha_1\alpha_2\dots\alpha_n\rangle \\
+ & \dots \\
= & \sum_{\alpha,\beta,\gamma,\delta}{}' \langle \alpha\beta|\hat{v}|\gamma\delta\rangle a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma |\alpha_1\alpha_2\dots\alpha_n\rangle
\end{aligned} \tag{3.48}
$$

Here we let $\sum'$ indicate that the sums running over α and β run over all single-particle states, while the summations γ and δ run over all pairs of single-particle states. We wish to remove this restriction and since

$$
\langle \alpha\beta|\hat{v}|\gamma\delta\rangle = \langle \beta\alpha|\hat{v}|\delta\gamma\rangle \tag{3.49}
$$

we get

$$
\sum_{\alpha\beta} \langle \alpha\beta|\hat{v}|\gamma\delta\rangle a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma = \sum_{\alpha\beta} \langle \beta\alpha|\hat{v}|\delta\gamma\rangle a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma \tag{3.50}
$$

$$
= \sum_{\alpha\beta} \langle \beta\alpha|\hat{v}|\delta\gamma\rangle a^\dagger_\beta a^\dagger_\alpha a_\gamma a_\delta \tag{3.51}
$$

where we have used the anti-commutation rules.

Changing the summation indices α and β in (3.51) we obtain

$$
\sum_{\alpha\beta} \langle \alpha\beta|\hat{v}|\gamma\delta\rangle a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma = \sum_{\alpha\beta} \langle \alpha\beta|\hat{v}|\delta\gamma\rangle a^\dagger_\alpha a^\dagger_\beta a_\gamma a_\delta \tag{3.52}
$$

From this it follows that the restriction on the summation over γ and δ can be removed if we multiply with a factor $\frac{1}{2}$, resulting in

$$
\hat{H}_I = \frac{1}{2} \sum_{\alpha\beta\gamma\delta} \langle \alpha\beta|\hat{v}|\gamma\delta\rangle a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma \tag{3.53}
$$

where we sum freely over all single-particle states α, β, γ og δ.

With this expression we can now verify that the second quantization form of $\hat{H}_I$ in Eq. (3.53) results in the same matrix between two anti-symmetrized two-particle states as its corresponding coordinate space representation. We have

$$
\langle \alpha_1\alpha_2|\hat{H}_I|\beta_1\beta_2\rangle = \frac{1}{2} \sum_{\alpha\beta\gamma\delta} \langle \alpha\beta|\hat{v}|\gamma\delta\rangle \langle 0|a_{\alpha_2} a_{\alpha_1} a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma a^\dagger_{\beta_1} a^\dagger_{\beta_2}|0\rangle. \tag{3.54}
$$

Using the commutation relations we get

$$
\begin{aligned}
& a_{\alpha_2} a_{\alpha_1} a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma a^\dagger_{\beta_1} a^\dagger_{\beta_2} \\
= & a_{\alpha_2} a_{\alpha_1} a^\dagger_\alpha a^\dagger_\beta (a_\delta \delta_{\gamma\beta_1} a^\dagger_{\beta_2} - a_\delta a^\dagger_{\beta_1} a_\gamma a^\dagger_{\beta_2}) \\
= & a_{\alpha_2} a_{\alpha_1} a^\dagger_\alpha a^\dagger_\beta (\delta_{\gamma\beta_1}\delta_{\delta\beta_2} - \delta_{\gamma\beta_1} a^\dagger_{\beta_2} a_\delta - a_\delta a^\dagger_{\beta_1} \delta_{\gamma\beta_2} + a_\delta a^\dagger_{\beta_1} a^\dagger_{\beta_2} a_\gamma) \\
= & a_{\alpha_2} a_{\alpha_1} a^\dagger_\alpha a^\dagger_\beta (\delta_{\gamma\beta_1}\delta_{\delta\beta_2} - \delta_{\gamma\beta_1} a^\dagger_{\beta_2} a_\delta \\
& - \delta_{\delta\beta_1}\delta_{\gamma\beta_2} + \delta_{\gamma\beta_2} a^\dagger_{\beta_1} a_\delta + a_\delta a^\dagger_{\beta_1} a^\dagger_{\beta_2} a_\gamma)
\end{aligned} \tag{3.55}
$$

The vacuum expectation value of this product of operators becomes

$$\begin{aligned} & \langle 0|a_{\alpha_2}a_{\alpha_1}a^{\dagger}_{\alpha}a^{\dagger}_{\beta}a_{\delta}a_{\gamma}a^{\dagger}_{\beta_1}a^{\dagger}_{\beta_2}|0\rangle \\ = & (\delta_{\gamma\beta_1}\delta_{\delta\beta_2}-\delta_{\delta\beta_1}\delta_{\gamma\beta_2})\langle 0|a_{\alpha_2}a_{\alpha_1}a^{\dagger}_{\alpha}a^{\dagger}_{\beta}|0\rangle \\ = & (\delta_{\gamma\beta_1}\delta_{\delta\beta_2}-\delta_{\delta\beta_1}\delta_{\gamma\beta_2})(\delta_{\alpha\alpha_1}\delta_{\beta\alpha_2}-\delta_{\beta\alpha_1}\delta_{\alpha\alpha_2}) \end{aligned} \tag{3.56}$$

Insertion of Eq. (3.56) in Eq. (3.54) results in

$$\begin{aligned} \langle\alpha_1\alpha_2|\hat{H}_I|\beta_1\beta_2\rangle = & \frac{1}{2}\left[\langle\alpha_1\alpha_2|\hat{v}|\beta_1\beta_2\rangle-\langle\alpha_1\alpha_2|\hat{v}|\beta_2\beta_1\rangle\right. \\ & \left.-\langle\alpha_2\alpha_1|\hat{v}|\beta_1\beta_2\rangle+\langle\alpha_2\alpha_1|\hat{v}|\beta_2\beta_1\rangle\right] \\ = & \langle\alpha_1\alpha_2|\hat{v}|\beta_1\beta_2\rangle-\langle\alpha_1\alpha_2|\hat{v}|\beta_2\beta_1\rangle \\ = & \langle\alpha_1\alpha_2|\hat{v}|\beta_1\beta_2\rangle_{\mathrm{AS}}. \end{aligned} \tag{3.57}$$

The two-body operator can also be expressed in terms of the anti-symmetrized matrix elements we discussed previously as

$$\begin{aligned} \hat{H}_I = & \frac{1}{2}\sum_{\alpha\beta\gamma\delta}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle a^{\dagger}_{\alpha}a^{\dagger}_{\beta}a_{\delta}a_{\gamma} \\ = & \frac{1}{4}\sum_{\alpha\beta\gamma\delta}\left[\langle\alpha\beta|\hat{v}|\gamma\delta\rangle-\langle\alpha\beta|\hat{v}|\delta\gamma\rangle\right]a^{\dagger}_{\alpha}a^{\dagger}_{\beta}a_{\delta}a_{\gamma} \\ = & \frac{1}{4}\sum_{\alpha\beta\gamma\delta}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}}a^{\dagger}_{\alpha}a^{\dagger}_{\beta}a_{\delta}a_{\gamma} \end{aligned} \tag{3.58}$$

The factors in front of the operator, either $\frac{1}{4}$ or $\frac{1}{2}$ tells whether we use antisymmetrized matrix elements or not.

We can now express the Hamiltonian operator for a many-fermion system in the occupation basis representation as

$$H=\sum_{\alpha,\beta}\langle\alpha|\hat{t}+\hat{u}_{\mathrm{ext}}|\beta\rangle a^{\dagger}_{\alpha}a_{\beta}+\frac{1}{4}\sum_{\alpha\beta\gamma\delta}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle a^{\dagger}_{\alpha}a^{\dagger}_{\beta}a_{\delta}a_{\gamma}. \tag{3.59}$$

This is the form we will use in the rest of these lectures, assuming that we work with anti-symmetrized two-body matrix elements.

3.3 Particle-hole formalism

The index conventions used here – i,j,k,l for hole states below the Fermi level, a,b,c,d for particle states above it, and p,q,r,s for either – were introduced in Section 2.6. We now give them their proper setting: the reference determinant becomes a new vacuum, and the operators are redefined with respect to it.

Second quantization is a useful and elegant formalism for constructing many-body states and quantum mechanical operators. One can express and translate many physical processes into simple pictures such as Feynman diagrams. Expecation values of many-body states are also easily calculated. However, although the equations are seemingly easy to set up, from a practical point of view, that is the solution of Schroedinger's equation, there is no particular gain. The many-body equation is equally hard to solve, irrespective of representation. The cliche that there is no free lunch brings us down to earth again. Note however that a transformation to a particular basis, for cases where the interaction obeys specific symmetries, can ease the solution of Schroedinger's equation.

But there is at least one important case where second quantization comes to our rescue. It is namely easy to introduce another reference state than the pure vacuum $|0\rangle$, where all single-particle states are active. With many particles present it is often useful to introduce another reference state than the vacuum state$|0\rangle$.

We will label this state $|c\rangle$ (c for core) and as we will see it can reduce considerably the complexity and thereby the dimensionality of the many-body problem. It allows us to sum up to infinite order specific many-body correlations. The particle-hole representation is one of these handy representations.

In the original particle representation these states are products of the creation operators $a^\dagger_{\alpha_i}$ acting on the true vacuum $|0\rangle$. Following Eq. (3.3) we have

$$|\alpha_1\alpha_2\dots\alpha_{n-1}\alpha_n\rangle = a^\dagger_{\alpha_1}a^\dagger_{\alpha_2}\dots a^\dagger_{\alpha_{n-1}}a^\dagger_{\alpha_n}|0\rangle \tag{3.60}$$

$$|\alpha_1\alpha_2\dots\alpha_{n-1}\alpha_n\alpha_{n+1}\rangle = a^\dagger_{\alpha_1}a^\dagger_{\alpha_2}\dots a^\dagger_{\alpha_{n-1}}a^\dagger_{\alpha_n}a^\dagger_{\alpha_{n+1}}|0\rangle \tag{3.61}$$

$$|\alpha_1\alpha_2\dots\alpha_{n-1}\rangle = a^\dagger_{\alpha_1}a^\dagger_{\alpha_2}\dots a^\dagger_{\alpha_{n-1}}|0\rangle \tag{3.62}$$

If we use Eq. (3.60) as our new reference state, we can simplify considerably the representation of this state

$$|c\rangle \equiv |\alpha_1\alpha_2\dots\alpha_{n-1}\alpha_n\rangle = a^\dagger_{\alpha_1}a^\dagger_{\alpha_2}\dots a^\dagger_{\alpha_{n-1}}a^\dagger_{\alpha_n}|0\rangle \tag{3.63}$$

The new reference states for the $n+1$ and $n-1$ states can then be written as

$$|\alpha_1\alpha_2\dots\alpha_{n-1}\alpha_n\alpha_{n+1}\rangle = (-1)^n a^\dagger_{\alpha_{n+1}}|c\rangle \equiv (-1)^n|\alpha_{n+1}\rangle_c \tag{3.64}$$

$$|\alpha_1\alpha_2\dots\alpha_{n-1}\rangle = (-1)^{n-1}a_{\alpha_n}|c\rangle \equiv (-1)^{n-1}|\alpha_{n-1}\rangle_c \tag{3.65}$$

The first state has one additional particle with respect to the new vacuum state $|c\rangle$ and is normally referred to as a one-particle state or one particle added to the many-body reference state. The second state has one particle less than the reference vacuum state $|c\rangle$ and is referred to as a one-hole state. When dealing with a new reference state it is often convenient to introduce new creation and annihilation operators since we have from Eq. (3.65)

$$a_\alpha|c\rangle \neq 0 \tag{3.66}$$

since α is contained in $|c\rangle$, while for the true vacuum we have $a_\alpha|0\rangle = 0$ for all α.

The new reference state leads to the definition of new creation and annihilation operators which satisfy the following relations

$$b_\alpha|c\rangle = 0 \tag{3.67}$$

$$\{b^\dagger_\alpha, b^\dagger_\beta\} = \{b_\alpha, b_\beta\} = 0$$

$$\{b^\dagger_\alpha, b_\beta\} = \delta_{\alpha\beta} \tag{3.68}$$

We assume also that the new reference state is properly normalized

$$\langle c|c\rangle = 1 \tag{3.69}$$

The physical interpretation of these new operators is that of so-called quasiparticle states. This means that a state defined by the addition of one extra particle to a reference state $|c\rangle$ may not necesseraly be interpreted as one particle coupled to a core. We define now new creation operators that act on a state α creating a new quasiparticle state

$$b^\dagger_\alpha|c\rangle = \begin{cases} a^\dagger_\alpha|c\rangle = |\alpha\rangle, & \alpha > F \\ \\ a_\alpha|c\rangle = |\alpha^{-1}\rangle, & \alpha \leq F \end{cases} \tag{3.70}$$

where F is the Fermi level representing the last occupied single-particle orbit of the new reference state $|c\rangle$.

The annihilation is the hermitian conjugate of the creation operator

$$b_\alpha = (b^\dagger_\alpha)^\dagger,$$

resulting in

$$b_\alpha^\dagger = \begin{cases} a_\alpha^\dagger & \alpha > F \\ \\ a_\alpha & \alpha \leq F \end{cases} \qquad b_\alpha = \begin{cases} a_\alpha & \alpha > F \\ \\ a_\alpha^\dagger & \alpha \leq F \end{cases} \tag{3.71}$$

With the new creation and annihilation operator we can now construct many-body quasiparticle states, with one-particle-one-hole states, two-particle-two-hole states etc in the same fashion as we previously constructed many-particle states. We can write a general particle-hole state as

$$|\beta_1\beta_2\dots\beta_{n_p}\gamma_1^{-1}\gamma_2^{-1}\dots\gamma_{n_h}^{-1}\rangle \equiv \underbrace{b_{\beta_1}^\dagger b_{\beta_2}^\dagger\dots b_{\beta_{n_p}}^\dagger}_{>F}\underbrace{b_{\gamma_1}^\dagger b_{\gamma_2}^\dagger\dots b_{\gamma_{n_h}}^\dagger}_{\leq F}|c\rangle \tag{3.72}$$

We can now rewrite our one-body and two-body operators in terms of the new creation and annihilation operators. The number operator becomes

$$\hat{N} = \sum_\alpha a_\alpha^\dagger a_\alpha = \sum_{\alpha>F} b_\alpha^\dagger b_\alpha + n_c - \sum_{\alpha\leq F} b_\alpha^\dagger b_\alpha \tag{3.73}$$

where n_c is the number of particle in the new vacuum state $|c\rangle$. The action of $\hat{N}$ on a many-body state results in

$$N|\beta_1\beta_2\dots\beta_{n_p}\gamma_1^{-1}\gamma_2^{-1}\dots\gamma_{n_h}^{-1}\rangle = (n_p+n_c-n_h)|\beta_1\beta_2\dots\beta_{n_p}\gamma_1^{-1}\gamma_2^{-1}\dots\gamma_{n_h}^{-1}\rangle \tag{3.74}$$

Here $n = n_p + n_c - n_h$ is the total number of particles in the quasi-particle state of Eq. (3.72). Note that $\hat{N}$ counts the total number of particles present

$$N_{qp} = \sum_\alpha b_\alpha^\dagger b_\alpha, \tag{3.75}$$

gives us the number of quasi-particles as can be seen by computing

$$N_{qp} = |\beta_1\beta_2\dots\beta_{n_p}\gamma_1^{-1}\gamma_2^{-1}\dots\gamma_{n_h}^{-1}\rangle = (n_p+n_h)|\beta_1\beta_2\dots\beta_{n_p}\gamma_1^{-1}\gamma_2^{-1}\dots\gamma_{n_h}^{-1}\rangle \tag{3.76}$$

where $n_{qp} = n_p + n_h$ is the total number of quasi-particles.

We express the one-body operator $\hat{H}_0$ in terms of the quasi-particle creation and annihilation operators, resulting in

$$\begin{aligned}\hat{H}_0 = & \sum_{\alpha\beta>F}\langle\alpha|\hat{h}_0|\beta\rangle b_\alpha^\dagger b_\beta + \sum_{\alpha>F,\beta\leq F}\left[\langle\alpha|\hat{h}_0|\beta\rangle b_\alpha^\dagger b_\beta^\dagger + \langle\beta|\hat{h}_0|\alpha\rangle b_\beta b_\alpha\right] \\ + & \sum_{\alpha\leq F}\langle\alpha|\hat{h}_0|\alpha\rangle - \sum_{\alpha\beta\leq F}\langle\beta|\hat{h}_0|\alpha\rangle b_\alpha^\dagger b_\beta \end{aligned} \tag{3.77}$$

The first term gives contribution only for particle states, while the last one contributes only for holestates. The second term can create or destroy a set of quasi-particles and the third term is the contribution from the vacuum state $|c\rangle$.

Before we continue with the expressions for the two-body operator, we introduce a nomenclature we will use for the rest of this text. It is inspired by the notation used in quantum chemistry. We reserve the labels $i,j,k,\dots$ for hole states and $a,b,c,\dots$ for states above F, viz. particle states. This means also that we will skip the constraint $\leq F$ or $> F$ in the summation symbols. Our operator $\hat{H}_0$ reads now

$$\begin{aligned}\hat{H}_0 = & \sum_{ab}\langle a|\hat{h}|b\rangle b_a^\dagger b_b + \sum_{ai}\left[\langle a|\hat{h}|i\rangle b_a^\dagger b_i^\dagger + \langle i|\hat{h}|a\rangle b_i b_a\right] \\ + & \sum_i\langle i|\hat{h}|i\rangle - \sum_{ij}\langle j|\hat{h}|i\rangle b_i^\dagger b_j \end{aligned} \tag{3.78}$$

The two-particle operator in the particle-hole formalism is more complicated since we have to translate four indices $\alpha\beta\gamma\delta$ to the possible combinations of particle and hole states. When performing the commutator algebra we can regroup the operator in five different terms

$$\hat{H}_I = \hat{H}_I^{(a)} + \hat{H}_I^{(b)} + \hat{H}_I^{(c)} + \hat{H}_I^{(d)} + \hat{H}_I^{(e)} \tag{3.79}$$

Using anti-symmetrized matrix elements, bthe term $\hat{H}_I^{(a)}$ is

$$\hat{H}_I^{(a)} = \frac{1}{4}\sum_{abcd} \langle ab|\hat{V}|cd\rangle b_a^\dagger b_b^\dagger b_d b_c \tag{3.80}$$

The next term $\hat{H}_I^{(b)}$ reads

$$\hat{H}_I^{(b)} = \frac{1}{4}\sum_{abci} \left(\langle ab|\hat{V}|ci\rangle b_a^\dagger b_b^\dagger b_i^\dagger b_c + \langle ai|\hat{V}|cb\rangle b_a^\dagger b_i b_b b_c \right) \tag{3.81}$$

This term conserves the number of quasiparticles but creates or removes a three-particle-one-hole state. For $\hat{H}_I^{(c)}$ we have

$$\begin{aligned}\hat{H}_I^{(c)} = \quad & \frac{1}{4}\sum_{abij} \left(\langle ab|\hat{V}|ij\rangle b_a^\dagger b_b^\dagger b_j^\dagger b_i^\dagger + \langle ij|\hat{V}|ab\rangle b_a b_b b_j b_i \right) + \\ & \frac{1}{2}\sum_{abij} \langle ai|\hat{V}|bj\rangle b_a^\dagger b_j^\dagger b_b b_i + \frac{1}{2}\sum_{abi} \langle ai|\hat{V}|bi\rangle b_a^\dagger b_b.\end{aligned} \tag{3.82}$$

The first line stands for the creation of a two-particle-two-hole state, while the second line represents the creation to two one-particle-one-hole pairs while the last term represents a contribution to the particle single-particle energy from the hole states, that is an interaction between the particle states and the hole states within the new vacuum state. The fourth term reads

$$\begin{aligned}\hat{H}_I^{(d)} = \quad & \frac{1}{4}\sum_{aijk} \left(\langle ai|\hat{V}|jk\rangle b_a^\dagger b_k^\dagger b_j^\dagger b_i + \langle ji|\hat{V}|ak\rangle b_k^\dagger b_j b_i b_a \right) + \\ & \frac{1}{4}\sum_{aij} \left(\langle ai|\hat{V}|ji\rangle b_a^\dagger b_j^\dagger + \langle ji|\hat{V}|ai\rangle - \langle ji|\hat{V}|ia\rangle b_j b_a \right).\end{aligned} \tag{3.83}$$

The terms in the first line stand for the creation of a particle-hole state interacting with hole states, we will label this as a two-hole-one-particle contribution. The remaining terms are a particle-hole state interacting with the holes in the vacuum state. Finally we have

$$\hat{H}_I^{(e)} = \frac{1}{4}\sum_{ijkl} \langle kl|\hat{V}|ij\rangle b_i^\dagger b_j^\dagger b_l b_k + \frac{1}{2}\sum_{ijk} \langle ij|\hat{V}|kj\rangle b_k^\dagger b_i + \frac{1}{2}\sum_{ij} \langle ij|\hat{V}|ij\rangle \tag{3.84}$$

The first terms represents the interaction between two holes while the second stands for the interaction between a hole and the remaining holes in the vacuum state. It represents a contribution to single-hole energy to first order. The last term collects all contributions to the energy of the ground state of a closed-shell system arising from hole-hole correlations.

3.4 Summarizing and defining a normal-ordered Hamiltonian

Everything collected here has an exact counterpart in Chapter 2. The Slater determinant is Eq. (2.14); the antisymmetrised two-body matrix element is Eq. (2.36); and the energy functional that follows from the normal-ordered Hamiltonian is Eq. (2.37), whose minimisation gives the Hartree-Fock equations (2.41). The difference is that the derivations there ran over permutations, while here they will run over contractions.

$$\Phi_{AS}(\alpha_1, \dots, \alpha_A; x_1, \dots x_A) = \frac{1}{\sqrt{A}} \sum_{\hat{P}} (-1)^P \hat{P} \prod_{i=1}^{A} \psi_{\alpha_i}(x_i),$$

which is equivalent with $|\alpha_1 \dots \alpha_A\rangle = a^\dagger_{\alpha_1} \dots a^\dagger_{\alpha_A}|0\rangle$. We have also

$$a^\dagger_p|0\rangle = |p\rangle, \quad a_p|q\rangle = \delta_{pq}|0\rangle$$

$$\delta_{pq} = \left\{a_p, a^\dagger_q\right\},$$

and

$$0 = \left\{a^\dagger_p, a_q\right\} = \left\{a_p, a_q\right\} = \left\{a^\dagger_p, a^\dagger_q\right\}$$

$$|\Phi_0\rangle = |\alpha_1 \dots \alpha_A\rangle, \quad \alpha_1, \dots, \alpha_A \leq \alpha_F$$

$$\left\{a^\dagger_p, a_q\right\} = \delta_{pq}, p, q \leq \alpha_F$$

$$\left\{a_p, a^\dagger_q\right\} = \delta_{pq}, p, q > \alpha_F$$

with $i, j, \dots \leq \alpha_F, \quad a, b, \dots > \alpha_F, \quad p, q, \dots -$ any

$$a_i|\Phi_0\rangle = |\Phi_i\rangle, \qquad a^\dagger_a|\Phi_0\rangle = |\Phi^a\rangle$$

and

$$a^\dagger_i|\Phi_0\rangle = 0 \qquad a_a|\Phi_0\rangle = 0$$

The one-body operator is defined as

$$\hat{F} = \sum_{pq} \langle p|\hat{f}|q\rangle a^\dagger_p a_q$$

while the two-body opreator is defined as

$$\hat{V} = \frac{1}{4} \sum_{pqrs} \langle pq|\hat{v}|rs\rangle_{AS} a^\dagger_p a^\dagger_q a_s a_r$$

where we have defined the antisymmetric matrix elements

$$\langle pq|\hat{v}|rs\rangle_{AS} = \langle pq|\hat{v}|rs\rangle - \langle pq|\hat{v}|sr\rangle.$$

We can also define a three-body operator

$$\hat{V}_3 = \frac{1}{36} \sum_{pqrstu} \langle pqr|\hat{v}_3|stu\rangle_{AS} a^\dagger_p a^\dagger_q a^\dagger_r a_u a_t a_s$$

with the antisymmetrized matrix element

$$\langle pqr|\hat{v}_3|stu\rangle_{AS} = \langle pqr|\hat{v}_3|stu\rangle + \langle pqr|\hat{v}_3|tus\rangle + \langle pqr|\hat{v}_3|ust\rangle - \langle pqr|\hat{v}_3|sut\rangle - \langle pqr|\hat{v}_3|tsu\rangle - \langle pqr|\hat{v}_3|uts\rangle. \tag{3.85}$$

3.5 Normal ordering

The manipulations of the previous sections all follow the same pattern. Take a product of creation and annihilation operators, push the annihilation operators to the right using the anticommutation relations until they meet the vacuum and give zero, and collect the Kronecker deltas that fall out along the way. For two operators this is immediate. For the four operators of an overlap between two-particle states it is already an exercise in care, and for the eight or more operators that arise in perturbation theory and coupled-cluster theory it is hopeless. Wick's theorem organises the whole calculation once and for all. We need two definitions first.

A product of creation and annihilation operators is in *normal-ordered* form when all creation operators stand to the left of all annihilation operators. For an arbitrary product we define the normal-ordered product

$$N\left[xyz\cdots w\right] = (-1)^{\kappa}\Big(\text{the same operators, all creation operators to the left,}\Big) \tag{3.86}$$

where κ counts the interchanges of neighbouring operators needed to bring the product into that form. Every interchange of two fermion operators costs a minus sign, and the factor $(-1)^{\kappa}$ records them. Two elementary cases are

$$N\left[a_p a_q^{\dagger}\right] = -a_q^{\dagger} a_p, \qquad N\left[a_p^{\dagger} a_q\right] = a_p^{\dagger} a_q, \tag{3.87}$$

the first requiring one interchange and the second none. The ordering of the creation operators among themselves, and of the annihilation operators among themselves, is a matter of convention as long as the same sign rule is applied consistently.

The one property that makes normal ordering useful follows immediately from $a_p|0\rangle = 0$ and $\langle 0|a_p^{\dagger} = 0$:

$$\boxed{\langle 0|N\left[xyz\cdots w\right]|0\rangle = 0} \qquad \text{for every non-empty product.} \tag{3.88}$$

Whatever else it contains, a normal-ordered product has an annihilation operator on the far right or a creation operator on the far left, and either one kills the vacuum. A normal-ordered product therefore contributes nothing to a vacuum expectation value. The whole problem of evaluating $\langle 0|xyz\cdots w|0\rangle$ reduces to finding the pieces that are *not* normal-ordered.

3.6 Contractions

The *contraction* of two operators is defined as the difference between their product and their normal-ordered product,

$$\overbracket{x\,y} \;\equiv\; xy - N\left[xy\right], \tag{3.89}$$

and we indicate it by a line joining the two operators. Since the normal-ordered piece has vanishing vacuum expectation value by Eq. (3.88), a contraction is just a number,

$$\overbracket{x\,y} = \langle 0|xy|0\rangle. \tag{3.90}$$

There are four possible pairs of operators, and only one of them survives:

$$\overbracket{a_p\, a_q^{\dagger}} = \langle 0|a_p a_q^{\dagger}|0\rangle = \delta_{pq}, \tag{3.91}$$

$$\overbracket{a_p\, a_q} = \langle 0|a_p a_q|0\rangle = 0, \tag{3.92}$$

$$\overbracket{a_q^{\dagger}\, a_p} = \langle 0|a_q^{\dagger} a_p|0\rangle = 0, \tag{3.93}$$

$$\overbracket{a_p^{\dagger}\, a_q^{\dagger}} = \langle 0|a_p^{\dagger} a_q^{\dagger}|0\rangle = 0. \tag{3.94}$$

Only an annihilation operator standing to the *left* of a creation operator gives a non-vanishing contraction. This single fact carries all the bookkeeping that follows: it is the statement that the vacuum contains no particles, and every zero in the calculations below can be traced back to it.

When the two contracted operators are not adjacent, the contraction carries the sign of the interchanges needed to bring them together. For a set of contractions the total sign is

$$(-1)^{\nu}, \qquad \nu = \text{number of crossings of the contraction lines,} \tag{3.95}$$

since each crossing corresponds to one interchange of neighbouring fermion operators. We shall use this rule constantly, and it is verified numerically for arbitrary strings in `wick.py`.

3.7 Two operators

We now build the theorem up from the simplest cases, using nothing more than the anticommutation relations. For two operators the content of the theorem is the definition (3.89) rearranged,

$$xy = N[xy] + \overbracket{x\,y}. \tag{3.96}$$

Take $x = a_p$ and $y = a_q^\dagger$. The anticommutation relation $\{a_p, a_q^\dagger\} = \delta_{pq}$ gives

$$a_p a_q^\dagger = \delta_{pq} - a_q^\dagger a_p = \delta_{pq} + N\left[a_p a_q^\dagger\right], \tag{3.97}$$

where the second step is Eq. (3.87). Comparing with Eq. (3.96) identifies the contraction as δ_{pq}, which is Eq. (3.91). Taking the vacuum expectation value and using Eq. (3.88) recovers the overlap of two single-particle states,

$$\langle p|q\rangle = \langle 0|a_p a_q^\dagger|0\rangle = \delta_{pq}. \tag{3.98}$$

The three remaining combinations $a_p a_q$, $a_p^\dagger a_q$ and $a_p^\dagger a_q^\dagger$ are already normal-ordered up to a sign, so their contractions vanish and so do their vacuum expectation values.

3.8 Three operators

An odd number of operators can never be paired up completely. Whatever contractions we carry out, at least one operator is always left over, and a single uncontracted operator inside a vacuum expectation value gives

$$\langle 0|a_p|0\rangle = \langle 0|a_p^\dagger|0\rangle = 0. \tag{3.99}$$

Hence

$$\langle 0|xyz|0\rangle = 0 \tag{3.100}$$

for every choice of the three operators, as one may also check directly by anticommuting. Take $x = a_p$, $y = a_q^\dagger$, $z = a_r^\dagger$:

$$a_p a_q^\dagger a_r^\dagger = \left(\delta_{pq} - a_q^\dagger a_p\right) a_r^\dagger = \delta_{pq} a_r^\dagger - a_q^\dagger\left(\delta_{pr} - a_r^\dagger a_p\right), \tag{3.101}$$

and every surviving term contains a single unpaired creation operator acting on the vacuum, so the vacuum expectation value vanishes.

The result is worth stating explicitly because it is used constantly: *any* vacuum expectation value of an odd number of creation and annihilation operators is zero. The same argument reappears in the proof of the theorem, where it disposes of half the terms in the induction step.

3.9 Four operators

Four operators is the first genuinely interesting case, and it is the one that appears whenever we compute the overlap of two two-particle states,

$$|rs\rangle = a_r^\dagger a_s^\dagger|0\rangle, \qquad |pq\rangle = a_p^\dagger a_q^\dagger|0\rangle, \qquad \langle rs|pq\rangle = \langle 0|a_s a_r a_p^\dagger a_q^\dagger|0\rangle. \tag{3.102}$$

By anticommutation.

We evaluate it the elementary way first, so that the answer can be checked against the theorem afterwards. Anticommuting a_r past $a_p^\dagger$ with $a_r a_p^\dagger = \delta_{rp} - a_p^\dagger a_r$,

$$a_s a_r a_p^\dagger a_q^\dagger = a_s \left(\delta_{rp} - a_p^\dagger a_r\right) a_q^\dagger = \delta_{rp}\, a_s a_q^\dagger - a_s a_p^\dagger a_r a_q^\dagger. \tag{3.103}$$

The first term is handled by Eq. (3.97),

$$\delta_{rp}\, a_s a_q^\dagger = \delta_{rp}\left(\delta_{sq} - a_q^\dagger a_s\right) \;\longrightarrow\; \delta_{rp}\delta_{sq}, \tag{3.104}$$

the arrow denoting that we take the vacuum expectation value and drop the normal-ordered remainder. For the second term we anticommute twice more,

$$-a_s a_p^\dagger a_r a_q^\dagger = -a_s a_p^\dagger \left(\delta_{rq} - a_q^\dagger a_r\right) = -\delta_{rq}\, a_s a_p^\dagger + a_s a_p^\dagger a_q^\dagger a_r, \tag{3.105}$$

and the last term has an annihilation operator on the far right, so it dies on the vacuum. The surviving piece is

$$-\delta_{rq}\, a_s a_p^\dagger = -\delta_{rq}\left(\delta_{sp} - a_p^\dagger a_s\right) \;\longrightarrow\; -\delta_{rq}\delta_{sp}. \tag{3.106}$$

Collecting the two contributions,

$$\boxed{\langle rs|pq\rangle = \langle 0|a_s a_r a_p^\dagger a_q^\dagger|0\rangle = \delta_{rp}\delta_{sq} - \delta_{sp}\delta_{rq}} \tag{3.107}$$

Two-particle states are orthonormal up to precisely the antisymmetry expected of fermions: the overlap is the antisymmetrised product of single-particle overlaps.

By contractions.

The same answer can be read off without any algebra at all. Four operators can be paired in three ways. The first pairs the two annihilation operators with each other and the two creation operators with each other,

$$a_s\, a_r\, a_p^\dagger\, a_q^\dagger \;=\; 0, \tag{3.108}$$

which vanishes by Eqs. (3.92) and (3.94). The second is the nested pairing, with no crossings and hence a plus sign,

$$a_s\, a_r\, a_p^\dagger\, a_q^\dagger \;=\; +\delta_{sq}\delta_{rp}. \tag{3.109}$$

The third has the two lines crossing once, and therefore a minus sign by Eq. (3.95); we draw one line above and one below,

$$a_s\, a_r\, a_p^\dagger\, a_q^\dagger \;=\; -\delta_{sp}\delta_{rq}. \tag{3.110}$$

Adding the three gives Eq. (3.107) again.

Three pairings, one of which vanishes on inspection and two of which are read straight off: this is the economy the theorem buys. The program `wick.py` carries out both routes for arbitrary strings and confirms that they agree. For the eight-operator strings of Section 3.13 the anticommutation recursion takes 129 steps, while the theorem enumerates 105 pairings of which only a handful survive.

3.10 Wick's theorem

The pattern of the previous three sections generalises. *Wick's theorem* states that an arbitrary product of creation and annihilation operators can be written as the normal-ordered product plus the sum of all possible contractions,

$$\begin{aligned} xyz\cdots w = {} & N[xyz\cdots w] \\ & + \sum_{(1)} N[xyz\cdots w] \qquad \text{(all single contractions)} \\ & + \sum_{(2)} N[xyz\cdots w] \qquad \text{(all double contractions)} \\ & + \cdots + \sum_{[M/2]} N[xyz\cdots w]\,, \end{aligned} \tag{3.111}$$

where $\sum_{(m)}$ runs over all ways of choosing m disjoint pairs of operators to contract, each term carrying the sign (3.95), and $[M/2]$ is the largest number of disjoint pairs that M operators admit.

The theorem becomes a computational tool the moment we take a vacuum expectation value. Every term in Eq. (3.111) that still contains an uncontracted operator is normal-ordered and therefore vanishes by Eq. (3.88). Only the *fully* contracted terms survive:

$$\boxed{\langle 0|xyz\cdots w|0\rangle = \sum_{[M/2]} N[xyz\cdots w] \quad \text{(fully contracted terms only)}} \tag{3.112}$$

For M even there are $(M-1)!!$ such terms – one, three, fifteen and one hundred and five for $M = 2,4,6,8$ – and most of them vanish on inspection because one of their contractions is of the type (3.92)–(3.94). For M odd there are none, which is Eq. (3.100) again.

The practical consequence is the one anticipated in Section 2.16. Because an n-body operator can connect only Slater determinants differing by at most n single-particle states, a configuration-interaction Hamiltonian built from one- and two-body operators is extremely sparse: of the $\binom{n}{N}^2$ possible matrix elements only a vanishing fraction are non-zero. That sparsity is precisely what the Lanczos algorithm of Section 1.26 exploits, and it is why the Hamiltonian never has to be stored.

3.11 Proof of Wick's theorem

The proof is by induction on the number of operators. The case $M = 2$ is Eq. (3.96), established directly from the anticommutation relation. The induction step requires one preparatory result, which does all the real work.

A lemma: appending one operator on the right.

Let $x, y, z, \ldots, w$ be a set of operators and let Ω be one further creation or annihilation operator. Then

$$N[xyz\cdots w]\,\Omega = N[xyz\cdots w\,\Omega] + \sum_{(1)\Omega} N[xyz\cdots w\,\Omega]\,, \tag{3.113}$$

where $\sum_{(1)\Omega}$ runs over all single contractions *involving* Ω, that is over the contractions of Ω with each of $x, y, z, \ldots, w$ in turn.

Proof of the lemma.

There are four cases to consider, and three of them are immediate.

(i) *Ω is an annihilation operator.* Then $N[xyz\cdots w]\,\Omega$ is already in normal-ordered form, since appending an annihilation operator on the far right cannot disturb the ordering, and so the left-hand side equals $N[xyz\cdots w\,\Omega]$. At the same time every contraction with Ω vanishes: a non-zero contraction requires an annihilation operator to the *left* of a creation operator, and here Ω is itself an annihilation operator standing to the right of everything. Both extra terms are absent and the lemma holds trivially.

(ii) *All of $x, y, \ldots, w$ are annihilation operators and Ω is an annihilation operator.* This is a special case of (i), and the lemma is again trivially satisfied.

(iii) *Ω is a creation operator.* Here we need only prove the lemma when all of $x, y, \ldots, w$ are annihilation operators. The reason is that a creation operator inside the normal-ordered product already stands to the left of Ω's eventual position, and its contraction with Ω vanishes by Eq. (3.94),

$$\overbracket{a_i^\dagger\, a_j^\dagger} = 0.$$

Creation operators in the string thus contribute neither to the reordering nor to the sum of contractions, and may be carried along untouched.

(iv) It remains to anticommute Ω leftwards through the annihilation operators $x, y, \ldots, w$, one at a time. Each step uses

$$a_j\,\Omega = \delta_{j\Omega} - \Omega\, a_j,$$

and produces two pieces: a Kronecker delta, which is precisely the contraction of a_j with Ω carrying the sign of the operators Ω has had to cross, and a term in which Ω has moved one place to the left. Iterating until Ω has passed all of them, the accumulated delta terms are exactly the sum $\sum_{(1)\Omega} N[xyz\cdots w\,\Omega]$, and the final term, in which Ω stands to the left of every annihilation operator, is exactly $N[xyz\cdots w\,\Omega]$. This is Eq. (3.113). □

The lemma at work.

For $M = 1$ the lemma reads

$$N\,[a_1]\,a_\ell^\dagger = N\left[a_1 a_\ell^\dagger\right] + N\left[\overbracket{a_1\, a_\ell^\dagger}\right] = -\langle 0|a_\ell^\dagger a_1|0\rangle + \delta_{1\ell}, \tag{3.114}$$

which is just Eq. (3.97) once more. For $M = 2$,

$$a_0\,N\,[a_1]\,a_\ell^\dagger = N\left[a_0 a_1 a_\ell^\dagger\right] + N\left[a_0\,\overbracket{a_1\, a_\ell^\dagger}\right] + N\left[\overbracket{a_0\, a_1\, a_\ell^\dagger}\right], \tag{3.115}$$

that is $N[a_0 a_1 a_\ell^\dagger] + \delta_{1\ell} N[a_0] + \delta_{0\ell} N[a_1]$ up to the signs of the crossings. Generally,

$$a_0\,N\,[a_1 a_2 \cdots a_M]\,a_\ell^\dagger = \sum_{(1)\Omega} N\left[a_0 a_1 \cdots a_M a_\ell^\dagger\right] + N\left[a_0 a_1 \cdots a_M a_\ell^\dagger\right]. \tag{3.116}$$

The induction step.

Assume Wick's theorem, Eq. (3.111), holds for a product of M operators. Multiply both sides from the right by one further operator Ω and apply the lemma to every term. The first term gives

$$N\,[xyz\cdots w]\,\Omega = N\,[xyz\cdots w\,\Omega] + \sum_{(1)\Omega} N\,[xyz\cdots w\,\Omega]\,, \tag{3.117}$$

the m-fold contracted terms give

$$\sum_{(m)} N[xyz\cdots w]\,\Omega = \sum_{(m)\neq\Omega} N[xyz\cdots w\,\Omega] + \sum_{(m)\Omega} N[xyz\cdots w\,\Omega]\,, \tag{3.118}$$

where $\sum_{(m)\neq\Omega}$ collects the m-fold contractions that do not involve Ω and $\sum_{(m)\Omega}$ those in which one of the m contractions ends on Ω. But an m-fold contraction of the $M+1$ operators either involves Ω or it does not, so the two sums recombine,

$$\sum_{(m)\neq\Omega} + \sum_{(m)\Omega} = \sum_{(m)} \qquad \text{over all } M+1 \text{ operators.} \tag{3.119}$$

Collecting the terms by the number of contractions therefore reproduces Eq. (3.111) with M replaced by $M+1$,

$$xyz\cdots w\,\Omega = N[xyz\cdots w\,\Omega] + \sum_{(1)} N[\cdots] + \sum_{(2)} N[\cdots] + \cdots + \sum_{[(M+1)/2]} N[\cdots]\,. \tag{3.120}$$

This completes the induction. □

A remark on parity.

If $M+1$ is odd, the last sum $\sum_{[(M+1)/2]}$ still leaves one operator uncontracted, since an odd number of operators cannot be paired up completely. On taking the vacuum expectation value these terms vanish by Eq. (3.99), and we recover the result of Section 3.8: vacuum expectation values of odd strings are zero. For $M+1$ even the last sum consists of the fully contracted terms, and Eq. (3.112) follows.

3.12 Wick's generalised theorem

In practice we rarely meet a bare product of operators. What we meet is a product of groups, each of which is *already* in normal-ordered form: the bra, the operator, and the ket. Wick's generalised theorem states that for such a product

$$N[A_1A_2\cdots]\,N[B_1B_2\cdots]\,N[C_1C_2\cdots] = N[A_1A_2\cdots B_1B_2\cdots C_1C_2\cdots] + \sum_{\text{all}} N[A_1A_2\cdots C_1C_2\cdots]\,, \tag{3.121}$$

where the sum runs over all contractions *between operators belonging to different normal-ordered groups*. Contractions within a single group are absent, precisely because that group is already normal-ordered and its internal contractions vanish by Eqs. (3.92)–(3.94).

This is a substantial saving. In the examples of the next section the naive count of pairings runs into the hundreds, while the number of contractions between different groups is a handful.

3.13 Applications: matrix elements between two-particle states

We close with the two calculations that motivated the whole construction. Both use the generalised theorem, and in both the bra, the operator and the ket are separately normal-ordered.

A one-body operator.

With

$$\hat{O}^{(1)} = \sum_{pq} \langle p|\hat{o}^{(1)}|q\rangle\, a_p^\dagger a_q, \tag{3.122}$$

and the two-particle states $|ij\rangle = a_i^\dagger a_j^\dagger|0\rangle$, $\langle ij| = \langle 0|a_j a_i$, we need

$$\langle ij|\hat{O}^{(1)}|k\ell\rangle = \sum_{pq} \langle p|\hat{o}^{(1)}|q\rangle\, \langle 0|\; \underbrace{a_j a_i}_{\text{bra}}\; \underbrace{a_p^\dagger a_q}_{\text{operator}}\; \underbrace{a_k^\dagger a_\ell^\dagger}_{\text{ket}}\; |0\rangle. \tag{3.123}$$

Only contractions between different groups contribute, and each must join an annihilation operator to a creation operator standing to its right. One such term is

$$a_j\, a_i\, a_p^\dagger\, a_q\, a_k^\dagger\, a_\ell^\dagger \;=\; \delta_{ip}\, \delta_{qk}\, \delta_{j\ell}, \tag{3.124}$$

which contributes $\delta_{j\ell}\,\langle i|\hat{o}^{(1)}|k\rangle$ after the sums over p and q are carried out. Collecting the four such terms gives the familiar result

$$\langle ij|\hat{O}^{(1)}|k\ell\rangle = \delta_{j\ell}\langle i|\hat{o}^{(1)}|k\rangle - \delta_{jk}\langle i|\hat{o}^{(1)}|\ell\rangle - \delta_{i\ell}\langle j|\hat{o}^{(1)}|k\rangle + \delta_{ik}\langle j|\hat{o}^{(1)}|\ell\rangle. \tag{3.125}$$

The matrix element vanishes whenever the two Slater determinants differ by more than one single-particle state, since then no assignment of the deltas can be satisfied. This is the Condon-Slater rule of Section 2.6, now obtained without a single explicit permutation.

The two-body interaction.

With the normal-ordered interaction

$$\hat{H}_I = \frac{1}{2}\sum_{pqrs} \langle pq|v|rs\rangle\, a_p^\dagger a_q^\dagger a_s a_r, \tag{3.126}$$

the diagonal matrix element in the state $|ij\rangle$ is

$$\langle ij|\hat{H}_I|ij\rangle = \frac{1}{2}\sum_{pqrs} \langle pq|v|rs\rangle\, \langle 0|\; \underbrace{a_j a_i}_{\text{bra}}\; \underbrace{a_p^\dagger a_q^\dagger a_s a_r}_{\text{operator}}\; \underbrace{a_i^\dagger a_j^\dagger}_{\text{ket}}\; |0\rangle. \tag{3.127}$$

This is an eight-operator string. Enumerating its 105 pairings by hand would be unpleasant, but the generalised theorem restricts us to contractions between the three groups, and of those only four survive. Two of them are

$$a_j\, a_i\, a_p^\dagger\, a_q^\dagger\, a_s\, a_r\, a_i^\dagger\, a_j^\dagger \;=\; \delta_{ip}\delta_{jq}\delta_{ri}\delta_{sj}, \tag{3.128}$$

giving $\langle ij|v|ij\rangle$, and the one obtained by exchanging the roles of r and s, which crosses one more line and therefore enters with the opposite sign, giving $-\langle ij|v|ji\rangle$. The two remaining terms are these two with i and j interchanged; they are equal to the first two and cancel the factor $\frac{1}{2}$. The result is

$$\boxed{\langle ij|\hat{H}_I|ij\rangle = \langle ij|v|ij\rangle - \langle ij|v|ji\rangle = \langle ij|v|ij\rangle_{\text{AS}}} \tag{3.129}$$

which is exactly the antisymmetrised matrix element obtained in Chapter 2 by writing out the antisymmetriser and sorting through permutations. The two derivations agree, as they must; the point is that this one required no permutations at all, only the observation that a contraction is non-zero when an annihilation operator stands to the left of a creation operator.

The program `wick.py` performs every calculation of these sections in both ways – by brute-force anticommutation and by summing contractions with their signs – and verifies that they agree, for strings of two, four, six and eight operators. It also reproduces Eq. (3.129) by enumerating which of the (p,q,r,s) assignments survive.

3.14 Operators in second quantization

What follows is the bridge from the formalism to a running code. The Hamiltonian matrix built here is the sparse, never-stored matrix that the Lanczos algorithm of Section 1.26 was designed for: only the product $\boldsymbol{Hv}$ is needed, and the Condon-Slater rules of Section 3.13 say which of its elements can be non-zero. The bit representation below is how that product is evaluated in practice.

In the build-up of a shell-model or FCI code that is meant to tackle large dimensionalities is the action of the Hamiltonian $\hat{H}$ on a Slater determinant represented in second quantization as

$$|\alpha_1 \dots \alpha_n\rangle = a^\dagger_{\alpha_1} a^\dagger_{\alpha_2} \dots a^\dagger_{\alpha_n} |0\rangle.$$

The time consuming part stems from the action of the Hamiltonian on the above determinant,

$$\left(\sum_{\alpha\beta} \langle\alpha|t+u|\beta\rangle a^\dagger_\alpha a_\beta + \frac{1}{4}\sum_{\alpha\beta\gamma\delta} \langle\alpha\beta|\hat{v}|\gamma\delta\rangle a^\dagger_\alpha a^\dagger_\beta a_\delta a_\gamma\right) a^\dagger_{\alpha_1} a^\dagger_{\alpha_2} \dots a^\dagger_{\alpha_n}|0\rangle.$$

A practically useful way to implement this action is to encode a Slater determinant as a bit pattern.

Assume that we have at our disposal n different single-particle orbits $\alpha_0, \alpha_2, \dots, \alpha_{n-1}$ and that we can distribute among these orbits $N \le n$ particles.

A Slater determinant can then be coded as an integer of n bits. As an example, if we have $n = 16$ single-particle states $\alpha_0, \alpha_1, \dots, \alpha_{15}$ and $N = 4$ fermions occupying the states α_3, α_6, α_{10} and α_{13} we could write this Slater determinant as

$$\Phi_\Lambda = a^\dagger_{\alpha_3} a^\dagger_{\alpha_6} a^\dagger_{\alpha_{10}} a^\dagger_{\alpha_{13}} |0\rangle.$$

The unoccupied single-particle states have bit value 0 while the occupied ones are represented by bit state 1. In the binary notation we would write this 16 bits long integer as

$$\begin{array}{cccccccccccccccc} \alpha_0 & \alpha_1 & \alpha_2 & \alpha_3 & \alpha_4 & \alpha_5 & \alpha_6 & \alpha_7 & \alpha_8 & \alpha_9 & \alpha_{10} & \alpha_{11} & \alpha_{12} & \alpha_{13} & \alpha_{14} & \alpha_{15} \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 0 \end{array}$$

which translates into the decimal number

$$2^3 + 2^6 + 2^{10} + 2^{13} = 9288.$$

We can thus encode a Slater determinant as a bit pattern.

With N particles that can be distributed over n single-particle states, the total number of Slater determinats (and defining thereby the dimensionality of the system) is

$$\dim(\mathscr{H}) = \binom{n}{N}.$$

The total number of bit patterns is 2^n.

We assume again that we have at our disposal n different single-particle orbits $\alpha_0, \alpha_2, \dots, \alpha_{n-1}$ and that we can distribute among these orbits $N \le n$ particles. The ordering among these states is important as it defines the order of the creation operators. We will write the determinant

$$\Phi_\Lambda = a^\dagger_{\alpha_3} a^\dagger_{\alpha_6} a^\dagger_{\alpha_{10}} a^\dagger_{\alpha_{13}} |0\rangle,$$

in a more compact way as

$$\Phi_{3,6,10,13} = |0001001000100100\rangle.$$

The action of a creation operator is thus

$$a^{\dagger}_{\alpha_4}\Phi_{3,6,10,13} = a^{\dagger}_{\alpha_4}|0001001000100100\rangle = a^{\dagger}_{\alpha_4}a^{\dagger}_{\alpha_3}a^{\dagger}_{\alpha_6}a^{\dagger}_{\alpha_{10}}a^{\dagger}_{\alpha_{13}}|0\rangle,$$

which becomes

$$-a^{\dagger}_{\alpha_3}a^{\dagger}_{\alpha_4}a^{\dagger}_{\alpha_6}a^{\dagger}_{\alpha_{10}}a^{\dagger}_{\alpha_{13}}|0\rangle = -|0001101000100100\rangle.$$

Similarly

$$a^{\dagger}_{\alpha_6}\Phi_{3,6,10,13} = a^{\dagger}_{\alpha_6}|0001001000100100\rangle = a^{\dagger}_{\alpha_6}a^{\dagger}_{\alpha_3}a^{\dagger}_{\alpha_6}a^{\dagger}_{\alpha_{10}}a^{\dagger}_{\alpha_{13}}|0\rangle,$$

which becomes

$$-a^{\dagger}_{\alpha_4}(a^{\dagger}_{\alpha_6})^2a^{\dagger}_{\alpha_{10}}a^{\dagger}_{\alpha_{13}}|0\rangle = 0!$$

This gives a simple recipe:

- If one of the bits b_j is 1 and we act with a creation operator on this bit, we return a null vector
- If $b_j = 0$, we set it to 1 and return a sign factor $(-1)^l$, where l is the number of bits set before bit j.

Consider the action of $a^{\dagger}_{\alpha_2}$ on various slater determinants:

$$\begin{aligned}
a^{\dagger}_{\alpha_2}\Phi_{00111} &= a^{\dagger}_{\alpha_2}|00111\rangle &&= 0\times|00111\rangle\\
a^{\dagger}_{\alpha_2}\Phi_{01011} &= a^{\dagger}_{\alpha_2}|01011\rangle &&= (-1)\times|01111\rangle\\
a^{\dagger}_{\alpha_2}\Phi_{01101} &= a^{\dagger}_{\alpha_2}|01101\rangle &&= 0\times|01101\rangle\\
a^{\dagger}_{\alpha_2}\Phi_{01110} &= a^{\dagger}_{\alpha_2}|01110\rangle &&= 0\times|01110\rangle\\
a^{\dagger}_{\alpha_2}\Phi_{10011} &= a^{\dagger}_{\alpha_2}|10011\rangle &&= (-1)\times|10111\rangle\\
a^{\dagger}_{\alpha_2}\Phi_{10101} &= a^{\dagger}_{\alpha_2}|10101\rangle &&= 0\times|10101\rangle\\
a^{\dagger}_{\alpha_2}\Phi_{10110} &= a^{\dagger}_{\alpha_2}|10110\rangle &&= 0\times|10110\rangle\\
a^{\dagger}_{\alpha_2}\Phi_{11001} &= a^{\dagger}_{\alpha_2}|11001\rangle &&= (+1)\times|11101\rangle\\
a^{\dagger}_{\alpha_2}\Phi_{11010} &= a^{\dagger}_{\alpha_2}|11010\rangle &&= (+1)\times|11110\rangle
\end{aligned}$$

What is the simplest way to obtain the phase when we act with one annihilation(creation) operator on the given Slater determinant representation?

We have an SD representation

$$\Phi_{\Lambda} = a^{\dagger}_{\alpha_0}a^{\dagger}_{\alpha_3}a^{\dagger}_{\alpha_6}a^{\dagger}_{\alpha_{10}}a^{\dagger}_{\alpha_{13}}|0\rangle,$$

in a more compact way as

$$\Phi_{0,3,6,10,13} = |1001001000100100\rangle.$$

The action of

$$a^{\dagger}_{\alpha_4}a_{\alpha_0}\Phi_{0,3,6,10,13} = a^{\dagger}_{\alpha_4}|0001001000100100\rangle = a^{\dagger}_{\alpha_4}a^{\dagger}_{\alpha_3}a^{\dagger}_{\alpha_6}a^{\dagger}_{\alpha_{10}}a^{\dagger}_{\alpha_{13}}|0\rangle,$$

which becomes

$$-a^{\dagger}_{\alpha_3}a^{\dagger}_{\alpha_4}a^{\dagger}_{\alpha_6}a^{\dagger}_{\alpha_{10}}a^{\dagger}_{\alpha_{13}}|0\rangle = -|0001101000100100\rangle.$$

The action

$$a_{\alpha_0}\Phi_{0,3,6,10,13} = |0001001000100100\rangle,$$

can be obtained by subtracting the logical sum (AND operation) of $\Phi_{0,3,6,10,13}$ and a word which represents only α_0, that is

$$|1000000000000000\rangle,$$

from $\Phi_{0,3,6,10,13} = |1001001000100100\rangle$.

This operation gives $|0001001000100100\rangle$.

Similarly, we can form $a^{\dagger}_{\alpha_4}a_{\alpha_0}\Phi_{0,3,6,10,13}$, say, by adding $|0000100000000000\rangle$ to $a_{\alpha_0}\Phi_{0,3,6,10,13}$, first checking that their logical sum is zero in order to make sure that orbital α_4 is not already occupied.

It is trickier however to get the phase $(-1)^l$. One possibility is as follows

- Let S_1 be a word that represents the $1-$bit to be removed and all others set to zero.

In the previous example $S_1 = |1000000000000000\rangle$

- Define S_2 as the similar word that represents the bit to be added, that is in our case

$S_2 = |0000100000000000\rangle$.

- Compute then $S = S_1 - S_2$, which here becomes

$$S = |0111000000000000\rangle$$

- Perform then the logical AND operation of S with the word containing

$$\Phi_{0,3,6,10,13} = |1001001000100100\rangle,$$

which results in $|0001000000000000\rangle$. Counting the number of $1-$bits gives the phase. Here you need however an algorithm for bitcounting. Several efficient ones available.

Chapter 4
Physical systems and models

The three preceding chapters built a formalism. This one puts it to work on four systems.

The models collected here share a property that makes them invaluable: each is simple enough to be solved exactly, at least for small numbers of particles, yet each contains a genuine many-body mechanism. The Lipkin model has a collective mode driven by a two-body interaction that scatters pairs across a gap; the pairing model has superfluid correlations; the Fermi-Hubbard model has the competition between kinetic delocalisation and on-site repulsion that drives the Mott transition; and the Calogero-Sutherland model is exactly solvable for arbitrary particle number, with a ground state that is a pure Jastrow function. Because the exact answer is available, these models are the benchmarks against which every approximate method in the rest of this book is measured.

They also share a structure. Every one of them is written as

$$\hat{H} = \sum_{pq} \langle p|\hat{h}_0|q\rangle a_p^\dagger a_q + \frac{1}{4}\sum_{pqrs} \langle pq|\hat{v}|rs\rangle_{\mathrm{AS}} a_p^\dagger a_q^\dagger a_s a_r, \tag{4.1}$$

the general second-quantised Hamiltonian of Chapter 3, with a particular and very restricted choice of matrix elements. What differs between them is which single-particle states there are and which matrix elements are allowed to be non-zero. The symmetries that follow from those restrictions are what make the models tractable, and identifying them is the first step in every case.

Throughout, the detailed algebra is relegated to the exercises at the end of the chapter, where full solutions are given. The main text states each model, identifies its symmetries, and reports what the exact solution looks like. All numbers quoted are produced by `models.py`.

4.1 The Lipkin model

The Lipkin-Meshkov-Glick model [3] describes N fermions distributed over two levels, each of degeneracy Ω. The levels carry the quantum number $\sigma = \pm 1$: the upper has $\sigma = +1$ and energy $+\varepsilon/2$, the lower has $\sigma = -1$ and energy $-\varepsilon/2$. The substates of each level are labelled $p = 1, \ldots, \Omega$, so a single-particle state is $|p\sigma\rangle = a_{p\sigma}^\dagger |0\rangle$.

The Hamiltonian is

$$\hat{H} = \hat{H}_0 + \hat{H}_1 + \hat{H}_2, \tag{4.2}$$

with

$$\hat{H}_0 = \frac{1}{2}\varepsilon \sum_{p\sigma} \sigma\, a^\dagger_{p\sigma} a_{p\sigma}, \tag{4.3}$$

$$\hat{H}_1 = \frac{1}{2} V \sum_{pp'\sigma} a^\dagger_{p\sigma} a^\dagger_{p'\sigma} a_{p'-\sigma} a_{p-\sigma}, \tag{4.4}$$

$$\hat{H}_2 = \frac{1}{2} W \sum_{pp'\sigma} a^\dagger_{p\sigma} a^\dagger_{p'-\sigma} a_{p'\sigma} a_{p-\sigma}. \tag{4.5}$$

The operator $\hat{H}_1$ moves a *pair* of particles from one level to the other, with strength V; $\hat{H}_2$ is a spin-exchange term of strength W, which moves a pair from $(p\sigma, p'-\sigma)$ to $(p-\sigma, p'\sigma)$. Neither changes the substate labels p, and this is the crucial restriction: the interaction is completely degenerate in p.

Quasispin.

The degeneracy in p means the model has an $SU(2)$ symmetry. Define the *quasispin* operators

$$\hat{J}_\pm = \sum_p a^\dagger_{p\pm} a_{p\mp}, \qquad \hat{J}_z = \frac{1}{2}\sum_{p\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma}, \qquad \hat{J}^2 = \hat{J}_+\hat{J}_- + \hat{J}_z^2 - \hat{J}_z. \tag{4.6}$$

Exercise 4.6 shows, using only the anticommutation relations of Chapter 3, that these satisfy the angular-momentum algebra

$$[\hat{J}_z, \hat{J}_\pm] = \pm \hat{J}_\pm, \qquad [\hat{J}_+, \hat{J}_-] = 2\hat{J}_z, \qquad [\hat{J}^2, \hat{J}_\pm] = [\hat{J}^2, \hat{J}_z] = 0, \tag{4.7}$$

and that the Hamiltonian collapses to

$$\boxed{\hat{H} = \varepsilon \hat{J}_z + \frac{1}{2} V \left(\hat{J}_+^2 + \hat{J}_-^2\right) + \frac{1}{2} W \left(-\hat{N} + \hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+\right)} \tag{4.8}$$

with $\hat{N} = \sum_{p\sigma} a^\dagger_{p\sigma} a_{p\sigma}$ the number operator. Every trace of the substate index p has disappeared.

The consequence is decisive. Since $[\hat{H}, \hat{J}^2] = 0$ and $[\hat{H}, \hat{N}] = 0$, both J and N are good quantum numbers, and the Hamiltonian is block diagonal in J. A Hilbert space of dimension $\binom{2\Omega}{N}$ collapses into blocks of dimension $2J+1$. For $N = 4$ particles at half filling the largest block is $J = 2$, of dimension five: a 70-dimensional problem has become a 5×5 matrix.

The $J = 2$ block.

The states are labelled $|J, J_z\rangle$ with $J_z = -J, \ldots, J$. The lowest, with all four spins down, is

$$|2, -2\rangle = a^\dagger_{1-} a^\dagger_{2-} a^\dagger_{3-} a^\dagger_{4-} |0\rangle, \tag{4.9}$$

and the remaining four follow by applying $\hat{J}_+$ with the standard normalisations

$$\hat{J}_\pm |J, J_z\rangle = \sqrt{J(J+1) - J_z(J_z \pm 1)}\, |J, J_z \pm 1\rangle. \tag{4.10}$$

Working out the matrix elements of Eq. (4.8) – again Exercise 4.6 – gives for $\varepsilon = 2$, $V = -1/3$, $W = -1/4$

$$\boldsymbol{H}^{(1)}_{J=2} = \begin{bmatrix} -4 & 0 & -\sqrt{6}/3 & 0 & 0 \\ 0 & -11/4 & 0 & -1 & 0 \\ -\sqrt{6}/3 & 0 & -1 & 0 & -\sqrt{6}/3 \\ 0 & -1 & 0 & 5/4 & 0 \\ 0 & 0 & -\sqrt{6}/3 & 0 & 4 \end{bmatrix}, \tag{4.11}$$

whose eigenvalues are −4.21288, −2.98607, −0.91914, 1.48607 and 4.13201. The ground state is

$$|\psi_0\rangle = 0.96735|2,-2\rangle + 0.25221|2,0\rangle + 0.02507|2,2\rangle, \qquad E_0 = -4.21288. \tag{4.12}$$

Note the structure: the interaction connects only states differing by two units of J_z, because $\hat{J}_\pm^2$ moves a pair. The matrix therefore splits further into even and odd J_z.

Increasing the interaction to $V = -4/3$, $W = -1$ gives eigenvalues −7.75122, −7.47214, −1.55581, 1.47214, 5.30704 and the ground state

$$|\psi_0\rangle = 0.64268|2,-2\rangle + 0.73816|2,0\rangle + 0.20515|2,2\rangle, \qquad E_0 = -7.75122. \tag{4.13}$$

In the first case the unperturbed configuration $|2,-2\rangle$ carries 94% of the probability and a single Slater determinant is a good approximation. In the second the interaction matrix elements are comparable to the single-particle spacing, $|2,0\rangle$ has become the dominant component, and no single determinant will do. This is the same story the occupation numbers of Section 1.31 told, in a model small enough to see the whole of it.

Why this model matters. The Lipkin model was invented to test many-body methods, and it still is. Because the exact answer is a 5×5 eigenvalue problem while the interaction can be tuned from weak to strong, it exposes exactly where an approximation breaks down: Hartree-Fock theory undergoes a spurious phase transition at a critical coupling, the random-phase approximation predicts an excitation energy that goes imaginary there, and coupled-cluster theory tracks the exact result much further. Every one of these statements can be checked against Eq. (4.12) and its analogues.

Mapping to qubits.

The Lipkin model is a natural first target for quantum algorithms, and how one maps it onto qubits is worth a moment. There are three obvious schemes.

The *occupation mapping* assigns one qubit to each single-particle state (p,σ), the qubit being $|1\rangle$ or $|0\rangle$ according as that state is occupied or empty. It costs 2Ω qubits.

The *level mapping* exploits the fact that there are only two levels. Restricting to half filling, $N = \Omega$, each doublet $((p,+1),(p,-1))$ holds exactly one particle, so one qubit per doublet suffices: $|0\rangle$ for the particle in the upper state and $|1\rangle$ for the lower. This costs Ω qubits, half of the occupation mapping. It has a second advantage: moving a pair changes the state of two doublets, so any operator in the Hamiltonian is at most a two-qubit gate.

The *spin mapping* goes further and assigns a qubit to each state $|J,J_z\rangle$. For a given J there are only $2J+1$ of these, so with $J_{\max} = N/2$ the largest register is $N+1$ qubits. The price is that one must run a separate circuit for each J and take the lowest of the resulting energies.

Under the level mapping the quasispin operators become sums of Pauli matrices. Writing $\hat{J}_z = \sum_p j_z^{(p)}$ and $\hat{J}_\pm = \sum_p j_\pm^{(p)}$ with

$$j_z^{(p)} = \frac{1}{2}\sum_\sigma \sigma a_{p\sigma}^\dagger a_{p\sigma}, \qquad j_\pm^{(p)} = a_{p\pm}^\dagger a_{p\mp}, \tag{4.14}$$

each doublet operator is a single-qubit operator, and the correspondence with the Pauli matrices of Section 1.7 is

$$j_z^{(p)} \longleftrightarrow \tfrac{1}{2}Z_p, \qquad j_\pm^{(p)} \longleftrightarrow \tfrac{1}{2}\left(X_p \pm iY_p\right). \tag{4.15}$$

The Hamiltonian (4.8) therefore becomes a sum of Pauli strings of weight at most two – exactly the form a variational quantum eigensolver requires. The circuits themselves belong to the quantum-computing part of this book; what matters here is that the mapping costs Ω qubits and two-qubit interactions only, which is a direct consequence of the $SU(2)$ structure identified above.

4.2 The pairing model

The pairing model has appeared repeatedly already – as the test case for the Lanczos algorithm in Section 1.26, for the Schmidt decomposition in Section 1.30, and as the three-level exercise of week 35. We now set it out properly.

The single-particle space consists of L doubly degenerate, equally spaced levels labelled $p = 1,2,\ldots,L$ with spin $\sigma = \pm 1$. The Hamiltonian is

$$\hat{H} = \hat{H}_0 + \hat{V}, \qquad \hat{H}_0 = \xi \sum_{p\sigma} (p-1)\, a^\dagger_{p\sigma} a_{p\sigma}, \qquad \hat{V} = -\frac{1}{2} g \sum_{pq} a^\dagger_{p+} a^\dagger_{p-} a_{q-} a_{q+}. \tag{4.16}$$

The spacing ξ is set to unity without loss of generality. The interaction has a single term and a single strength g: it destroys a pair of particles in level q with opposite spins and creates one in level p. Nothing else happens. In the language of Eq. (4.1), only the matrix elements $\langle p+\, p- |\hat{v}| q+\, q-\rangle$ are non-zero, and they are all equal.

Symmetries.

Exercise 4.6 shows that both $\hat{H}_0$ and $\hat{V}$ commute with the spin projection and the total spin,

$$\hat{S}_z = \frac{1}{2} \sum_{p\sigma} \sigma a^\dagger_{p\sigma} a_{p\sigma}, \qquad \hat{S}^2 = \hat{S}_z^2 + \frac{1}{2}\left(\hat{S}_+ \hat{S}_- + \hat{S}_- \hat{S}_+\right), \qquad \hat{S}_\pm = \sum_p a^\dagger_{p\pm} a_{p\mp}, \tag{4.17}$$

so the Hamiltonian is block diagonal in S and S_z. We work in the $S = 0$ block. There it is natural to introduce the pair operators

$$\hat{P}^+_p = a^\dagger_{p+} a^\dagger_{p-}, \qquad \hat{P}^-_p = a_{p-} a_{p+}, \tag{4.18}$$

in terms of which

$$\hat{H} = \sum_{p\sigma} (p-1) a^\dagger_{p\sigma} a_{p\sigma} - \frac{1}{2} g \sum_{pq} \hat{P}^+_p \hat{P}^-_q. \tag{4.19}$$

The pair creation operators commute among themselves, $[\hat{P}^+_p, \hat{P}^+_q] = 0$, which is why they behave like bosons and why the model describes a superfluid. They are *not* bosons, however: $(\hat{P}^+_p)^2 = 0$, the Pauli principle asserting itself.

The seniority-zero space.

Because the interaction only moves pairs, it never breaks one. If we start from a state in which every occupied level carries a full pair, we stay in that space forever. A basis state is then simply a choice of which n of the L levels carry a pair, and the dimension is $\binom{L}{n}$ rather than $\binom{2L}{2n}$. For $L = 4$ and $n = 2$ this is 6 instead of 28; for $L = 12$ and $n = 6$ it is 924 instead of 2704156. This is the space in which Table 1.3 was computed.

In that basis the Hamiltonian is easy to write down: the diagonal is $2\sum_{p\in \mathrm{occ}}(p-1) - \frac{1}{2} g n$, and any two configurations differing by the position of a single pair are connected by $-\frac{1}{2} g$.

Table 4.1 gives the exact ground-state energy for four particles in four levels as a function of g, together with the energy of the unperturbed configuration. At $g = 0$ the reference determinant is exact. For small $|g|$ the correlation energy grows quadratically, as second-order perturbation theory requires, and for $g = 1$ it has reached 36% of the reference energy.

Why this model matters. The pairing model is the schematic version of superfluidity and superconductivity, and it is the standard proving ground for the hierarchy of many-body methods. Because the

g	E_{ref}	E_0 (exact)	E_{corr}
−1.00	3.000000	2.77987014	−0.22012986
−0.50	2.500000	2.43688426	−0.06311574
0.00	2.000000	2.00000000	0.00000000
0.50	1.500000	1.41677428	−0.08322572
1.00	1.000000	0.63554847	−0.36445153

Table 4.1 The pairing model with four doubly degenerate levels and two pairs, $\xi = 1$. E_{ref} is the energy of the configuration with both pairs in the two lowest levels; E_0 is the exact ground-state energy in the six-dimensional seniority-zero space.

exact answer is available for any g, one can plot Hartree-Fock, second- and third-order perturbation theory, coupled cluster with doubles, and full configuration interaction on the same axes and watch each one fail at a different coupling. The second midterm of FYS4480 does exactly this, and the exercises at the end of this chapter set up the ingredients.

4.3 The Fermi-Hubbard model

The two models so far live in an abstract single-particle space with no geometry. The Fermi-Hubbard model introduces a lattice, and with it the competition that dominates the physics of correlated electrons: kinetic energy wants to delocalise the particles, the interaction wants to keep them apart.

Electrons hop between neighbouring sites of a lattice and pay an energy U whenever two of them, necessarily of opposite spin, occupy the same site,

$$\boxed{\hat{H} = -\sum_{\langle ij\rangle\sigma} t_{ij}\left(c^\dagger_{i\sigma}c_{j\sigma} + \text{h.c.}\right) + U\sum_i \hat{n}_{i\uparrow}\hat{n}_{i\downarrow}} \tag{4.20}$$

with $\hat{n}_{i\sigma} = c^\dagger_{i\sigma}c_{i\sigma}$. This is again Eq. (4.1): the one-body part is the hopping matrix, and the only non-vanishing two-body matrix element is the on-site one.

The single-particle basis is now the set of lattice sites, so the creation operators carry a position label. Fermionic antisymmetry is enforced by the Jordan-Wigner string: $c^\dagger_i c_j$ acting on an occupation-number state picks up the parity of the occupied sites strictly between i and j, which is exactly the phase computed by the bit-counting algorithm at the end of Chapter 3.

Symmetries and sector dimensions.

The Hamiltonian conserves $N_\uparrow$ and $N_\downarrow$ separately, so we may diagonalise in a fixed sector of dimension

$$\dim = \binom{N}{N_\uparrow}\binom{N}{N_\downarrow}. \tag{4.21}$$

Better still, the hopping term acts on the two spin species independently, so the sector factorises as a tensor product and the kinetic part of the Hamiltonian is

$$\boldsymbol{K}_\uparrow \otimes \boldsymbol{I}_\downarrow + \boldsymbol{I}_\uparrow \otimes \boldsymbol{K}_\downarrow, \tag{4.22}$$

the tensor-product construction of Section 1.10 used in earnest. The interaction is diagonal in the occupation basis. On a periodic chain translational invariance gives a further reduction into momentum sectors.

From metal to Mott insulator.

Table 4.2 follows the half-filled four-site ring as U grows. At $U = 0$ the model is exact free fermions: the single-particle energies are $\varepsilon_k = -2t\cos k$ with $k = 0, \pm\pi/2, \pi$, and filling the two lowest levels for each spin gives $E_0 = -4t$, which the diagonalisation reproduces. As U increases, the double occupancy falls steadily – the electrons avoid each other – and the ground-state energy rises towards zero.

U/t	E_0/t	$E_0/(Nt)$	$\sum_i \langle n_{i\uparrow} n_{i\downarrow} \rangle$
0	−4.00000000	−1.000000	1.000000
2	−2.82842712	−0.707107	0.453082
4	−2.10274848	−0.525687	0.287325
8	−1.32023496	−0.330059	0.129992
16	−0.72286432	−0.180716	0.042033

Table 4.2 The half-filled four-site Hubbard ring, $N_\uparrow = N_\downarrow = 2$, sector dimension 36. The last column is the total double occupancy in the ground state.

The strong-coupling limit.

When $U \gg t$ the doubly occupied states are pushed far up in energy and can be eliminated perturbatively. What remains is a model of localised spins,

$$\hat{H}_{tJ} = \hat{P}_G \left[-\sum_{\langle ij \rangle \sigma} t_{ij} \left(c^\dagger_{i\sigma} c_{j\sigma} + \text{h.c.} \right) \right] \hat{P}_G + \sum_{\langle ij \rangle} J_{ij} \left[\boldsymbol{S}_i \cdot \boldsymbol{S}_j - \tfrac{1}{4} \hat{P}_G \hat{n}_i \hat{n}_j \hat{P}_G \right], \qquad J_{ij} = \frac{4t_{ij}^2}{U}, \tag{4.23}$$

where $\hat{P}_G$ is the Gutzwiller projector removing all doubly occupied sites. At half filling there is no empty site to hop into, the kinetic term is frozen, and the t-J model reduces to the Heisenberg antiferromagnet.

This is a prediction we can check. For the four-site ring the Heisenberg part contributes $-2J$ and the constant $-\frac{1}{4} n_i n_j$ term on four bonds contributes $-J$, so $E_0 \to -3J = -12t^2/U$. Table 4.3 shows the ratio E_0/J converging to -3.

U/t	E_0/t	$E_0/J, \quad J = 4t^2/U$
8	−1.3202349583	−2.64047
16	−0.7228643224	−2.89146
32	−0.3714104223	−2.97128
64	−0.1870445461	−2.99271
128	−0.0936928521	−2.99817

Table 4.3 The half-filled four-site Hubbard ring at strong coupling. The ratio converges to -3, confirming the mapping onto the Heisenberg model with $J = 4t^2/U$.

Why this model matters. The Hubbard model is the minimal model of strong correlation. It is believed to contain the essential physics of the cuprate superconductors, it is the standard benchmark for quantum Monte Carlo, tensor-network and machine-learning wave functions, and it is among the first targets for quantum simulation with cold atoms and with quantum computers. It is also the model in which the exponential wall of Section 2.16 bites hardest: the sector dimension (4.21) for a 4×4 lattice at half filling is already 1.66×10^8, and the two-dimensional model remains unsolved.

4.4 The Calogero-Sutherland model

The three models above are solvable because a symmetry cuts the Hilbert space down to a manageable size. The Calogero-Sutherland model [4,5] is solvable for a different and much rarer reason: it is *integrable*, and its ground state is known in closed form for any number of particles.

The Calogero model describes N particles on a line in a harmonic trap, interacting through an inverse-square potential,

$$\hat{H} = -\frac{\hbar^2}{2m}\sum_{i=1}^{N}\frac{\partial^2}{\partial x_i^2} + \frac{m\omega^2}{2}\sum_{i=1}^{N} x_i^2 + \sum_{i<j}\frac{g}{(x_i - x_j)^2}. \tag{4.24}$$

It is conventional to parametrise the coupling as

$$g = \frac{\hbar^2}{m}\lambda(\lambda - 1), \tag{4.25}$$

so that λ measures the interaction strength. The Sutherland model is the same system on a ring of circumference L, with the inverse square replaced by the chord distance,

$$\hat{H} = -\frac{\hbar^2}{2m}\sum_{i}\frac{\partial^2}{\partial x_i^2} + \frac{\hbar^2}{m}\left(\frac{\pi}{L}\right)^2\sum_{i<j}\frac{\lambda(\lambda - 1)}{\sin^2(\pi(x_i - x_j)/L)}. \tag{4.26}$$

The exact ground state.

In units $\hbar = m = 1$ the ground state of Eq. (4.24) is

$$\boxed{\Psi_0(x_1,\ldots,x_N) = \prod_{i<j}|x_i - x_j|^{\lambda}\,\exp\left(-\frac{\omega}{2}\sum_i x_i^2\right)} \tag{4.27}$$

with energy

$$E_0 = \omega\left[\frac{N}{2} + \frac{\lambda N(N-1)}{2}\right]. \tag{4.28}$$

Exercise 4.6 verifies this by direct substitution; the inverse-square form of the potential is precisely what is needed for the cross terms generated by the Laplacian to cancel.

The factor $\prod_{i<j}|x_i - x_j|^{\lambda}$ is a *Jastrow factor*, and it is the reason this model deserves a place in a book about many-body methods. It is a correlated wave function of exactly the type that variational Monte Carlo optimises numerically – but here it is exact, for every N and every λ. The model is therefore the sharpest available test of a Jastrow ansatz: any method that cannot reproduce Eq. (4.27) is missing something structural.

Two limits are instructive. At $\lambda = 0$ the coupling g vanishes and the particles are free bosons in a trap, $E_0 = \omega N/2$. At $\lambda = 1$ the coupling vanishes again, but now

$$\prod_{i<j}|x_i - x_j| \tag{4.29}$$

is the absolute value of the Vandermonde determinant of Section 1.20, which is precisely the Slater determinant of N free fermions in a harmonic trap. The Calogero model interpolates continuously between bosons and fermions, and for non-integer λ it describes particles obeying *fractional statistics*.

A numerical check.

For $N = 2$ the centre-of-mass and relative motions separate. With $x = x_1 - x_2$ the relative Hamiltonian is

$$\hat{h}_{\mathrm{rel}} = -\frac{d^2}{dx^2} + \frac{\omega^2}{4}x^2 + \frac{\lambda(\lambda-1)}{x^2}, \tag{4.30}$$

whose exact ground-state energy is $\omega(\lambda + \frac{1}{2})$. Discretising on a grid and diagonalising with the machinery of Section 1.27 reproduces this to better than 10^{-5} for λ between 1 and 3, as `models.py` confirms. The inverse-square barrier keeps the wave function away from the origin, so the boundary condition $\Psi \to 0$ as $x \to 0$ is automatic for $\lambda > 0$.

Why this model matters. The Calogero-Sutherland model sits at the junction of several subjects. Its ground state is a Laughlin-type wave function, $\prod_{i<j}(z_i - z_j)^m e^{-\Sigma|z_i|^2/4}$, connecting it to the fractional quantum Hall effect; its excitation spectrum is that of free particles with fractional statistics; and the probability density $|\Psi_0|^2 \propto \prod_{i<j}|x_i - x_j|^{2\lambda} e^{-\omega\Sigma x_i^2}$ is exactly the eigenvalue distribution of a random matrix ensemble, with $\beta = 2\lambda$ the Dyson index. For our purposes it is above all a benchmark with a known answer for arbitrary N – something none of the other models in this chapter can offer.

4.5 What the four models have in common

Table 4.4 collects the four. The pattern is worth stating explicitly, because it is the pattern of the whole subject.

Model	**Single-particle space**	**Interaction**	**Symmetry used**	**Exactly solvable**
Lipkin	two levels, degeneracy Ω	pair transfer, spin exchange	$SU(2)$ quasispin	yes, block by block
Pairing	L doubly degenerate levels	pair transfer	S, S_z, seniority	yes, small L
Hubbard	lattice sites	on-site U	$N_\uparrow$, $N_\downarrow$, momentum	1D only
Calogero	continuum, N particles	inverse square	integrability	yes, any N

Table 4.4 The four models of this chapter. Each is the general Hamiltonian (4.1) with a drastically restricted set of two-body matrix elements, and in each case a symmetry reduces the dimension of the problem.

In every case the route is the same. Write the Hamiltonian in second quantisation; identify the operators that commute with it; use them to block diagonalise; and solve the blocks. The first step is Chapter 3, the second is a commutator calculation of the kind Wick's theorem makes routine, the third is the statement that a symmetry gives a basis in which the matrix is block diagonal (Section 1.11), and the fourth is the eigenvalue problem of Sections 1.24 and 1.26.

What separates these models from a realistic system is only the last step of the reduction. In a real nucleus or a real molecule the symmetries reduce the dimension by orders of magnitude but not to five; the matrix is sparse but enormous; and one must fall back on the approximate methods that occupy the rest of this book. The value of the models in this chapter is that they let us watch those approximations succeed and fail against an answer we already know.

4.6 Exercises

Exercise 1: the Lipkin model in the quasispin basis

a) Show that the quasispin operators (4.6) obey the angular-momentum algebra (4.7), using only the anti-commutation relations of Chapter 3.
b) Show that the Lipkin Hamiltonian (4.2) can be rewritten as Eq. (4.8).
c) Show that $[\hat{H},\hat{J}^2]=0$ and $[\hat{H},\hat{N}]=0$, and explain what this implies for the structure of the Hamiltonian matrix.
d) Construct the five states $|2,J_z\rangle$ for four particles, and compute the matrix (4.11). Diagonalise it and compare with Eq. (4.12).

Answer to a).

Insert the definitions and use $\{a_l,a_k\}=\{a_l^\dagger,a_k^\dagger\}=0$ and $\{a_l^\dagger,a_k\}=\delta_{lk}$. For the first commutator,

$$[\hat{J}_z,\hat{J}_\pm]=\frac{1}{2}\sum_{pp'\sigma}\sigma\left(a_{p\sigma}^\dagger a_{p\sigma}a_{p'\pm}^\dagger a_{p'\mp}-a_{p'\pm}^\dagger a_{p'\mp}a_{p\sigma}^\dagger a_{p\sigma}\right). \tag{4.31}$$

Anticommute the operators of the second product into the order of the first. Using $a_{p'\mp}a_{p\sigma}^\dagger=\delta_{pp'}\delta_{\mp\sigma}-a_{p\sigma}^\dagger a_{p'\mp}$ and then $a_{p\sigma}a_{p'\pm}^\dagger=\delta_{pp'}\delta_{\pm\sigma}-a_{p'\pm}^\dagger a_{p\sigma}$, the four-operator terms cancel in pairs and what survives is

$$[\hat{J}_z,\hat{J}_\pm]=\frac{1}{2}\sum_{p\sigma}\sigma\left(\delta_{\pm\sigma}a_{p\sigma}^\dagger a_{p\mp}-\delta_{\mp\sigma}a_{p\pm}^\dagger a_{p\sigma}\right)=\pm\frac{1}{2}\sum_{p}\left(a_{p\pm}^\dagger a_{p\mp}+a_{p\pm}^\dagger a_{p\mp}\right)=\pm\hat{J}_\pm, \tag{4.32}$$

where the two Kronecker deltas each pick out one value of σ, and the factor $\sigma=\pm1$ supplies the sign. The same manipulation gives $[\hat{J}_+,\hat{J}_-]=2\hat{J}_z$, and $[\hat{J}^2,\hat{J}_\pm]=[\hat{J}^2,\hat{J}_z]=0$ then follows from the definition $\hat{J}^2=\hat{J}_+\hat{J}_-+\hat{J}_z^2-\hat{J}_z$ exactly as for ordinary angular momentum.

Answer to b).

The one-body part is immediate: comparing Eqs. (4.3) and (4.6) gives $\hat{H}_0=\varepsilon\hat{J}_z$.

For $\hat{H}_1$, first anticommute the two annihilation operators, $a_{p'-\sigma}a_{p-\sigma}=-a_{p-\sigma}a_{p'-\sigma}$, and then move $a_{p-\sigma}$ past $a_{p'\sigma}^\dagger$ using $a_{p-\sigma}a_{p'\sigma}^\dagger=\delta_{pp'}\delta_{-\sigma\sigma}-a_{p'\sigma}^\dagger a_{p-\sigma}$. Since $\delta_{-\sigma\sigma}=0$, the delta term drops out and

$$\hat{H}_1=\frac{1}{2}V\sum_{pp'\sigma}a_{p\sigma}^\dagger a_{p-\sigma}\,a_{p'\sigma}^\dagger a_{p'-\sigma}. \tag{4.33}$$

The two factors are now independent sums over p and p'. Writing out $\sigma=\pm$ separately,

$$\hat{H}_1=\frac{1}{2}V\left[\left(\sum_p a_{p+}^\dagger a_{p-}\right)\left(\sum_{p'}a_{p'+}^\dagger a_{p'-}\right)+\left(\sum_p a_{p-}^\dagger a_{p+}\right)\left(\sum_{p'}a_{p'-}^\dagger a_{p'+}\right)\right]=\frac{1}{2}V\left(\hat{J}_+^2+\hat{J}_-^2\right). \tag{4.34}$$

For $\hat{H}_2$ the same two steps give

$$\hat{H}_2=\frac{1}{2}W\left(-\sum_{p\sigma}a_{p\sigma}^\dagger a_{p\sigma}+\sum_{pp'\sigma}a_{p\sigma}^\dagger a_{p-\sigma}a_{p'-\sigma}^\dagger a_{p'\sigma}\right), \tag{4.35}$$

where this time $\delta_{-\sigma-\sigma}=1$ and the delta term survives as $-\hat{N}$. Splitting the remaining sum over σ as before identifies it as $\hat{J}_+\hat{J}_-+\hat{J}_-\hat{J}_+$, so $\hat{H}_2=\frac{1}{2}W(-\hat{N}+\hat{J}_+\hat{J}_-+\hat{J}_-\hat{J}_+)$. Adding the three pieces gives Eq. (4.8).

Answer to c).

Written as in Eq. (4.8), $\hat{H}$ is built entirely from $\hat{J}_z$, $\hat{J}_\pm$ and $\hat{N}$. Since $\hat{J}^2$ commutes with each of $\hat{J}_z$ and $\hat{J}_\pm$ by part a), and with $\hat{N}$ trivially, it commutes with $\hat{H}$. For $\hat{N}$: the one-body term conserves particle number, and every term of the interaction contains two creation and two annihilation operators, so $[\hat{H},\hat{N}]=0$.

Both J and N are therefore good quantum numbers. In a basis $|J,J_z\rangle$ the Hamiltonian matrix is block diagonal, one block for each J, of dimension $2J+1$. Within a block, $\hat{J}_z$ is diagonal and $\hat{J}_\pm^2$ connects J_z to $J_z\pm 2$, so the block splits once more into even and odd J_z. A problem of nominal dimension $\binom{2\Omega}{N}$ has been reduced to a set of matrices of dimension at most $N/2+1$.

Answer to d).

Start from $|2,-2\rangle$, Eq. (4.9), and apply Eq. (4.10). With $J=2$ and $J_z=-2$ the factor is $\sqrt{6-2}=2$, so

$$|2,-1\rangle = \frac{1}{2}\hat{J}_+|2,-2\rangle = \frac{1}{2}\Big(a^\dagger_{1+}a^\dagger_{2-}a^\dagger_{3-}a^\dagger_{4-} + a^\dagger_{1-}a^\dagger_{2+}a^\dagger_{3-}a^\dagger_{4-} + a^\dagger_{1-}a^\dagger_{2-}a^\dagger_{3+}a^\dagger_{4-} + a^\dagger_{1-}a^\dagger_{2-}a^\dagger_{3-}a^\dagger_{4+}\Big)|0\rangle, \tag{4.36}$$

one term for each substate that can be flipped. Applying $\hat{J}_+$ again with the factor $\sqrt{6-0}=\sqrt{6}$ gives $|2,0\rangle$ as the normalised sum of the six ways of flipping two of the four substates, and so on up to $|2,2\rangle$ with all four spins up.

The matrix elements follow from Eq. (4.8). The diagonal is

$$\langle 2,J_z|\hat{H}|2,J_z\rangle = \varepsilon J_z + \frac{W}{2}\Big(-N+2\big[J(J+1)-J_z^2\big]\Big), \tag{4.37}$$

since $\hat{J}_+\hat{J}_- + \hat{J}_-\hat{J}_+ = 2(\hat{J}^2-\hat{J}_z^2)$. The off-diagonal elements come from $\frac{1}{2}V(\hat{J}_+^2+\hat{J}_-^2)$,

$$\langle 2,J_z+2|\hat{H}|2,J_z\rangle = \frac{V}{2}\sqrt{J(J+1)-J_z(J_z+1)}\,\sqrt{J(J+1)-(J_z+1)(J_z+2)}. \tag{4.38}$$

With $\varepsilon=2$, $V=-1/3$, $W=-1/4$ and $N=4$ this gives Eq. (4.11); for instance the $(-2,0)$ element is $\frac{1}{2}V\sqrt{4}\sqrt{6}=-\sqrt{6}/3$, and the $J_z=-1$ diagonal element is $-2+\frac{1}{2}(-\frac{1}{4})(-4+2[6-1]) = -2-\frac{3}{4}=-\frac{11}{4}$. Diagonalising reproduces Eq. (4.12).

Exercise 2: the pairing model

a) Show that $\hat{H}_0$ and $\hat{V}$ of Eq. (4.16) commute with $\hat{S}_z$ and $\hat{S}^2$ of Eq. (4.17).
b) Show that the Hamiltonian can be written in the pair form (4.19), and that the pair creation operators commute among themselves while $(\hat{P}_p^+)^2=0$.
c) Construct the Hamiltonian matrix for four particles in the four lowest levels, with no broken pairs and $S=0$, and find its eigenvalues as a function of g.

Answer to a).

$\hat{H}_0$ is a sum of number operators $a^\dagger_{p\sigma}a_{p\sigma}$, and $\hat{S}_z$ is a weighted sum of the same operators, so the two commute term by term. For $\hat{V}$, note that $a^\dagger_{p+}a^\dagger_{p-}$ creates one particle with $\sigma=+1$ and one with $\sigma=-1$, changing S_z by $\frac{1}{2}(+1)+\frac{1}{2}(-1)=0$; likewise $a_{q-}a_{q+}$ destroys such a pair. Hence $[\hat{V},\hat{S}_z]=0$.

For $\hat{S}^2$ it is enough to show $[\hat{V},\hat{S}_\pm]=0$. The operator $\hat{S}_+=\sum_p a^\dagger_{p+}a_{p-}$ acting on a fully paired level gives zero, because the state $|p+\rangle$ is already occupied and $(a^\dagger_{p+})^2=0$; acting on an empty level it also gives zero. Since $\hat{V}$ maps the fully paired space into itself, $\hat{S}_+\hat{V}$ and $\hat{V}\hat{S}_+$ both annihilate it, and the commutator vanishes. Both S and S_z are therefore good quantum numbers and the Hamiltonian is block diagonal in them.

Answer to b).

Insert the definitions (4.18) into Eq. (4.16):

$$-\frac{1}{2}g\sum_{pq}a^\dagger_{p+}a^\dagger_{p-}a_{q-}a_{q+} = -\frac{1}{2}g\sum_{pq}\hat{P}^+_p\hat{P}^-_q, \tag{4.39}$$

which is Eq. (4.19).

For the commutator, $\hat{P}^+_p\hat{P}^+_q = a^\dagger_{p+}a^\dagger_{p-}a^\dagger_{q+}a^\dagger_{q-}$. Moving $a^\dagger_{q+}a^\dagger_{q-}$ to the front requires four interchanges of neighbouring creation operators, each contributing a minus sign, so the total sign is $(-1)^4 = +1$ and $[\hat{P}^+_p, \hat{P}^+_q] = 0$. The pair operators therefore behave like bosons under exchange.

They are not bosons, however. Since $(a^\dagger_{p+})^2 = 0$ by the Pauli principle,

$$\left(\hat{P}^+_p\right)^2 = a^\dagger_{p+}a^\dagger_{p-}a^\dagger_{p+}a^\dagger_{p-} = -\left(a^\dagger_{p+}\right)^2\left(a^\dagger_{p-}\right)^2 = 0. \tag{4.40}$$

A level can hold one pair but not two. This is the hard-core constraint that distinguishes the pairing model from a gas of free bosons, and it is what makes the model non-trivial.

Answer to c).

With four levels and two pairs there are $\binom{4}{2} = 6$ basis states, which we label by the occupied levels: (12), (13), (14), (23), (24), (34). The diagonal element of a configuration is $2\sum_{p\in\mathrm{occ}}(p-1) - g$, the last term coming from the $p = q$ terms of the interaction with two pairs present. Every pair of configurations differing by the position of one pair is connected by $-g/2$; configurations differing in both pairs are not connected. Ordering the basis as above,

$$\boldsymbol{H} = \begin{bmatrix} 2-g & -g/2 & -g/2 & -g/2 & -g/2 & 0 \\ -g/2 & 4-g & -g/2 & -g/2 & 0 & -g/2 \\ -g/2 & -g/2 & 6-g & 0 & -g/2 & -g/2 \\ -g/2 & -g/2 & 0 & 6-g & -g/2 & -g/2 \\ -g/2 & 0 & -g/2 & -g/2 & 8-g & -g/2 \\ 0 & -g/2 & -g/2 & -g/2 & -g/2 & 10-g \end{bmatrix}. \tag{4.41}$$

Diagonalising for a range of g gives Table 4.1. Note that the (13) and (23) configurations are degenerate at $g = 0$ – both have unperturbed energy 6 – and that the correlation energy is negative for both signs of g, since the interaction can always lower the energy by mixing configurations.

Exercise 3: the Hubbard model

a) Show that the Hubbard Hamiltonian (4.20) conserves $N_\uparrow$ and $N_\downarrow$ separately, and hence S_z.
b) Diagonalise the $U = 0$ Hubbard ring analytically and verify the entry $E_0 = -4t$ of Table 4.2.
c) Derive the effective spin Hamiltonian in the limit $U \gg t$ at half filling, and predict the ratio E_0/J for the four-site ring.

Answer to a).

The hopping term contains $c_{i\sigma}^{\dagger}c_{j\sigma}$ with the *same* σ on both operators, so it conserves the number of particles of each spin separately. The interaction is a product of number operators and conserves everything. Hence $[\hat{H},\hat{N}_{\uparrow}]=[\hat{H},\hat{N}_{\downarrow}]=0$, and since $\hat{S}_z=\frac{1}{2}(\hat{N}_{\uparrow}-\hat{N}_{\downarrow})$, also $[\hat{H},\hat{S}_z]=0$. This is what allows the diagonalisation to be carried out in a fixed sector of dimension (4.21).

Answer to b).

At $U=0$ the Hamiltonian is a sum of independent one-body problems. On a ring of N sites with periodic boundary conditions the hopping matrix is diagonalised by plane waves $c_{k\sigma}^{\dagger}=N^{-1/2}\sum_j e^{ikj}c_{j\sigma}^{\dagger}$ with $k=2\pi n/N$, giving

$$\varepsilon_k=-2t\cos k. \tag{4.42}$$

For $N=4$ the allowed momenta are $k=0,\pm\pi/2,\pi$ with energies $-2t$, 0, 0, $+2t$. Filling the two lowest for each spin gives $-2t+0=-2t$ per spin species, hence $E_0=-4t$, which is the first line of Table 4.2. The degeneracy at $k=\pm\pi/2$ means the $U=0$ ground state is not unique; switching on U lifts the degeneracy.

Answer to c).

At half filling and $U\gg t$ every site carries exactly one electron in the ground state, and states with a doubly occupied site lie an energy U higher. Treat the hopping as a perturbation. A single hop creates a doubly occupied site and an empty one, costing U; hopping back restores a singly occupied configuration. Second-order perturbation theory therefore gives an energy gain $-2t^2/U$ for each such virtual excursion, and the process is possible only if the two spins involved are antiparallel, since the Pauli principle forbids two electrons of the same spin on one site.

Collecting the terms gives the Heisenberg Hamiltonian

$$\hat{H}_{\text{eff}}=J\sum_{\langle ij\rangle}\left(\boldsymbol{S}_i\cdot\boldsymbol{S}_j-\frac{\hat{n}_i\hat{n}_j}{4}\right),\qquad J=\frac{4t^2}{U}, \tag{4.43}$$

which is the half-filled limit of the t-J model, Eq. (4.23). For the four-site ring the Heisenberg ground-state energy is $-2J$, and the constant $-\frac{1}{4}n_in_j$ contributes $-\frac{1}{4}$ per bond on four bonds, that is $-J$. Hence

$$E_0\longrightarrow -3J=-\frac{12t^2}{U},\qquad \text{so}\qquad \frac{E_0}{J}\longrightarrow -3, \tag{4.44}$$

which is what Table 4.3 confirms numerically.

Exercise 4: the Calogero ground state

a) Verify by direct substitution that Eq. (4.27) is an eigenstate of Eq. (4.24) with energy (4.28).
b) Show that at $\lambda=1$ the ground state is the Slater determinant of N free fermions in a harmonic trap.
c) Separate the $N=2$ problem and verify the relative energy $\omega(\lambda+\frac{1}{2})$.

Answer to a).

Work in units $\hbar=m=1$ and write $\Psi_0=e^{-\Phi}$ with

$$\Phi = \frac{\omega}{2}\sum_i x_i^2 - \lambda\sum_{i<j}\ln|x_i - x_j|. \tag{4.45}$$

For any such form,

$$-\frac{1}{2}\sum_i \frac{\partial^2\Psi_0}{\partial x_i^2} = \frac{1}{2}\left[\sum_i \frac{\partial^2\Phi}{\partial x_i^2} - \sum_i\left(\frac{\partial\Phi}{\partial x_i}\right)^2\right]\Psi_0. \tag{4.46}$$

The derivatives are

$$\frac{\partial\Phi}{\partial x_i} = \omega x_i - \lambda\sum_{j\neq i}\frac{1}{x_i - x_j}, \qquad \frac{\partial^2\Phi}{\partial x_i^2} = \omega + \lambda\sum_{j\neq i}\frac{1}{(x_i - x_j)^2}. \tag{4.47}$$

Squaring the first,

$$\sum_i\left(\frac{\partial\Phi}{\partial x_i}\right)^2 = \omega^2\sum_i x_i^2 - 2\lambda\omega\sum_i\sum_{j\neq i}\frac{x_i}{x_i - x_j} + \lambda^2\sum_i\sum_{j\neq i}\sum_{k\neq i}\frac{1}{(x_i - x_j)(x_i - x_k)}. \tag{4.48}$$

Two simplifications now do all the work. In the second term, pairing (i,j) with (j,i) gives $x_i/(x_i - x_j) + x_j/(x_j - x_i) = 1$, so the double sum equals $N(N-1)/2$ times two, that is $-2\lambda\omega \cdot N(N-1)/2$. In the third term the pieces with $j = k$ give $\lambda^2\sum_{i\neq j}(x_i - x_j)^{-2}$, while the pieces with $j \neq k$ cancel identically by the three-term identity

$$\frac{1}{(x_i - x_j)(x_i - x_k)} + \frac{1}{(x_j - x_i)(x_j - x_k)} + \frac{1}{(x_k - x_i)(x_k - x_j)} = 0. \tag{4.49}$$

Assembling everything, the inverse-square terms combine as

$$\frac{1}{2}\left[\lambda\sum_{i\neq j}\frac{1}{(x_i - x_j)^2} - \lambda^2\sum_{i\neq j}\frac{1}{(x_i - x_j)^2}\right] = -\lambda(\lambda - 1)\sum_{i<j}\frac{1}{(x_i - x_j)^2}, \tag{4.50}$$

which cancels the interaction term of the Hamiltonian exactly; the $\omega^2 x_i^2$ terms cancel the trap; and what is left is the constant

$$E_0 = \frac{1}{2}\left[N\omega + \lambda\omega N(N-1)\right] = \omega\left[\frac{N}{2} + \frac{\lambda N(N-1)}{2}\right], \tag{4.51}$$

which is Eq. (4.28). The cancellation in Eq. (4.50) works only because the interaction falls off as the inverse *square*; this is the sense in which the model is fine-tuned to be solvable.

Answer to b).

At $\lambda = 1$ the coupling $g = \lambda(\lambda - 1)$ vanishes and the particles are free, but the wave function

$$\Psi_0 = \prod_{i<j}|x_i - x_j|\, e^{-\omega\sum_i x_i^2/2} \tag{4.52}$$

is not the bosonic ground state. Up to an overall sign it is the Vandermonde determinant of Section 1.20,

$$\prod_{i<j}(x_j - x_i) = \det\left[x_i^{j-1}\right], \tag{4.53}$$

and multiplying each column by the Gaussian and taking suitable linear combinations turns $x_i^{j-1}e^{-\omega x_i^2/2}$ into the Hermite functions of Section 1.27. The result is precisely the Slater determinant of the N lowest harmonic oscillator orbitals. The energy check is immediate: Eq. (4.28) with $\lambda = 1$ gives $E_0 = \omega N^2/2 = \omega\sum_{n=0}^{N-1}(n + \frac{1}{2})$, the sum of the N lowest oscillator energies.

For non-integer λ the wave function is neither symmetric nor antisymmetric, and the particles obey fractional statistics interpolating between the two.

Answer to c).

Introduce $X=(x_1+x_2)/2$ and $x=x_1-x_2$. The kinetic energy separates as $-\frac{1}{4}\partial_X^2-\partial_x^2$ and the trap as $\omega^2X^2+\frac{1}{4}\omega^2x^2$, so with $\hbar=m=1$

$$\hat{H}=\left[-\frac{1}{4}\frac{\partial^2}{\partial X^2}+\omega^2X^2\right]+\left[-\frac{\partial^2}{\partial x^2}+\frac{\omega^2}{4}x^2+\frac{\lambda(\lambda-1)}{x^2}\right]. \tag{4.54}$$

The centre-of-mass part is a harmonic oscillator with ground-state energy $\omega/2$. The relative part is Eq. (4.30); its ground state is $x^\lambda e^{-\omega x^2/4}$ with energy $\omega(\lambda+\frac{1}{2})$, as substitution confirms. The total is $\omega/2+\omega(\lambda+\frac{1}{2})=\omega(1+\lambda)$, which agrees with Eq. (4.28) for $N=2$. Solving the relative problem numerically on a grid, as in Section 1.27, reproduces $\omega(\lambda+\frac{1}{2})$ to better than 10^{-5}; see `models.py`.

Exercise 5: numerical explorations

a) Using `models.py`, plot the Lipkin ground-state energy and the weight of $|2,-2\rangle$ in the ground state as functions of V at fixed ε and W. At roughly what value of V/ε does the single-configuration picture break down?
b) Repeat the pairing calculation of Table 4.1 for six and eight levels. How does the correlation energy at $g=1$ scale with the number of levels?
c) Compute the double occupancy of the Hubbard ring as a function of U/t and identify the crossover. Compare a six-site ring with the four-site one.
d) For the pairing model with twelve levels and six pairs, compare the exact ground-state energy from full diagonalisation with the Lanczos result of Table 1.3, and with the Schmidt truncation of Table 1.7. How many Schmidt states are needed for the energy to be correct to six digits?

Part II
Wave function based methods

Chapter 5
Full configuration interaction theory

We will start our discussions of many-body methods with what is called full configuration interaction theory, or just FCI. The reason for this is simple. If the effective degrees of freedom which are relevant for describing a physical system give a sufficiently good reproduction of the static properties (and later we will add dynamical properties) of a many-body system, FCI gives us all relevant eigenpairs (eigenvalues and eigenstates) and allows us to compute other expectation values than the energy levels.

Although it would have been natural to start with mean-field methods (FCI is often labelled as a post Hartree-Fock method), we have chosen a slightly different route. The reason for this is that FCI, within a restricted effective space, provides us with the exact eigenvalues and can thus serve as a benchmark for the approximative methods discussed in the coming chapters, from mean field methods to the Tamm-Dancoff (TDA) and random-phase (RPA) approximations, variants of many-body perturbation theory and coupled-cluster theory, Greens' function approaches and other. We will use FCI to illustrate how these other methods can be interpreeted in terms of specific truncations/approximations of FCI.

We start thus with, what this authour likes to claim as the simplest possible approach to many-body problems, an approach which involves us being able to set up a many-body Hamiltonian matrix (within an effective and reduced Hilbert space), and then simply diagonalize. The overarching principles and the underlying mathematics are straightforward, although the actual implementations may easily lead to endless frustrations and sleepless nights. Let us get started.

5.1 Slater determinants as a basis

The simplest many-body wave function is a product,

$$\Psi(x_1,x_2,\dots,x_N) \approx \phi_1(x_1)\phi_2(x_2)\cdots\phi_N(x_N), \tag{5.1}$$

because we are really only good at thinking about one particle at a time. Product functions are easy to work with: if the single-particle states are orthonormal the products are too, and matrix elements factorise into one-dimensional integrals. The price is the complete absence of correlations, which must then be built up by superposing many, many products. This is the trade-off that runs through the whole subject – a compact representation with difficult integrals, or easy integrals and very many states.

Antisymmetry turns the product into a determinant, and Chapter 2 showed that the two elementary properties of determinants – a sign change under the interchange of two columns, and vanishing when two rows coincide – are exactly the Pauli principle:

- no two particles at the same place (two equal columns);
- no two particles in the same state (two equal rows).

Beyond $N = 4$ the coordinate-space determinant becomes unwieldy, which is why Chapter 3 replaced it by the occupation representation. There a Slater determinant is a string of creation operators acting on the vacuum, each index appearing at most once because $a_i^\dagger a_i^\dagger = 0$, and antisymmetry is carried by the algebra rather than by the notation.

We take that formalism as given. Writing the reference determinant as

$$|\Phi_0\rangle = \left(\prod_{i\le F} \hat{a}_i^\dagger\right)|0\rangle, \tag{5.2}$$

the excited determinants are

$$|\Phi_i^a\rangle = \hat{a}_a^\dagger \hat{a}_i|\Phi_0\rangle, \qquad |\Phi_{ij}^{ab}\rangle = \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i|\Phi_0\rangle, \tag{5.3}$$

and in general

$$|\Phi_{ijk\dots}^{abc\dots}\rangle = \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_c^\dagger \cdots \hat{a}_k \hat{a}_j \hat{a}_i|\Phi_0\rangle, \tag{5.4}$$

with the hole indices i,j,k below the Fermi level and the particle indices a,b,c above it, following the convention of Section 2.6. We call such a determinant an np-nh excitation, or a *configuration*.

5.2 The configuration-interaction expansion

The exact ground state is an arbitrary superposition of all of these,

$$|\Psi_0\rangle = C_0|\Phi_0\rangle + \sum_{ai} C_i^a|\Phi_i^a\rangle + \sum_{abij} C_{ij}^{ab}|\Phi_{ij}^{ab}\rangle + \cdots = \left(C_0 + \hat{C}\right)|\Phi_0\rangle, \tag{5.5}$$

where we have introduced the *correlation operator*

$$\hat{C} = \sum_{ai} C_i^a \hat{a}_a^\dagger \hat{a}_i + \sum_{abij} C_{ij}^{ab} \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i + \cdots \tag{5.6}$$

The normalisation of $|\Psi_0\rangle$ is at our disposal, and C_0 is by hypothesis non-zero, so we may set $C_0 = 1$ with proportional changes in all the other coefficients. With this *intermediate normalisation*,

$$\langle\Psi_0|\Phi_0\rangle = \langle\Phi_0|\Phi_0\rangle = 1, \qquad |\Psi_0\rangle = \left(1+\hat{C}\right)|\Phi_0\rangle. \tag{5.7}$$

It is convenient to compress the notation. Writing H for a set of $0,1,\dots,n$ hole indices and P for a set of $0,1,\dots,n$ particle indices,

$$|\Psi_0\rangle = \sum_{PH} C_H^P|\Phi_H^P\rangle = \left(\sum_{PH} C_H^P \hat{A}_H^P\right)|\Phi_0\rangle, \tag{5.8}$$

with $\hat{A}_H^P$ the operator that creates the corresponding excitation. Unit normalisation reads $\sum_{PH}|C_H^P|^2 = 1$, and the energy is

$$E = \langle\Psi_0|\hat{H}|\Psi_0\rangle = \sum_{PP'HH'} C_H^{*P}\langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}. \tag{5.9}$$

If the expansion runs over *all* determinants that can be built from the chosen single-particle basis, this is *full configuration interaction*. The only approximation left is the truncation of the single-particle space itself.

5.3 The eigenvalue problem

Equation (5.9) is solved by diagonalisation, and it is worth seeing why the diagonalisation is not merely convenient but is the variational answer. We minimise the energy subject to normalisation,

$$\delta\left[\langle\Psi_0|\hat{H}|\Psi_0\rangle-\lambda\langle\Psi_0|\Psi_0\rangle\right]=0, \tag{5.10}$$

with λ a Lagrange multiplier. Carrying out the variation,

$$\sum_{P'H'}\left\{\delta[C_H^{*P}]\langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}+C_H^{*P}\langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle\delta[C_{H'}^{P'}]-\lambda\left(\delta[C_H^{*P}]C_{H'}^{P'}+C_H^{*P}\delta[C_{H'}^{P'}]\right)\right\}=0. \tag{5.11}$$

Since $\delta[C_H^{*P}]$ and $\delta[C_{H'}^{P'}]$ are complex conjugates, it is necessary and sufficient that the quantities multiplying $\delta[C_H^{*P}]$ vanish, giving

$$\sum_{P'H'}\langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}=\lambda\, C_H^P \tag{5.12}$$

for every set PH. Multiplying by C_H^{*P} and summing identifies $\lambda=E$, so the multiplier is the energy and

$$\boxed{\sum_{P'H'}\langle\Phi_H^P|\hat{H}-E|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}=0} \tag{5.13}$$

This is the full CI equation. The same result follows more directly by writing $(\hat{H}-E)|\Psi_0\rangle=0$ and projecting successively onto every $\langle\Phi_H^P|$.

Equation (5.13) is a Hermitian eigenvalue problem, and the diagonalisation is a Rayleigh-Ritz variational calculation in the space spanned by the chosen determinants: the lowest eigenvalue is the best upper bound to E_0 obtainable in that space, and it can only improve as the space grows. This is exactly the property that made the Lanczos algorithm of Section 1.26 so well suited to the problem – the lowest Ritz value there is the same variational bound, computed in a Krylov space instead of a space of determinants.

Everything now depends on being able to build and diagonalise the matrix $\langle\Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle$. The matrix elements are supplied by the Condon-Slater rules of Section 3.13, and the diagonalisation by the algorithms of Chapter 1: Householder and QL for small matrices, Lanczos for large ones.

5.4 The structure of the Hamiltonian matrix

Suppose we have six fermions below the Fermi level, so that at most $6p$-$6h$ excitations are possible. Ordering the determinants by excitation level, the Hamiltonian matrix has the block structure

$$\begin{array}{c|ccccccc}
 & 0p0h & 1p1h & 2p2h & 3p3h & 4p4h & 5p5h & 6p6h \\
\hline
0p0h & \times & \times & \times & 0 & 0 & 0 & 0 \\
1p1h & \times & \times & \times & \times & 0 & 0 & 0 \\
2p2h & \times & \times & \times & \times & \times & 0 & 0 \\
3p3h & 0 & \times & \times & \times & \times & \times & 0 \\
4p4h & 0 & 0 & \times & \times & \times & \times & \times \\
5p5h & 0 & 0 & 0 & \times & \times & \times & \times \\
6p6h & 0 & 0 & 0 & 0 & \times & \times & \times
\end{array} \tag{5.14}$$

The zeros are not accidental. A two-body operator connects Slater determinants differing by at most two single-particle states, which is the last of the Condon-Slater rules of Section 3.13. The matrix is therefore *banded* in the excitation level, with bandwidth two, and overwhelmingly sparse: for six particles the fraction of non-zero blocks is already small, and it falls as the space grows. Chapter 1 called this the structure that Lanczos exploits; here we see where it comes from.

In a Hartree-Fock basis.

If the single-particle states are the solutions of the Hartree-Fock equations rather than the eigenstates of $\hat{h}_0$, one more set of elements vanishes. Brillouin's theorem, met in Section 3.13, states that

$$\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = 0, \tag{5.15}$$

so the reference determinant does not couple to the singles at all,

$$\begin{array}{c|ccccccc}
 & 0p0h & 1p1h & 2p2h & 3p3h & 4p4h & 5p5h & 6p6h \\
\hline
0p0h & \tilde{\times} & 0 & \tilde{\times} & 0 & 0 & 0 & 0 \\
1p1h & 0 & \tilde{\times} & \tilde{\times} & \tilde{\times} & 0 & 0 & 0 \\
2p2h & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} & 0 & 0 \\
3p3h & 0 & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} & 0 \\
4p4h & 0 & 0 & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} \\
5p5h & 0 & 0 & 0 & \tilde{\times} & \tilde{\times} & \tilde{\times} & \tilde{\times} \\
6p6h & 0 & 0 & 0 & 0 & \tilde{\times} & \tilde{\times} & \tilde{\times}
\end{array} \tag{5.16}$$

where the tildes are a reminder that a unitary transformation of the single-particle basis has changed every matrix element. The correlation energy is then carried entirely by the doubles at leading order, which is the starting point for both many-body perturbation theory and coupled-cluster theory.

The programme `fci.py` builds this structure explicitly. For the pairing model with four levels and four particles the full space has $\binom{8}{4} = 70$ determinants, distributed as 1, 16, 36, 16, 1 over the excitation levels 0 to 4, and the computed block table is

$$\begin{array}{c|ccccc}
 & 0p0h & 1p1h & 2p2h & 3p3h & 4p4h \\
\hline
0p0h & \times & 0 & \times & 0 & 0 \\
1p1h & 0 & \times & 0 & \times & 0 \\
2p2h & \times & 0 & \times & 0 & \times \\
3p3h & 0 & \times & 0 & \times & 0 \\
4p4h & 0 & 0 & \times & 0 & \times
\end{array} \tag{5.17}$$

Even the odd blocks are empty here, because the pairing interaction moves particles two at a time and can never change the number of broken pairs. The symmetry has done more work than Brillouin's theorem would.

5.5 The exponential wall

Full configuration interaction is exact, and that is precisely its problem. With N particles distributed over n single-particle states the number of Slater determinants is

$$\dim \mathscr{H} = \binom{n}{N}, \tag{5.18}$$

which is the count of bit patterns with N bits set out of n, in the representation of Section 3.14. This is a combinatorial expression, but combinatorial growth turns into exponential growth as soon as the particle number scales with the size of the basis. At half filling, $N = n/2$, Stirling's formula gives

$$\binom{n}{n/2} \sim \frac{2^n}{\sqrt{\pi n/2}} \sim \frac{2^n}{\sqrt{n}}, \tag{5.19}$$

so the dimension doubles every time one single-particle state is added. The program `fci.py` tabulates both sides:

n	$\binom{n}{n/2}$	$2^n/\sqrt{n}$
10	2.520×10^2	3.238×10^2
20	1.848×10^5	2.345×10^5
40	1.378×10^{11}	1.738×10^{11}
80	1.075×10^{23}	1.352×10^{23}
160	9.205×10^{46}	1.155×10^{47}

The estimate is accurate to within a factor of order one over five decades of n and twenty-five orders of magnitude of dimension.

A realistic example.

Consider oxygen-16 and let the neutrons occupy the four lowest major oscillator shells, the $0s$, $0p$, $1s0d$ and $1p0f$ shells, which together hold $2+6+12+20=40$ single-particle states for each species. The eight neutrons can be distributed in

$$\binom{40}{8} = 7.690 \times 10^7 \tag{5.20}$$

ways, and since the protons can be distributed independently the full proton-neutron space has

$$\binom{40}{8}^2 = 5.914 \times 10^{15} \tag{5.21}$$

Slater determinants. Rotational and parity symmetry reduce this by orders of magnitude, but adding a single further major shell, or relaxing the truncation, restores the growth immediately. Direct diagonalisation by Householder and QL, Sections 1.24 and 1.25, is limited to matrices of dimension around 10^5 because it stores and modifies the whole matrix. Lanczos, Section 1.26, needs only the product $\boldsymbol{Hv}$ and reaches perhaps 10^{10} on a large machine. Beyond that no rearrangement of the linear algebra helps: the expansion itself must be truncated.

The same wall in the models of Chapter 4.

Each of the models of the previous chapter meets Eq. (5.19) in its own way.

- The *Lipkin model* with N particles in two N-fold degenerate levels has $\binom{2N}{N}$ determinants in total and $\binom{N}{N/2} \sim 2^N/\sqrt{N}$ in the maximal-quasispin multiplet. That the SU(2) structure of Section 4.1 collapses the problem to an $(N+1)$-dimensional matrix is a symmetry accident, not a general escape.
- The *pairing model* of Section 4.2 has $\binom{2n}{N}$ determinants for n doubly degenerate levels, of which only $\binom{n}{N/2}$ have seniority zero. Again a symmetry – here the conservation of the number of broken pairs – takes the square root of the dimension, and again it is special to the model.
- The *Hubbard model* of Section 4.3 has a local Hilbert space of dimension four, so L sites give 4^L states in total. At fixed particle number this becomes $\binom{L}{N_\uparrow}\binom{L}{N_\downarrow}$, Eq. (4.21), which is again exponential at any extensive filling.
- A *spin chain*, the limit of the Hubbard model at large U, has no combinatorial disguise at all: $\mathscr{H} = \bigotimes_{i=1}^{L} \mathbb{C}^2$ has dimension 2^L by construction.

The pattern is worth stating plainly. Combinatorial counting and tensor-product structure are two faces of the same growth; the tensor product is fundamental, the binomial coefficient is what survives after a conservation law has been imposed. A conserved quantity can remove a factor, but never the exponential.

Why the wall is not the end of the story. The dimension of $\mathscr{H}$ counts the coefficients an *arbitrary* state would need. Physical ground states are not arbitrary. For local Hamiltonians with a gap, the entanglement entropy across a cut grows with the area of the cut rather than with the volume it encloses, and states obeying such an area law are described to high accuracy by a number of parameters polynomial in L. The Schmidt decomposition of Section 1.30 is the one-cut version of this statement, and the singular values decaying rapidly is exactly what makes the truncation work. Applying it at every bond of a chain gives a matrix product state,

$$|\psi\rangle = \sum_{s_1 \dots s_L} A_1^{s_1} A_2^{s_2} \cdots A_L^{s_L} |s_1 s_2 \dots s_L\rangle, \tag{5.22}$$

whose cost is polynomial in the bond dimension D; this is the density-matrix renormalisation group. The same idea appears elsewhere in this book in different clothing: a neural-network quantum state replaces the exponentially many amplitudes by a polynomially parameterised function of the occupation pattern, and quantum computing replaces them by n qubits whose Hilbert space is exponential by construction. Full CI is the reference against which all of these compressions are measured, which is why we treat it first.

5.6 Full CI for the pairing model

The pairing Hamiltonian of Section 4.2,

$$\hat{H} = \xi \sum_{p\sigma} (p-1) \hat{a}^\dagger_{p\sigma} \hat{a}_{p\sigma} - \frac{1}{2} g \sum_{pq} \hat{a}^\dagger_{p+} \hat{a}^\dagger_{p-} \hat{a}_{q-} \hat{a}_{q+}, \tag{5.23}$$

is the ideal first test. It is small enough to diagonalise in the full space, and Chapter 4 already gave the exact answer in the seniority-zero subspace, so the two calculations can be checked against each other.

Take four doubly degenerate levels and four particles, $\xi = 1$. The full space has $\binom{8}{4} = 70$ determinants, distributed over the excitation levels as 1, 16, 36, 16, 1. The seniority-zero subspace used in Chapter 4 has only $\binom{4}{2} = 6$ states. Diagonalising the 70×70 matrix at $g = 1$ gives $E_0 = 0.63554847$, identical to the value in Table 4.1 to all digits shown. The determinants with broken pairs are present in the basis but carry no amplitude in the ground state, because the interaction cannot create a broken pair from a paired configuration. The symmetry that we imposed by hand in Chapter 4 is recovered automatically by the diagonalisation.

This is the general situation. A symmetry of $\hat{H}$ block-diagonalises the FCI matrix, and one may either exploit it, gaining a smaller matrix and a labelled spectrum, or ignore it, gaining a simpler code at the cost of diagonalising a larger matrix. Production codes exploit it; the program accompanying this chapter does not, precisely so that the two answers can be compared.

5.7 Full CI for the Hubbard model

The Hubbard model of Section 4.3 is the harder test, and the more instructive one. We take a ring of $L = 6$ sites at half filling, $N_\uparrow = N_\downarrow = 3$, and work in the momentum basis, where the hopping term is diagonal,

$$\hat{H} = \sum_{k\sigma} \varepsilon_k \hat{c}^\dagger_{k\sigma} \hat{c}_{k\sigma} + \frac{U}{L} \sum_{kk'q} \hat{c}^\dagger_{k+q\uparrow} \hat{c}^\dagger_{k'-q\downarrow} \hat{c}_{k'\downarrow} \hat{c}_{k\uparrow}, \qquad \varepsilon_k = -2t\cos k, \tag{5.24}$$

with $k = 2\pi m/L$. In this basis the non-interacting ground state is a single determinant – the Fermi sea – so the machinery of Section 5.2 applies directly with $|\Phi_0\rangle$ the filled Fermi sea. The dimension is $\binom{6}{3}^2 = 400$, with the excitation levels populated as 1, 18, 99, 164, 99, 18, 1.

The interesting quantity is how much of the correlation energy survives truncation as U/t grows:

U/t	E_{ref}	CIS	CID	CISD	FCI
1	-6.500000	-6.50000000	-6.59980159	-6.59989778	-6.60115829
2	-5.000000	-5.00000000	-5.38877703	-5.39026458	-5.40945685
4	-2.000000	-2.00000000	-3.41192866	-3.43018837	-3.66870618
8	4.000000	2.48338852	-0.36779217	-1.05692183	-2.04813089

The doubles recover 98.66% of the correlation energy at $U/t = 1$ but only 72.22% at $U/t = 8$. Two features deserve comment.

First, the reference does not couple to the singles at all: the interaction conserves total momentum, and a single excitation changes it, so $\langle \Phi_0|\hat{H}|\Phi_i^a\rangle = 0$ exactly, for every U. This is Brillouin's theorem, Eq. (5.15), enforced here by a symmetry rather than by a variational condition – the plane-wave determinant is the Hartree-Fock solution of a translationally invariant problem. The CIS matrix is therefore block diagonal, and its lowest eigenvalue is the smaller of the reference energy and the lowest eigenvalue of the singles block alone. For $U/t \leq 4$ the reference wins and CIS reproduces it exactly. At $U/t = 8$ it does not: the singles block, strongly mixed by the large interaction, has an eigenvalue at 2.483, below the reference at 4.000. The number in the table is that eigenvalue, and the state it belongs to is orthogonal to $|\Phi_0\rangle$. What CIS reports as a ground state is by then not a correlated version of the reference at all.

Second, the deterioration with U/t is the signature of *strong correlation*. At small U the ground state is a single determinant with small corrections and any method built on that picture works. At large U the ground state is a superposition of very many determinants with comparable weight, the reference is a poor starting point, and a truncation at any fixed excitation level fails. This is where the tensor-network and quantum-computing approaches of the later chapters become attractive, and it is why the Hubbard model at intermediate U/t remains a benchmark problem.

5.8 Truncating the expansion

Since the full expansion is out of reach for all but the smallest systems, we truncate it. The natural truncation is by excitation level: keep all determinants up to n particle-hole pairs and discard the rest. This gives the familiar hierarchy

$$|\Psi_{\mathrm{CIS}}\rangle = \left(1+\hat{C}_1\right)|\Phi_0\rangle, \tag{5.25}$$

$$|\Psi_{\mathrm{CID}}\rangle = \left(1+\hat{C}_2\right)|\Phi_0\rangle, \tag{5.26}$$

$$|\Psi_{\mathrm{CISD}}\rangle = \left(1+\hat{C}_1+\hat{C}_2\right)|\Phi_0\rangle, \tag{5.27}$$

and so on to CISDT, CISDTQ, until at $\hat{C}_N$ we are back at full CI. Each of these is again a variational calculation in a subspace, so each lowest eigenvalue is an upper bound to E_0 and the bounds decrease monotonically as the truncation is relaxed. The cost, for a basis of n states and N particles, is roughly $\binom{N}{k}\binom{n-N}{k}$ determinants at excitation level k, that is polynomial of degree $2k$ – which is why CISD, at $\mathscr{O}(n^6)$, is affordable and CISDTQ, at $\mathscr{O}(n^{10})$, is not.

For the pairing model with four levels and four particles the truncated spaces have dimensions 1, 17, 53, 69, 70, and the energies are

g	E_{ref}	CIS	CID	CISD	FCI
0.25	1.750000	1.75000000	1.73050700	1.73050700	1.73045985
0.50	1.500000	1.50000000	1.41759645	1.41759645	1.41677428
1.00	1.000000	1.00000000	0.64890655	0.64890655	0.63554847
2.00	0.000000	0.00000000	-1.35208670	-1.35208670	-1.48965216

The singles change nothing at all, and CISD coincides with CID: the pairing interaction moves pairs and cannot break them, so it has no matrix element between the reference and a $1p$-$1h$ state, nor between a $1p$-$1h$ state and the reference through any intermediate. The doubles carry essentially the whole correlation energy – 99.8% at $g = 0.25$, still 96.3% at $g = 1$, but only 90.8% at $g = 2$. The pattern is the same as in the Hubbard model: truncation at fixed excitation level works while the reference is good and degrades as the coupling grows.

5.9 Size consistency

There is a defect of truncated CI that is more serious than the loss of accuracy, because it does not go away as the basis is improved. Consider two identical systems A and B, infinitely far apart and non-interacting. The Hamiltonian is $\hat{H}_A + \hat{H}_B$ and the exact ground state is the product $|\Psi_A\rangle|\Psi_B\rangle$, so the exact energy must satisfy

$$E(AB) = E(A) + E(B) = 2E(A). \tag{5.28}$$

A method that reproduces this is called *size consistent*.

Full CI is size consistent, since the product state lies in the full space. Truncated CI is not. The reason is immediate: if A and B are each doubly excited, the combined state is a *quadruple* excitation of the combined reference, and CID has no room for it. The product of two doubles is a quadruple, and truncating at a fixed excitation level is not a multiplicatively closed operation.

The program `fci.py` demonstrates this on two non-interacting copies of a two-level, two-particle pairing system:

g	$2E(A)$	$E(AB)$, FCI	$E(AB)$, CID	CID error
0.25	-0.26556444	-0.26556444	-0.26550480	5.96×10^{-5}
0.50	-0.56155281	-0.56155281	-0.56066017	8.93×10^{-4}
1.00	-1.23606798	-1.23606798	-1.22474487	1.13×10^{-2}

Full CI passes to machine precision; CID fails, and the failure grows with the coupling. With M subsystems instead of two the error grows with M, so the energy per particle computed with truncated CI drifts as the system is made larger. For an extended system – nuclear matter, the electron gas, a large molecule – this is fatal.

The repair is to make the ansatz multiplicative rather than additive. Replacing $1 + \hat{C}$ by $\exp(\hat{T})$ with $\hat{T} = \hat{T}_1 + \hat{T}_2$ gives

$$|\Psi_{\mathrm{CCSD}}\rangle = e^{\hat{T}_1 + \hat{T}_2}|\Phi_0\rangle = \left(1 + \hat{T}_1 + \hat{T}_2 + \tfrac{1}{2}\hat{T}_2^2 + \cdots\right)|\Phi_0\rangle, \tag{5.29}$$

in which the quadruple excitation $\frac{1}{2}\hat{T}_2^2$ appears automatically, with its coefficient fixed by the doubles rather than determined independently. For two non-interacting subsystems $e^{\hat{T}} = e^{\hat{T}_A}e^{\hat{T}_B}$ and size consistency is exact. This is coupled-cluster theory, and Eq. (5.29) is the reason it, rather than truncated CI, is the workhorse of modern many-body physics. We shall develop it properly in a later chapter; the point here is that its central idea is a response to a defect that full CI lets us see clearly.

5.10 Truncations and the other many-body methods

We can now do what the introduction to this chapter promised, and read the standard many-body methods as truncations of Eq. (5.13). Write the projected equations explicitly. Multiplying $(\hat{H} - E)|\Psi_0\rangle = 0$ from the left by $\langle\Phi_0|$ and by $\langle\Phi_H^P|$ gives

$$E = \langle \Phi_0|\hat{H}|\Phi_0\rangle + \sum_{P'H'\neq 0} \langle \Phi_0|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}, \tag{5.30}$$

$$\left(E - \langle \Phi_H^P|\hat{H}|\Phi_H^P\rangle\right) C_H^P = \sum_{P'H'\neq PH} \langle \Phi_H^P|\hat{H}|\Phi_{H'}^{P'}\rangle C_{H'}^{P'}. \tag{5.31}$$

The first equation says that the correlation energy comes entirely from the coupling of the reference to the doubles, since only those have a non-zero matrix element with $|\Phi_0\rangle$. The second is a set of coupled equations for the coefficients, which can be solved by iteration: guess the C's, compute the right-hand side, update, repeat. Every method in the remaining chapters is a prescription for which terms of Eq. (5.31) to keep and how to solve it.

- **Hartree-Fock theory** keeps only C_0 and instead varies the single-particle basis so that $\langle \Phi_0|\hat{H}|\Phi_i^a\rangle = 0$ – the Brillouin condition of Eq. (5.15). It is the optimisation of the reference rather than of the expansion, and it fixes the basis in which every subsequent truncation is made. In the language of Eq. (5.16), Hartree-Fock chooses the unitary transformation that empties the $0p0h$-$1p1h$ block.
- **The Tamm-Dancoff approximation** restricts the space to the $1p$-$1h$ block alone and diagonalises it. It is CIS applied to excited states, and it gives the excitation spectrum of the reference at the cheapest non-trivial level.
- **The random-phase approximation** enlarges TDA by allowing the ground state to contain $2p$-$2h$ correlations, so that the excitation operator carries both $1p$-$1h$ and $1h$-$1p$ pieces. It is a partial summation of the doubles block, performed to all orders in the coupling but only for a selected class of terms.
- **Many-body perturbation theory** solves Eq. (5.31) by expanding in powers of the interaction rather than by iterating to convergence. Setting E on the left-hand side to the reference energy and $C_{H'}^{P'}$ on the right to zero gives the first-order coefficients; substituting these back gives the second-order energy, and so on. The order in perturbation theory is the number of times the right-hand side has been substituted into itself.
- **Coupled-cluster theory** keeps the same truncation in excitation level as CISD but writes the wave function multiplicatively, Eq. (5.29). The equations look like Eq. (5.31) with additional nonlinear terms, and it is those nonlinear terms that restore size consistency.

Seen this way the methods are not a list of unrelated techniques but a set of answers to one question: given that we cannot keep all of Eq. (5.5), which parts do we keep, and do we keep them additively or multiplicatively? Full CI is the fixed point that lets us measure each answer. Whenever a model is small enough for Eq. (5.13) to be solved exactly – the pairing model, the Lipkin model, a short Hubbard ring – we should solve it, and use the result to calibrate the approximation before applying it to a system where no calibration is possible.

5.11 Practical considerations

Three remarks on implementation, which anticipate the code discussed in the companion notebook.

Never store the matrix.

The FCI matrix is sparse, with the band structure of Eq. (5.14), and for interesting dimensions it cannot be stored. The Lanczos algorithm of Section 1.26 needs only the product $\boldsymbol{Hv}$, and that product is computed by looping over the determinants and over the non-zero matrix elements of $\hat{H}$, using the bit representation of Section 3.14 to find which determinant each term produces and with which sign. This is the single most important structural choice in an FCI code.

Reorthogonalise.

The Lanczos vectors lose orthogonality as soon as an eigenvalue has converged, and spurious copies of converged eigenvalues appear. Section 1.26 discussed the remedies; for FCI the usual choice is full or selective reorthogonalisation, at the price of storing the Lanczos vectors.

Use the symmetries.

Angular momentum, parity, particle number, total momentum and spin projection all commute with $\hat{H}$ and block-diagonalise the matrix. Working in a single symmetry block reduces the dimension by orders of magnitude and, just as importantly, labels the states. The pairing calculation of Section 5.6 illustrated both options – 70 determinants without the symmetry, 6 with it, the same ground-state energy.

5.12 Summary

Full configuration interaction is the statement that a many-body state is a superposition of all Slater determinants that the single-particle basis allows, together with the observation that finding the best superposition is a Hermitian eigenvalue problem, Eq. (5.13). Everything else in this chapter follows from that. The Condon-Slater rules make the matrix banded in the excitation level; a Hartree-Fock basis empties one more block; the combinatorics of Eq. (5.18) makes the matrix too large to build beyond a few tens of single-particle states; and the truncations that this forces upon us are precisely the approximations that the following chapters elevate into methods.

The program `fci.py` in `BookMaterial/Programs/` carries out every calculation quoted above: the pairing model in the full 70-dimensional space and in the seniority-zero subspace, the block structure of the Hamiltonian, the CIS/CID/CISD/FCI comparison for pairing and for the Hubbard ring, the size-consistency test, and the dimension tables of Section 5.5. The accompanying notebook `fci.ipynb` runs the same code cell by cell.

5.13 Exercises

Exercise 1: counting determinants

a) Show that the number of Slater determinants with N particles in n single-particle states that are k particle-hole excitations of a given reference is $\binom{N}{k}\binom{n-N}{k}$, and verify that $\sum_k \binom{N}{k}\binom{n-N}{k} = \binom{n}{N}$.
b) Evaluate this for $n = 8$, $N = 4$ and compare with the distribution $1, 16, 36, 16, 1$ quoted in Section 5.4.
c) Use Stirling's formula to derive Eq. (5.19), and determine the largest n for which $\binom{n}{n/2}$ stays below 10^{10}.

Answer to a).

Choosing which k of the N occupied states to empty can be done in $\binom{N}{k}$ ways, and which k of the $n-N$ empty states to fill in $\binom{n-N}{k}$ ways; the two choices are independent. Every determinant with N particles arises in this way exactly once, for exactly one value of k, namely the number of occupied states it does not share with the reference. Summing over k must therefore give the total, which is Vandermonde's identity

$$\sum_k \binom{N}{k}\binom{n-N}{k} = \sum_k \binom{N}{N-k}\binom{n-N}{k} = \binom{n}{N}. \tag{5.32}$$

Answer to b).

With $n = 8$ and $N = 4$ the products are $\binom{4}{k}^2 = 1, 16, 36, 16, 1$ for $k = 0, \ldots, 4$, summing to $70 = \binom{8}{4}$. This is exactly the distribution that `fci.py` finds by classifying the 70 determinants of the pairing model by excitation level.

Answer to c).

Stirling's formula $m! \sim \sqrt{2\pi m}\,(m/e)^m$ gives

$$\binom{n}{n/2} = \frac{n!}{[(n/2)!]^2} \sim \frac{\sqrt{2\pi n}\,(n/e)^n}{2\pi(n/2)\,\left[(n/2e)^{n/2}\right]^2} = \frac{2^n}{\sqrt{\pi n/2}}, \tag{5.33}$$

which is Eq. (5.19) up to the constant $\sqrt{\pi/2} \approx 1.25$. Solving $\binom{n}{n/2} = 10^{10}$ numerically: $\binom{36}{18} = 9.08 \times 10^9$ and $\binom{38}{19} = 3.49 \times 10^{10}$, so $n = 36$ is the last even value below the limit. Every further pair of single-particle states multiplies the dimension by about four.

Exercise 2: which blocks vanish

Let $\hat{H} = \hat{H}_0 + \hat{H}_I$ with $\hat{H}_I$ a two-body operator.

a) Show that $\langle \Phi_H^P | \hat{H} | \Phi_{H'}^{P'} \rangle = 0$ whenever the two determinants differ by more than two single-particle states, and deduce the band structure of Eq. (5.14).
b) Suppose $\hat{H}$ also contains a genuine three-body force. How does the band structure change, and what does that do to the cost of a matrix-vector product?
c) For the pairing Hamiltonian, Eq. (5.23), show that every block connecting an even to an odd excitation level vanishes, as in Eq. (5.17).

Answer to a).

Write both determinants in the occupation representation of Section 3.14. A two-body operator is a product of two creation and two annihilation operators, so it can change at most two occupation numbers from one to zero and two from zero to one. If the two determinants differ in more than four occupation numbers – that is, by more than two single-particle states – then in the Wick expansion of Section 3.13 at least one operator is left uncontracted in every term, and the matrix element is zero. Since an *np*-*nh* and an *mp*-*mh* determinant differ by at least $|n - m|$ single-particle states, the element vanishes for $|n - m| > 2$. The matrix is therefore banded with bandwidth two in the excitation level, which is Eq. (5.14).

Answer to b).

A three-body operator changes up to three occupations of each kind, so the bandwidth becomes three and the $0p0h$ row now reaches the $3p3h$ block. The number of non-zero elements per row grows from $\mathcal{O}(N^2(n-N)^2)$ to $\mathcal{O}(N^3(n-N)^3)$, so a matrix-vector product costs a further factor $N(n-N)$ – typically two to three orders of magnitude. This is why three-body forces, though physically important in nuclear physics, are usually normal-ordered with respect to the reference determinant and then truncated at the two-body level, a procedure whose justification rests on Section 3.5.

Answer to c).

The interaction $-\frac{1}{2}g\sum_{pq}\hat{a}^\dagger_{p+}\hat{a}^\dagger_{p-}\hat{a}_{q-}\hat{a}_{q+}$ removes a complete pair from level q and creates a complete pair in level p. Define the seniority s as the number of unpaired particles. Both the one-body term and the interaction leave s unchanged, so $[\hat{H},\hat{s}]=0$ and $\hat{H}$ is block diagonal in seniority. An excitation of odd order relative to the paired reference must leave at least two particles unpaired, so it has $s\geq 2$, whereas the reference and all even excitations reached from it by moving whole pairs have $s=0$. Every even-odd matrix element therefore connects different seniorities and vanishes. Numerically the 70×70 matrix splits into a six-dimensional $s=0$ block, a $s=2$ block and so on, and the ground state lies in the first.

Exercise 3: full CI for the pairing model

Take four doubly degenerate levels with single-particle energies $\varepsilon_p=(p-1)\xi$, $p=1,\ldots,4$, four particles, $\xi=1$, and the Hamiltonian of Eq. (5.23).

a) Set up the six seniority-zero configurations and write down the 6×6 Hamiltonian matrix.
b) Diagonalise it for $g=-1,-0.5,0,0.5,1$ and reproduce Table 4.1.
c) Repeat the calculation in the full 70-dimensional space and confirm that the ground-state energy is unchanged. Inspect the eigenvector and verify that every determinant with a broken pair has zero amplitude.
d) Plot the correlation energy against g and show that it is quadratic for small $|g|$, as second-order perturbation theory requires.

Answer to a).

Label a seniority-zero configuration by the pair of levels that are occupied. With four levels and two pairs there are $\binom{4}{2}=6$ of them,

$$|12\rangle,\ |13\rangle,\ |14\rangle,\ |23\rangle,\ |24\rangle,\ |34\rangle, \tag{5.34}$$

where $|pq\rangle$ has pairs in levels p and q. The diagonal element of $|pq\rangle$ is $2\xi[(p-1)+(q-1)]-g$: the factor two because each level holds two particles, and $-g$ because the $p=q$ terms of the interaction give $-\frac{1}{2}g$ for each of the two occupied pairs. A single application of the interaction moves *one* pair, so two configurations are connected by $-\frac{1}{2}g$ if and only if they share exactly one occupied level; the three pairs of configurations that share no level, $|12\rangle$–$|34\rangle$, $|13\rangle$–$|24\rangle$ and $|14\rangle$–$|23\rangle$, are not connected at all. With $\xi=1$ and the ordering of Eq. (5.34),

$$\boldsymbol{H}=\begin{pmatrix} 2-g & -g/2 & -g/2 & -g/2 & -g/2 & 0 \\ -g/2 & 4-g & -g/2 & -g/2 & 0 & -g/2 \\ -g/2 & -g/2 & 6-g & 0 & -g/2 & -g/2 \\ -g/2 & -g/2 & 0 & 6-g & -g/2 & -g/2 \\ -g/2 & 0 & -g/2 & -g/2 & 8-g & -g/2 \\ 0 & -g/2 & -g/2 & -g/2 & -g/2 & 10-g \end{pmatrix}. \tag{5.35}$$

Answer to b).

Diagonalising Eq. (5.35) reproduces Table 4.1; at $g=1$ the lowest eigenvalue is 0.63554847 and the reference energy is $E_{\text{ref}}=H_{11}=1$.

Answer to c).

The class `PairingFCI` in `fci.py` builds the full space with no seniority restriction. Its dimension is $\binom{8}{4} = 70$, the excitation levels are populated as $1, 16, 36, 16, 1$, and the lowest eigenvalue at $g = 1$ is 0.63554847, identical to b) to every digit. Projecting the ground-state eigenvector onto the 64 determinants with at least one broken pair gives zero to machine precision, as Exercise 2c) predicts.

Answer to d).

The reference is $|12\rangle$. Second-order perturbation theory in the interaction gives

$$E^{(2)} = \sum_{\Phi \neq \Phi_0} \frac{|\langle \Phi_0 | \hat{H}_I | \Phi \rangle|^2}{E_0^{(0)} - E_\Phi^{(0)}} = \frac{g^2}{4}\left(\frac{1}{-2} + \frac{1}{-4} + \frac{1}{-4} + \frac{1}{-6}\right) = -\frac{7}{24}g^2, \tag{5.36}$$

using the four configurations reachable from $|12\rangle$ by moving one pair, with unperturbed excitation energies $2, 4, 4, 6$. A log-log plot of $|E_{\mathrm{corr}}|$ against g has slope 2 for $|g| \lesssim 0.2$ and bends away above that, where the higher orders take over. At $g = 0.25$, Eq. (5.36) gives -0.01823 against the exact -0.01954; at $g = 0.5$ it gives -0.0729 against -0.0832, and at $g = 1$ only -0.2917 against -0.3645.

Exercise 4: truncations and size consistency

a) Using the code of the previous exercise, restrict the basis to excitation levels ≤ 1 and ≤ 2 and reproduce the CIS and CID columns of Section 5.8. Explain why CIS returns exactly the reference energy.
b) Build a system of two identical, non-interacting two-level pairing systems. Show that full CI gives exactly twice the energy of one subsystem while CID does not, and tabulate the error for $g = 0.25, 0.5, 1$.
c) Identify the configuration that CID is missing, and show that including quadruple excitations restores the exact result.
d) Show that the ansatz $\exp(\hat{T}_2)|\Phi_0\rangle$, with the same number of independent parameters as CID, is size consistent for this system.

Answer to a).

In a truncated space the eigenvalue problem is Eq. (5.13) restricted to the corresponding rows and columns. The dimensions for the cuts at $0, 1, 2, 3, 4$ are $1, 17, 53, 69, 70$. CIS returns the reference energy because $\langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = 0$: a single excitation breaks a pair and so changes the seniority, which the pairing Hamiltonian conserves (Exercise 2c). The reference is therefore decoupled from the entire singles block and remains an exact eigenstate of the truncated matrix. Its energy is also the lowest diagonal element, and the singles block lies far above it for the couplings considered, so it is the CIS ground state.

Answer to b).

Let each subsystem have two levels at 0 and ξ, holding one pair. The combined system then has four levels with energies $0, \xi, 0, \xi$ and two pairs, one per subsystem, with the interaction acting only within a subsystem. The class `NonInteractingCopies` gives

g	$2E(A)$	$E(AB)_{\mathrm{FCI}}$	$E(AB)_{\mathrm{CID}}$	CID error
0.25	-0.26556444	-0.26556444	-0.26550480	5.96×10^{-5}
0.50	-0.56155281	-0.56155281	-0.56066017	8.93×10^{-4}
1.00	-1.23606798	-1.23606798	-1.22474487	1.13×10^{-2}

(5.37)

Full CI is exact to machine precision; the CID error grows by roughly an order of magnitude each time g is doubled.

Answer to c).

Writing $\hat{A}_2^X$ for the operator that excites the pair of subsystem X, the exact product state is

$$\left(1+c\hat{A}_2^A\right)\left(1+c\hat{A}_2^B\right)|\Phi_0\rangle = \left(1+c\hat{A}_2^A+c\hat{A}_2^B+c^2\hat{A}_2^A\hat{A}_2^B\right)|\Phi_0\rangle. \tag{5.38}$$

The last term excites both subsystems at once and is a $4p$-$4h$ excitation of the combined reference; it is absent from CID. Adding the quadruples – which for this small system means going to the full space – recovers the exact energy, and the coefficient of the quadruple comes out equal to c^2, the product of the two double amplitudes.

Answer to d).

With $\hat{T}_2 = t\left(\hat{A}_2^A+\hat{A}_2^B\right)$ the two terms commute, since they act on disjoint sets of orbitals, so

$$e^{\hat{T}_2} = e^{t\hat{A}_2^A}e^{t\hat{A}_2^B} = \left(1+t\hat{A}_2^A\right)\left(1+t\hat{A}_2^B\right), \tag{5.39}$$

the last step because $(\hat{A}_2^X)^2 = 0$ – one cannot excite the same pair twice. The exponential ansatz therefore generates exactly the product state of Eq. (5.38) with one parameter per subsystem, the same count as CID, and the energy is additive by construction. This is coupled cluster with doubles in miniature, and it shows that the cure for size inconsistency costs nothing in parameters – only a change from a linear to a multiplicative ansatz.

Exercise 5: the Hubbard ring by full CI

Take a ring of $L=6$ sites at half filling with $t=1$, in the momentum basis of Eq. (5.24).

a) Verify that the dimension is $\binom{6}{3}^2 = 400$ and that the excitation levels are populated as 1, 18, 99, 164, 99, 18, 1.
b) Compute the exact ground-state energy for $U/t = 1,2,4,8$ and compare with the CID result.
c) Show that $\langle\Phi_0|\hat{H}|\Phi_i^a\rangle = 0$ for every single excitation, and explain this using conservation of total momentum.
d) At $U/t=8$ the CIS energy lies *below* the reference energy even though the reference does not couple to the singles. Explain how this is possible and what it says about the reliability of a variational bound.
e) Compare the exact energies at large U with the Heisenberg limit of Section 4.3.

Answer to a).

There are $\binom{6}{3} = 20$ ways to place three electrons of each spin, and the two species are independent, so the dimension is 400. Counting for each determinant the number of occupied momenta outside the Fermi sea, separately for each spin, and adding the two counts gives $1,18,99,164,99,18,1$, which sums to 400. The distribution is symmetric because a kp-kh excitation of the Fermi sea is a $(6-k)p$-$(6-k)h$ excitation of the completely inverted determinant.

Answer to b).

The values are those of Section 5.7. Relative to the reference, the doubles recover 98.66%, 94.95%, 84.61% and 72.22% of the correlation energy at $U/t = 1,2,4,8$.

Answer to c).

The interaction is $\frac{U}{L}\sum_{kk'q} \hat{c}^{\dagger}_{k+q\uparrow}\hat{c}^{\dagger}_{k'-q\downarrow}\hat{c}_{k'\downarrow}\hat{c}_{k\uparrow}$, in which the momenta of the two created operators sum to those of the two annihilated ones; total crystal momentum is therefore conserved. A single excitation $i \to a$ changes the total momentum by $k_a - k_i \neq 0$ and so cannot be reached from the Fermi sea. The one-body part is diagonal in k and cannot connect them either. Evaluating the couplings numerically gives zero to machine precision for every U. Note that this is Brillouin's theorem, Eq. (5.15), obtained here from a symmetry rather than from a variational condition: for a translationally invariant problem the plane-wave determinant *is* the Hartree-Fock solution.

Answer to d).

Because the coupling vanishes, the CIS matrix is block diagonal: a 1×1 block holding E_{ref} and an 18×18 singles block. Its lowest eigenvalue is the smaller of the two numbers. At $U/t \leq 4$ the reference wins and CIS reproduces it exactly. At $U/t = 8$ the singles block, strongly mixed by the large interaction, has an eigenvalue at 2.483, which lies below $E_{\text{ref}} = 4.000$, and the variational principle selects the corresponding state – one that is exactly orthogonal to $|\Phi_0\rangle$. The number is a legitimate upper bound to E_0, but it is not a correlated correction to the reference; the calculation has silently changed which state it describes. The moral is that a lower variational energy does not by itself mean a better description of the state one set out to compute, and that the overlap with the reference should always be monitored alongside the energy.

Answer to e).

For $U/t \gg 1$ the half-filled Hubbard model maps onto a Heisenberg chain with $J = 4t^2/U$, Eq. (4.23), whose ground-state energy per site tends to $-J(\ln 2 - \frac{1}{4})$ in the thermodynamic limit. At $U/t = 8$ and $L = 6$ this estimate gives about -1.3 against the exact -2.05. The discrepancy is expected: the mapping neglects terms of order t^4/U^3, the ring is short, and $U/t = 8$ is not yet deep in the strong-coupling regime. Repeating the comparison at $U/t = 20$ and 40 shows the two converging.

Exercise 6: the exponential wall in practice

a) A shell-model code stores the Hamiltonian matrix in double precision. For a dimension of 10^6 with on average 10^3 non-zero elements per row, and a 4-byte column index per element, how much memory does the sparse matrix need? Compare with storing the full matrix.
b) How large a single-particle space can be treated at half filling if a Lanczos code can handle 10^{10} determinants?
c) A matrix product state with bond dimension D on L sites has about LD^2d parameters, with d the local dimension. For a spin chain with $d = 2$, $L = 100$ and $D = 200$, compare this with 2^L.
d) Discuss what must be true of the physical state for the comparison in c) to be a fair one.

Answer to a).

The sparse matrix holds $10^6 \times 10^3 = 10^9$ elements at $8+4=12$ bytes each, or 12 GB – large but routine. The full matrix would need 8×10^{12} bytes, 8 TB, and diagonalising it by Householder and QL would take $\mathscr{O}(10^{18})$ floating-point operations. This is the practical content of Section 5.11: the dense algorithms of Sections 1.24 and 1.25 are unusable here, and even the sparse matrix is best not stored at all but regenerated on the fly inside the matrix-vector product.

Answer to b).

From Eq. (5.19), $2^n/\sqrt{n} = 10^{10}$ gives $n \approx 38$, and the exact condition $\binom{n}{n/2} \leq 10^{10}$ gives $n = 36$. A Lanczos calculation at the limit of current hardware therefore reaches roughly 36–40 single-particle states at half filling. Away from half filling the binomial coefficient is much smaller and the reach is greater, which is why calculations near closed shells are so much easier than calculations in the middle of a shell.

Answer to c).

The matrix product state has about $100 \times 200^2 \times 2 = 8 \times 10^6$ parameters, against $2^{100} = 1.27 \times 10^{30}$ amplitudes for a general state – a compression by twenty-four orders of magnitude.

Answer to d).

The comparison is fair only if the state of interest is actually representable at that bond dimension. The Schmidt spectrum across every bond must decay fast enough that keeping D singular values costs little, which is the truncation error of the Eckart-Young theorem in Section 1.30 applied bond by bond. This holds for the ground states of gapped one-dimensional local Hamiltonians, which obey an area law. For critical chains the required D grows polynomially with L; for volume-law states – generic excited states, or the state produced by a quantum quench after a time proportional to L – no polynomial D suffices and the exponential wall stands undiminished. The same caveat applies to every other compressed representation, including the neural-network and quantum-circuit ansaetze of the later chapters: each buys its efficiency by restricting the class of states it can describe, and full configuration interaction remains the only method that makes no such restriction at all.

Chapter 6
Mean-field approaches

Chapter 5 showed that the exact answer is a superposition of an exponentially large number of Slater determinants, and that we cannot afford it. This chapter takes the opposite tack. Instead of adding determinants we keep exactly one and ask which one is best. The answer is Hartree-Fock theory, and it is the foundation on which every method in the remaining chapters is built: perturbation theory expands about it, coupled cluster exponentiates corrections to it, and the truncated configuration interaction of Section 5.8 is measured against it.

Chapter 2 already wrote down the Hartree-Fock equations, Eq. (2.41), and deferred their derivation. We supply it here, in two independent ways – by varying the expansion coefficients of the orbitals, and by varying the determinant itself in second quantization. The second route requires Thouless' theorem, which characterises what a "nearby determinant" means, and Thouless' theorem in turn rests on the Baker-Campbell-Hausdorff expansion. That expansion is worth a section of its own, because the same algebra reappears in the Trotter splitting used for time evolution and for quantum simulation later in the book. Armed with it we can ask not merely whether the Hartree-Fock solution is stationary but whether it is a minimum, and we close with the one many-body system for which the whole programme can be carried through analytically: the infinite homogeneous electron gas.

6.1 Why a mean field?

Hartree-Fock theory is an algorithm for finding an approximate ground state of a given Hamiltonian. It has two ingredients. The first is a single-particle basis defined by its own eigenvalue problem,

$$\hat{h}^{\mathrm{HF}}\psi_\alpha = \varepsilon_\alpha \psi_\alpha, \qquad \hat{h}^{\mathrm{HF}} = \hat{t} + \hat{u}_{\mathrm{ext}} + \hat{u}^{\mathrm{HF}}, \tag{6.1}$$

in which $\hat{u}^{\mathrm{HF}}$ is a one-body potential still to be determined. The second is the prescription for determining it: choose $\hat{u}^{\mathrm{HF}}$ so that

$$E^{\mathrm{HF}} = \langle \Phi_0 | \hat{H} | \Phi_0 \rangle \tag{6.2}$$

is stationary, with $|\Phi_0\rangle$ a single Slater determinant built from the N lowest solutions of Eq. (6.1). The variational principle of Section 2.7 then guarantees $E^{\mathrm{HF}} \geq E_0$, so whatever we obtain is an upper bound on the exact ground-state energy.

We shall see that the resulting one-body operator is precisely the operator $\hat{f}$ that appears when the Hamiltonian is normal-ordered with respect to the reference determinant. For a specific matrix element,

$$\langle p|\hat{h}^{\mathrm{HF}}|q\rangle = \langle p|\hat{f}|q\rangle = \langle p|\hat{t} + \hat{u}_{\mathrm{ext}}|q\rangle + \sum_{i \leq F} \langle pi|\hat{v}|qi\rangle_{\mathrm{AS}}, \tag{6.3}$$

so that

$$\langle p|\hat{u}^{\mathrm{HF}}|q\rangle = \sum_{i \leq F} \langle pi|\hat{v}|qi\rangle_{\mathrm{AS}}. \tag{6.4}$$

The sum runs over all single-particle states below the Fermi level, and this is the whole content of the mean-field idea: the two-body interaction, summed over the other fermions, creates an effective one-body field in which a given fermion moves.

Two features of Eq. (6.4) deserve emphasis. It brings an explicit *medium dependence*, since the potential felt by a particle depends on which states are occupied; and it brings an explicit dependence on the interaction, so different interactions give different mean fields even at the same density. For atoms and molecules there is an external potential $\hat{u}_{\text{ext}}$ – the attraction of the nuclei. For nuclei there is none: a nucleus is a self-bound system, held together by the interaction between its own constituents, and the entire one-body field must be generated by Eq. (6.4).

6.2 Variational calculus and Lagrange multipliers

Before deriving the Hartree-Fock equations we recall the machinery, in its simplest setting. The calculus of variations deals with problems in which the quantity to be made stationary is an integral,

$$E[\Phi] = \int_a^b f\left(\Phi(x), \frac{\partial \Phi}{\partial x}, x\right) dx. \tag{6.5}$$

The difficulty is that although f is a known function of Φ, $\partial\Phi/\partial x$ and x, the dependence of Φ on x is not known: the limits are fixed but the path is not. In our case the unknowns are the single-particle wave functions, and the constraints are the integrals expressing their orthonormality. Constraints of this kind are handled with Lagrange multipliers.

Take one particle in three dimensions, with

$$E = \int \psi^* \hat{H} \psi \, dxdydz, \qquad \int \psi^* \psi \, dxdydz = 1, \qquad \hat{H} = -\frac{1}{2}\nabla^2 + V. \tag{6.6}$$

Integrating the kinetic term by parts, with ψ vanishing at the boundaries,

$$\int \psi^* \left(-\frac{1}{2}\nabla^2\right) \psi \, dxdydz = \int \frac{1}{2} \nabla\psi^* \cdot \nabla\psi \, dxdydz, \tag{6.7}$$

so that the constrained variation reads

$$\delta \left\{ \int \left(\frac{1}{2}\nabla\psi^* \cdot \nabla\psi + V\psi^*\psi - \lambda\psi^*\psi \right) dxdydz \right\} = 0. \tag{6.8}$$

Writing the integrand as

$$f = \frac{1}{2}\left(\psi_x^*\psi_x + \psi_y^*\psi_y + \psi_z^*\psi_z\right) + V\psi^*\psi - \lambda\psi^*\psi, \tag{6.9}$$

with subscripts denoting partial derivatives, the Euler-Lagrange equation for ψ^*,

$$\frac{\partial f}{\partial \psi^*} - \frac{\partial}{\partial x}\frac{\partial f}{\partial \psi_x^*} - \frac{\partial}{\partial y}\frac{\partial f}{\partial \psi_y^*} - \frac{\partial}{\partial z}\frac{\partial f}{\partial \psi_z^*} = 0, \tag{6.10}$$

gives

$$-\frac{1}{2}\nabla^2\psi + V\psi = \lambda\psi. \tag{6.11}$$

The multiplier is the energy, and the stationarity condition is nothing but the Schrödinger equation. This is the pattern we shall meet again: a constrained variation produces an eigenvalue problem, and the multiplier enforcing normalisation turns out to be the eigenvalue. It happened already in Section 5.3, where the same manipulation produced the full configuration-interaction equation, and it will happen once more below.

6.3 Determinants under a linear change of basis

One property of determinants is needed before we can vary the coefficients, and it is worth isolating because it explains why the whole construction is consistent.

A determinant is linear in each column separately. For a 2×2 example,

$$\begin{vmatrix} \alpha_1 b_{11} + \alpha_2 b_{12} & a_{12} \\ \alpha_1 b_{21} + \alpha_2 b_{22} & a_{22} \end{vmatrix} = \alpha_1 \begin{vmatrix} b_{11} & a_{12} \\ b_{21} & a_{22} \end{vmatrix} + \alpha_2 \begin{vmatrix} b_{12} & a_{12} \\ b_{22} & a_{22} \end{vmatrix}, \tag{6.12}$$

and in general, for an $n \times n$ matrix in which one column is a linear combination $\sum_k c_k b_{jk}$,

$$\det\left(\dots \; \textstyle\sum_k c_k b_{jk} \; \dots\right) = \sum_{k=1}^{n} c_k \det\left(\dots \; b_{jk} \; \dots\right), \tag{6.13}$$

where the omitted columns are the same on both sides. Applying this to *every* column in turn gives the product rule

$$\det(\boldsymbol{BC}) = \det(\boldsymbol{B}) \det(\boldsymbol{C}), \tag{6.14}$$

which we already used in Section 2.3. The reader is encouraged to check the case $n = 2$ explicitly.

Now let the new orbitals be expanded in a fixed orthonormal basis, as in Eq. (2.17),

$$\psi_p(x) = \sum_\lambda C_{p\lambda} \phi_\lambda(x). \tag{6.15}$$

The Slater determinant built from the ψ's is

$$\frac{1}{\sqrt{N!}} \begin{vmatrix} \sum_\lambda C_{p\lambda} \phi_\lambda(x_1) & \dots & \sum_\lambda C_{p\lambda} \phi_\lambda(x_N) \\ \vdots & & \vdots \\ \sum_\lambda C_{t\lambda} \phi_\lambda(x_1) & \dots & \sum_\lambda C_{t\lambda} \phi_\lambda(x_N) \end{vmatrix} = \det(\boldsymbol{C}) \; \Phi(\phi), \tag{6.16}$$

with $\Phi(\phi)$ the determinant of the original basis functions. If $\boldsymbol{C}$ is unitary then $|\det(\boldsymbol{C})| = 1$ and the two determinants differ by a phase only. Two consequences follow, and both are used constantly.

First, orthonormality is preserved. If $\boldsymbol{v}_j^T \boldsymbol{v}_i = \delta_{ij}$ and $\boldsymbol{w}_i = \boldsymbol{U}\boldsymbol{v}_i$ with $\boldsymbol{U}$ unitary, then

$$\boldsymbol{w}_j^\dagger \boldsymbol{w}_i = \boldsymbol{v}_j^\dagger \boldsymbol{U}^\dagger \boldsymbol{U} \boldsymbol{v}_i = \boldsymbol{v}_j^\dagger \boldsymbol{v}_i = \delta_{ij}, \tag{6.17}$$

which is Section 1.9 restated. So if $\langle \alpha | \beta \rangle = \delta_{\alpha\beta}$ then $\langle p | q \rangle = \delta_{pq}$, and a unitary change of coefficients never spoils the basis we are working in.

Second, a Slater determinant is invariant, up to a phase, under any unitary mixing of its occupied orbitals. The determinant does not know which particular orbitals were used to build it, only which N-dimensional subspace they span. This is why the Hartree-Fock solution is properly a statement about a subspace, and why the individual orbitals are determined only up to such a mixing – the freedom that is fixed, conventionally, by demanding that the Fock matrix be diagonal.

Where this comes from. Every ingredient of this section already appeared in Chapter 1: multilinearity and the product rule in Section 2.3, unitary transformations in Section 1.9, and the practical evaluation of $\det(\boldsymbol{C})$ by LU decomposition in Section 1.15. The last is not an idle remark: in variational Monte Carlo the ratio of two such determinants is evaluated millions of times, which is why Section 2.11 developed the Sherman-Morrison update instead.

6.4 The Hartree-Fock equations by varying the coefficients

We can now carry out the derivation that Chapter 2 postponed. The energy functional of a single determinant, Eq. (2.37), is

$$E[\Phi] = \sum_{\mu=1}^{N} \langle\mu|\hat{h}_0|\mu\rangle + \frac{1}{2}\sum_{\mu=1}^{N}\sum_{\nu=1}^{N} \langle\mu\nu|\hat{v}|\mu\nu\rangle_{\mathrm{AS}}, \tag{6.18}$$

and inserting Eq. (6.15) turns it into a function of the coefficients,

$$E[\Phi^{\mathrm{HF}}] = \sum_{i=1}^{N}\sum_{\alpha\beta} C_{i\alpha}^{*}C_{i\beta}\langle\alpha|\hat{h}_0|\beta\rangle + \frac{1}{2}\sum_{i,j=1}^{N}\sum_{\alpha\beta\gamma\delta} C_{i\alpha}^{*}C_{j\beta}^{*}C_{i\gamma}C_{j\delta}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}}. \tag{6.19}$$

Greek indices run over the whole fixed basis, Latin indices over the N occupied orbitals.

The coefficients are not free: orthonormality of the new orbitals requires

$$\langle i|j\rangle = \delta_{ij} = \sum_{\alpha\beta} C_{i\alpha}^{*}C_{i\beta}\langle\alpha|\beta\rangle = \sum_{\alpha} C_{i\alpha}^{*}C_{i\alpha}, \tag{6.20}$$

so we introduce one multiplier ε_i per occupied orbital and minimise

$$F[\Phi^{\mathrm{HF}}] = E[\Phi^{\mathrm{HF}}] - \sum_{i=1}^{N}\varepsilon_i\sum_{\alpha} C_{i\alpha}^{*}C_{i\alpha}. \tag{6.21}$$

The coefficients and their complex conjugates may be varied independently – the real and imaginary parts are two independent real variations, and it is equivalent and more convenient to use C and C^* – so we impose

$$\frac{\partial}{\partial C_{i\alpha}^{*}}\left[E[\Phi^{\mathrm{HF}}] - \sum_{j}\varepsilon_j\sum_{\beta} C_{j\beta}^{*}C_{j\beta}\right] = 0. \tag{6.22}$$

Differentiating Eq. (6.19) is straightforward. The one-body term contributes $\sum_\beta C_{i\beta}\langle\alpha|\hat{h}_0|\beta\rangle$. In the two-body term $C_{i\alpha}^*$ appears twice, once as $C_{i\alpha}^*$ and once as $C_{j\beta}^*$ with $j=i$; the two contributions are equal by the antisymmetry of $\langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}}$ under the simultaneous interchange of both pairs, and they cancel the factor $\frac{1}{2}$. The result is

$$\sum_{\beta} C_{i\beta}\langle\alpha|\hat{h}_0|\beta\rangle + \sum_{j=1}^{N}\sum_{\beta\gamma\delta} C_{j\beta}^{*}C_{j\delta}C_{i\gamma}\langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}} = \varepsilon_i^{\mathrm{HF}}C_{i\alpha}. \tag{6.23}$$

Relabelling the dummy indices so that the coefficient $C_{i\beta}$ can be factored out of both terms,

$$\sum_{\beta}\left\{\langle\alpha|\hat{h}_0|\beta\rangle + \sum_{j=1}^{N}\sum_{\gamma\delta} C_{j\gamma}^{*}C_{j\delta}\langle\alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}\right\}C_{i\beta} = \varepsilon_i^{\mathrm{HF}}C_{i\alpha}. \tag{6.24}$$

Defining the Hartree-Fock, or Fock, matrix

$$h_{\alpha\beta}^{\mathrm{HF}} = \langle\alpha|\hat{h}_0|\beta\rangle + \sum_{j=1}^{N}\sum_{\gamma\delta} C_{j\gamma}^{*}C_{j\delta}\langle\alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}, \tag{6.25}$$

the condition becomes

$$\boxed{\sum_{\beta} h_{\alpha\beta}^{\mathrm{HF}}C_{i\beta} = \varepsilon_i^{\mathrm{HF}}C_{i\alpha}} \tag{6.26}$$

This is Eq. (2.41), now derived. It is a standard Hermitian eigenvalue problem, solvable by the Householder reduction and QL iteration of Sections 1.24 and 1.25, or by Lanczos, Section 1.26, when the basis is large – with the one complication that the matrix depends on its own eigenvectors.

If the fixed basis is chosen as the eigenbasis of $\hat{h}_0$, which is the usual choice, the first term is diagonal,

$$h^{\mathrm{HF}}_{\alpha\beta} = \varepsilon_\alpha \delta_{\alpha\beta} + \sum_{j=1}^{N} \sum_{\gamma\delta} C^*_{j\gamma} C_{j\delta} \langle \alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}, \tag{6.27}$$

and Eq. (6.26) can be written compactly as $\boldsymbol{h}^{\mathrm{HF}}\boldsymbol{C} = \boldsymbol{\varepsilon}^{\mathrm{HF}}\boldsymbol{C}$.

An important practical point is visible in Eq. (6.27): no integrals need to be recomputed during the iteration. The matrix elements $\langle\alpha|\hat{h}_0|\beta\rangle$ and $\langle\alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}$ are tabulated once and for all; every subsequent iteration is a contraction of those tables with the current coefficients, followed by a diagonalisation. The cost of storing them is the n^4 scaling discussed in Section 2.10, and it is the reason the factorisations of Section 1.32 matter.

6.5 The density matrix and the self-consistent field algorithm

The double sum over the occupied orbitals in Eq. (6.27) depends on the coefficients only through the combination

$$\rho_{\gamma\delta} = \sum_{i=1}^{N} \langle\gamma|i\rangle\langle i|\delta\rangle = \sum_{i=1}^{N} C_{i\gamma} C^*_{i\delta}, \tag{6.28}$$

the one-body *density matrix*, which is precisely the object introduced in Section 1.31. It is the projector onto the occupied subspace, $\boldsymbol{\rho}^2 = \boldsymbol{\rho}$ and $\operatorname{tr}\boldsymbol{\rho} = N$, which is the formal statement of the invariance found in Section 6.3: the Fock matrix depends on the occupied subspace, not on the particular orbitals chosen to span it. With it,

$$h^{\mathrm{HF}}_{\alpha\beta} = \varepsilon_\alpha \delta_{\alpha\beta} + \sum_{\gamma\delta} \rho_{\gamma\delta} \langle \alpha\gamma|\hat{v}|\beta\delta\rangle_{\mathrm{AS}}. \tag{6.29}$$

Building $\boldsymbol{\rho}$ once per iteration and contracting it with the stored two-body table is the efficient way to organise the calculation.

The algorithm is then an iteration of the kind discussed in Section 1.17, with a diagonalisation replacing a single sweep.

1. Start from a guess, conventionally $C^{(0)}_{i\alpha} = \delta_{i\alpha}$, which makes the reference determinant the one built from the N lowest eigenstates of $\hat{h}_0$. Random orthonormal vectors work too, and are sometimes preferable because they avoid convergence onto a symmetry-constrained saddle point.
2. Build $\boldsymbol{\rho}$ from Eq. (6.28) and $\boldsymbol{h}^{\mathrm{HF}}$ from Eq. (6.29).
3. Diagonalise, obtaining $C^{(1)}_{i\alpha}$ and $\varepsilon^{\mathrm{HF}(1)}_i$.
4. Repeat from step 2 until the single-particle energies stop moving,

$$\frac{1}{m}\sum_{p}\left|\varepsilon^{(n)}_p - \varepsilon^{(n-1)}_p\right| \leq \lambda, \tag{6.30}$$

with m the number of single-particle states and λ a prescribed tolerance, typically 10^{-8} or smaller.

The program `hartreefock.py` implements exactly this loop. For the system of Section 2.9 – spinless fermions in a one-dimensional oscillator trap with a softened Coulomb repulsion, in a basis of eight oscillator orbitals – it reproduces Table 2.2, and the gain over the unoptimised determinant grows with the particle number:

N	iterations	E(reference)	E(Hartree-Fock)	gain
2	17	2.69018342	2.66308878	-2.71×10^{-2}
3	17	6.46801021	6.39016133	-7.78×10^{-2}
4	17	11.77084712	11.62305385	-1.48×10^{-1}

Brillouin's theorem.

At convergence the Fock matrix is diagonal in its own eigenbasis, so its occupied-unoccupied block vanishes,

$$f_{ai} = 0 \qquad \Longleftrightarrow \qquad \langle \Phi_0 | \hat{H} | \Phi_i^a \rangle = 0. \tag{6.31}$$

The program reports $\max|f_{ai}| = 8.7 \times 10^{-11}$ for $N = 4$, at the prescribed tolerance. This is Brillouin's theorem, met first in Section 3.13 and used in Section 5.4 to empty the $0p0h$–$1p1h$ block of the configuration-interaction matrix. We can now see what it means: the Hartree-Fock solution is the choice of single-particle basis that decouples the reference from all single excitations. Everything left over is carried by the doubles and higher, which is why Eq. (5.30) contained only doubles and why both perturbation theory and coupled-cluster theory start there.

6.6 Hartree-Fock in second quantization

The derivation of Section 6.4 varied the orbitals. We now vary the determinant, which is more abstract but far better suited to the question of stability, and which is the natural language given Chapter 3.

Our ansatz is the determinant

$$|\Phi_0\rangle = |c\rangle = \hat{a}_i^\dagger \hat{a}_j^\dagger \cdots \hat{a}_l^\dagger |0\rangle, \tag{6.32}$$

and we seek $\hat{u}^{\mathrm{HF}}$ such that $E_0^{\mathrm{HF}} = \langle c|\hat{H}|c\rangle$ is a local minimum. The stationarity condition is that the energy does not change to first order under any variation of the determinant,

$$\langle \delta c | \hat{H} | c \rangle = 0 \tag{6.33}$$

for every admissible $|\delta c\rangle$. To make this precise we must know what the admissible variations are – that is, what the neighbourhood of a Slater determinant looks like inside the space of Slater determinants. That is the content of Thouless' theorem.

6.7 Thouless' theorem

Theorem 6.1 (Thouless). *Any Slater determinant $|c'\rangle$ that is not orthogonal to a given determinant $|c\rangle = \prod_{i=1}^{n} \hat{a}_{\alpha_i}^\dagger |0\rangle$ can be written as*

$$|c'\rangle = \exp\left\{ \sum_{a>F} \sum_{i \le F} C_{ai}\, \hat{a}_a^\dagger \hat{a}_i \right\} |c\rangle. \tag{6.34}$$

In words: every determinant in the neighbourhood of $|c\rangle$ is obtained from it by an exponentiated linear combination of one-particle-one-hole excitations. This is a strong statement. It says that the manifold of Slater determinants is parametrised, near any point, by the $(n-N) \times N$ amplitudes C_{ai} and by nothing else – so the Hartree-Fock variation has exactly that many degrees of freedom.

Step one: the exponential collapses.

The exponential of a sum of operators factorises into a product of exponentials only when the operators commute. When they do not, the Baker-Campbell-Hausdorff formula of Section 6.8 supplies the correction. Here we are lucky: for particle indices $a, b > F$ and hole indices $i, j \le F$, the two index sets are disjoint, and a short calculation with the anticommutation relations of Chapter 3 gives

$$\left[\hat{a}_a^\dagger \hat{a}_i,\, \hat{a}_b^\dagger \hat{a}_j \right] = 0. \tag{6.35}$$

Indeed, since $i \neq b$ we may write $\hat{a}_i\hat{a}_b^\dagger = -\hat{a}_b^\dagger\hat{a}_i$, and likewise $\hat{a}_j\hat{a}_a^\dagger = -\hat{a}_a^\dagger\hat{a}_j$; both orderings then reduce to $-\hat{a}_a^\dagger\hat{a}_b^\dagger\hat{a}_i\hat{a}_j$ and the commutator vanishes. Writing

$$\hat{T} = \sum_{i\leq F}\hat{T}_i, \qquad \hat{T}_i = \sum_{a>F} C_{ai}\hat{a}_a^\dagger\hat{a}_i, \tag{6.36}$$

we therefore have $e^{\hat{T}} = \prod_i e^{\hat{T}_i}$. Moreover each factor truncates, because

$$\hat{T}_i^2 = \sum_{ab>F} C_{ai}C_{bi}\,\hat{a}_a^\dagger\hat{a}_i\hat{a}_b^\dagger\hat{a}_i = 0, \tag{6.37}$$

since the rightmost $\hat{a}_i$ empties the hole i and the second one then annihilates the state – an instance of $(\hat{a}_i)^n = 0$ for $n > 1$. Hence

$$|c'\rangle = \prod_{i\leq F}\left\{1 + \sum_{a>F} C_{ai}\hat{a}_a^\dagger\hat{a}_i\right\}|c\rangle. \tag{6.38}$$

Note carefully that the product in Eq. (6.38) is *not* the same as $1+\hat{T}$: the cross terms $\hat{T}_i\hat{T}_j$ with $i \neq j$ survive and describe $2p$-$2h$ admixtures. Dropping them is legitimate only for infinitesimal amplitudes, which is exactly the situation in the stability analysis below.

Step two: the product is a determinant.

Insert $|c\rangle = \hat{a}_{i_1}^\dagger\hat{a}_{i_2}^\dagger\cdots\hat{a}_{i_n}^\dagger|0\rangle$ into Eq. (6.38). The factor labelled i_1 acts only on $\hat{a}_{i_1}^\dagger$, the factor labelled i_2 only on $\hat{a}_{i_2}^\dagger$, and so on, so the product reorganises into

$$\begin{aligned} |c'\rangle &= \left(1 + \sum_{a>F} C_{ai_1}\hat{a}_a^\dagger\hat{a}_{i_1}\right)\hat{a}_{i_1}^\dagger\left(1 + \sum_{a>F} C_{ai_2}\hat{a}_a^\dagger\hat{a}_{i_2}\right)\hat{a}_{i_2}^\dagger\cdots|0\rangle \\ &= \prod_{i\leq F}\left(\hat{a}_i^\dagger + \sum_{a>F} C_{ai}\hat{a}_a^\dagger\right)|0\rangle. \end{aligned} \tag{6.39}$$

Defining a new creation operator

$$\hat{b}_i^\dagger = \hat{a}_i^\dagger + \sum_{a>F} C_{ai}\hat{a}_a^\dagger, \tag{6.40}$$

we have $|c'\rangle = \prod_i \hat{b}_i^\dagger|0\rangle$, which has exactly the form of a Slater determinant. The rotated state is a single determinant, not a superposition: this is the whole point.

Step three: it is the general determinant.

Equation (6.40) looks less general than it should, because the sum runs only over states above the Fermi level. Consider therefore the most general one,

$$\tilde{b}_i^\dagger = \sum_p f_{ip}\hat{a}_p^\dagger, \qquad |\tilde{c}\rangle = \prod_i \tilde{b}_i^\dagger|0\rangle, \tag{6.41}$$

with f unitary, and require only that $\langle c|\tilde{c}\rangle \neq 0$. Expanding,

$$\langle c|\tilde{c}\rangle = \langle 0|\hat{a}_{i_n}\cdots\hat{a}_{i_1}\left(\sum_p f_{i_1p}\hat{a}_p^\dagger\right)\cdots\left(\sum_t f_{i_nt}\hat{a}_t^\dagger\right)|0\rangle = \det\left(f_{ip}\right)_{i,p\leq F}, \tag{6.42}$$

the determinant of the occupied block of f: only the terms with p below the Fermi level survive the vacuum expectation value, and summing over the permutations reconstructs the determinant. The hypothesis

$\langle c|\tilde{c}\rangle \neq 0$ says exactly that this determinant is non-zero, so the occupied block is invertible, and intermediate normalisation lets us rescale to $\det(f_{ip}) = 1$. With f^{-1} available we can form

$$\sum_i f_{ki}^{-1} \tilde{b}_i^\dagger = \sum_i \sum_p f_{ki}^{-1} f_{ip} \hat{a}_p^\dagger = \hat{a}_k^\dagger + \sum_{p>F} \left(\sum_{i \leq F} f_{ki}^{-1} f_{ip} \right) \hat{a}_p^\dagger, \tag{6.43}$$

where the terms with $p \leq F$ have collapsed to δ_{kp}. Setting $C_{pk} = \sum_{i \leq F} f_{ki}^{-1} f_{ip}$ returns Eq. (6.40) exactly. Since a unitary mixing of the $\tilde{b}^\dagger$ changes the determinant only by a factor, as Section 6.3 showed, $|\tilde{c}\rangle$ and $|c'\rangle$ are the same state. The theorem is proved.

Numerical check.

The program `hartreefock.py` verifies each step explicitly for six orbitals and three particles, with random amplitudes of size 0.3: $\max|[\hat{T}_i, \hat{T}_j]| = 2.8 \times 10^{-17}$, $\max|\hat{T}_i^2| = 0$, $|c'\rangle$ agrees with $\prod_i \hat{b}_i^\dagger|0\rangle$ to 5.6×10^{-17}, the overlap $\langle c|c'\rangle$ is exactly one, and $\det(f_{ip}) = 1$. The difference between $e^{\hat{T}}|c\rangle$ and $(1+\hat{T})|c\rangle$ falls by a factor of four each time the amplitudes are halved, confirming that the discarded piece is second order.

What it buys us.

Thouless' theorem lets us write any nearby determinant as

$$|c'\rangle = |c\rangle + |\delta c\rangle, \qquad |c'\rangle \simeq \left(1 + \sum_{ai} \delta C_{ai} \hat{a}_a^\dagger \hat{a}_i \right) |c\rangle \tag{6.44}$$

to first order in the amplitudes. The stationarity condition (6.33) then reads

$$\sum_{ai} \delta C_{ai}^* \langle c|\hat{a}_i^\dagger \hat{a}_a \hat{H}|c\rangle = 0 \quad \text{for arbitrary } \delta C \qquad \Longrightarrow \qquad \langle \Phi_i^a|\hat{H}|\Phi_0\rangle = 0, \tag{6.45}$$

which is Brillouin's theorem again, Eq. (6.31). The two derivations therefore agree: varying the orbitals and varying the determinant give the same equations. What the second route adds is the ability to go to second order, which is the subject of Section 6.10.

6.8 The Baker-Campbell-Hausdorff expansion

Thouless' theorem needed only the trivial case of the following result, but the general case is used repeatedly later – in the similarity transformation of coupled-cluster theory, in the interaction picture, and in the Trotter splitting of the next section – so we develop it here.

For two operators X and Y that do not commute, $e^X e^Y \neq e^{X+Y}$. The Baker-Campbell-Hausdorff formula states that the product is nevertheless a single exponential, $e^X e^Y = e^Z$, with

$$Z = X + Y + \frac{1}{2}[X,Y] + \frac{1}{12}\Big([X,[X,Y]] + [Y,[Y,X]] \Big) - \frac{1}{24}\big[Y,[X,[X,Y]]\big] + \cdots \tag{6.46}$$

Every term beyond the first is a nested commutator of X and Y. No ordinary products appear, which is the essential structural statement: if X and Y belong to a Lie algebra, so does Z. The coefficients $\frac{1}{2}, \frac{1}{12}, \frac{1}{24}, \ldots$ involve Bernoulli numbers and were first given in general by Dynkin.

Derivation of the first terms.

Expand both sides in powers and match order by order. With $Z = X+Y+A_2+A_3+\cdots$, where A_n collects the terms of total degree n,

$$e^X e^Y = I + (X+Y) + \frac{1}{2}\left(X^2+2XY+Y^2\right) + \mathscr{O}(3), \qquad e^Z = I + Z + \frac{1}{2}Z^2 + \mathscr{O}(3). \tag{6.47}$$

At first order, A_1 is absent and $Z^{(1)} = X+Y$. At second order the left-hand side supplies $\frac{1}{2}(X^2+2XY+Y^2)$ while $\frac{1}{2}(X+Y)^2 = \frac{1}{2}(X^2+XY+YX+Y^2)$; the difference is $\frac{1}{2}(XY-YX) = \frac{1}{2}[X,Y]$, which must therefore be supplied by A_2,

$$A_2 = \frac{1}{2}[X,Y]. \tag{6.48}$$

The same procedure at third order, using the Jacobi identity $[X,[Y,W]]+[Y,[W,X]]+[W,[X,Y]] = 0$ to reduce the independent structures, gives

$$A_3 = \frac{1}{12}\big[X,[X,Y]\big] - \frac{1}{12}\big[Y,[X,Y]\big] = \frac{1}{12}\Big([X,[X,Y]]+[Y,[Y,X]]\Big). \tag{6.49}$$

When the series terminates.

If $[X,Y]$ commutes with both X and Y – in particular if it is a multiple of the identity, $[X,Y] = cI$ – then every higher commutator vanishes and

$$e^X e^Y = \exp\left(X+Y+\tfrac{1}{2}[X,Y]\right) \tag{6.50}$$

is exact. This is the case for canonically conjugate operators, $[\hat{x},\hat{p}] = i\hbar$, from which the Weyl form of the displacement operator follows, and for the ladder operators of the harmonic oscillator, $[\hat{a},\hat{a}^\dagger] = 1$. It is also, trivially, the case in Thouless' theorem, where by Eq. (6.35) the commutator is zero and the series stops at the first term.

How good is a truncation?

For the two nilpotent matrices $X = \left(\begin{smallmatrix}0 & 0.1\\ 0 & 0\end{smallmatrix}\right)$ and $Y = \left(\begin{smallmatrix}0 & 0\\ 0.1 & 0\end{smallmatrix}\right)$, `hartreefock.py` gives

order	terms kept	$\Vert e^X e^Y - e^{Z_n}\Vert$
1	$X+Y$	7.071×10^{-3}
2	$+\frac{1}{2}[X,Y]$	2.379×10^{-4}
3	$+\frac{1}{12}\big([X,[X,Y]]+[Y,[Y,X]]\big)$	1.190×10^{-5}
4	$-\frac{1}{24}\big[Y,[X,[X,Y]]\big]$	4.757×10^{-7}

Each additional commutator gains between one and two orders of magnitude for amplitudes of this size. The series is asymptotic rather than convergent in general, and the gain per order shrinks as the operators grow; the practical lesson is that BCH is a small-parameter expansion, and the small parameter is the size of the commutator relative to the operators themselves.

6.9 The Trotter-Suzuki splitting

The most common use of Eq. (6.46) is in reverse. Suppose we can exponentiate H_1 and H_2 separately but not their sum – the standard situation, with H_1 the kinetic energy, diagonal in momentum space, and H_2 the potential, diagonal in coordinate space. Setting $X = -iH_1\Delta$ and $Y = -iH_2\Delta$ in Eq. (6.46),

$$e^{-iH_1\Delta}e^{-iH_2\Delta} = \exp\left(-i(H_1+H_2)\Delta - \frac{1}{2}[H_1,H_2]\Delta^2 + \mathscr{O}(\Delta^3)\right), \tag{6.51}$$

so a single split step is correct to first order in Δ, with an error governed by the commutator. Dividing the evolution into N steps of length t/N gives the Lie-Trotter product formula

$$e^{-i(H_1+H_2)t} = \lim_{N\to\infty}\left(e^{-iH_1t/N}e^{-iH_2t/N}\right)^N, \tag{6.52}$$

whose error at finite N is $\mathscr{O}(t^2/N)$.

Symmetrising the step cancels the leading commutator. The second-order, or Strang, splitting

$$S_2(\Delta) = e^{-iH_1\Delta/2}e^{-iH_2\Delta}e^{-iH_1\Delta/2} = e^{-i(H_1+H_2)\Delta} + \mathscr{O}(\Delta^3) \tag{6.53}$$

has a local error governed by the double commutators $[H_1,[H_1,H_2]]$ and $[H_2,[H_1,H_2]]$, hence a global error $\mathscr{O}(t^3/N^2)$. Suzuki's recursive construction generalises this to any even order, at the cost of more exponentials per step and, from fourth order on, of steps with negative coefficients.

For a single qubit with $\hat{H} = \sigma_x + \sigma_z$, so that $[\sigma_x,\sigma_z] = -2i\sigma_y \neq 0$, and $t = 1$, `hartreefock.py` finds

N	first order	ratio	second order	ratio
1	1.130×10^{0}	—	4.436×10^{-1}	—
2	5.125×10^{-1}	2.21	1.006×10^{-1}	4.41
4	2.493×10^{-1}	2.06	2.443×10^{-2}	4.12
8	1.238×10^{-1}	2.01	6.061×10^{-3}	4.03
16	6.177×10^{-2}	2.00	1.513×10^{-3}	4.01
32	3.087×10^{-2}	2.00	3.779×10^{-4}	4.00
64	1.543×10^{-2}	2.00	9.448×10^{-5}	4.00

Doubling N halves the first-order error and quarters the second-order one, exactly as the leading terms of Eq. (6.46) predict. The ratios approach their asymptotic values from above, which is the usual signature of higher-order terms still contributing at small N.

Why this belongs in a chapter on mean fields. The immediate reason is that Thouless' theorem is a statement about an exponential of one-body operators, and the reason it is so simple is that those operators commute. The longer-range reason is that the same expansion controls three later developments. In coupled-cluster theory the similarity-transformed Hamiltonian $e^{-\hat{T}}\hat{H}e^{\hat{T}}$ is expanded by the Baker-Hausdorff lemma $e^{-X}Ye^{X} = Y - [X,Y] + \frac{1}{2!}[X,[X,Y]] - \cdots$, and terminates exactly at the fourth commutator because $\hat{H}$ contains at most two-body terms. In time-dependent Hartree-Fock and in real-time methods generally, the propagator is built by splitting. And in quantum computing the Trotter formula is how a Hamiltonian is turned into a circuit: each exponential $e^{-iH_j\Delta}$ becomes a gate, Eq. (6.52) fixes how many of them are needed, and the trade-off between circuit depth and accuracy is precisely the trade-off between N and the error above. A first-order scheme needs $N = \mathscr{O}(t^2/\varepsilon)$ steps for an accuracy ε; a second-order scheme needs only $N = \mathscr{O}(t^{3/2}/\sqrt{\varepsilon})$.

6.10 Stability of the Hartree-Fock solution

The variational condition (6.45) guarantees only that $\langle c|\hat{H}|c\rangle$ is *stationary*. A stationary point may be a minimum, a maximum or a saddle, and which one it is matters: a Hartree-Fock solution sitting on a saddle point is a poor reference for perturbation theory or coupled cluster, because the corrections it generates are organised about the wrong state. Thouless' theorem lets us settle the question by going to second order.

We ask whether

$$\frac{\langle c'|\hat{H}|c'\rangle}{\langle c'|c'\rangle} \;\geq\; \langle c|\hat{H}|c\rangle = E_0 \tag{6.54}$$

for every nearby determinant $|c'\rangle = |c\rangle + |\delta c\rangle$. By Thouless' theorem,

$$|c'\rangle = \exp\left\{\sum_{a>F}\sum_{i\leq F}\delta C_{ai}\hat{a}_a^\dagger\hat{a}_i\right\}|c\rangle = \left\{1+\sum_{ai}\delta C_{ai}\hat{a}_a^\dagger\hat{a}_i + \frac{1}{2!}\sum_{abij}\delta C_{ai}\delta C_{bj}\hat{a}_a^\dagger\hat{a}_i\hat{a}_b^\dagger\hat{a}_j+\cdots\right\}|c\rangle, \tag{6.55}$$

with the amplitudes δC small. With intermediate normalisation $\langle c'|c\rangle = 1$ the norm is

$$\langle c'|c'\rangle = 1+\sum_{a>F}\sum_{i\leq F}|\delta C_{ai}|^2+\mathscr{O}(\delta C^3). \tag{6.56}$$

The energy to second order.

Expanding the numerator and using the Hartree-Fock condition (6.45) to kill the terms linear in δC,

$$\begin{aligned}\langle c'|\hat{H}|c'\rangle = \langle c|\hat{H}|c\rangle &+ \sum_{ab>F}\sum_{ij\leq F}\delta C_{ai}^*\delta C_{bj}\langle c|\hat{a}_i^\dagger\hat{a}_a\hat{H}\hat{a}_b^\dagger\hat{a}_j|c\rangle\\ &+\frac{1}{2!}\sum_{ab>F}\sum_{ij\leq F}\delta C_{ai}\delta C_{bj}\langle c|\hat{H}\hat{a}_a^\dagger\hat{a}_i\hat{a}_b^\dagger\hat{a}_j|c\rangle\\ &+\frac{1}{2!}\sum_{ab>F}\sum_{ij\leq F}\delta C_{ai}^*\delta C_{bj}^*\langle c|\hat{a}_j^\dagger\hat{a}_b\hat{a}_i^\dagger\hat{a}_a\hat{H}|c\rangle+\cdots\end{aligned} \tag{6.57}$$

Each of these expectation values is evaluated with Wick's theorem, Section 3.10. The first of them is the $1p$-$1h$ matrix element computed in Section 3.13,

$$\begin{aligned}\langle c|\{\hat{a}_i^\dagger\hat{a}_a\}\hat{H}\{\hat{a}_b^\dagger\hat{a}_j\}|c\rangle = &\sum_{pq}\langle p|\hat{h}_0|q\rangle\langle c|\{\hat{a}_i^\dagger\hat{a}_a\}\{\hat{a}_p^\dagger\hat{a}_q\}\{\hat{a}_b^\dagger\hat{a}_j\}|c\rangle\\ &+\frac{1}{4}\sum_{pqrs}\langle pq|\hat{v}|rs\rangle\langle c|\{\hat{a}_i^\dagger\hat{a}_a\}\{\hat{a}_p^\dagger\hat{a}_q^\dagger\hat{a}_s\hat{a}_r\}\{\hat{a}_b^\dagger\hat{a}_j\}|c\rangle,\end{aligned} \tag{6.58}$$

and collecting the surviving contractions gives

$$E_0\sum_{ai}|\delta C_{ai}|^2+\sum_{ai}|\delta C_{ai}|^2(\varepsilon_a-\varepsilon_i)-\sum_{ijab}\langle aj|\hat{v}|bi\rangle\,\delta C_{ai}^*\delta C_{bj}. \tag{6.59}$$

The third line of Eq. (6.57) is the Hermitian conjugate of the second, since $\langle a|\hat{A}|b\rangle = \left(\langle b|\hat{A}^\dagger|a\rangle\right)^*$ and $\hat{H}$ is Hermitian, and the second evaluates to

$$\frac{1}{2}\sum_{ijab}\langle ab|\hat{v}|ij\rangle\,\delta C_{ai}^*\delta C_{bj}^*. \tag{6.60}$$

The stability matrix.

Define the two antisymmetrised blocks

$$A_{ai,bj} = -\langle aj|\hat{v}|bi\rangle_{\mathrm{AS}}, \qquad B_{ai,bj} = \langle ab|\hat{v}|ij\rangle_{\mathrm{AS}}, \tag{6.61}$$

together with the diagonal matrix of unperturbed excitation energies

$$\Delta_{ai,bj} = (\varepsilon_a - \varepsilon_i)\,\delta_{ab}\delta_{ij}. \tag{6.62}$$

Then Eqs. (6.59) and (6.60) combine into

$$\langle c'|\hat{H}|c'\rangle = \left(1 + \sum_{ai} |\delta C_{ai}|^2\right)\langle c|\hat{H}|c\rangle + \sum_{ai} |\delta C_{ai}|^2\left(\varepsilon_a^{\mathrm{HF}} - \varepsilon_i^{\mathrm{HF}}\right) + \sum_{ijab} A_{ai,bj}\,\delta C_{ai}^*\delta C_{bj}$$
$$+ \frac{1}{2}\sum_{ijab} B_{ai,bj}^*\,\delta C_{ai}\delta C_{bj} + \frac{1}{2}\sum_{ijab} B_{ai,bj}\,\delta C_{ai}^*\delta C_{bj}^* + \mathscr{O}(\delta C^3), \tag{6.63}$$

so that, dividing by the norm (6.56),

$$\frac{\langle c'|\hat{H}|c'\rangle}{\langle c'|c'\rangle} = E_0 + \frac{\Delta E}{1 + \sum_{ai} |\delta C_{ai}|^2}, \qquad \Delta E = \frac{1}{2}\langle \chi|\hat{M}|\chi\rangle, \tag{6.64}$$

with the composite vector and matrix

$$\chi = \left[\delta C\ \delta C^*\right]^T, \qquad \hat{M} = \begin{pmatrix} \Delta + A & B \\ B^* & \Delta + A^* \end{pmatrix}. \tag{6.65}$$

The conclusion is now immediate. The denominator in Eq. (6.64) is positive, so the Hartree-Fock solution is a local minimum if and only if

$$\boxed{\Delta E = \frac{1}{2}\langle \chi|\hat{M}|\chi\rangle \ \geq\ 0 \quad \text{for every } \chi} \tag{6.66}$$

that is, if and only if every eigenvalue of $\hat{M}$ is non-negative. A necessary but not sufficient condition, obtained by restricting χ to the upper block, is that

$$(\varepsilon_a - \varepsilon_i)\,\delta_{ab}\delta_{ij} + A_{ai,bj} \ \geq\ 0 \tag{6.67}$$

as a matrix. This is cheap to test and is the usual first check.

The stability matrix is the RPA matrix. The upper-left block $\Delta + A$ is exactly the matrix diagonalised in the Tamm-Dancoff approximation: it is the Hamiltonian restricted to the $1p$-$1h$ space, measured from the reference energy. The full $\hat{M}$, with its off-diagonal B blocks, is the matrix of the random-phase approximation. So the statement “the Hartree-Fock solution is stable” and the statement “the RPA frequencies are real” are one and the same, and an imaginary RPA frequency is not a numerical accident but a signal that the mean field itself is wrong. This is why Chapter 5 could describe TDA and RPA as truncations of full configuration interaction and why they belong immediately after this chapter: they are the second derivative of the Hartree-Fock energy surface.

6.11 An explicit instability: the Lipkin model

The Lipkin model of Section 4.1 is the standard illustration, because everything can be computed in closed form. Take $W = 0$, so that

$$\hat{H} = \varepsilon\hat{J}_z + \frac{V}{2}\left(\hat{J}_+^2 + \hat{J}_-^2\right), \tag{6.68}$$

with N particles in two N-fold degenerate levels separated by ε. The natural mean-field ansatz rotates every single-particle state by the same angle α in the two-level space,

$$\hat{b}_p^\dagger = \cos\left(\tfrac{\alpha}{2}\right)\hat{a}_{p-}^\dagger + \sin\left(\tfrac{\alpha}{2}\right)\hat{a}_{p+}^\dagger, \tag{6.69}$$

which is a Thouless rotation with $C_{ai} = \tan(\alpha/2)\,\delta_{pq}$. The resulting energy is the classic result

$$\frac{E(\alpha)}{N} = -\frac{\varepsilon}{2}\cos\alpha - \frac{\varepsilon\chi}{4}\sin^2\alpha, \qquad \chi = \frac{|V|(N-1)}{\varepsilon}, \tag{6.70}$$

where χ is the dimensionless coupling. Differentiating,

$$\frac{1}{N}\frac{dE}{d\alpha} = \frac{\varepsilon}{2}\sin\alpha\,(1-\chi\cos\alpha)\,, \tag{6.71}$$

so there are two stationary points: the *spherical* solution $\alpha = 0$, which exists always, and a *deformed* solution $\cos\alpha = 1/\chi$, which exists only for $\chi \geq 1$. The second derivative at $\alpha = 0$ is

$$\frac{1}{N}\left.\frac{d^2E}{d\alpha^2}\right|_{\alpha=0} = \frac{\varepsilon}{2}\,(1-\chi)\,, \tag{6.72}$$

which changes sign at $\chi = 1$. The spherical solution turns from a minimum into a maximum exactly where the deformed solution appears, which is the signature of a continuous symmetry-breaking transition.

The stability matrix says the same thing without any reference to the parametrisation (6.69). Because the interaction in Eq. (6.68) changes J_z by two units, it has no matrix element within the $1p$-$1h$ space, so $A = 0$ and $\Delta + A = \varepsilon\boldsymbol{I}$. The block B has matrix elements $\pm V$ connecting different pairs, and its largest eigenvalue is $|V|(N-1)$. The lowest eigenvalue of $\hat{M}$ is therefore

$$\lambda_{\min} = \varepsilon - |V|(N-1) = \varepsilon\,(1-\chi)\,, \tag{6.73}$$

which is exactly Eq. (6.72) up to the factor $N/2$. The program `hartreefock.py` builds $\hat{M}$ numerically from the Hamiltonian in the determinant basis, for $N = 4$ and $\varepsilon = 1$:

χ	$\lambda_{\min}(\hat{M})$	$\alpha_{\min}$	E_{mf}	E_{exact}	solution
0.25	0.75	0.000000	−2.000000	−2.020726	spherical
0.50	0.50	0.000000	−2.000000	−2.081666	spherical
0.90	0.10	0.000000	−2.000000	−2.253886	spherical
1.00	0.00	0.000000	−2.000000	−2.309401	critical
1.50	−0.50	0.841069	−2.166667	−2.645751	deformed
2.00	−1.00	1.047198	−2.500000	−3.055050	deformed

Table 6.1 Stability of the spherical Hartree-Fock solution of the Lipkin model with $N = 4$, $\varepsilon = 1$ and $W = 0$. The lowest eigenvalue of the stability matrix crosses zero exactly at $\chi = 1$, where the deformed minimum appears. E_{mf} is the mean-field energy at the minimum and E_{exact} the exact ground-state energy.

Two lessons are worth drawing. The first is that a solution can satisfy the Hartree-Fock equations perfectly and still be useless: for $\chi > 1$ the spherical state is a genuine stationary point, obeys Brillouin's theorem, and is a saddle. The second is that the deformed solution buys a lower energy by breaking a symmetry the exact ground state does not break – at $\chi = 2$ it gains 0.5 but is still 0.55 above the exact answer. Symmetry breaking in a mean field is a device for capturing correlations cheaply, and the symmetry must eventually be restored, by projection or by a method that never broke it.

6.12 When the mean field does nothing

It is instructive to see the opposite extreme. Apply the same machinery to the pairing model of Section 4.2, with four doubly degenerate levels and four particles. The program reports

g	E(reference)	E(Hartree-Fock)	E(exact)	iterations
0.25	1.75000000	1.75000000	1.73045985	2
0.50	1.50000000	1.50000000	1.41677428	2
1.00	1.00000000	1.00000000	0.63554847	2

Hartree-Fock gains nothing at all. The reason is the seniority conservation of Section 5.6: the pairing interaction moves complete pairs and cannot connect the reference to a $1p$-$1h$ state, so $\langle \Phi_0|\hat{H}|\Phi_i^a\rangle = 0$ identically, the Fock matrix is diagonal from the first iteration, and the self-consistency loop terminates without having moved. The entire correlation energy – 36% of the reference energy at $g = 1$ – must come from the doubles, which is precisely what Table 4.1 and Section 5.8 showed.

The moral is that a mean field is only as useful as the one-body physics the interaction actually contains. Where an interaction can be well approximated by an average field – atoms, molecules, closed-shell nuclei – Hartree-Fock does most of the work. Where the interaction is intrinsically a pair-correlation effect – pairing, superconductivity, the strongly coupled Hubbard model of Section 5.7 – it does none, and one must either generalise the mean field, as Bardeen, Cooper and Schrieffer did by allowing the number of particles to fluctuate, or go beyond it.

6.13 The infinite homogeneous electron gas

The electron gas is very likely the only realistic model of an interacting many-particle system for which the Hartree-Fock equations can be solved in closed form. Better still, the total energy and several other properties follow analytically to first order in the interaction, so the whole chapter can be tested against an exact mean-field answer.

The model is defined by three assumptions:

- the electrons feel no external force except the attraction of a uniform, stationary background of positive charge;
- the system as a whole is neutral;
- there are N electrons in a cubic box of side L and volume $\Omega = L^3$, containing a uniform positive charge of density Ne/Ω.

It is a very good approximation for the valence electrons of a metal, and the three-dimensional gas is the cornerstone of the local-density approximation of density-functional theory. Two-dimensional electron fluids occur on metal and liquid-helium surfaces and at metal-oxide-semiconductor interfaces. For all its simplicity, the correlations of this system have resisted a satisfactory first-principles description by anything except quantum Monte Carlo; the diffusion Monte Carlo calculations of Ceperley, and of Ceperley and Tanatar, remain the benchmark in three and in two dimensions. At low density the electrons localise into a lattice – Wigner crystallisation, a direct consequence of the long range of the repulsion – while at high density the system behaves as a liquid. The infinite range of the Coulomb interaction also generates finite-size effects that are absent in nuclear or neutron matter, and that any finite calculation must handle with care.

6.13.1 Plane waves and periodic boundary conditions

The system is homogeneous, so the single-particle states are plane waves normalised in the box,

$$\psi_{\boldsymbol{k}\sigma}(\boldsymbol{r}) = \frac{1}{\sqrt{\Omega}} e^{i\boldsymbol{k}\cdot\boldsymbol{r}}\, \xi_\sigma, \qquad \xi_{+1/2} = \begin{pmatrix}1\\0\end{pmatrix}, \quad \xi_{-1/2} = \begin{pmatrix}0\\1\end{pmatrix}, \tag{6.74}$$

and periodic boundary conditions restrict the wave numbers to

$$k_i = \frac{2\pi n_i}{L}, \qquad i = x, y, z, \qquad n_i = 0, \pm 1, \pm 2, \ldots \tag{6.75}$$

The limit $L \to \infty$ is taken after the expectation values have been computed. The kinetic energy operator is diagonal,

$$\hat{T} = \sum_{\boldsymbol{k}\sigma} \frac{\hbar^2 k^2}{2m} \hat{a}^\dagger_{\boldsymbol{k}\sigma} \hat{a}_{\boldsymbol{k}\sigma}, \tag{6.76}$$

which is the same structure as the momentum-space Hubbard model of Section 5.7: the plane-wave determinant is automatically the Hartree-Fock solution of a translationally invariant problem, since momentum conservation forces $f_{ai} = 0$. Everything in this section is therefore a calculation *in* the Hartree-Fock basis, not a search for it.

The full Hamiltonian has three pieces,

$$\hat{H} = \hat{H}_{\mathrm{el}} + \hat{H}_b + \hat{H}_{\mathrm{el}-b}, \tag{6.77}$$

namely the electrons,

$$\hat{H}_{\mathrm{el}} = \sum_{i=1}^{N} \frac{p_i^2}{2m} + \frac{e^2}{2} \sum_{i \neq j} \frac{e^{-\mu |\boldsymbol{r}_i - \boldsymbol{r}_j|}}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}, \tag{6.78}$$

the background with itself,

$$\hat{H}_b = \frac{e^2}{2} \iint d\boldsymbol{r} d\boldsymbol{r}' \frac{n(\boldsymbol{r}) n(\boldsymbol{r}') e^{-\mu |\boldsymbol{r} - \boldsymbol{r}'|}}{|\boldsymbol{r} - \boldsymbol{r}'|}, \tag{6.79}$$

with $n(\boldsymbol{r}) = N/\Omega$, and the electrons with the background,

$$\hat{H}_{\mathrm{el}-b} = -\frac{e^2}{2} \sum_{i=1}^{N} \int d\boldsymbol{r} \frac{n(\boldsymbol{r}) e^{-\mu |\boldsymbol{r} - \boldsymbol{x}_i|}}{|\boldsymbol{r} - \boldsymbol{x}_i|}. \tag{6.80}$$

The factor $e^{-\mu r}$ is a convergence device: each integral is computed at finite μ, and the limit $\mu \to 0$ is taken at the end. Individually the three terms diverge as μ^{-2},

$$\hat{H}_b = \frac{e^2}{2} \frac{N^2}{\Omega} \frac{4\pi}{\mu^2}, \qquad \hat{H}_{\mathrm{el}-b} = -e^2 \frac{N^2}{\Omega} \frac{4\pi}{\mu^2}, \tag{6.81}$$

and the direct term of the electron-electron repulsion supplies exactly the missing $+\frac{1}{2}$. The three divergences cancel, and what survives is the kinetic energy and the exchange term. This cancellation is special to the Coulomb interaction and to a neutral system; for a short-ranged interaction, nuclear forces for example, both direct and exchange terms must be kept. What remains is

$$\hat{H} = \hat{H}_0 + \hat{H}_I, \qquad \hat{H}_0 = \sum_{\boldsymbol{k}\sigma} \frac{\hbar^2 k^2}{2m} \hat{a}^\dagger_{\boldsymbol{k}\sigma} \hat{a}_{\boldsymbol{k}\sigma}, \tag{6.82}$$

$$\hat{H}_I = \frac{e^2}{2\Omega} \sum_{\sigma_1 \sigma_2} \sum_{\boldsymbol{q} \neq 0, \boldsymbol{k}, \boldsymbol{p}} \frac{4\pi}{q^2} \hat{a}^\dagger_{\boldsymbol{k}+\boldsymbol{q}, \sigma_1} \hat{a}^\dagger_{\boldsymbol{p}-\boldsymbol{q}, \sigma_2} \hat{a}_{\boldsymbol{p}\sigma_2} \hat{a}_{\boldsymbol{k}\sigma_1}, \tag{6.83}$$

with the $\boldsymbol{q} = 0$ term – the one that diverged – removed by the background.

6.13.2 The interaction in momentum space

Equation (6.83) deserves a derivation, since the same steps apply to any interaction depending only on the relative distance. Starting from

$$\hat{V} = \frac{1}{2} \sum_{\substack{\sigma_p\sigma_q\\ \sigma_r\sigma_s}} \sum_{\substack{\boldsymbol{k}_p\boldsymbol{k}_q\\ \boldsymbol{k}_r\boldsymbol{k}_s}} \langle \boldsymbol{k}_p\sigma_p, \boldsymbol{k}_q\sigma_q|\hat{v}|\boldsymbol{k}_r\sigma_r, \boldsymbol{k}_s\sigma_s\rangle\, \hat{a}^\dagger_{\boldsymbol{k}_p\sigma_p}\hat{a}^\dagger_{\boldsymbol{k}_q\sigma_q}\hat{a}_{\boldsymbol{k}_s\sigma_s}\hat{a}_{\boldsymbol{k}_r\sigma_r}, \tag{6.84}$$

insert the plane waves (6.74). The spin functions are orthonormal and $\hat{v}$ is spin independent, so

$$\langle \boldsymbol{k}_p\sigma_p, \boldsymbol{k}_q\sigma_q|\hat{v}|\boldsymbol{k}_r\sigma_r, \boldsymbol{k}_s\sigma_s\rangle = \frac{\delta_{\sigma_p\sigma_r}\delta_{\sigma_q\sigma_s}}{\Omega^2} \iint e^{-i\boldsymbol{k}_p\cdot\boldsymbol{r}_p} e^{-i\boldsymbol{k}_q\cdot\boldsymbol{r}_q}\, v(r)\, e^{i\boldsymbol{k}_r\cdot\boldsymbol{r}_p} e^{i\boldsymbol{k}_s\cdot\boldsymbol{r}_q} d\boldsymbol{r}_p d\boldsymbol{r}_q. \tag{6.85}$$

Change variables to $\boldsymbol{r} = \boldsymbol{r}_p - \boldsymbol{r}_q$ at fixed $\boldsymbol{r}_q$; the limits are unchanged, and the exponentials separate,

$$= \frac{\delta_{\sigma_p\sigma_r}\delta_{\sigma_q\sigma_s}}{\Omega^2} \int v(r) e^{i(\boldsymbol{k}_r-\boldsymbol{k}_p)\cdot\boldsymbol{r}} d\boldsymbol{r} \int e^{i(\boldsymbol{k}_s-\boldsymbol{k}_q+\boldsymbol{k}_r-\boldsymbol{k}_p)\cdot\boldsymbol{r}_q} d\boldsymbol{r}_q. \tag{6.86}$$

The second integral is a Kronecker delta enforcing momentum conservation,

$$\langle \boldsymbol{k}_p\sigma_p, \boldsymbol{k}_q\sigma_q|\hat{v}|\boldsymbol{k}_r\sigma_r, \boldsymbol{k}_s\sigma_s\rangle = \frac{\delta_{\sigma_p\sigma_r}\delta_{\sigma_q\sigma_s}}{\Omega}\, \delta_{\boldsymbol{k}_p+\boldsymbol{k}_q,\boldsymbol{k}_r+\boldsymbol{k}_s} \int v(r) e^{i(\boldsymbol{k}_r-\boldsymbol{k}_p)\cdot\boldsymbol{r}} d\boldsymbol{r}. \tag{6.87}$$

The matrix element vanishes unless $\boldsymbol{k}_p + \boldsymbol{k}_q = \boldsymbol{k}_r + \boldsymbol{k}_s$. Using this to eliminate one summation variable and writing $\boldsymbol{p} = \boldsymbol{k}_p + \boldsymbol{k}_q - \boldsymbol{k}_r$, $\boldsymbol{k} = \boldsymbol{k}_r$ and $\boldsymbol{q} = \boldsymbol{k}_p - \boldsymbol{k}_r$,

$$\hat{V} = \frac{1}{2\Omega} \sum_{\sigma\sigma'} \sum_{\boldsymbol{kpq}} \left[\int v(r) e^{-i\boldsymbol{q}\cdot\boldsymbol{r}} d\boldsymbol{r}\right] \hat{a}^\dagger_{\boldsymbol{k}+\boldsymbol{q},\sigma}\hat{a}^\dagger_{\boldsymbol{p}-\boldsymbol{q},\sigma'}\hat{a}_{\boldsymbol{p}\sigma'}\hat{a}_{\boldsymbol{k}\sigma}. \tag{6.88}$$

For the screened Coulomb interaction the bracket is $4\pi e^2/(\mu^2+q^2)$, which becomes $4\pi e^2/q^2$ as $\mu \to 0$, giving Eq. (6.83). The structure – a momentum transfer $\boldsymbol{q}$ shared between two particles – is exactly the Hubbard interaction of Eq. (5.24), with $4\pi e^2/q^2$ in place of the constant U.

6.13.3 The reference energy

With $|\Phi_0\rangle$ the filled Fermi sea, the kinetic energy follows from the contractions of Section 3.13,

$$\langle\Phi_0|\hat{T}|\Phi_0\rangle = \sum_{i\le F} \frac{\hbar^2 k_i^2}{2m}\cdot 2 = \frac{2\Omega}{(2\pi)^3}\frac{\hbar^2}{2m}\int_0^{k_F} 4\pi k^4 dk = \frac{\hbar^2\Omega}{10\pi^2 m} k_F^5, \tag{6.89}$$

the factor two being the spin degeneracy. The same density of states fixes the particle number,

$$N = \frac{2\Omega}{(2\pi)^3}\cdot\frac{4}{3}\pi k_F^3 = \frac{\Omega}{3\pi^2}k_F^3 \qquad \Longrightarrow \qquad k_F = \left(\frac{3\pi^2 N}{\Omega}\right)^{1/3}, \tag{6.90}$$

so that

$$\frac{\langle\Phi_0|\hat{T}|\Phi_0\rangle}{N} = \frac{3}{5}\frac{\hbar^2 k_F^2}{2m} = \frac{\hbar^2(3\pi^2)^{5/3}}{10\pi^2 m}\rho^{2/3}, \qquad \rho = \frac{N}{\Omega}. \tag{6.91}$$

For the interaction, only two classes of contraction survive: either $\boldsymbol{k}+\boldsymbol{q} = \boldsymbol{p}$ with $\sigma = \sigma'$, or $\boldsymbol{q} = 0$. The first is the exchange term, the second the direct one, and

$$\langle\Phi_0|\hat{V}|\Phi_0\rangle = \frac{1}{2\Omega}\left(N^2\int v(r)d\boldsymbol{r} - N\sum_{\boldsymbol{q}}\int v(r)e^{-i\boldsymbol{q}\cdot\boldsymbol{r}}d\boldsymbol{r}\right), \tag{6.92}$$

valid for any interaction depending only on r. For the electron gas the first term is cancelled by the background, as we saw, and only the exchange term remains.

6.13.4 The single-particle energy

The Hartree-Fock single-particle energy is the kinetic energy plus the exchange self-energy,

$$\varepsilon_k^{\mathrm{HF}} = \frac{\hbar^2 k^2}{2m} - \frac{e^2}{\Omega^2} \sum_{p \le k_F} \int d\boldsymbol{r}\, e^{i(\boldsymbol{p}-\boldsymbol{k})\cdot\boldsymbol{r}} \int d\boldsymbol{r}'\, \frac{e^{i(\boldsymbol{k}-\boldsymbol{p})\cdot\boldsymbol{r}'}}{|\boldsymbol{r}-\boldsymbol{r}'|}, \tag{6.93}$$

and we shall show that it evaluates to

$$\boxed{\varepsilon_k^{\mathrm{HF}} = \frac{\hbar^2 k^2}{2m} - \frac{e^2 k_F}{2\pi}\left[2 + \frac{k_F^2 - k^2}{k k_F} \ln\left|\frac{k+k_F}{k-k_F}\right|\right].} \tag{6.94}$$

Insert the convergence factor $e^{-\mu|\boldsymbol{r}-\boldsymbol{r}'|}$ and replace the sum by an integral, $\sum_{\boldsymbol{p}} \to \Omega(2\pi)^{-3}\int d\boldsymbol{p}$,

$$\lim_{\mu\to 0} \frac{e^2}{\Omega(2\pi)^3} \iint d\boldsymbol{r} d\boldsymbol{r}'\, \frac{e^{-\mu|\boldsymbol{r}-\boldsymbol{r}'|}}{|\boldsymbol{r}-\boldsymbol{r}'|} \int d\boldsymbol{p}\, e^{i(\boldsymbol{p}-\boldsymbol{k})\cdot(\boldsymbol{r}-\boldsymbol{r}')}. \tag{6.95}$$

Change variables to $\boldsymbol{x} = \boldsymbol{r} - \boldsymbol{r}'$ and $\boldsymbol{y} = \boldsymbol{r}'$. The integrand no longer depends on $\boldsymbol{y}$, whose integral gives Ω, and the $\boldsymbol{x}$ integral in spherical coordinates is

$$\int d\boldsymbol{x}\, e^{i(\boldsymbol{p}-\boldsymbol{k})\cdot\boldsymbol{x}}\, \frac{e^{-\mu x}}{x} = 4\pi \int_0^\infty \frac{\sin(|\boldsymbol{p}-\boldsymbol{k}|x)}{|\boldsymbol{p}-\boldsymbol{k}|} e^{-\mu x} dx = \frac{4\pi}{\mu^2 + |\boldsymbol{p}-\boldsymbol{k}|^2}. \tag{6.96}$$

The limit $\mu \to 0$ is now harmless, and we are left with

$$\frac{e^2}{2\pi^2} \int_{p \le k_F} \frac{d\boldsymbol{p}}{|\boldsymbol{p}-\boldsymbol{k}|^2} = \frac{e^2}{\pi} \int_0^{k_F} p^2 dp \int_{-1}^{1} \frac{du}{p^2 + k^2 - 2pku}, \tag{6.97}$$

having set $u = \cos\theta$. The angular integral is elementary,

$$\int_{-1}^{1} \frac{du}{p^2 + k^2 - 2pku} = \frac{1}{pk} \ln\left|\frac{p+k}{p-k}\right|, \tag{6.98}$$

so the exchange term becomes

$$\frac{e^2}{k\pi} \int_0^{k_F} p \ln\left|\frac{p+k}{p-k}\right| dp = \frac{e^2}{k\pi}\left[k k_F + \frac{k_F^2 - k^2}{2} \ln\left|\frac{k+k_F}{k-k_F}\right|\right], \tag{6.99}$$

where the remaining integral is done by substituting $x = p + k$ and $y = p - k$. Rearranging gives Eq. (6.94).

6.13.5 Dimensionless form, band width and effective mass

Introduce the radius r_0 of the sphere whose volume is the volume per electron,

$$\rho = \frac{k_F^3}{3\pi^2} = \frac{3}{4\pi r_0^3} \qquad \Longrightarrow \qquad k_F = \frac{(9\pi/4)^{1/3}}{r_0} = \frac{1.9192}{r_0}, \tag{6.100}$$

the Bohr radius $a_0 = \hbar^2/me^2$ and the dimensionless density parameter $r_s = r_0/a_0$, which lies between about 2 and 6 for most metals. With $x = k/k_F$ and

$$F(x) = \frac{1}{2} + \frac{1 - x^2}{4x} \ln\left|\frac{1+x}{1-x}\right|, \tag{6.101}$$

Equation (6.94) becomes

$$\varepsilon_k^{\mathrm{HF}} = \frac{\hbar^2 k^2}{2m} - \frac{2e^2 k_F}{\pi} F(x), \qquad \frac{\varepsilon_k^{\mathrm{HF}}}{\varepsilon_0^F} = x^2 - \frac{4}{\pi k_F a_0} F(x) = x^2 - 0.663\, r_s\, F(x), \tag{6.102}$$

where $\varepsilon_0^F = \hbar^2 k_F^2/2m$ is the free energy at the Fermi surface. The function F is smooth except at $x = 1$, where it takes the value $\frac{1}{2}$ but has an infinite derivative; at $x = 0$ it equals 1. The program `hartreefock.py` tabulates the result for $r_s = 4$:

$x = k/k_F$	$F(x)$	$\varepsilon_k^{\mathrm{HF}}/\varepsilon_0^F$	$\varepsilon_k^{(0)}/\varepsilon_0^F$
0.00	1.000000	−2.652000	0.0000
0.25	0.978899	−2.533540	0.0625
0.50	0.911980	−2.168570	0.2500
0.75	0.783779	−1.516081	0.5625
1.00	0.500000	−0.326000	1.0000
1.25	0.252812	0.892042	1.5625
1.50	0.164700	1.813214	2.2500
2.00	0.088020	3.766570	4.0000

Table 6.2 The Hartree-Fock single-particle energy of the three-dimensional electron gas at $r_s = 4$, compared with the free-particle energy. The exchange term lowers every state and lowers the bottom of the band far more than the top.

Band width.

The width of the occupied band is

$$\Delta\varepsilon^{\mathrm{HF}} = \varepsilon_{k_F}^{\mathrm{HF}} - \varepsilon_0^{\mathrm{HF}}. \tag{6.103}$$

At $r_s = 4$ the two entries of Table 6.2 give -0.326 and -2.652, hence $\Delta\varepsilon^{\mathrm{HF}} = 2.326$ in units of ε_0^F. In general, using $F(1) = \frac{1}{2}$ and $F(0) = 1$,

$$\Delta\varepsilon^{\mathrm{HF}} = 1 + \frac{0.663}{2} r_s = 1 + 0.3315\, r_s, \tag{6.104}$$

which reproduces 2.326 at $r_s = 4$. The free gas has a band width of exactly one, so exchange more than doubles it – an effect of the right sign but, as experiment shows, considerably too large.

Effective mass and level density.

Expanding about the Fermi surface,

$$\varepsilon_k^{\mathrm{HF}} = \varepsilon_{k_F}^{\mathrm{HF}} + \left(\frac{\partial \varepsilon_k^{\mathrm{HF}}}{\partial k}\right)_{k_F} (k - k_F) + \cdots \tag{6.105}$$

and comparing with the free-particle expansion $\varepsilon_k^{(0)} = \hbar^2 k_F^2/2m + (\hbar^2 k_F/m)(k - k_F) + \cdots$ defines the effective mass

$$m_{\mathrm{HF}}^* \equiv \hbar^2 k_F \left(\frac{\partial \varepsilon_k^{\mathrm{HF}}}{\partial k}\right)_{k_F}^{-1} = \frac{2m}{2 - 0.663\, r_s\, F'(1)}. \tag{6.106}$$

Here is the difficulty: $F'(x)$ diverges logarithmically as $x \to 1$, so the slope of $\varepsilon_k^{\mathrm{HF}}$ is infinite at the Fermi surface and $m^*_{\mathrm{HF}} \to 0$. Approaching from below, the program finds $m^*/m = 0.256, 0.185, 0.144, 0.118, 0.100$ at $1-x = 10^{-2}, \ldots, 10^{-6}$: the ratio creeps towards zero like $1/\ln(1-x)$. Since the level density is

$$n(\varepsilon) = \frac{\Omega k^2}{2\pi^2}\left(\frac{\partial \varepsilon}{\partial k}\right)^{-1}, \tag{6.107}$$

it vanishes at the Fermi surface.

This is qualitatively wrong. Metals have a finite density of states at the Fermi level – it is what gives them a linear specific heat and a finite Pauli susceptibility. The culprit is the $1/q^2$ singularity of the bare Coulomb interaction at small momentum transfer, which Hartree-Fock treats without any screening. In a real metal the other electrons rearrange to screen it, and the effective interaction falls off exponentially beyond the screening length. Summing the diagrams that produce that screening is precisely what the random-phase approximation does, and it restores a finite effective mass. So the electron gas illustrates both what a mean field does well and where it must fail, and it points directly to the next method.

6.13.6 The energy per electron

Collecting the kinetic energy (6.91) and the exchange term, and measuring energies in Rydberg, $e^2/2a_0 = 13.606$ eV, the Hartree-Fock energy per electron is

$$\boxed{\frac{E_0}{N} = \frac{e^2}{2a_0}\left[\frac{2.21}{r_s^2} - \frac{0.916}{r_s}\right].} \tag{6.108}$$

The two coefficients are not empirical:

$$\frac{3}{5}\left(\frac{9\pi}{4}\right)^{2/3} = 2.2099, \qquad \frac{3}{2\pi}\left(\frac{9\pi}{4}\right)^{1/3} = 0.9163, \tag{6.109}$$

as `hartreefock.py` confirms. The first term is the kinetic energy, which scales as $\rho^{2/3} \propto r_s^{-2}$ and dominates at high density; the second is exchange, which scales as $\rho^{1/3} \propto r_s^{-1}$ and wins at low density. Because they carry opposite signs and different powers, their sum has a minimum:

$$\frac{d}{dr_s}\left[\frac{2.21}{r_s^2} - \frac{0.916}{r_s}\right] = 0 \qquad \Longrightarrow \qquad r_s = \frac{2\times 2.21}{0.916} = 4.83, \tag{6.110}$$

at which $E_0/N = -0.0949$ Ry $= -1.29$ eV. The values along the curve are

r_s	E_0/N [Ry]
1.00	1.294000
2.00	0.094500
3.00	−0.059778
4.00	−0.090875
4.83	−0.094916
6.00	−0.091278
8.00	−0.079969

The electron gas is therefore *bound* already at the Hartree-Fock level, at a density lying squarely in the range observed in the alkali metals. This is a real success: exchange alone, with no correlation whatsoever, explains why a metal is stable against expansion. The binding energy is too small by roughly a factor of two – the missing part is the correlation energy, which for this system is about -0.06 Ry near equilibrium – and the equilibrium density is somewhat too low, but the mechanism is right.

6.14 Summary

Hartree-Fock theory is the variational principle of Section 2.7 applied to the smallest interesting trial space: a single Slater determinant, with the orbitals free. The stationarity condition is a nonlinear eigenvalue problem, Eq. (6.26), solved by iterating a linear one – the self-consistent field. Its solution is characterised by Brillouin's theorem, $\langle \Phi_0|\hat{H}|\Phi_i^a\rangle = 0$, which is exactly the statement that the reference decouples from all single excitations and which is why every post-Hartree-Fock method begins at the doubles.

Thouless' theorem makes the geometry precise: the neighbourhood of a determinant, inside the manifold of determinants, is parametrised by $e^{\hat{T}}$ with $\hat{T}$ a linear combination of one-particle-one-hole operators. Because those operators commute, the exponential collapses, and the Baker-Campbell-Hausdorff series – which controls the general case and reappears in the Trotter splitting and in coupled-cluster theory – reduces to its first term. Going to second order in $\hat{T}$ produces the stability matrix (6.65), whose positivity decides whether the stationary point is a minimum, and which is the same matrix that the random-phase approximation diagonalises.

The two model calculations bracket the range of behaviour. For the pairing model the mean field does nothing at all, since the interaction conserves seniority. For the Lipkin model it does too much: beyond $\chi = 1$ the symmetric solution goes unstable and a symmetry-broken one takes over, buying energy at the cost of a broken symmetry that must later be restored. Between them lies the electron gas, where the whole programme goes through analytically, predicts a bound system at a realistic density, and yet gets the effective mass qualitatively wrong for want of screening.

The program `hartreefock.py` carries out every calculation of this chapter, and the notebook `hartreefock.ipynb` runs it cell by cell.

6.15 Exercises

Exercise 1: determinants and unitary transformations

a) Verify Eq. (6.12) by direct expansion, and then prove Eq. (6.13) for a general n.
b) Use a) to prove the product rule $\det(\boldsymbol{BC}) = \det(\boldsymbol{B})\det(\boldsymbol{C})$ for $n = 2$, and convince yourself that the argument generalises.
c) Show that a Slater determinant is invariant, up to a phase, under any unitary mixing of its occupied orbitals, and that the one-body density matrix (6.28) is strictly invariant.
d) Show that $\boldsymbol{\rho}^2 = \boldsymbol{\rho}$ and $\operatorname{tr}\boldsymbol{\rho} = N$, and explain what these two statements say about the occupied subspace.

Answer to a).

Expanding the left-hand side of Eq. (6.12),

$$(\alpha_1 b_{11} + \alpha_2 b_{12})a_{22} - (\alpha_1 b_{21} + \alpha_2 b_{22})a_{12} = \alpha_1(b_{11}a_{22} - b_{21}a_{12}) + \alpha_2(b_{12}a_{22} - b_{22}a_{12}), \tag{6.111}$$

which is the right-hand side. For general n, use the Leibniz formula $\det(\boldsymbol{A}) = \sum_\pi \operatorname{sgn}(\pi)\prod_j a_{\pi(j)j}$. Exactly one factor in each product comes from column j; substituting $a_{ij} \to \sum_k c_k b_{ik}$ in that column and interchanging the two sums gives $\sum_k c_k \sum_\pi \operatorname{sgn}(\pi) b_{\pi(j)k} \prod_{l\neq j} a_{\pi(l)l}$, which is Eq. (6.13).

Answer to b).

Write $\boldsymbol{A} = \boldsymbol{BC}$, so that column j of $\boldsymbol{A}$ is $\sum_k C_{kj}\boldsymbol{b}_k$ with $\boldsymbol{b}_k$ the columns of $\boldsymbol{B}$. Applying Eq. (6.13) to both columns of a 2×2 matrix,

$$\det(\boldsymbol{A}) = \sum_{k_1 k_2} C_{k_1 1} C_{k_2 2} \det\left(\boldsymbol{b}_{k_1}\ \boldsymbol{b}_{k_2}\right). \tag{6.112}$$

Terms with $k_1 = k_2$ vanish, and the two surviving terms combine to $(C_{11}C_{22} - C_{21}C_{12})\det(\boldsymbol{b}_1\ \boldsymbol{b}_2) = \det(\boldsymbol{C})\det(\boldsymbol{B})$. For general n the same expansion leaves only the $n!$ terms with distinct k's, each carrying the sign of the corresponding permutation, and the sum is again $\det(\boldsymbol{C})\det(\boldsymbol{B})$.

Answer to c).

Let $\psi_i' = \sum_j U_{ij}\psi_j$ with $\boldsymbol{U}$ unitary and i, j running over the occupied orbitals. By Eq. (6.16) the new determinant is $\det(\boldsymbol{U})$ times the old one, and $|\det(\boldsymbol{U})| = 1$ for a unitary matrix, so the two states differ by a phase and describe the same physics. The density matrix is $\boldsymbol{\rho}' = \boldsymbol{U}^\dagger\boldsymbol{\rho}\boldsymbol{U}$ evaluated on the occupied block; since $\boldsymbol{\rho}$ is the projector onto the span of the occupied orbitals and $\boldsymbol{U}$ merely rotates within that span, $\boldsymbol{\rho}$ is unchanged exactly, not merely up to a phase.

Answer to d).

With orthonormal orbitals, $\rho_{\gamma\delta} = \sum_i C_{i\gamma}C^*_{i\delta}$ gives

$$(\boldsymbol{\rho}^2)_{\gamma\delta} = \sum_{ij\lambda} C_{i\gamma}C^*_{i\lambda}C_{j\lambda}C^*_{j\delta} = \sum_{ij}\delta_{ij}C_{i\gamma}C^*_{j\delta} = \rho_{\gamma\delta}, \tag{6.113}$$

and $\operatorname{tr}\boldsymbol{\rho} = \sum_i\sum_\gamma |C_{i\gamma}|^2 = N$. A Hermitian idempotent is a projector, and its trace is the dimension of the space it projects onto. So $\boldsymbol{\rho}$ is the projector onto the N-dimensional occupied subspace, and its eigenvalues are N ones and the rest zeros. Together these say that a Hartree-Fock solution is a choice of subspace, and that only for a Slater determinant are the occupation numbers exactly zero and one – for a correlated state the eigenvalues of $\boldsymbol{\rho}$ are the fractional natural occupations of Section 1.31.

Exercise 2: the Hartree-Fock equations

a) Carry out the differentiation in Eq. (6.22) in full, and confirm that the factor $\frac{1}{2}$ in the two-body term is cancelled by the two ways in which $C^*_{i\alpha}$ can appear.
b) Show that the Hartree-Fock energy is *not* the sum of the single-particle energies, and derive the correct relation

$$E^{\mathrm{HF}} = \sum_{i=1}^{N}\varepsilon_i^{\mathrm{HF}} - \frac{1}{2}\sum_{i,j=1}^{N}\langle ij|\hat{v}|ij\rangle_{\mathrm{AS}}.$$

c) Interpret $\varepsilon_i^{\mathrm{HF}}$ physically by computing the energy needed to remove a particle from orbital i, assuming the remaining orbitals do not relax. (This is Koopmans' theorem.)
d) Show directly from Eq. (6.25) that at self-consistency $f_{ai} = 0$, and hence that $\langle\Phi_0|\hat{H}|\Phi_i^a\rangle = 0$.

Answer to a).

The one-body term of Eq. (6.19) is linear in $C^*_{i\alpha}$ and differentiates to $\sum_\beta C_{i\beta}\langle\alpha|\hat{h}_0|\beta\rangle$. In the two-body term $C^*_{i\alpha}$ can be either the first starred factor, with the sum over j free, or the second, with the roles of the index pairs exchanged. Relabelling the dummy indices of the second contribution and using

$$\langle\alpha\beta|\hat{v}|\gamma\delta\rangle_{\mathrm{AS}} = \langle\beta\alpha|\hat{v}|\delta\gamma\rangle_{\mathrm{AS}} \tag{6.114}$$

shows the two are identical, so together they cancel the $\frac{1}{2}$ and give the second term of Eq. (6.23).

Answer to b).

Multiply Eq. (6.26) by $C^*_{i\alpha}$ and sum over α and over the occupied i:

$$\sum_{i=1}^{N} \varepsilon_i^{\mathrm{HF}} = \sum_{i=1}^{N} \langle i|\hat{h}_0|i\rangle + \sum_{i,j=1}^{N} \langle ij|\hat{v}|ij\rangle_{\mathrm{AS}}. \tag{6.115}$$

Comparing with Eq. (6.18), which carries a factor $\frac{1}{2}$ on the two-body term, gives the stated relation. The interaction energy is counted twice in the sum of single-particle energies – once for each partner – and half of it must be removed. This is the standard double-counting correction, and forgetting it is one of the commonest errors in a first Hartree-Fock code.

Answer to c).

Removing a particle from orbital i without relaxing the others changes the energy by

$$E^{\mathrm{HF}}(N) - E^{\mathrm{HF}}(N-1) = \langle i|\hat{h}_0|i\rangle + \sum_{j\neq i} \langle ij|\hat{v}|ij\rangle_{\mathrm{AS}} = \varepsilon_i^{\mathrm{HF}}, \tag{6.116}$$

using $\langle ii|\hat{v}|ii\rangle_{\mathrm{AS}} = 0$. So each Hartree-Fock eigenvalue is, in the frozen-orbital approximation, minus the ionisation energy from that orbital. This is Koopmans' theorem. It is only an approximation, because in reality the remaining orbitals relax – an effect that lowers the energy of the $(N-1)$-particle system and therefore makes the true ionisation energy smaller than $-\varepsilon_i^{\mathrm{HF}}$. Correlation works in the opposite direction, and for atoms the two errors partly cancel, which is why Koopmans' theorem is more accurate than it has any right to be.

Answer to d).

At self-consistency the coefficients diagonalise the Fock matrix, so in the Hartree-Fock basis $f_{pq} = \varepsilon_p \delta_{pq}$ and in particular $f_{ai} = 0$ for $a > F \geq i$. By the Condon-Slater rules of Section 3.13, the matrix element of $\hat{H}$ between the reference and a single excitation is

$$\langle\Phi_0|\hat{H}|\Phi_i^a\rangle = \langle a|\hat{h}_0|i\rangle + \sum_{j\leq F} \langle aj|\hat{v}|ij\rangle_{\mathrm{AS}} = f_{ai} = 0. \tag{6.117}$$

This is the identity that makes the Hartree-Fock determinant the natural reference for everything that follows.

Exercise 3: Thouless' theorem

a) Prove Eq. (6.35) from the anticommutation relations, being explicit about where the disjointness of the particle and hole index sets is used.
b) Show that $\hat{T}_i^2 = 0$ and hence that $e^{\hat{T}} = \prod_i (1+\hat{T}_i)$. Explain why this is *not* the same as $1+\hat{T}$, and identify what the difference describes.

c) Verify all of this numerically with `hartreefock.py` for six orbitals and three particles, and check that the discarded piece scales as the square of the amplitudes.
d) Show that $\langle c|\tilde{c}\rangle = \det(f_{ip})$ over the occupied block, and explain why the hypothesis $\langle c|\tilde{c}\rangle \neq 0$ is exactly what is needed for the proof to go through.

Answer to a).

Take $a,b > F$ and $i,j \leq F$, so that all four indices in a given product are distinct except possibly $a = b$ or $i = j$. Since $i \neq b$, $\hat{a}_i\hat{a}_b^\dagger = -\hat{a}_b^\dagger\hat{a}_i$, whence

$$\hat{a}_a^\dagger\hat{a}_i\hat{a}_b^\dagger\hat{a}_j = -\hat{a}_a^\dagger\hat{a}_b^\dagger\hat{a}_i\hat{a}_j. \tag{6.118}$$

Since $j \neq a$, the same step on the reversed product gives

$$\hat{a}_b^\dagger\hat{a}_j\hat{a}_a^\dagger\hat{a}_i = -\hat{a}_b^\dagger\hat{a}_a^\dagger\hat{a}_j\hat{a}_i = -\hat{a}_a^\dagger\hat{a}_b^\dagger\hat{a}_i\hat{a}_j, \tag{6.119}$$

where the second equality interchanges both the two creation operators and the two annihilation operators, contributing two signs that cancel. The two products are equal, so the commutator vanishes. The disjointness of the index sets is what allowed both anticommutations to be performed without generating a delta function; this is why the result is special to particle-hole operators and fails for general $\hat{a}_p^\dagger\hat{a}_q$.

Answer to b).

$\hat{T}_i^2$ contains $\hat{a}_i\cdots\hat{a}_i$, and by Eq. (6.118) the two annihilation operators can be brought adjacent; $\hat{a}_i^2 = 0$, so $\hat{T}_i^2 = 0$ and $e^{\hat{T}_i} = 1+\hat{T}_i$. Combining with a) gives $e^{\hat{T}} = \prod_i e^{\hat{T}_i} = \prod_i(1+\hat{T}_i)$. Expanding the product,

$$\prod_i(1+\hat{T}_i) = 1+\sum_i \hat{T}_i + \sum_{i<j}\hat{T}_i\hat{T}_j + \cdots = 1+\hat{T}+\mathscr{O}(\hat{T}^2), \tag{6.120}$$

so the difference from $1+\hat{T}$ consists of the cross terms $\hat{T}_i\hat{T}_j$ with $i \neq j$, which are $2p$-$2h$, $3p$-$3h$, $\ldots$ excitations. They are second and higher order in the amplitudes, which is why the linearisation is exact for an infinitesimal variation and why the stability analysis of Section 6.10, which works to second order in the energy but first order in the state, is consistent.

Answer to c).

The class `ThoulessRotation` reports $\max|[\hat{T}_i,\hat{T}_j]| = 2.8\times10^{-17}$, $\max|\hat{T}_i^2| = 0$ exactly, agreement between $e^{\hat{T}}|c\rangle$ and $\prod_i \hat{b}_i^\dagger|0\rangle$ to 5.6×10^{-17}, an overlap $\langle c|c'\rangle = 1$ and $\det(f) = 1$ over the occupied block. Scaling the amplitudes by s and comparing $e^{s\hat{T}}|c\rangle$ with $(1+s\hat{T})|c\rangle$ gives errors 3.68×10^{-2}, 9.20×10^{-3}, 2.30×10^{-3}, 5.75×10^{-4} and 1.44×10^{-4} for $s = 0.4, 0.2, 0.1, 0.05, 0.025$: each halving of s divides the error by four, confirming the $\mathscr{O}(s^2)$ scaling.

Answer to d).

Expanding Eq. (6.41) and taking the overlap with $\langle c| = \langle 0|\hat{a}_{i_n}\cdots\hat{a}_{i_1}$, only terms in which every $\hat{a}_p^\dagger$ carries an index below the Fermi level survive, and each must be contracted with a distinct $\hat{a}_{i_k}$. Summing over the assignments with their signs reconstructs $\sum_\pi \operatorname{sgn}(\pi)\prod_k f_{i_k\pi(i_k)} = \det(f_{ip})$ over the occupied block. The proof needs f restricted to that block to be invertible, since f^{-1} appears in Eq. (6.43); a non-vanishing determinant is precisely invertibility. Geometrically, if $\langle c|\tilde{c}\rangle = 0$ the two determinants have no overlap and

$|\tilde{c}\rangle$ is not in the neighbourhood of $|c\rangle$ at all – one cannot reach it by an infinitesimal rotation, and Thouless' theorem makes no claim about it.

Exercise 4: BCH and Trotter

a) Derive Eq. (6.48) by matching the power series of $e^X e^Y$ and e^Z to second order.
b) Show that if $[X,Y] = cI$ then Eq. (6.50) is exact, and use it to derive the Weyl form $e^{ia\hat{x}/\hbar}e^{ib\hat{p}/\hbar} = e^{-iab/2\hbar}e^{i(a\hat{x}+b\hat{p})/\hbar}$.
c) Use Eq. (6.46) to show that the leading error of the first-order splitting is $-\frac{1}{2}[H_1,H_2]\Delta^2$, and verify that the symmetric formula (6.53) cancels it. Which commutators govern the leading error of S_2?
d) Prove the Baker-Hausdorff lemma $e^{-X}Ye^{X} = Y - [X,Y] + \frac{1}{2!}[X,[X,Y]] - \cdots$ and explain why the corresponding series in coupled-cluster theory terminates.
e) Reproduce the two tables of Sections 6.8 and 6.9 with `hartreefock.py`, and estimate how many Trotter steps a first- and a second-order scheme need for an accuracy $\varepsilon = 10^{-3}$ at $t = 1$.

Answer to a).

See Eq. (6.47) and the discussion following it. The essential observation is that $\frac{1}{2}(X+Y)^2$ symmetrises the cross term into $\frac{1}{2}(XY+YX)$, whereas $e^X e^Y$ produces XY with unit weight; the deficit is $\frac{1}{2}(XY-YX)$.

Answer to b).

If $[X,Y] = cI$ then $[X,[X,Y]] = [Y,[X,Y]] = 0$ and every term of Eq. (6.46) beyond $\frac{1}{2}[X,Y]$ contains at least one such double commutator. For $X = ia\hat{x}/\hbar$ and $Y = ib\hat{p}/\hbar$ we have $[X,Y] = -(ab/\hbar^2)[\hat{x},\hat{p}] = -iab/\hbar$, so

$$e^X e^Y = \exp\left(\frac{i}{\hbar}(a\hat{x}+b\hat{p}) - \frac{iab}{2\hbar}\right) = e^{-iab/2\hbar}\, e^{i(a\hat{x}+b\hat{p})/\hbar}. \tag{6.121}$$

The same argument applied to $[\hat{a},\hat{a}^\dagger] = 1$ gives the normal-ordered form of the coherent-state displacement operator.

Answer to c).

Setting $X = -iH_1\Delta$, $Y = -iH_2\Delta$ in Eq. (6.46), the second term is $\frac{1}{2}[X,Y] = -\frac{1}{2}[H_1,H_2]\Delta^2$, which is Eq. (6.51). For the symmetric product, write $S_2(\Delta) = e^{X/2}e^{Y}e^{X/2}$; applying BCH twice, the $[X,Y]$ contributions from the two half-steps enter with opposite signs and cancel exactly. What survives at the next order is a combination of $[H_1,[H_1,H_2]]$ and $[H_2,[H_1,H_2]]$ multiplying Δ^3. This is also visible from symmetry alone: $S_2(-\Delta) = S_2(\Delta)^{-1}$, so the error must be an odd function of Δ and cannot contain a Δ^2 term.

Answer to d).

Define $G(\lambda) = e^{-\lambda X}Ye^{\lambda X}$. Then $G'(\lambda) = e^{-\lambda X}[Y,X]e^{\lambda X} = -e^{-\lambda X}[X,Y]e^{\lambda X}$, and iterating, $G^{(n)}(0) = (-1)^n \mathrm{ad}_X^n(Y)$ with $\mathrm{ad}_X(Y) = [X,Y]$. Taylor-expanding about $\lambda = 0$ and setting $\lambda = 1$ gives the lemma. In coupled-cluster theory $X = \hat{T}$ is a sum of excitation operators, all of which commute with one another, and $Y = \hat{H}$ contains at most two-body terms. Each commutator with $\hat{T}$ must contract at least one annihilation operator of $\hat{H}$, and $\hat{H}$ has only four, so the series terminates exactly after the fourth commutator – which is why coupled cluster is a finite, not an asymptotic, expansion.

Answer to e).

The tables are reproduced by `demo_bch()` and `demo_trotter()`. Reading off the first-order column, the error is about $1.0/N$, so $\varepsilon = 10^{-3}$ requires $N \approx 10^3$ steps. The second-order error is about $0.39/N^2$, giving $N \approx 20$. The second-order scheme costs about 50% more exponentials per step, so it is cheaper by more than an order of magnitude here – and the advantage grows as the accuracy demand tightens, which is why practical quantum-simulation circuits are never built at first order.

Exercise 5: stability and the Lipkin model

a) Derive Eq. (6.56) for the norm of the rotated determinant, keeping track of which terms are second order.
b) Starting from Eq. (6.63), obtain Eq. (6.64) and the stability matrix (6.65). Why does χ contain both δC and δC^*?
c) For the Lipkin model with $W = 0$, show that $A = 0$ and that the largest eigenvalue of B is $|V|(N-1)$, and hence recover Eq. (6.73).
d) Derive the mean-field energy (6.70) from the rotation (6.69), and locate both stationary points.
e) Reproduce Table 6.1 and comment on the gap between $E_{\rm mf}$ and $E_{\rm exact}$ on either side of $\chi = 1$.

Answer to a).

Expanding Eq. (6.55) and taking the overlap with itself, the zeroth-order term gives 1; the cross terms between the constant and the linear piece vanish, since $\langle c|\hat{a}_a^\dagger \hat{a}_i|c\rangle = 0$ for $a > F \geq i$; and the linear piece with itself gives $\sum_{abij} \delta C_{ai}^* \delta C_{bj} \langle \Phi_i^a | \Phi_j^b \rangle = \sum_{ai} |\delta C_{ai}|^2$ by the orthonormality of the excited determinants. Terms involving the quadratic piece of the expansion are of third order or higher when paired with the linear one, and of fourth order with themselves.

Answer to b).

Subtract $(1 + \sum |\delta C|^2) E_0$ from Eq. (6.63) and divide by the norm; what is left over the norm is ΔE. Collecting the terms, those proportional to $\delta C^* \delta C$ assemble into $\chi^\dagger$ times the diagonal blocks $\Delta + A$, and those proportional to $\delta C \delta C$ and $\delta C^* \delta C^*$ into the off-diagonal blocks B and B^*. The doubled vector χ is needed precisely because the quadratic form is not Hermitian in δC alone: the terms $B\, \delta C^* \delta C^*$ pair two conjugated amplitudes, and only by treating δC and δC^* as independent components of a single vector can the whole expression be written as a Hermitian form. This doubling is exactly the structure of the RPA equations, where it produces the pairs of solutions $\pm\omega$.

Answer to c).

With $W = 0$ the interaction is $(V/2)(\hat{J}_+^2 + \hat{J}_-^2)$, which changes J_z by ± 2, so it has no matrix element within the $1p$-$1h$ space and $A_{ai,bj} = 0$; the diagonal block is $\Delta = \varepsilon \boldsymbol{I}$, since every particle-hole excitation costs ε. For B, the operator $\hat{J}_-^2$ connects the reference to a $2p$-$2h$ state in which levels p and q are both excited, giving $B_{ai,bj} = V$ for $p \neq q$ and zero for $p = q$; that is, $\boldsymbol{B} = V(\boldsymbol{J} - \boldsymbol{I})$ within each spin block, with $\boldsymbol{J}$ the matrix of ones. Its eigenvalues are $V(N-1)$, once, and $-V$, $N-1$ times. Because $\hat{M}$ has the form $\left(\begin{smallmatrix} \varepsilon \boldsymbol{I} & \boldsymbol{B} \\ \boldsymbol{B} & \varepsilon \boldsymbol{I} \end{smallmatrix}\right)$, its eigenvalues are $\varepsilon \pm b$ with b an eigenvalue of $\boldsymbol{B}$, and the smallest is $\varepsilon - |V|(N-1) = \varepsilon(1-\chi)$.

Answer to d).

Under the rotation (6.69) the quasispin vector rotates rigidly, $\langle \hat{J}_z \rangle = -\frac{N}{2}\cos\alpha$ and $\langle \hat{J}_x \rangle = \frac{N}{2}\sin\alpha$. The one-body term gives $-\frac{N\varepsilon}{2}\cos\alpha$ directly. For the interaction, $\hat{J}_+^2 + \hat{J}_-^2 = 2(\hat{J}_x^2 - \hat{J}_y^2)$, and evaluating this in the rotated determinant – where the N particles are uncorrelated, so that $\langle \hat{J}_x^2 \rangle = \langle \hat{J}_x \rangle^2 + \mathscr{O}(N)$ – gives $-\frac{N\varepsilon\chi}{4}\sin^2\alpha$ with $\chi = |V|(N-1)/\varepsilon$, the factor $N-1$ rather than N arising because a particle does not interact with itself. Setting the derivative (6.71) to zero gives $\sin\alpha = 0$ or $\cos\alpha = 1/\chi$, the second having a solution only when $\chi \geq 1$.

Answer to e).

The table is produced by `demo_stability()`. Below $\chi = 1$ the mean-field energy is flat at $-N\varepsilon/2 = -2$ while the exact energy falls steadily, so the error grows: the mean field is stable but increasingly inadequate, and the missing energy is pure correlation. At $\chi = 1.5$ and 2.0 the deformed solution recovers part of it – 0.167 and 0.5 respectively – but remains 0.48 and 0.56 above the exact answer. The deformed state breaks the symmetry of $\hat{H}$ under the rotation generated by $\hat{J}_z$ parity, which the exact ground state respects, so part of the remaining error is spurious and would be removed by projecting the broken symmetry back out.

Exercise 6: the electron gas

a) Derive Eq. (6.88) for a general interaction $v(r)$, and specialise it to the screened Coulomb interaction.
b) Show that $\hat{H}_b$ and $\hat{H}_{\text{el}-b}$ take the forms (6.81), and that together with the direct term of the electron-electron repulsion they cancel in the thermodynamic limit.
c) Carry out the steps from Eq. (6.93) to Eq. (6.94).
d) Compute the Hartree-Fock effective mass m^*_{HF} and the level density $n(\varepsilon_F^{\text{HF}})$, and comment on the result.
e) Show that the energy per electron is Eq. (6.108), plot it, and explain why the system is bound. Compute the pressure $P = -(\partial E/\partial \Omega)_N$ and the bulk modulus $B = -\Omega(\partial P/\partial \Omega)_N$, and comment.

Answer to a).

See Eqs. (6.85)–(6.88). For the screened Coulomb interaction the Fourier transform is

$$\int \frac{e^2 e^{-\mu r}}{r} e^{-i\boldsymbol{q}\cdot\boldsymbol{r}} d\boldsymbol{r} = \frac{4\pi e^2}{\mu^2 + q^2} \xrightarrow[\mu\to 0]{} \frac{4\pi e^2}{q^2}, \tag{6.122}$$

which is Eq. (6.96) again. The q^{-2} divergence at small momentum transfer is the signature of the infinite range of the interaction, and it is the source of every difficulty in this section.

Answer to b).

Both are elementary integrals of $e^{-\mu r}/r$ over the box with a constant density $n = N/\Omega$: $\int d\boldsymbol{r}\, e^{-\mu r}/r = 4\pi/\mu^2$, whence $\hat{H}_b = \frac{e^2}{2} n^2 \Omega \cdot 4\pi/\mu^2 = \frac{e^2}{2}\frac{N^2}{\Omega}\frac{4\pi}{\mu^2}$ and $\hat{H}_{\text{el}-b} = -e^2 \frac{N^2}{\Omega}\frac{4\pi}{\mu^2}$, twice as large and of opposite sign. The direct term of Eq. (6.92) is $\frac{N^2}{2\Omega}\int v(r) d\boldsymbol{r} = \frac{e^2}{2}\frac{N^2}{\Omega}\frac{4\pi}{\mu^2}$, and the three sum to zero. Physically: a neutral system has no monopole, so the $\boldsymbol{q} = 0$ component of the interaction must drop out, and it is only the separation into three individually divergent pieces that made the cancellation look accidental.

Answer to c).

See Eqs. (6.95)–(6.99). The last integral is

$$\int_0^{k_F} p \ln\left|\frac{p+k}{p-k}\right| dp = kk_F + \frac{k_F^2-k^2}{2}\ln\left|\frac{k+k_F}{k-k_F}\right|, \tag{6.123}$$

which may be checked by differentiating with respect to k_F, or numerically.

Answer to d).

Differentiating Eq. (6.102),

$$\frac{m^*_{\mathrm{HF}}}{m} = \frac{2}{2-0.663\, r_s F'(1)}, \qquad F'(x) \sim -\frac{1}{2}\ln\left|\frac{2}{1-x}\right| \;\text{ as } x \to 1. \tag{6.124}$$

Since $F'(1) = -\infty$, the denominator diverges and $m^*_{\mathrm{HF}} \to 0$; the program gives 0.256, 0.185, 0.144, 0.118 and 0.100 at $1-x = 10^{-2}, \ldots, 10^{-6}$, decreasing like the reciprocal logarithm. By Eq. (6.107) the level density is proportional to m^* and therefore vanishes at the Fermi surface. This contradicts experiment: the electronic specific heat of a metal is linear in T with a coefficient proportional to $n(\varepsilon_F)$, and it is measured to be finite. The failure is traceable to the unscreened $1/q^2$, and it is repaired by the random-phase approximation, which replaces it by a screened interaction with a finite range.

Answer to e).

The kinetic term is Eq. (6.91); writing $k_F = (9\pi/4)^{1/3}/r_0$ and measuring in Rydberg,

$$\frac{\langle \hat{T}\rangle}{N} = \frac{3}{5}\frac{\hbar^2 k_F^2}{2m} = \frac{e^2}{2a_0}\frac{3}{5}\left(\frac{9\pi}{4}\right)^{2/3}\frac{1}{r_s^2} = \frac{e^2}{2a_0}\frac{2.21}{r_s^2}, \tag{6.125}$$

and the exchange term evaluates similarly to $-0.916/r_s$ in the same units. The system is bound because the two terms carry opposite signs and different powers of r_s: the kinetic repulsion, $\propto r_s^{-2}$, dominates at high density and the exchange attraction, $\propto r_s^{-1}$, at low density, so their sum has a minimum, at $r_s = 4.83$ with $E_0/N = -0.0949$ Ry. For the pressure, with $\Omega \propto N r_0^3$ so that $\partial/\partial\Omega = -(r_s/3\Omega)\,\partial/\partial r_s$,

$$P = -\left(\frac{\partial E}{\partial \Omega}\right)_N = \frac{\rho\, r_s}{3}\frac{d}{dr_s}\left(\frac{E_0}{N}\right) = \frac{e^2\rho}{2a_0}\frac{1}{3}\left[-\frac{4.42}{r_s^2} + \frac{0.916}{r_s}\right], \tag{6.126}$$

which vanishes exactly at the equilibrium r_s, as it must, is negative (the gas would contract) at higher density and positive at lower. The bulk modulus $B = -\Omega(\partial P/\partial\Omega)_N$ is positive at the minimum, confirming mechanical stability, and its measured values in the alkali metals are of the right order but consistently smaller than the Hartree-Fock prediction – correlation softens the gas.

Chapter 7
Mean-field applications: TDA and RPA

Chapter 6 ended with a matrix. Expanding the Hartree-Fock energy to second order in the amplitudes of a Thouless rotation produced the stability matrix (6.65), and we remarked in passing that it is the matrix of the random-phase approximation. This chapter makes good on that remark. We shall see that the same two blocks A and B, arranged Hermitian, decide whether the mean field is a minimum, and arranged non-Hermitian, give the excitation energies of the system. Stability and spectroscopy are two readings of one calculation.

The vehicle is a schematic model with two competing interactions – a pairing term that scatters whole pairs between levels, and a particle-hole term that breaks them. At zero particle-hole strength it reduces exactly to the pairing model of Section 4.2, so every number can be checked against Table 4.1; switching the particle-hole term on makes the particle-hole channel non-trivial and lets us watch two mean fields compete. We solve the model exactly by the methods of Chapter 5, then in the Tamm-Dancoff approximation, then in the random-phase approximation on the Hartree-Fock state, and finally in the quasiparticle random-phase approximation built on a BCS vacuum. Comparing four approximations against one exact answer, in a system small enough that nothing is hidden, is the point of the exercise.

7.1 The pairing plus particle-hole model

We take L doubly degenerate, equally spaced single-particle levels $p = 1, \ldots, L$, each carrying a spin label $\sigma = \pm$, occupied by N spin-$\frac{1}{2}$ fermions. The Hamiltonian is $\hat{H} = \hat{H}_0 + \hat{V}_{\text{pair}} + \hat{V}_{\text{ph}}$ with

$$\hat{H}_0 = \xi \sum_{p\sigma} (p-1)\, \hat{a}^\dagger_{p\sigma} \hat{a}_{p\sigma}, \tag{7.1}$$

$$\hat{V}_{\text{pair}} = -\frac{g}{2} \sum_{pq} \hat{a}^\dagger_{p+} \hat{a}^\dagger_{p-} \hat{a}_{q-} \hat{a}_{q+}, \tag{7.2}$$

$$\hat{V}_{\text{ph}} = -\frac{f}{2} \sum_{pqr} \left(\hat{a}^\dagger_{p+} \hat{a}^\dagger_{p-} \hat{a}_{q-} \hat{a}_{r+} + \text{h.c.} \right). \tag{7.3}$$

The one-body term gives level p the energy $\xi(p-1)$; we set $\xi = 1$ throughout. The pairing term, met already in Section 4.2, moves a complete pair – both spin components of one level – from level q to level p. The particle-hole term removes a spin-down particle from level q and a spin-up particle from level r and deposits a pair at level p; when $q \neq r$ it therefore *breaks* a pair, which is exactly what the pairing term cannot do.

Both terms conserve N_+ and N_- separately, so $S_z = (N_+ - N_-)/2$ is a good quantum number and the low-lying spectrum lives in the balanced sector $S_z = 0$. For $N = L = 4$ the full space has $\binom{8}{4} = 70$ determinants, of which 36 have $S_z = 0$.

Two limits organise everything that follows.

At $f = 0$.

The model is the pairing model of Chapter 4. The seniority – the number of unpaired particles – is conserved, the ground state has seniority zero, and the exact energies are those of Table 4.1. The program `rpa.py` reproduces them:

g	E_{ref}	E_0 (this chapter)	Table 4.1
−1.00	3.000000	2.77987014	2.77987014
−0.50	2.500000	2.43688426	2.43688426
0.00	2.000000	2.00000000	2.00000000
0.50	1.500000	1.41677428	1.41677428
1.00	1.000000	0.63554847	0.63554847

At $g = 0$.

Only the pair-breaking term survives, and the seniority is maximally violated. For intermediate f/g the two mechanisms compete, and the interesting question – the one the whole chapter is about – is which mean field to build the fluctuations on.

7.2 Exact diagonalisation

The exact solution is Chapter 5 applied without modification. Determinants are bit strings, as in Section 3.14; the Hamiltonian is built by applying each term of Eqs. (7.1)–(7.3) to every basis state and routing the result back to its index with the accumulated fermion sign; and the lowest eigenvalues follow from the algorithms of Chapter 1 – a dense solve when the matrix is small enough, Lanczos when it is not.

Restricting to $S_z = 0$ already cuts the dimension from 70 to 36, an instance of the general remark of Section 5.11 that symmetries are worth exploiting. For $g = 0.5$ and $f = 0.05g$ the lowest exact eigenvalues are

$$E_0 = 1.347133, \quad E_1 = 2.710867, \quad E_2 = 2.711098, \quad E_3 = 3.413340, \tag{7.4}$$

with the near-degeneracy of E_1 and E_2 split only by the small particle-hole term. These are the numbers every approximation in this chapter will be measured against.

7.3 The Hartree-Fock reference

Both approximations to come are built on a stationary mean field, so we solve the Hartree-Fock equations of Section 6.4 first. The antisymmetrised two-body matrix elements are read off the two-particle sector of $\hat{V} = \hat{H} - \hat{H}_0$,

$$\bar{v}_{abcd} = \left(\hat{a}_a^\dagger \hat{a}_b^\dagger |0\rangle\right)^\dagger \hat{V} \left(\hat{a}_c^\dagger \hat{a}_d^\dagger |0\rangle\right), \tag{7.5}$$

and the self-consistent field of Section 6.5 does the rest:

g	f	E_{HF}	iterations	Hartree-Fock energies
0.70	0.00	1.30000	2	−0.350, −0.350, 0.650, 0.650, …
0.70	0.05	1.19709	20	−0.404, −0.404, 0.600, 0.600, …
1.00	0.50	−0.26976	40	−1.505, −1.505, 0.119, 0.119, …

The first row is the case of Section 6.12 again: with pure pairing the filled Fermi sea is already the Hartree-Fock solution, the iteration converges immediately, and the only effect of the interaction is to lower each occupied level by $g/2$ while leaving the empty ones alone. The particle-hole term does connect the reference to a $1p$-$1h$ state, so the mean field must rotate the orbitals until Brillouin's theorem, Eq. (6.31), is restored, and the iteration becomes non-trivial.

7.4 The equations of motion

Rather than derive TDA and RPA separately, we obtain both from a single principle. Suppose the exact eigenstates satisfy $\hat{H}|\nu\rangle = E_\nu|\nu\rangle$ and define, for each excited state, an operator $\hat{Q}^\dagger_\nu$ with

$$|\nu\rangle = \hat{Q}^\dagger_\nu|0\rangle, \qquad \hat{Q}_\nu|0\rangle = 0, \tag{7.6}$$

where $|0\rangle$ is the exact ground state. Such an operator always exists – $\hat{Q}^\dagger_\nu = |\nu\rangle\langle 0|$ will do – and it satisfies

$$\left[\hat{H},\hat{Q}^\dagger_\nu\right]|0\rangle = \omega_\nu \hat{Q}^\dagger_\nu|0\rangle, \qquad \omega_\nu = E_\nu - E_0. \tag{7.7}$$

Taking the matrix element of Eq. (7.7) with an arbitrary variation $\delta\hat{Q}$ and using $\hat{Q}_\nu|0\rangle = 0$ to symmetrise, one arrives at the *equation-of-motion* form

$$\boxed{\langle 0|\left[\delta\hat{Q},\left[\hat{H},\hat{Q}^\dagger_\nu\right]\right]|0\rangle = \omega_\nu\,\langle 0|\left[\delta\hat{Q},\hat{Q}^\dagger_\nu\right]|0\rangle.} \tag{7.8}$$

This is still exact. It becomes an approximation as soon as we (i) restrict the operators $\hat{Q}^\dagger$ to a manageable family, and (ii) replace the unknown exact ground state $|0\rangle$ by something we can compute. Every method in this chapter is a choice of those two things.

The double commutator on the left has a virtue worth noticing: it is symmetric in the two operators, so each of them can be evaluated with fewer contractions than the raw product would need. This is why the RPA matrices are traditionally written as double commutators, and it is how the program computes them – literally, as expectation values of $[\hat{O}^\dagger_K,[\hat{H},\hat{O}_L]]$ in the chosen reference, with nothing transcribed by hand.

7.5 The Tamm-Dancoff approximation

The simplest choice is to build the excitation from one particle-hole pair only, and to take the Hartree-Fock determinant for the ground state:

$$\hat{Q}^\dagger_\nu = \sum_{mi} X^\nu_{mi}\,\hat{a}^\dagger_m \hat{a}_i, \qquad |0\rangle \longrightarrow |\mathrm{HF}\rangle, \tag{7.9}$$

with $i,j,\dots$ labelling occupied and $m,n,\dots$ unoccupied Hartree-Fock orbitals. Since $\hat{a}^\dagger_m\hat{a}_i$ does annihilate the determinant when acting to the left, condition (7.6) holds exactly, and the right-hand side of Eq. (7.8) reduces to $\delta_{mn}\delta_{ij}$. What remains is an ordinary Hermitian eigenvalue problem,

$$\sum_{nj} A_{mi,nj} X^\nu_{nj} = \omega_\nu X^\nu_{mi}, \tag{7.10}$$

with

$$A_{mi,nj} = \langle \mathrm{HF}|\left[\hat{a}^\dagger_i\hat{a}_m,\left[\hat{H},\hat{a}^\dagger_n\hat{a}_j\right]\right]|\mathrm{HF}\rangle = (\varepsilon_m - \varepsilon_i)\,\delta_{mn}\delta_{ij} + \bar{v}_{mjin}. \tag{7.11}$$

There is a plainer way to say what this is. The matrix A is nothing but the Hamiltonian restricted to the space of $1p$-$1h$ determinants, measured from the reference energy,

$$A_{mi,nj} = \langle \Phi_i^m | \hat{H} | \Phi_j^n \rangle - E_{\mathrm{HF}}\, \delta_{mn} \delta_{ij}. \tag{7.12}$$

The program checks this identity directly and finds agreement to 3×10^{-15}. So the Tamm-Dancoff approximation is exactly the configuration-interaction truncation of Section 5.8 applied to excited states: keep the reference and all single excitations, diagonalise, and read off the excitation energies. It is CIS, no more and no less, and it inherits both the variational property – every TDA root is an upper bound to the corresponding exact excitation energy within that space – and the defect, that the ground state is left uncorrelated.

7.6 The random-phase approximation

The Tamm-Dancoff ground state is a single determinant, which we know is wrong. The random-phase approximation improves it by allowing the phonon to *de*-excite as well as excite,

$$\hat{Q}_\nu^\dagger = \sum_{mi} \left(X_{mi}^\nu \hat{a}_m^\dagger \hat{a}_i - Y_{mi}^\nu \hat{a}_i^\dagger \hat{a}_m \right). \tag{7.13}$$

The second term does not annihilate a Slater determinant acting to the right, so the condition $\hat{Q}_\nu |0\rangle = 0$ can no longer be satisfied by the Hartree-Fock state. It defines instead a new, correlated ground state $|\mathrm{RPA}\rangle$ – one that we never construct explicitly, but whose existence is the whole content of the approximation.

To make progress we need the norm on the right of Eq. (7.8), which requires the commutator of two particle-hole operators,

$$\left[\hat{a}_i^\dagger \hat{a}_m, \hat{a}_n^\dagger \hat{a}_j \right] = \delta_{mn} \hat{a}_i^\dagger \hat{a}_j - \delta_{ij} \hat{a}_n^\dagger \hat{a}_m. \tag{7.14}$$

This is an operator, not a number. The *quasiboson approximation* replaces it by its expectation value in the reference determinant,

$$\left[\hat{a}_i^\dagger \hat{a}_m, \hat{a}_n^\dagger \hat{a}_j \right] \longrightarrow \langle \mathrm{HF} | \left[\hat{a}_i^\dagger \hat{a}_m, \hat{a}_n^\dagger \hat{a}_j \right] | \mathrm{HF} \rangle = \delta_{mn} \delta_{ij}, \tag{7.15}$$

which is to say that particle-hole pairs are treated as bosons. It is exact when the occupations are exactly zero and one and nothing else is going on; it fails to the extent that the true ground state has particles above the Fermi level, and it is the origin of every pathology of the method.

With Eq. (7.15), Eq. (7.8) becomes the *RPA eigenvalue problem*

$$\boxed{\begin{pmatrix} A & B \\ -B^* & -A^* \end{pmatrix} \begin{pmatrix} X^\nu \\ Y^\nu \end{pmatrix} = \omega_\nu \begin{pmatrix} X^\nu \\ Y^\nu \end{pmatrix},} \tag{7.16}$$

with A as before and

$$B_{mi,nj} = -\langle \mathrm{HF} | \left[\hat{a}_i^\dagger \hat{a}_m, \left[\hat{H}, \hat{a}_j^\dagger \hat{a}_n \right] \right] | \mathrm{HF} \rangle = \bar{v}_{mnij}. \tag{7.17}$$

The program verifies both closed forms against the double commutators to better than 10^{-13}. Setting $B = 0$ discards the Y amplitudes and returns the Tamm-Dancoff equation, so TDA is RPA with the ground-state correlations switched off.

Equation (7.16) is not Hermitian. Its eigenvalues come in pairs $\pm\omega_\nu$ – if (X, Y) solves it with ω, then (Y^*, X^*) solves it with $-\omega$ – and they need not be real. The physical solutions are the positive ones, normalised by

$$\sum_{mi} \left(|X_{mi}^\nu|^2 - |Y_{mi}^\nu|^2 \right) = 1, \tag{7.18}$$

a metric that is indefinite rather than positive, which is precisely why real eigenvalues are not guaranteed.

7.7 RPA and the stability of the mean field

We can now close the loop with Chapter 6. The blocks A and B of Eqs. (7.11) and (7.17) are, term for term, the blocks of the stability matrix (6.65); only the arrangement differs,

$$\hat{M}_{\text{stab}} = \begin{pmatrix} A & B \\ B^* & A^* \end{pmatrix} \text{ (Hermitian)}, \qquad \hat{M}_{\text{RPA}} = \begin{pmatrix} A & B \\ -B^* & -A^* \end{pmatrix} \text{ (not Hermitian)}. \tag{7.19}$$

The relation between them is exact and easy to state. Writing $\boldsymbol{\eta} = \operatorname{diag}(\mathbf{1}, -\mathbf{1})$ for the metric of Eq. (7.18), we have $\hat{M}_{\text{RPA}} = \boldsymbol{\eta}\,\hat{M}_{\text{stab}}$. If $\hat{M}_{\text{stab}}$ is positive definite it has a Hermitian square root, and

$$\hat{M}_{\text{stab}}^{1/2}\,\boldsymbol{\eta}\,\hat{M}_{\text{stab}}^{1/2} \tag{7.20}$$

is Hermitian and similar to $\hat{M}_{\text{RPA}}$, so the RPA frequencies are real. Conversely, if $\hat{M}_{\text{stab}}$ has a negative eigenvalue the argument fails and imaginary roots appear. Hence

$$\boxed{\text{the RPA frequencies are real} \iff \hat{M}_{\text{stab}} \geq 0 \iff \text{the mean field is a local minimum.}} \tag{7.21}$$

The Lipkin model of Sections 4.1 and 6.11 shows this happening. Feeding the same blocks into both arrangements:

χ	$\lambda_{\min}(\hat{M}_{\text{stab}})$	imaginary roots	lowest real RPA root
0.25	0.75	0	0.968246
0.50	0.50	0	0.866025
0.90	0.10	0	0.435890
1.00	0.00	0	0.000000
1.50	−0.50	2	—
2.00	−1.00	2	—

Table 7.1 The Lipkin model with $N = 4$, $\varepsilon = 1$, $W = 0$. The lowest eigenvalue of the Hermitian stability matrix and the RPA spectrum built from the same blocks cross zero together at $\chi = 1$. Beyond that the collective root is imaginary and no excitation energy can be quoted.

The collective root goes *soft* – falls to zero – exactly at $\chi = 1$, where the deformed solution of Section 6.11 appears, and becomes imaginary beyond it. This is the general pattern, and it is one of the most useful things RPA does: a vanishing frequency announces a mean-field phase transition, and an imaginary one says the reference is a saddle point. The excitation spectrum and the stability analysis are the same calculation read two ways.

For the pairing plus particle-hole model the particle-hole stability matrix never goes negative – $\lambda_{\min}$ stays at or above 1 for every coupling we consider, including $g = 4$, $f = 2$. The particle-hole channel is simply not where this model becomes collective. Its instability is in the pairing channel, and reaching it requires a different mean field, which is the subject of Sections 7.10 and 7.11.

7.8 The RPA ground state and its correlation energy

Because $\hat{Q}_\nu|\text{RPA}\rangle = 0$ with $\hat{Q}$ containing the de-excitation piece, the RPA ground state contains particle-hole pairs. One can show, and Exercise 7.14 asks for the argument, that $|\text{RPA}\rangle$ has the form

$$|\mathrm{RPA}\rangle = \mathscr{N} \exp\left\{ \frac{1}{2} \sum_{mnij} Z_{mi,nj}\, \hat{a}_m^\dagger \hat{a}_i \hat{a}_n^\dagger \hat{a}_j \right\} |\mathrm{HF}\rangle, \qquad \boldsymbol{Z} = \boldsymbol{Y}^* \left(\boldsymbol{X}^*\right)^{-1}, \tag{7.22}$$

an exponential of $2p$-$2h$ operators acting on the Hartree-Fock determinant. The reader will recognise the structure of Thouless' theorem, Eq. (6.34), with two-body operators in place of one-body ones, and the structure of the coupled-cluster ansatz, Eq. (5.29), with the doubles amplitude fixed by the RPA amplitudes rather than solved for independently. RPA is a doubles theory in disguise.

Its energy relative to Hartree-Fock is

$$E_{\mathrm{corr}}^{\mathrm{RPA}} = \frac{1}{2} \left(\sum_{\nu} \omega_\nu - \mathrm{Tr} A \right), \tag{7.23}$$

a remarkably compact result: the correlation energy is half the difference between the sum of the RPA frequencies and the sum of the TDA ones. Since each RPA root lies below the corresponding TDA root, the correction is always negative – RPA always lowers the energy. For $g = 0.7$, $f = 0.05$ the program gives $\mathrm{Tr} A = 38.423313$ and $\sum_\nu \omega_\nu = 37.824314$, hence $E_{\mathrm{corr}} = -0.299500$.

7.9 Results in the particle-hole channel

We can now compare. The table gives, for four combinations of the two couplings, the Hartree-Fock energy, the RPA-corrected ground-state energy, the exact ground-state energy, and the lowest excitation energy in the three treatments.

g	f	E_{HF}	$E_{\mathrm{HF}} + E_{\mathrm{corr}}$	E_{exact}	ω_1^{TDA}	ω_1^{RPA}	$\omega_1^{\mathrm{exact}}$
0.50	0.025	1.44927	1.29898	1.34713	1.2748	1.2448	1.3637
0.70	0.050	1.19709	0.89759	0.96976	1.3992	1.3412	1.5972
1.00	0.050	0.89709	0.36862	0.45058	1.5492	1.4488	1.9425
1.00	0.500	−0.26976	−1.73461	−1.69174	1.8111	1.6445	2.8596

Table 7.2 The pairing plus particle-hole model with $N = L = 4$, $\xi = 1$. Hartree-Fock, RPA and exact ground-state energies, and the lowest excitation energy in the Tamm-Dancoff and random-phase approximations compared with the exact one.

Three things are worth reading off.

First, RPA always lowers the energy below Hartree-Fock, by construction, and here it *overshoots* the exact answer in every row. This is the standard behaviour: the quasiboson approximation, by treating particle-hole pairs as bosons that can be created without limit, over-counts the ground-state correlations. The overshoot grows with the coupling, from 0.048 at $g = 0.5$ to 0.043 at $g = 1$, $f = 0.5$ – modest here, but in a strongly collective system RPA can overbind badly and eventually collapse altogether.

Second, $\omega_1^{\mathrm{RPA}} < \omega_1^{\mathrm{TDA}}$ in every row. The B block always pushes the collective root down, for the same reason it lowers the ground state: it adds correlations to the ground state, and the excitation energy is measured from a lower floor.

Third, both approximations put the lowest excitation well below the exact one, and the gap widens as the pairing strength grows – at $g = 1$, $f = 0.5$ the exact first excitation is 2.86 against an RPA value of 1.64. This is not a failure of arithmetic but of the choice of channel. The particle-hole phonon knows nothing about pairing correlations, and in this model the low-lying spectrum is shaped by them.

A closed form.

For pure pairing the particle-hole problem collapses. All sixteen particle-hole pairs are degenerate, the matrices reduce to multiples of simple structures, and one finds

$$\omega^{\mathrm{TDA}} = \xi + \frac{g}{2}, \qquad \omega^{\mathrm{RPA}} = \sqrt{\xi^2 + g\xi} = \sqrt{\left(\xi + \tfrac{g}{2}\right)^2 - \left(\tfrac{g}{2}\right)^2}, \tag{7.24}$$

the second being the classic $\sqrt{A^2 - B^2}$ of a two-level RPA problem with $A = \xi + g/2$ and $B = g/2$. The program confirms both to eight digits:

g	TDA	$\xi + g/2$	RPA	$\sqrt{1+g}$
0.5	1.25000000	1.25000000	1.22474487	1.22474487
1.0	1.50000000	1.50000000	1.41421356	1.41421356
2.0	2.00000000	2.00000000	1.73205081	1.73205081
3.0	2.50000000	2.50000000	2.00000000	2.00000000
5.0	3.50000000	3.50000000	2.44948974	2.44948974

Equation (7.24) is worth staring at. The RPA root is real for all $g > -\xi$, so the particle-hole channel of the pure pairing model never goes unstable, however strong the pairing. The instability is elsewhere, and finding it requires a mean field that can break a different symmetry.

7.10 BCS: the pairing mean field

The pairing interaction is $-G\sum_{pq} \hat{P}_p^\dagger \hat{P}_q$ with $\hat{P}_p^\dagger = \hat{a}_{p+}^\dagger \hat{a}_{p-}^\dagger$ and $G = g/2$. Its natural mean field is not of Hartree-Fock type at all, because the object that acquires an expectation value is $\langle \hat{P}_p^\dagger \rangle$, which does not conserve particle number. The trial state is the BCS wave function

$$|\mathrm{BCS}\rangle = \prod_p \left(u_p + v_p \hat{a}_{p+}^\dagger \hat{a}_{p-}^\dagger \right) |0\rangle, \qquad u_p^2 + v_p^2 = 1, \tag{7.25}$$

a product over levels in which each level is empty with amplitude u_p and occupied by a pair with amplitude v_p. It is a superposition of states with different particle numbers, so the number is fixed only on average, by minimising $\langle \mathrm{BCS}|\hat{H} - \lambda \hat{N}|\mathrm{BCS}\rangle$ with λ adjusted until $\langle \hat{N} \rangle = N$.

Because the trial state is a product over levels, every expectation value is available in closed form. Writing $u_p = \cos\theta_p$, $v_p = \sin\theta_p$,

$$\langle \hat{N} \rangle = 2\sum_p v_p^2, \qquad \langle \hat{H}_0 \rangle = 2\sum_p \varepsilon_p v_p^2, \tag{7.26}$$

and, using $\langle \hat{P}_p^\dagger \hat{P}_q \rangle = u_p v_p u_q v_q$ for $p \neq q$ and v_p^2 for $p = q$,

$$\langle \hat{H} \rangle = 2\sum_p \varepsilon_p v_p^2 - S\left[\sum_p v_p^2 + \sum_{p \neq q} u_p v_p u_q v_q \right], \qquad S = \frac{g}{2} + f. \tag{7.27}$$

The program checks Eq. (7.27) against the expectation value computed directly in the 2^{2L}-dimensional Fock space and finds agreement to 5×10^{-16}.

The particle-hole term as extra pairing.

Equation (7.27) contains a small surprise. Only the $q = r$ part of $\hat{V}_{\mathrm{ph}}$ survives in the BCS state – for $q \neq r$ the operator $\hat{a}_{q-}\hat{a}_{r+}$ leaves two broken pairs, and the BCS state has none – and $\hat{a}_{q-}\hat{a}_{q+} = \hat{P}_q$ exactly. Seen from the BCS vacuum, therefore, the particle-hole term *is* a pairing term, and the whole problem depends on g and f only through the combination $g+2f$. The program confirms it:

g	f	$g+2f$	E_{BCS}	gap Δ
1.4	0.0	1.4	0.515684	0.789218
1.2	0.1	1.4	0.515684	0.789218
0.8	0.3	1.4	0.515684	0.789218
0.0	0.7	1.4	0.515684	0.789218

This is a clean illustration of a general point: a mean field sees only that part of the interaction which can be written as a one-body field in its own channel. Hartree-Fock and BCS decompose the same $\hat{V}$ differently, and each is blind to what the other sees.

The pairing transition.

With $\Delta = S\sum_p u_p v_p$ the pairing gap, the solution of the BCS equations for $f = 0$ is

g	Δ	λ	v_p^2
0.40	0.00000	1.2500	1.000, 1.000, 0.000, 0.000
0.80	0.00000	1.2500	1.000, 1.000, 0.000, 0.000
1.00	0.00000	1.2500	1.000, 1.000, 0.000, 0.000
1.20	0.46716	1.2000	0.984, 0.925, 0.075, 0.016
1.60	1.05608	1.1000	0.934, 0.783, 0.217, 0.066
2.00	1.54088	1.0000	0.887, 0.709, 0.291, 0.113

Table 7.3 The BCS solution of the pairing model with $L = 4$ levels and $N = 4$ particles. Below $g_c = 1.07037$ the only solution is the normal Fermi sea; above it a gap opens and the occupations become fractional.

Below a critical coupling the only solution is the normal Fermi sea, with $\Delta = 0$ and sharp occupations; above it a gap opens and the occupation numbers spread. The condition for a non-trivial solution, obtained by linearising the gap equation about $\Delta = 0$, is

$$1 = 2S\left[\frac{1}{3\xi+S} + \frac{1}{\xi+S}\right], \tag{7.28}$$

whose root is $S_c = 0.53519$, so that $g_c = 1.07037$ at $f = 0$ – exactly the value the program locates by bisection.

Two comments. The transition is *sharp* even though the system is finite, but that sharpness is an artefact of the mean field: the exact ground-state energy of Table 4.1 is perfectly smooth through g_c, and nothing whatever happens there. Mean-field phase transitions in finite systems are approximations to smooth crossovers. And above g_c the reference has stopped being an eigenstate of $\hat{N}$, which is a broken symmetry, with all that this implies for the fluctuations built on top of it.

7.11 Quasiparticle RPA

The BCS state is a vacuum, not for particles, but for the Bogoliubov quasiparticles

$$\hat{\alpha}^{\dagger}_{p+} = u_p \hat{a}^{\dagger}_{p+} - v_p \hat{a}_{p-}, \qquad \hat{\alpha}^{\dagger}_{p-} = u_p \hat{a}^{\dagger}_{p-} + v_p \hat{a}_{p+}, \tag{7.29}$$

which satisfy $\hat{\alpha}_{p\sigma}|\mathrm{BCS}\rangle = 0$. The program verifies this to 3×10^{-17}. These are the natural elementary excitations of the paired system, and the quasiparticle random-phase approximation builds collective modes out of *pairs* of them,

$$\hat{Q}^{\dagger}_{\nu} = \sum_{a<b} \left(X^{\nu}_{ab}\, \hat{\alpha}^{\dagger}_{a} \hat{\alpha}^{\dagger}_{b} - Y^{\nu}_{ab}\, \hat{\alpha}_{b} \hat{\alpha}_{a} \right). \tag{7.30}$$

The structure of Eq. (7.8) is unchanged; only the reference and the elementary operators are different, so the same routine produces

$$A_{ab,cd} = \langle \mathrm{BCS}| \left[\hat{\alpha}_b \hat{\alpha}_a, \left[\hat{H}', \hat{\alpha}^{\dagger}_{c} \hat{\alpha}^{\dagger}_{d} \right] \right] |\mathrm{BCS}\rangle, \qquad B_{ab,cd} = -\langle \mathrm{BCS}| \left[\hat{\alpha}_b \hat{\alpha}_a, \left[\hat{H}', \hat{\alpha}_d \hat{\alpha}_c \right] \right] |\mathrm{BCS}\rangle, \tag{7.31}$$

and the same eigenvalue problem (7.16). For $L = 4$ there are $2L = 8$ quasiparticle states and hence $2L(2L-1)/2 = 28$ two-quasiparticle operators.

Two features require attention, and they are the physics of this section.

7.11.1 The spurious mode

The BCS state breaks the $U(1)$ symmetry generated by $\hat{N}$. Whenever a mean field breaks a continuous symmetry of the Hamiltonian, the generator of that symmetry produces an exact zero mode of the RPA equations – a Goldstone mode. The argument is short: since $[\hat{H}, \hat{N}] = 0$, the amplitudes

$$X^{\mathrm{sp}}_{K} = \langle \mathrm{BCS}| \left[\hat{O}_K, \hat{N} \right] |\mathrm{BCS}\rangle, \qquad Y^{\mathrm{sp}}_{K} = \langle \mathrm{BCS}| \left[\hat{O}^{\dagger}_{K}, \hat{N} \right] |\mathrm{BCS}\rangle \tag{7.32}$$

satisfy $\hat{M}_{\mathrm{RPA}}\, (X^{\mathrm{sp}}, Y^{\mathrm{sp}})^T = 0$. Physically the mode is not an excitation at all: it is the rotation of the BCS phase, which costs nothing and merely moves the system to a state with a different average particle number.

The standard device is to build the QRPA with the *cranked* Hamiltonian

$$\hat{H}' = \hat{H} - \lambda \hat{N}, \tag{7.33}$$

using the same λ that the BCS equations extremise, which places the spurious root at zero where it can be discarded. In finite arithmetic it does not land exactly on zero, and because the RPA metric is singular there it may emerge as a small *imaginary* pair rather than a small real one. This is worth emphasising, since it is easily mistaken for an instability:

g	Δ	$\lvert\omega_{\mathrm{sp}}\rvert$	overlap with $\hat{N}$	imaginary roots
1.20	0.46716	1.14×10^{-4}	1.000000	0
1.40	0.78922	1.83×10^{-4}	1.000000	0
1.60	1.05608	2.59×10^{-4}	1.000000	0
1.80	1.30338	1.64×10^{-4}	1.000000	0
2.00	1.54088	2.77×10^{-4}	1.000000	0

Table 7.4 The spurious mode of the QRPA for the pairing model, $f = 0$. The root nearest zero has unit overlap with the amplitudes (7.32) built from the number operator, identifying it beyond doubt. Once it is removed, no imaginary eigenvalues remain at any coupling.

The overlap with the number-operator amplitudes is one to six decimal places, which identifies the near-zero root beyond any doubt. Once it is removed, the QRPA matrix has no imaginary eigenvalues anywhere in the superfluid regime. The lesson generalises: before declaring a mean field unstable on the strength of an imaginary RPA root, check whether the root is a spurious mode in disguise. The test is cheap – compute the amplitudes (7.32) for each broken symmetry and take overlaps.

7.11.2 The mode content

The second feature is that quasiparticles mix particle number, so a two-quasiparticle phonon need not conserve it. Building the phonon operator explicitly, applying it to the BCS vacuum and taking the expectation value of $\hat{N}$ separates the modes cleanly:

ω_ν	$\langle \Delta\hat{N} \rangle$	character
$g=2.0$, $f=0$; *exact excitations* 3.3274, 3.3274, 3.4897, 4.2536		
2.6015	0.0000	excitation of the N-particle system
2.9212	0.0000	excitation of the N-particle system (fourfold)
3.7879	-0.3566	pairing vibration
3.7879	$+0.3566$	pairing vibration

Table 7.5 QRPA phonons at $g=2$, classified by the average particle number they carry. Modes with $\langle \Delta\hat{N} \rangle = 0$ are excitations within the N-particle system; the others connect towards $N \pm 2$ and are the pairing vibrations. The shift is not quantised because the BCS vacuum is not an eigenstate of $\hat{N}$.

Only the $\langle \Delta\hat{N} \rangle = 0$ modes are directly comparable with the exact spectrum of the N-particle system; the others describe transitions to the neighbouring even systems and would be compared with $E_0(N \pm 2)$. Notice that even the comparable ones lie well below the exact excitations – 2.60 and 2.92 against 3.33 and 3.49 – the same downward bias RPA showed in the particle-hole channel.

7.12 Everything against everything

We can now put the four treatments side by side. For pure pairing, where the exact answers are those of Chapter 4:

The table repays reading across.

Hartree-Fock does nothing at all, exactly as Section 6.12 predicted: $E_{\mathrm{HF}} = 2 - g$ is the unperturbed reference energy for every g. The whole correlation energy has to come from the fluctuations.

Particle-hole RPA supplies almost all of it, and tracks the exact curve remarkably closely – -0.61230 against -0.565660 at $g = 1.6$ – crossing it between $g = 1.6$ and $g = 2$. It is slightly under-bound at weak coupling and slightly over-bound at strong coupling. That a particle-hole theory should do this well on a pure pairing interaction is a reminder that the labels “particle-hole” and “pairing” refer to how a calculation is organised, not to a partition of the physics.

BCS improves on Hartree-Fock only above g_c, and then only modestly. A mean field that merely smears the occupation numbers cannot reproduce a genuinely correlated ground state; at $g = 2$ it recovers 0.50 of the required 1.49.

QRPA on top of BCS overshoots badly, and Section 7.11.2 says why: the two-quasiparticle space contains pairing vibrations reaching towards $N \pm 2$, and the standard formula (7.23) counts every root. Restoring particle number by projection is necessary before BCS plus QRPA becomes quantitative for a system

g	E_{exact}	E_{HF}	$E_{\text{HF}}+E_{\text{corr}}^{\text{RPA}}$	E_{BCS}	$E_{\text{BCS}}+E_{\text{corr}}^{\text{QRPA}}$
0.40	1.548001	1.60000	1.51748	1.60000	1.46366
0.60	1.277449	1.40000	1.22390	1.40000	1.08964
0.80	0.973552	1.20000	0.90179	1.20000	0.62540
1.00	0.635548	1.00000	0.55459	1.00000	−0.01157
1.20	0.264504	0.80000	0.18506	0.78522	−0.40860
1.40	−0.137035	0.60000	−0.20453	0.51568	−0.67091
1.60	−0.565660	0.40000	−0.61230	0.20452	−1.04351
2.00	−1.489652	0.00000	−1.47622	−0.50353	−1.94223

Table 7.6 Ground-state energies for the pure pairing model, $N = L = 4$, $\xi = 1$. The Hartree-Fock energy is the unperturbed reference, since the pairing interaction cannot couple it to a $1p$-$1h$ state. The BCS energy departs from it only above $g_c = 1.07$.

this small. In a heavy nucleus, with many levels and a small relative number fluctuation, the same machinery is far better behaved – which is why it remains a standard tool there.

What to take away. Both approximations of this chapter are the *same* approximation applied to different reference states and different elementary operators: linearise the equations of motion, replace the commutator of two elementary excitations by its expectation value in the reference, and diagonalise. Everything that follows – the $\pm\omega$ pairing of the roots, the indefinite metric, the possibility of imaginary frequencies, the correlation-energy formula, the appearance of Goldstone modes whenever the reference breaks a symmetry – follows from that one step, and is independent of which channel one is working in. The quality of the answer, however, is not: it depends entirely on whether the reference has already captured the physics, and the two halves of Table 7.6 show the same interaction treated well and badly by that criterion.

7.13 Summary

The Tamm-Dancoff approximation diagonalises the Hamiltonian in the $1p$-$1h$ space; it is the excited-state version of the CIS truncation of Chapter 5, and its matrix A is the Hamiltonian restricted to that space, measured from the reference. The random-phase approximation adds the de-excitation amplitudes Y, which correlates the ground state, lowers every root, and turns the Hermitian eigenvalue problem into the non-Hermitian one of Eq. (7.16). The blocks A and B that appear there are precisely the blocks of Chapter 6's stability matrix, and the RPA frequencies are real exactly when the mean field is a local minimum: excitation spectrum and stability analysis are one calculation.

Applied to a model with competing pairing and particle-hole interactions, the approximations behave as their construction demands. Particle-hole RPA is stable at every coupling and does surprisingly well on the ground-state energy, while systematically underestimating the excitation energies. BCS finds the pairing instability that the particle-hole channel cannot see, at the price of breaking particle-number conservation, and QRPA built on it carries a spurious Goldstone mode which must be identified and removed – by its overlap with the number-operator amplitudes, not by inspecting the spectrum alone.

The program `rpa.py` performs every calculation quoted here, and the notebook `tdarpa.ipynb` runs it cell by cell. All the matrices are built as literal double commutators, so the same routine serves the particle-hole and the quasiparticle case, and the textbook formulas (7.11) and (7.17) are checked against them rather than assumed.

7.14 Exercises

Exercise 1: the model and its limits

a) Show that $\hat{V}_{\text{pair}}$ and $\hat{V}_{\text{ph}}$ both conserve N_+ and N_- separately, and hence that S_z is a good quantum number. What is the dimension of the $S_z = 0$ sector for $N = L = 4$?
b) Show that at $f = 0$ the seniority is conserved, and that at $f \neq 0$ it is not. Which matrix elements does the particle-hole term supply that the pairing term does not?
c) Verify with `rpa.py` that the $f = 0$ spectrum reproduces Table 4.1, and that the ground state at $f = 0$ lies entirely in the seniority-zero subspace.
d) For $g = 0.5$, $f = 0.05g$ the two lowest excited states are split by only 2×10^{-4}. Explain the near-degeneracy.

Answer to a).

Every term of Eq. (7.2) contains one $\hat{a}^\dagger_{p+}$ and one $\hat{a}_{q+}$, so N_+ is unchanged, and likewise for N_-. In Eq. (7.3) the operator $\hat{a}^\dagger_{p+}\hat{a}^\dagger_{p-}\hat{a}_{q-}\hat{a}_{r+}$ again creates and destroys one particle of each spin. Hence $[\hat{H}, \hat{N}_\pm] = 0$ and $S_z = (N_+ - N_-)/2$ is conserved. For $N = L = 4$ the balanced sector requires two spin-up and two spin-down particles among four levels each, so its dimension is $\binom{4}{2}^2 = 36$, against $\binom{8}{4} = 70$ for the full space.

Answer to b).

The pairing term moves both members of a pair together, so the number of unpaired particles – the seniority – is unchanged, as Exercise 5.13c) showed. The particle-hole term with $q \neq r$ removes a spin-down from level q and a spin-up from a *different* level r, leaving unpaired particles behind at both, and deposits a pair at p. It therefore changes the seniority by two. These are exactly the matrix elements that connect the seniority-zero subspace to the rest of the $S_z = 0$ space, and they are why the full 36-dimensional problem must be solved once $f \neq 0$.

Answer to c).

Running `demo_model()` prints the $f = 0$ energies alongside Table 4.1; they agree to all eight digits shown. Projecting the ground-state eigenvector onto the determinants with broken pairs gives zero to machine precision.

Answer to d).

At $f = 0$ the first excited state of the seniority-zero problem is doubly degenerate by the symmetry of the level scheme. The particle-hole term lifts the degeneracy, but only at order f, and $f = 0.025$ here; the splitting $2.711098 - 2.710867 = 2.3 \times 10^{-4}$ is of order f^2/ξ, consistent with the degeneracy being lifted at second order.

Exercise 2: from the equations of motion to TDA and RPA

a) Derive Eq. (7.8) from Eq. (7.7), stating where $\hat{Q}_\nu|0\rangle = 0$ is used.

b) Show that with the operator (7.9) and a Slater determinant reference, Eq. (7.8) reduces to the Hermitian problem (7.10), and evaluate $A_{mi,nj}$ using Wick's theorem.
c) Show that the Tamm-Dancoff matrix is the Hamiltonian restricted to the $1p$-$1h$ space, Eq. (7.12), and hence that TDA is CIS.
d) With the operator (7.13), derive Eq. (7.16) and identify precisely where the quasiboson approximation enters.
e) Show that if (X,Y) solves Eq. (7.16) with frequency ω, then (Y^*,X^*) solves it with $-\omega$.

Answer to a).

Take the matrix element of Eq. (7.7) with $\langle 0|\delta\hat{Q}\cdot|\rangle$: $\langle 0|\delta\hat{Q}[\hat{H},\hat{Q}_\nu^\dagger]|0\rangle = \omega_\nu\langle 0|\delta\hat{Q}\hat{Q}_\nu^\dagger|0\rangle$. Now use $\hat{Q}_\nu|0\rangle = 0$, and its adjoint $\langle 0|\hat{Q}_\nu^\dagger|\rangle = 0$, to add for free the terms $\langle 0|[\hat{H},\hat{Q}_\nu^\dagger]\delta\hat{Q}|0\rangle$ and $\omega_\nu\langle 0|\hat{Q}_\nu^\dagger\delta\hat{Q}|0\rangle$, which vanish because $\delta\hat{Q}$ is itself a de-excitation operator. Subtracting turns both sides into commutators and gives Eq. (7.8). The symmetrised form is preferable because the double commutator can be evaluated with fewer contractions, and because it is manifestly Hermitian.

Answer to b).

With $\delta\hat{Q} = \hat{a}_i^\dagger\hat{a}_m$ and $\hat{Q}^\dagger = \sum_{nj} X_{nj}\hat{a}_n^\dagger\hat{a}_j$ the right-hand side of Eq. (7.8) is $\sum_{nj} X_{nj}\langle \mathrm{HF}|[\hat{a}_i^\dagger\hat{a}_m,\hat{a}_n^\dagger\hat{a}_j]|\mathrm{HF}\rangle = X_{mi}$ by Eq. (7.14), exactly and without approximation, because the expectation value of $\hat{a}_i^\dagger\hat{a}_j$ in a determinant is δ_{ij} for occupied indices and zero otherwise. The left-hand side is $\sum_{nj} A_{mi,nj}X_{nj}$ with A given by Eq. (7.11); evaluating the double commutator with the generalised Wick theorem of Section 3.12 gives the one-body piece $(\varepsilon_m-\varepsilon_i)\delta_{mn}\delta_{ij}$ from $\hat{H}_0$ and the single surviving contraction $\bar{v}_{mjin}$ from the interaction.

Answer to c).

The $1p$-$1h$ determinants are orthonormal, and the Condon-Slater rules of Section 3.13 give

$$\langle \Phi_i^m|\hat{H}|\Phi_j^n\rangle = E_{\mathrm{HF}}\delta_{mn}\delta_{ij} + (\varepsilon_m-\varepsilon_i)\,\delta_{mn}\delta_{ij} + \bar{v}_{mjin}, \tag{7.34}$$

which is $E_{\mathrm{HF}}\delta + A_{mi,nj}$. Diagonalising A is therefore diagonalising $\hat{H}$ in the space spanned by $|\mathrm{HF}\rangle$ and all singles – CIS – and reading the eigenvalues relative to the reference. The reference itself decouples by Brillouin's theorem, which is why it does not appear in A. The program checks $\max|A-(\boldsymbol{H}_{1p1h}-E_{\mathrm{HF}})| = 3\times 10^{-15}$.

Answer to d).

Now $\delta\hat{Q}$ ranges over both $\hat{a}_i^\dagger\hat{a}_m$ and $\hat{a}_m^\dagger\hat{a}_i$, so Eq. (7.8) becomes a pair of coupled equations for X and Y. The left-hand sides give A and B as in Eqs. (7.11) and (7.17). The right-hand sides require $\langle \mathrm{RPA}|[\hat{a}_i^\dagger\hat{a}_m,\hat{a}_n^\dagger\hat{a}_j]|\mathrm{RPA}\rangle$ in the *correlated* ground state, which we do not have. Replacing it by its value in the Hartree-Fock determinant, Eq. (7.15), is the quasiboson approximation, and it is the only approximation made. It produces the norm matrix $\boldsymbol{\eta} = \mathrm{diag}(\mathbf{1},-\mathbf{1})$; the minus sign, from the second row, is what makes Eq. (7.16) non-Hermitian.

Answer to e).

Take the complex conjugate of Eq. (7.16) and interchange the two blocks of rows: $A^*X^* + B^*Y^* = \omega^*X^*$ and $-BY^* - AX^* = \omega^*Y^*$, that is

$$\begin{pmatrix} A & B \\ -B^* & -A^* \end{pmatrix} \begin{pmatrix} Y^* \\ X^* \end{pmatrix} = -\omega^* \begin{pmatrix} Y^* \\ X^* \end{pmatrix}. \tag{7.35}$$

So the spectrum is symmetric about zero. If ω is real this is the partner $-\omega$; if it is complex the four roots $\pm\omega, \pm\omega^*$ appear together, which is why imaginary frequencies arrive in pairs.

Exercise 3: RPA, stability and Thouless

a) Show that $\hat{M}_{\text{RPA}} = \boldsymbol{\eta}\hat{M}_{\text{stab}}$ with $\boldsymbol{\eta} = \text{diag}(\mathbf{1}, -\mathbf{1})$, and deduce Eq. (7.21).
b) Reproduce Table 7.1 for the Lipkin model and show that the collective root behaves as $\omega = \varepsilon\sqrt{1-\chi^2}$ near the transition.
c) Verify Eq. (7.24) for the pure pairing model by computing A and B analytically in the particle-hole channel.
d) Show that $\hat{Q}_\nu|\text{RPA}\rangle = 0$ is solved by Eq. (7.22), and compare its structure with Thouless' theorem and with the coupled-cluster ansatz.

Answer to a).

Multiplying $\hat{M}_{\text{stab}}$ on the left by $\boldsymbol{\eta}$ flips the sign of its second block row, turning (B^*, A^*) into $(-B^*, -A^*)$, which is $\hat{M}_{\text{RPA}}$. If $\hat{M}_{\text{stab}} > 0$ it has a positive-definite Hermitian square root $\boldsymbol{S}$, and $\boldsymbol{S}^{-1}\hat{M}_{\text{RPA}}\boldsymbol{S} = \boldsymbol{S}^{-1}\boldsymbol{\eta}\boldsymbol{S}^2\boldsymbol{S} = \boldsymbol{S}\boldsymbol{\eta}\boldsymbol{S}$, which is Hermitian. Similar matrices have the same spectrum, so the RPA eigenvalues are real. If $\hat{M}_{\text{stab}}$ has a negative eigenvalue the square root is no longer Hermitian and the argument collapses; the roots may then be imaginary, and for a single negative direction they are.

Answer to b).

Section 6.11 gives $\Delta + A = \varepsilon\boldsymbol{I}$ and $\boldsymbol{B} = V(\boldsymbol{J} - \boldsymbol{I})$, whose largest eigenvalue is $V(N-1) = \varepsilon\chi$. Since A and B commute in this model, the RPA roots are $\omega = \sqrt{A^2 - B^2}$ evaluated eigenvalue by eigenvalue, and the collective one is

$$\omega = \sqrt{\varepsilon^2 - \varepsilon^2\chi^2} = \varepsilon\sqrt{1-\chi^2}. \tag{7.36}$$

This vanishes at $\chi = 1$ and is imaginary beyond, reproducing Table 7.1: at $\chi = 0.9$ it gives $\sqrt{1-0.81} = 0.43589$, exactly the tabulated value. Note also the square root singularity: the mode softens with infinite slope as the transition is approached, which is the mean-field signature of a second-order transition.

Answer to c).

With pure pairing the Hartree-Fock orbitals are the original ones, the occupied levels shifted down by $g/2$, so $\varepsilon_m - \varepsilon_i$ for the lowest particle-hole pair is $\xi + g/2$. The interaction contributes $\bar{v}_{mjin}$ to A and $\bar{v}_{mnij}$ to B; for the pairing force only the terms in which m, n and i, j are spin partners of the same level survive, giving a diagonal $A = \xi + g/2$ and $B = g/2$ in the collective channel. Then $\omega = \sqrt{A^2 - B^2} = \sqrt{(\xi + g/2)^2 - (g/2)^2} = \sqrt{\xi^2 + g\xi}$, which is Eq. (7.24). Since $\xi^2 + g\xi > 0$ for all $g > -\xi$, the particle-hole channel never becomes unstable.

Answer to d).

Write $|\mathrm{RPA}\rangle = \mathscr{N} e^{\hat{S}}|\mathrm{HF}\rangle$ with $\hat{S} = \frac{1}{2}\sum Z_{mi,nj}\hat{a}_m^\dagger \hat{a}_i \hat{a}_n^\dagger \hat{a}_j$. Since the particle-hole operators commute among themselves in the quasiboson approximation, $[\hat{a}_i^\dagger \hat{a}_m, \hat{S}] = \sum_{nj} Z_{mi,nj}\hat{a}_n^\dagger \hat{a}_j$, and imposing $\hat{Q}_\nu e^{\hat{S}}|\mathrm{HF}\rangle = 0$ gives $Y^\nu = \boldsymbol{Z}\boldsymbol{X}^\nu$ in matrix form, hence $\boldsymbol{Z} = \boldsymbol{Y}^*(\boldsymbol{X}^*)^{-1}$. The structure is the same as Thouless' theorem, Eq. (6.34), with $2p$-$2h$ operators in place of the $1p$-$1h$ ones – so the RPA ground state stands to the doubles as a Slater determinant does to the singles. It is also the coupled-cluster doubles ansatz $e^{\hat{T}_2}|\Phi_0\rangle$ of Eq. (5.29), with the amplitudes fixed by the RPA equations rather than determined by their own nonlinear equations. RPA is thus an approximation to coupled-cluster doubles, and the comparison between the two is a standard way of exposing what the quasiboson approximation costs.

Exercise 4: BCS

a) Derive Eq. (7.27), being explicit about the $p = q$ and $p \neq q$ contributions.
b) Show that only the $q = r$ part of $\hat{V}_{\mathrm{ph}}$ has a non-zero expectation value in the BCS state, and hence that the problem depends on g and f only through $g + 2f$.
c) Derive the critical condition (7.28) and solve it for S_c and g_c.
d) Verify that the Bogoliubov quasiparticles (7.29) annihilate the BCS state and obey the fermion anticommutation relations.
e) Compare the BCS ground-state energy with the exact one through the transition and comment on the sharpness of the mean-field transition.

Answer to a).

$|\mathrm{BCS}\rangle$ is a product over levels, so $\langle \hat{N}\rangle = 2\sum_p v_p^2$ and $\langle \hat{H}_0\rangle = 2\sum_p \varepsilon_p v_p^2$ immediately. For the interaction, $\hat{P}_q$ acting on the product picks out the v_q term and leaves level q empty; $\hat{P}_p^\dagger$ in the bra does the same at p. For $p = q$ the two operations are on the same level and the overlap is v_p^2; for $p \neq q$ the surviving overlap has u_p at p and u_q at q, giving $u_p v_p u_q v_q$. Assembling, $\sum_{pq}\langle \hat{P}_p^\dagger \hat{P}_q\rangle = \sum_p v_p^2 + \sum_{p\neq q} u_p v_p u_q v_q$, and multiplying by $-S$ gives Eq. (7.27).

Answer to b).

$\hat{a}_{q-}\hat{a}_{r+}$ with $q \neq r$ leaves an unpaired particle at each of q and r; the resulting state has two broken pairs and is orthogonal to anything $\hat{P}_p^\dagger$ can produce from $|\mathrm{BCS}\rangle$, which contains only paired levels. For $q = r$, $\hat{a}_{q-}\hat{a}_{q+} = \hat{P}_q$ exactly, so the term reduces to $-\frac{f}{2}\sum_{pq}\hat{P}_p^\dagger \hat{P}_q$ and its Hermitian conjugate contributes an equal amount. The total is $-f\sum_{pq}\hat{P}_p^\dagger \hat{P}_q$, which added to $-\frac{g}{2}\sum_{pq}\hat{P}_p^\dagger \hat{P}_q$ gives the single strength $S = g/2 + f$, that is $g \to g + 2f$. The program confirms that $(g, f) = (1.4, 0)$, $(1.2, 0.1)$, $(0.8, 0.3)$ and $(0, 0.7)$ give identical energies and gaps.

Answer to c).

Linearise about $\Delta = 0$. There $v_p^2 = 1$ for the two lowest levels and zero above, so the self-energy shifts them to $\tilde{\varepsilon}_p = \varepsilon_p - S$ for $p \leq 2$ and leaves $\tilde{\varepsilon}_p = \varepsilon_p$ above. With $\varepsilon_p = 0, 1, 2, 3$ this gives $(-S, 1 - S, 2, 3)$, and the chemical potential sits midway between the highest occupied and lowest empty level, $\lambda = (3 - S)/2$. The linearised gap equation $1 = (S/2)\sum_p |\tilde{\varepsilon}_p - \lambda|^{-1}$ becomes

$$1 = \frac{S}{2}\left[\frac{2}{(3+S)/2}\cdot 1 + \frac{2}{(1+S)/2}\cdot 1\right]\cdot\frac{1}{2}\cdot 2 = 2S\left[\frac{1}{3+S}+\frac{1}{1+S}\right], \tag{7.37}$$

which is Eq. (7.28) with $\xi = 1$. Solving numerically gives $S_c = 0.53519$ and $g_c = 2S_c = 1.07037$ at $f = 0$, matching the bisection in the program.

Answer to d).

$\hat{\alpha}_{p+}|\mathrm{BCS}\rangle = (u_p\hat{a}_{p+} - v_p\hat{a}^{\dagger}_{p-})\prod_q(u_q + v_q\hat{P}^{\dagger}_q)|0\rangle$. Only the factor at level p matters; acting on $(u_p + v_p\hat{a}^{\dagger}_{p+}\hat{a}^{\dagger}_{p-})$ gives $u_pv_p\hat{a}^{\dagger}_{p-} - v_pu_p\hat{a}^{\dagger}_{p-} = 0$. For the anticommutators,

$$\left\{\hat{\alpha}_{p+},\hat{\alpha}^{\dagger}_{p+}\right\} = u_p^2\left\{\hat{a}_{p+},\hat{a}^{\dagger}_{p+}\right\} + v_p^2\left\{\hat{a}^{\dagger}_{p-},\hat{a}_{p-}\right\} = u_p^2 + v_p^2 = 1, \tag{7.38}$$

the cross terms vanishing because they pair two creation or two annihilation operators. The normalisation $u_p^2 + v_p^2 = 1$ is exactly what makes the transformation canonical. The program checks $\max\|\hat{\alpha}_a|\mathrm{BCS}\rangle\| = 3\times 10^{-17}$.

Answer to e).

From Table 7.6, $E_{\mathrm{BCS}} = E_{\mathrm{HF}}$ exactly below $g_c = 1.07$ and departs from it above. The exact energy, by contrast, runs smoothly through g_c with no feature of any kind: at $g = 1.0$, 1.2 and 1.4 it is 0.6355, 0.2645 and -0.1370, a perfectly regular sequence. The mean-field transition is therefore an artefact of restricting the trial state to a product form. What is physical is the crossover – the gradual build-up of pair correlations – and what is spurious is its sharpness, and with it the broken symmetry. For a large system the crossover narrows and the mean-field picture becomes accurate; for four particles it does not.

Exercise 5: QRPA and the spurious mode

a) Show that if $[\hat{H},\hat{F}] = 0$ while the reference is not an eigenstate of $\hat{F}$, then the amplitudes (7.32) built from $\hat{F}$ give an exact zero mode of the RPA equations.
b) Explain why cranking with $\hat{H}' = \hat{H} - \lambda\hat{N}$ places the spurious root at zero, and why the root may come out as a small *imaginary* pair rather than a small real one.
c) Reproduce Table 7.4 and confirm the unit overlap. What would you conclude if the overlap were small?
d) Classify the QRPA phonons at $g = 2$ by $\langle\Delta\hat{N}\rangle$, and explain why the shift is not quantised at ± 2.
e) Explain why $E_{\mathrm{corr}}^{\mathrm{QRPA}}$ overshoots so badly in Table 7.6, and suggest what would fix it.

Answer to a).

Because $[\hat{H},\hat{F}] = 0$, the operator $\hat{F}$ generates a family of states $e^{i\theta\hat{F}}|\mathrm{ref}\rangle$ all with the same energy. If the reference is not an eigenstate of $\hat{F}$ these are genuinely different states, so the energy surface has a flat direction, and the second derivative – which is what $\hat{M}_{\mathrm{stab}}$ is – has a zero eigenvalue along it. Algebraically, substituting the amplitudes (7.32) into the left-hand side of Eq. (7.8) gives $\langle\mathrm{ref}|[\delta\hat{Q},[\hat{H},\hat{F}]]|\mathrm{ref}\rangle = 0$ identically, so $\omega = 0$.

Answer to b).

The BCS state extremises $\hat{H}-\lambda\hat{N}$, not $\hat{H}$, so it is $\hat{H}'$ whose energy surface is flat at the BCS point and whose second derivative therefore has the exact zero eigenvalue. Building the QRPA from $\hat{H}$ instead displaces the spurious root to a small non-zero value and contaminates the physical ones. The root may emerge imaginary because the RPA metric $\boldsymbol{\eta}$ is singular in the spurious direction: there $\sum(|X|^2-|Y|^2)=0$, the mode cannot be normalised, and rounding errors of either sign are amplified into a small $\pm i\varepsilon$ pair. This is a property of the numerics, not of the physics.

Answer to c).

`demo_qrpa()` prints the table. The overlap between the near-zero eigenvector and the amplitudes (7.32) is 1.000000 throughout, so the root is unambiguously the number mode. Were the overlap small, the conclusion would be the opposite and far more interesting: a genuine soft mode, signalling that the BCS solution is about to become unstable against some other deformation, exactly as the Lipkin collective root does at $\chi=1$. The two cases look identical in the spectrum alone, which is why the overlap test matters.

Answer to d).

Running `classify_modes` gives $\langle\Delta\hat{N}\rangle=0$ to machine precision for the roots at 2.6015 and 2.9212, and ± 0.357 for the pair at 3.7879. The shift is not ± 2 because $|\mathrm{BCS}\rangle$ is not an eigenstate of $\hat{N}$: it is a wave packet in particle number, and $\langle\hat{N}\rangle$ in the phonon state measures the *mean* of a shifted packet, not a quantum number. What is exact and meaningful is the vanishing of the shift for the modes that stay within the N-particle system, which is enforced by symmetry. Number projection would restore the quantisation.

Answer to e).

Two effects compound. First, Eq. (7.23) sums over *all* roots, and the two-quasiparticle space contains pairing vibrations that describe the neighbouring $N\pm 2$ systems; their zero-point energy has no business in the correlation energy of the N-particle system. Second, the quasiboson approximation is worse here than in the particle-hole channel, because the BCS occupations are fractional and the commutator of two two-quasiparticle operators is correspondingly further from a c-number. The cure for the first is particle-number projection before or after variation; the cure for the second is a renormalised QRPA that keeps the leading correction to Eq. (7.15). Both are standard, and both are essential for a system with only four particles; in a heavy nucleus the relative number fluctuation $\Delta N/N$ is small and plain QRPA is adequate.

Exercise 6: numerical explorations

a) Using `rpa.py`, plot the exact, TDA and RPA lowest excitation energies against g for $f=0.05g$ and for $f=0.5g$, and comment on where the approximations fail.
b) Repeat Table 7.6 for $L=6$ and $N=6$. Does RPA improve or deteriorate as the system grows?
c) Compute the RPA correlation energy from Eq. (7.23) and, separately, as the difference between the exact and Hartree-Fock energies. Plot the ratio against g.
d) Find, by sweeping f at fixed g, the coupling at which the particle-hole stability matrix of this model first acquires a negative eigenvalue – or show that it never does within the range you can reach.
e) Restrict the QRPA correlation energy to the $\langle\Delta\hat{N}\rangle=0$ modes and compare with Table 7.6. How much of the overshoot does this remove?

Answer.

These are open-ended; a few pointers. In b) the RPA correlation energy is an extensive quantity while the quasiboson error per phonon is not, so RPA generally improves in relative terms as the level space grows – the opposite of truncated configuration interaction, whose size-consistency error grows (Section 5.9). In c) the ratio starts near one at small g and drifts past it as RPA begins to overshoot; the crossing point is a useful single number for characterising where the method turns over. In d) the answer for this model is that it never does within any reasonable range, since Eq. (7.24) is real for all $g > -\xi$ and the particle-hole term only stiffens the matrix; the instability is in the pairing channel, which requires the BCS reference to see. In e) removing the pairing-vibration roots takes out a large part of the overshoot but not all of it, since the quasiboson approximation remains; the residual is a fair estimate of what number projection alone cannot fix.

Chapter 8
Density functional theory

Every method so far has been a method for the wave function. Full configuration interaction stores its coefficients, Hartree-Fock optimises the orbitals it is built from, and the random-phase approximation describes the fluctuations around that. All of them run into the same wall: the object being computed lives in a space whose dimension grows exponentially with the number of particles, Section 5.5.

Density functional theory takes a different route. It replaces the wave function of N electrons – a function of $3N$ coordinates – by the electronic density, a function of three. This is not an approximation but, remarkably, an exact reformulation: Hohenberg and Kohn proved that the ground-state density determines the external potential, hence the Hamiltonian, hence everything. The catch, which we shall be careful to state precisely, is that the theorem asserts the existence of a universal functional without telling us its form.

The chapter has three parts. We first assemble the tools – the Born-Oppenheimer Hamiltonian and the reduced density matrices of Sections 1.31 and 2.9, now in coordinate space. We then prove the Hohenberg-Kohn theorems and examine what they do and do not deliver. Finally we build the Kohn-Sham scheme, which recovers most of what the mean-field methods of Chapter 6 do well while replacing the expensive part by a local functional, and we construct that functional explicitly from the electron gas of Section 6.13.

Density functional theory has been an extraordinary practical success. Most crystalline materials and a very large fraction of molecular structures have been described with it; Walter Kohn shared the 1998 Nobel Prize in Chemistry for it; and it is now standard in quantum chemistry, materials science, condensed-matter physics and, increasingly, low-energy nuclear physics. It is also, as we shall see, an approximation whose errors are of a quite different character from those of the wave-function methods, and understanding that difference is the point of putting it here.

8.1 The electronic Hamiltonian

Any piece of matter – a crystal, a molecule, a living cell – is a collection of nuclei and electrons held together by the Coulomb interaction, and is governed by

$$i\hbar\frac{\partial}{\partial t}\Phi(\boldsymbol{r};t) = \left(-\sum_i^N \frac{\hbar^2}{2m_i}\frac{\partial^2}{\partial \boldsymbol{r}_i^2} + \sum_{i<j}^N \frac{e^2 Z_i Z_j}{|\boldsymbol{r}_i - \boldsymbol{r}_j|}\right)\Phi(\boldsymbol{r};t). \tag{8.1}$$

Nothing else is needed in principle, and nothing is soluble in practice.

The first simplification is universal. Even the lightest nucleus is nearly two thousand times heavier than an electron, so on the time scale of electronic motion the nuclei may be treated as fixed. This is the *Born-Oppenheimer approximation*, and it turns Eq. (8.1) into a time-independent problem for the electrons alone, moving in the static field of the nuclei,

$$\left[\sum_i^N\left(-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial \boldsymbol{r}_i^2} + v_{\text{ext}}(\boldsymbol{r}_i)\right) + \sum_{i<j}^N v(r_{ij})\right]\Psi(\boldsymbol{r}) = E_0\Psi(\boldsymbol{r}), \tag{8.2}$$

with $r_{ij} = |\boldsymbol{r}_i - \boldsymbol{r}_j|$ and the external potential supplied by the nuclei,

$$v_{\text{ext}}(\boldsymbol{r}_i) = -\sum_j \frac{e^2 Z_j}{|\boldsymbol{r}_i - \boldsymbol{R}_j|}. \tag{8.3}$$

We have written a generic two-body interaction $v(r_{ij})$ rather than the bare Coulomb repulsion, because everything below applies unchanged to any interaction depending only on the relative distance – nuclear forces included, where v_{ext} is absent and the system is self-bound, as Section 6.1 noted.

Equation (8.2) is the Hamiltonian of Eq. (4.1) written in coordinate space. Only one thing about it will matter in what follows: the kinetic energy and the interaction are the *same for every system*, and the only thing that distinguishes one material from another is v_{ext}.

8.2 Reduced density matrices

The central object of this chapter is the electronic density. It is the diagonal of the one-body reduced density matrix, which we met in Section 1.31 as a matrix in a single-particle basis and now write in coordinate space.

8.2.1 The one-body density

Define the density operator

$$\hat{n}(\boldsymbol{r}) = \sum_{i=1}^{N} \delta(\boldsymbol{r} - \boldsymbol{r}_i), \qquad n(\boldsymbol{r}) = \frac{\langle \Psi | \hat{n}(\boldsymbol{r}) | \Psi \rangle}{\langle \Psi | \Psi \rangle}, \tag{8.4}$$

for any normalisable state Ψ. Equivalently, integrating out all coordinates but one,

$$n(\boldsymbol{r}) = N \int d^3\boldsymbol{r}_2 \cdots d^3\boldsymbol{r}_N \, |\Psi(\boldsymbol{r}, \boldsymbol{r}_2, \ldots, \boldsymbol{r}_N)|^2. \tag{8.5}$$

For a single Slater determinant built from orbitals ψ_i this collapses to

$$n(\boldsymbol{r}) = \sum_{i=1}^{N} |\psi_i(\boldsymbol{r})|^2, \tag{8.6}$$

a result already used in Section 6.5.

In second quantization the operator reads

$$\hat{n}(\boldsymbol{r}) = \sum_{\alpha\beta} \psi_\alpha^*(\boldsymbol{r}) \psi_\beta(\boldsymbol{r}) \, \hat{a}_\alpha^\dagger \hat{a}_\beta, \tag{8.7}$$

so that its expectation value in *any* state – a full configuration-interaction ground state $|\Psi_0\rangle = \sum_i C_{0i} |\Phi_i\rangle$ included – is

$$n(\boldsymbol{r}) = \sum_{ij} C_{0i}^* C_{0j} \sum_{\alpha\beta} \psi_\alpha^*(\boldsymbol{r}) \psi_\beta(\boldsymbol{r}) \langle \Phi_i | \hat{a}_\alpha^\dagger \hat{a}_\beta | \Phi_j \rangle. \tag{8.8}$$

Integrating over $\boldsymbol{r}$ and using orthonormality gives $\int n(\boldsymbol{r}) d^3\boldsymbol{r} = N$, as it must.

The more general object, needed below, is the one-body reduced density matrix off the diagonal,

$$\rho^{(1)}(x'; x) = N \int dx_2 \cdots dx_N \, \Psi^*(x', x_2, \ldots, x_N) \, \Psi(x, x_2, \ldots, x_N), \tag{8.9}$$

with x the combined space and spin coordinate. It is Hermitian, its diagonal is the density, $n(x) = \rho^{(1)}(x;x)$, and its eigenvalues are the occupation numbers of Section 1.31: at most one for fermions, and exactly N ones and the rest zeros when – and only when – Ψ is a single Slater determinant, in which case $\rho^{(1)}$ is the projector onto the occupied subspace.

8.2.2 The two-body density

Two-body operators need the pair density,

$$\rho^{(2)}(x_1,x_2;y_1,y_2) = \frac{N(N-1)}{2} \int dx_3 \cdots dx_N \, \Psi^*(y_1,y_2,x_3,\ldots,x_N)\, \Psi(x_1,x_2,x_3,\ldots,x_N), \tag{8.10}$$

the prefactor counting the $N(N-1)/2$ distinct pairs. Its diagonal $\rho^{(2)}(x_1,x_2)$ is the joint probability of finding one particle at x_1 and another at x_2, and it contains everything needed for the expectation value of any two-body operator – which is to say, everything needed for the energy. In second quantization,

$$\hat{\rho}^{(2)}(\boldsymbol{r}_1,\boldsymbol{r}_2) = \sum_{\alpha\beta\gamma\delta} \rho_{\alpha\gamma}(\boldsymbol{r}_1)\rho_{\beta\delta}(\boldsymbol{r}_2)\, \hat{a}^\dagger_\alpha \hat{a}^\dagger_\beta \hat{a}_\delta \hat{a}_\gamma, \qquad \rho_{\alpha\beta}(\boldsymbol{r}) = \psi^*_\alpha(\boldsymbol{r})\psi_\beta(\boldsymbol{r}). \tag{8.11}$$

8.2.3 Antisymmetry, and when the pair density factorises

Fermionic wave functions change sign under the exchange of any two particles,

$$\Psi(\ldots,x_i,\ldots,x_j,\ldots) = -\Psi(\ldots,x_j,\ldots,x_i,\ldots), \tag{8.12}$$

and this has a direct consequence for the pair density: it must vanish when $x_1 = x_2$, since two identical fermions cannot be at the same point in the same spin state. This is the Pauli principle expressed as a statement about $\rho^{(2)}$, and any approximate treatment that violates it is in trouble.

For a single Slater determinant the pair density is completely determined by the one-body density matrix. Applying Wick's theorem, Section 3.10, to $\langle \Phi_0 | \hat{a}^\dagger \hat{a}^\dagger \hat{a} \hat{a} | \Phi_0 \rangle$ leaves exactly two contractions, and

$$\boxed{\rho^{(2)}(x_1,x_2;y_1,y_2) = \rho^{(1)}(x_1;y_1)\rho^{(1)}(x_2;y_2) - \rho^{(1)}(x_1;y_2)\rho^{(1)}(x_2;y_1).} \tag{8.13}$$

The first term is the naive product – what one would write for independent particles – and the second is the exchange term that enforces antisymmetry. On the diagonal,

$$\rho^{(2)}(x_1,x_2) = n(x_1)n(x_2) - \left|\rho^{(1)}(x_1;x_2)\right|^2, \tag{8.14}$$

and setting $x_1 = x_2$ gives $n(x_1)^2 - n(x_1)^2 = 0$ exactly, as required.

Equation (8.13) is worth dwelling on, because it marks precisely the boundary of mean-field theory. For a Hartree product – a plain product of orbitals with no antisymmetry – one would have $\rho^{(2)} = n(x_1)n(x_2)$, but such a state is not a legitimate fermionic wave function. The closest legitimate object is a single determinant, for which Eq. (8.13) holds *exactly*, with the exchange term as the only two-particle correlation. Beyond that – for any genuinely correlated state – the pair density cannot be reduced to one-body quantities at all, and the difference is what we have been calling correlation throughout this book.

Where this leaves us. The energy of Eq. (8.2) needs only $\rho^{(1)}$ and $\rho^{(2)}$; the full wave function is superfluous. That observation is the origin of reduced-density-matrix methods, which try to determine $\rho^{(2)}$ directly. The difficulty is the *N-representability problem*: not every Hermitian, antisymmetric, correctly normalised $\rho^{(2)}$ comes from an actual N-fermion wave function, and characterising those that do is extremely hard. Density functional theory sidesteps this by going all the way down to $n(\boldsymbol{r})$, at the price of an unknown functional. The two difficulties – an unknown functional here, an unknown representability condition there – are two faces of the same problem, and neither has been solved.

8.3 The Hohenberg-Kohn theorems

We now come to the two results on which everything rests. Consider the family of Hamiltonians

$$\hat{H} = \hat{T} + \hat{V}_{\text{ext}} + \hat{V}, \qquad \hat{V}_{\text{ext}} \in \mathscr{V}_{\text{ext}}, \tag{8.15}$$

in which $\hat{T}$ and $\hat{V}$ are fixed once and for all – they are properties of electrons, not of materials – and only the external potential varies. Assume for now that the ground state is non-degenerate.

Theorem 8.1 (Hohenberg-Kohn I). *The external potential $v_{\text{ext}}(\boldsymbol{r})$ is determined, up to an additive constant, by the ground-state density $n_0(\boldsymbol{r})$.*

Theorem 8.2 (Hohenberg-Kohn II). *There exists a universal functional $F[n]$, the same for every external potential, such that*

$$E[n] = F[n] + \int d^3\boldsymbol{r}\, v_{\text{ext}}(\boldsymbol{r}) n(\boldsymbol{r}), \qquad F[n] = T[n] + V_{\text{int}}[n], \tag{8.16}$$

is minimised, over densities integrating to N, by the true ground-state density, and its minimum value is the ground-state energy E_0.

Together these say something startling. The density – three variables – determines the potential, hence the Hamiltonian, hence the full many-body wave function and every ground-state observable. A quantity that is measured routinely by X-ray diffraction contains, in principle, all the information in the $3N$-dimensional Ψ.

8.3.1 Proof of the first theorem

The proof is a two-line *reductio ad absurdum* and is worth giving in full, because its simplicity is itself informative.

Suppose two external potentials $v^{(1)}$ and $v^{(2)}$, differing by more than a constant, produced the same ground-state density $n(\boldsymbol{r})$. They have different Hamiltonians $\hat{H}^{(1)}, \hat{H}^{(2)}$ and, by the unique continuation theorem, different ground states $\Psi^{(1)} \neq \Psi^{(2)}$ with energies $E^{(1)}, E^{(2)}$.

Use $\Psi^{(2)}$ as a trial function for $\hat{H}^{(1)}$. The variational principle of Section 2.7 is strict for a non-degenerate ground state, so

$$E^{(1)} < \langle \Psi^{(2)} | \hat{H}^{(1)} | \Psi^{(2)} \rangle = \langle \Psi^{(2)} | \hat{H}^{(2)} | \Psi^{(2)} \rangle + \int \left[v^{(1)} - v^{(2)} \right] n(\boldsymbol{r}) d^3\boldsymbol{r} = E^{(2)} + \int \left[v^{(1)} - v^{(2)} \right] n. \tag{8.17}$$

The two Hamiltonians differ only in the external potential, and the density is by hypothesis the same, which is what allows the middle step. Interchanging the labels,

$$E^{(2)} < E^{(1)} + \int \left[v^{(2)} - v^{(1)} \right] n. \tag{8.18}$$

Adding Eqs. (8.17) and (8.18), the integrals cancel and we are left with

$$E^{(1)} + E^{(2)} < E^{(1)} + E^{(2)}, \tag{8.19}$$

which is absurd. Hence no two potentials differing by more than a constant can share a ground-state density.

The second theorem follows immediately. Since n_0 fixes v_{ext}, it fixes $\hat{H}$ and therefore $\Psi[n]$, so

$$F_{\text{HK}}[n] \equiv \langle \Psi[n] | \hat{T} + \hat{V} | \Psi[n] \rangle \tag{8.20}$$

is a well-defined functional of the density alone, and it is universal because $\hat{T}$ and $\hat{V}$ contain no reference to v_{ext}. The Rayleigh-Ritz principle applied within this family gives $E_0 = \min_n E[n]$.

8.3.2 What the theorems do not give

Three caveats deserve stating plainly, because they are the difference between a beautiful theorem and a working method.

They are existence theorems.

Nothing in the proof constructs $F[n]$, and nothing suggests it is simple. All the many-body difficulty of the preceding chapters is still there; it has been moved inside $F[n]$. This is why the practical history of density functional theory is a history of *approximate functionals*.

v-representability.

The argument assumes that the density under consideration is the ground-state density of *some* potential in the family. Not every well-behaved, positive, correctly normalised $n(\boldsymbol{r})$ is, and characterising those that are is unsolved. Levy and Lieb removed the difficulty by redefining the functional as a constrained search over wave functions,

$$F_{\text{LL}}[n] = \min_{\Psi \to n} \langle \Psi | \hat{T} + \hat{V} | \Psi \rangle, \tag{8.21}$$

the minimum running over all antisymmetric Ψ that yield the density n. This is defined for any N-representable density, agrees with F_{HK} where the latter exists, and extends the theory to degenerate ground states. It is also, of course, no more computable.

Ground states only.

The functional delivers E_0 and n_0 and nothing else. Excitation energies – the subject of Chapter 7 – are not obtained from the ground-state functional, and reaching them requires a separate construction such as time-dependent density functional theory. The Kohn-Sham eigenvalues introduced below are *not* excitation energies, a point we return to in Section 8.5.1.

8.3.3 The variational equation

Given $F[n]$, the ground state follows from minimising $E[n]$ subject to $\int n = N$. Introducing a Lagrange multiplier exactly as in Section 6.2,

$$\frac{\delta}{\delta n(\boldsymbol{r})} \left\{ E[n] - \mu \left(\int n \, d^3\boldsymbol{r} - N \right) \right\} = 0 \qquad \Longrightarrow \qquad \left. \frac{\delta E[n]}{\delta n(\boldsymbol{r})} \right|_{n=n_0} = \mu, \tag{8.22}$$

so that the functional derivative of the energy is constant throughout the system. That constant is the chemical potential, the Fermi level. The pattern is by now familiar: a constrained variation produces a stationarity condition, and the multiplier enforcing normalisation is the energy conjugate to the particle number.

8.4 Building a functional from the electron gas

The theorems tell us $F[n]$ exists. To make progress we need an approximation, and the oldest and still the most instructive strategy is to take a system we *can* solve exactly and use it locally.

That system is the homogeneous electron gas of Section 6.13. There we computed, for a uniform density n, both the kinetic energy per unit volume,

$$t_0[n] = \frac{3\hbar^2(3\pi^2)^{2/3}}{10m}\, n^{5/3}, \tag{8.23}$$

which is Eq. (6.91) divided by the volume, and the exchange energy per particle. If the density varies slowly enough, we may pretend that each small region behaves like a uniform gas at the local density and integrate,

$$\mathscr{T}_0[n] = \frac{3\hbar^2(3\pi^2)^{2/3}}{10m} \int d^3\boldsymbol{r}\, (n(\boldsymbol{r}))^{5/3}. \tag{8.24}$$

This is the *Thomas-Fermi* kinetic functional, and with the classical electrostatic (Hartree) energy

$$\mathscr{U}_{\mathrm{H}}[n] = \frac{e^2}{2} \iint d^3\boldsymbol{r} d^3\boldsymbol{r}'\, \frac{n(\boldsymbol{r})n(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|}, \tag{8.25}$$

it gives a complete, if crude, theory. Everything not accounted for is collected into a single remainder, the *exchange-correlation energy* $E_{\mathrm{xc}}[n]$, and the Hohenberg-Kohn functional is written

$$\mathscr{E}_{\mathrm{HK}}[n] = \mathscr{T}_0[n] + \int d^3\boldsymbol{r}\, v_{\mathrm{ext}}(\boldsymbol{r}) n(\boldsymbol{r}) + \mathscr{U}_{\mathrm{H}}[n] + E_{\mathrm{xc}}[n]. \tag{8.26}$$

The decomposition is a definition, not an approximation: whatever is missing is in E_{xc} by construction. It is useful only if that remainder is small, which is the whole gamble of the subject.

Why Thomas-Fermi is not enough.

Equation (8.24) has no orbitals in it, and minimising Eq. (8.26) directly over n gives an algebraic equation. It is enormously cheaper than anything in the preceding chapters. It also fails, and the failure is instructive rather than marginal. The program `dft.py` compares the exact kinetic energy of N non-interacting fermions in a one-dimensional trap with the local estimate:

N	T (exact)	T (Thomas-Fermi)	ratio
1	0.250000	0.302300	1.2092
2	1.000000	1.074844	1.0748
3	2.250000	2.340025	1.0400
4	4.000000	4.101805	1.0255
6	9.000000	9.120038	1.0133
8	16.000000	16.134286	1.0084

The error falls steadily as the particle number grows, which is exactly what the construction promises – the density becomes smoother on the scale of the local Fermi wavelength – but a one per cent error in the *largest* term of the energy is fatal when chemical accuracy is wanted. Worse, the Thomas-Fermi density has no shell structure at all: it is smooth and featureless, with none of the nodal structure that orbitals supply, and Teller proved that within the model no molecule is bound. Something has to be done about the kinetic energy specifically.

8.5 The Kohn-Sham construction

Kohn and Sham's idea was to keep orbitals for the one term that cannot be approximated locally, and only that term.

Introduce an auxiliary system of N *non-interacting* fermions moving in some effective potential $v_{\mathrm{KS}}(\boldsymbol{r})$, chosen so that its ground-state density equals the density of the real, interacting system. Its kinetic energy,

$$T_s[n] = -\frac{\hbar^2}{2m}\sum_{i=1}^{N}\int d^3\boldsymbol{r}\,\psi_i^*(\boldsymbol{r})\nabla^2\psi_i(\boldsymbol{r}), \tag{8.27}$$

is computed exactly from orbitals rather than approximated from the density, and it captures the shell structure that Thomas-Fermi threw away. It is not the true kinetic energy $T[n]$ of the interacting system, but the difference is small, and we simply redefine the remainder to absorb it,

$$E_{\mathrm{xc}}[n] \equiv (T[n] - T_s[n]) + (V_{\mathrm{int}}[n] - \mathscr{U}_{\mathrm{H}}[n])\,. \tag{8.28}$$

The total energy is then

$$E[n] = T_s[n] + \int d^3\boldsymbol{r}\,v_{\mathrm{ext}}n + \mathscr{U}_{\mathrm{H}}[n] + E_{\mathrm{xc}}[n]. \tag{8.29}$$

Varying Eq. (8.29) with respect to the orbitals, subject to orthonormality, and following exactly the steps of Section 6.4, gives the *Kohn-Sham equations*

$$\boxed{\left(-\frac{\hbar^2}{2m}\nabla^2 + v_{\mathrm{KS}}(\boldsymbol{r})\right)\psi_i(\boldsymbol{r}) = \varepsilon_i\psi_i(\boldsymbol{r}),} \tag{8.30}$$

with the Kohn-Sham potential

$$v_{\mathrm{KS}}(\boldsymbol{r}) = v_{\mathrm{ext}}(\boldsymbol{r}) + \int d^3\boldsymbol{r}'\,\frac{e^2 n(\boldsymbol{r}')}{|\boldsymbol{r}-\boldsymbol{r}'|} + v_{\mathrm{xc}}(\boldsymbol{r}), \qquad v_{\mathrm{xc}}(\boldsymbol{r}) = \frac{\delta E_{\mathrm{xc}}[n]}{\delta n(\boldsymbol{r})}, \tag{8.31}$$

and the density reconstructed from the occupied orbitals,

$$n(\boldsymbol{r}) = \sum_{i=1}^{N}|\psi_i(\boldsymbol{r})|^2. \tag{8.32}$$

Since v_{KS} depends on n and n on the orbitals, the equations are nonlinear and must be iterated to self-consistency – the same self-consistent field loop as Section 6.5, and for the same reason.

The total energy.

The sum of the Kohn-Sham eigenvalues is *not* the total energy, exactly as in Hartree-Fock, Exercise 6.15b). Since $\sum_i \varepsilon_i = T_s + \int v_{\mathrm{KS}} n$,

$$E = \sum_{i=1}^{N}\varepsilon_i - \mathscr{U}_{\mathrm{H}}[n] + E_{\mathrm{xc}}[n] - \int d^3\boldsymbol{r}\,v_{\mathrm{xc}}(\boldsymbol{r})n(\boldsymbol{r}), \tag{8.33}$$

the last three terms correcting the double counting of the Hartree and exchange-correlation contributions.

8.5.1 Kohn-Sham against Hartree-Fock

It is worth setting the two schemes side by side, since they look almost identical and differ in one decisive respect. The Hartree-Fock equation of Chapter 6 reads, in coordinate space,

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + v_{\mathrm{ext}}(\boldsymbol{r}) + \int d^3\boldsymbol{r}'\, w(\boldsymbol{r},\boldsymbol{r}')n(\boldsymbol{r}')\right)\phi_k(\boldsymbol{r}) \\ \underbrace{-\sum_{l=1}^{N}\int d^3\boldsymbol{r}'\,\phi_l^*(\boldsymbol{r}')w(\boldsymbol{r},\boldsymbol{r}')\phi_k(\boldsymbol{r}')\phi_l(\boldsymbol{r})}_{\text{exchange}} = \varepsilon_k\phi_k(\boldsymbol{r}), \tag{8.34}$$

whereas the Kohn-Sham equation is

$$\left(-\frac{\hbar^2}{2m}\nabla^2 + v_{\mathrm{ext}}(\boldsymbol{r}) + \int d^3\boldsymbol{r}'\, w(\boldsymbol{r},\boldsymbol{r}')n(\boldsymbol{r}') + \underbrace{v_{\mathrm{xc}}([n];\boldsymbol{r})}_{\text{exchange } \textit{and} \text{ correlation}}\right)\phi_k(\boldsymbol{r}) = \varepsilon_k\phi_k(\boldsymbol{r}). \tag{8.35}$$

The difference is the last term. Hartree-Fock's exchange is *non-local* – an integral operator, different for each orbital – and it is exact. Kohn-Sham's v_{xc} is a *local multiplicative potential*, the same for every orbital, and it is approximate; but it also contains correlation, which Hartree-Fock does not have at all. The trade is non-locality and exactness against locality, cheapness and a bit of correlation. Both scale formally as the cube of the number of basis functions, but the Kohn-Sham prefactor is far smaller because no four-index integrals are needed for v_{xc}.

What the Kohn-Sham orbitals are. They are the orbitals of a fictitious non-interacting system chosen to reproduce the right density. They are not approximations to the true one-electron wave functions, and the eigenvalues ε_i are not ionisation energies – Koopmans' theorem, Exercise 6.15c), applies to Hartree-Fock, not here. The one exception is the highest occupied level, which for the exact functional equals minus the exact ionisation energy. The systematic underestimation of band gaps by Kohn-Sham eigenvalue differences is not a numerical deficiency but a consequence of asking the orbitals a question they were not built to answer.

8.6 The local density approximation

Everything now rests on $E_{\mathrm{xc}}[n]$. The oldest approximation, and still the reference point for all the others, is the *local density approximation*: assume that the exchange-correlation energy density at a point depends only on the density at that point, and take that dependence from the uniform electron gas,

$$E_{\mathrm{xc}}^{\mathrm{LDA}}[n] = \int d^3\boldsymbol{r}\, n(\boldsymbol{r})\,\varepsilon_{\mathrm{xc}}(n(\boldsymbol{r}))\,, \tag{8.36}$$

where $\varepsilon_{\mathrm{xc}}(n)$ is the exchange-correlation energy per particle of a uniform gas of density n. The corresponding potential follows by differentiation,

$$v_{\mathrm{xc}}(\boldsymbol{r}) = \frac{d}{dn}\left[n\,\varepsilon_{\mathrm{xc}}(n)\right]_{n=n(\boldsymbol{r})}. \tag{8.37}$$

The exchange part is available in closed form. It is the exchange energy of Section 6.13.4 expressed per particle rather than per state,

$$\varepsilon_x(n) = -\frac{3}{4}\left(\frac{3}{\pi}\right)^{1/3} n^{1/3}, \qquad v_x(n) = \frac{4}{3}\varepsilon_x(n), \tag{8.38}$$

in Hartree atomic units. That this must reproduce the electron-gas result of Chapter 6 is a consistency check the program performs:

r_s	ε_x [Ha]	ε_x [Ry]	$-0.916/r_s$ [Ry]
1.0000	−0.458165	−0.916331	−0.916000
2.0000	−0.229083	−0.458165	−0.458000
4.0000	−0.114541	−0.229083	−0.229000
4.8253	−0.094951	−0.189901	−0.189833
6.0000	−0.076361	−0.152722	−0.152667

The coefficient comes out as 0.916331 against the 0.9163 of Eq. (6.109). This agreement is a tautology – the local approximation is *defined* to be exact for the uniform gas – and the interesting question is not whether it holds but why the approximation remains useful when the density is far from uniform.

The correlation part $\varepsilon_c(n)$ has no closed form. It is obtained from quantum Monte Carlo calculations of the uniform gas, the diffusion Monte Carlo results of Ceperley and Alder being the standard source, and then fitted to a convenient analytic form. This is worth noticing: the input to the most widely used approximation in computational materials science is a wave-function calculation of the kind developed in the preceding chapters.

8.6.1 The exchange hole

The reason a local approximation works at all becomes clearer if one looks at what exchange actually does. Divide the pair density (8.14) by the product of densities to define the pair correlation function g; for the uniform gas at the Hartree-Fock level it depends only on $y = k_F s$ and is known exactly,

$$g(y) = 1 - \frac{9}{2}\left[\frac{\sin y - y\cos y}{y^3}\right]^2. \tag{8.39}$$

The program tabulates it:

$k_F s$	$g(s)$	
0.00	0.500000	same spins never coincide
0.50	0.524471	
1.00	0.591838	
2.00	0.786732	
4.00	0.996208	
8.00	0.999920	the hole has healed

At coincidence $g(0) = \frac{1}{2}$: electrons of the same spin are completely excluded, those of opposite spin are uncorrelated at this level. Each electron therefore moves inside a hole in the density of its neighbours, and that hole obeys an exact sum rule – it contains precisely one missing electron,

$$n\int d^3\boldsymbol{s}\,[g(s)-1] = \frac{4}{3\pi}\int_0^\infty dy\, y^2\,[g(y)-1] = -1, \tag{8.40}$$

which the program verifies numerically to four digits.

This is the deeper reason the local approximation succeeds. The exchange-correlation energy is, to a good approximation, the electrostatic interaction of each electron with its own hole. That energy depends mostly on the *total charge* of the hole – which the sum rule fixes exactly – and only weakly on its shape. A functional that gets the shape badly wrong but the normalisation right can still give good energies, and this is what the local density approximation does.

8.6.2 Beyond LDA

The density in a real atom or molecule is anything but slowly varying, and the natural first correction is to let the functional see the gradient as well,

$$E_{\mathrm{xc}}^{\mathrm{GGA}}[n] = \int d^3\boldsymbol{r}\, n(\boldsymbol{r})\, \varepsilon_{\mathrm{xc}}(n(\boldsymbol{r}), |\nabla n(\boldsymbol{r})|)\,, \tag{8.41}$$

the *generalised gradient approximation*. A naive gradient expansion of the uniform gas actually makes things worse; what works is to construct the gradient dependence so as to preserve the exact constraints – the sum rule above, scaling relations, known limits – and this is how the widely used functionals were built. Beyond that lie meta-GGAs, which add the kinetic energy density, and hybrids, which mix in a fraction of exact Hartree-Fock exchange, Eq. (8.34), to repair the self-interaction error discussed next.

8.7 Kohn-Sham in practice

To see the machinery work, and fail, we apply it to a system we have solved twice already: N spinless fermions in a one-dimensional harmonic trap with a softened Coulomb repulsion, in a basis of eight oscillator orbitals. This is the system of Section 2.9 and of the Hartree-Fock calculation of Section 6.5, so the three treatments can be compared for the same Hamiltonian.

There is no tabulated exchange functional for a one-dimensional gas with this interaction, so the program builds one, which is instructive in itself. For a spinless one-dimensional gas the Fermi momentum is $k_F = \pi n$ and the one-body density matrix is $\rho^{(1)}(s) = \sin(k_F s)/\pi s$, so

$$\varepsilon_x(n) = -\frac{1}{n}\int_0^\infty ds\, v(s) \left[\frac{\sin(k_F s)}{\pi s}\right]^2, \tag{8.42}$$

which is evaluated numerically on a grid of densities, tabulated and interpolated. This is precisely how real functionals are made. The result:

n	$\varepsilon_x(n)$	$v_x(n)$
0.05	−0.17352608	−0.29740316
0.10	−0.27855522	−0.45906496
0.20	−0.42332628	−0.65789106
0.40	−0.59557914	−0.84878595
0.80	−0.75884371	−0.96587633
1.50	−0.86548980	−0.99703512
2.50	−0.91895548	−0.99989688

Solving Eq. (8.30) self-consistently in the oscillator basis then gives

Note first that the exact two-particle energy, 2.65923049, lies -3.86×10^{-3} below the Hartree-Fock value, which is the correlation energy quoted in Eq. (2.42) – the three chapters are consistent. Note next

N	E (Kohn-Sham)	E (Hartree-Fock)	E (exact)
1	0.54752674	0.50000000	0.50000000
2	2.72022768	2.66308878	2.65923049
3	6.45108052	6.39016133	—
4	11.68602229	11.62305385	—

Table 8.1 Kohn-Sham with a local exchange functional, Hartree-Fock and the exact answer for N spinless fermions in a one-dimensional trap with a softened Coulomb repulsion, in a basis of eight oscillator orbitals. The Hartree-Fock column reproduces Table 2.2; the exact value for $N = 2$ comes from full diagonalisation in the same basis.

that Kohn-Sham lies *above* both. This is not a failure of the variational principle, because we have used exchange only and put no correlation functional in at all; it is the error of treating exchange locally, compounded by a second defect that we now isolate.

8.7.1 The self-interaction error

Take a single particle. There is nothing for it to interact with, so the exact energy is the oscillator ground state, 0.5. Hartree-Fock gives exactly that: in Eq. (8.34) the direct term with $l = k$ is cancelled term by term by the exchange term with $l = k$, and a Slater determinant is rigorously free of self-interaction.

A local functional cannot do this, because it sees only the density, and the density of one electron is indistinguishable from the density of half of two. The program reports, for $N = 1$:

$$\mathscr{U}_{\mathrm{H}} = 0.60855522, \qquad E_x^{\mathrm{LDA}} = -0.56166291, \qquad \mathscr{U}_{\mathrm{H}} + E_x^{\mathrm{LDA}} = 0.04689232, \tag{8.43}$$

where the sum should vanish identically. The residue 0.047 is the *self-interaction error*, and it accounts for almost all of the 0.0475 by which Kohn-Sham misses the exact one-particle energy.

This single number explains a long list of known LDA failures: anions are overbound, reaction barriers too low, band gaps too small, and charge is over-delocalised, because an electron is spuriously repelled by itself and spreads out to reduce that repulsion. Self-interaction corrections remove it explicitly, orbital by orbital; hybrid functionals reduce it by mixing in a fraction of exact exchange, which is self-interaction free by construction.

8.7.2 How local is too local?

We can isolate the error of *locality* alone, cleanly, by evaluating on one and the same density both the exact exchange energy of the determinant and the local estimate:

N	E_x (exact)	E_x (local)	ratio
1	−0.60855522	−0.56166291	0.9229
2	−1.33077771	−1.27749456	0.9600
3	−2.11087395	−2.05548416	0.9738
4	−2.92523775	−2.86866697	0.9807

The local approximation recovers 92% of the exchange energy for one particle and 98% for four, improving steadily as the density smooths out on the scale of the local Fermi wavelength. This is the behaviour the construction promises, and the corresponding figure for real atoms in three dimensions is close to 90%

– an error of the order of ten per cent in a large term, which is exactly why gradient corrections were needed before density functional theory became quantitative for chemistry.

8.8 Summary

Density functional theory rests on an exact statement – the ground-state density determines the Hamiltonian – and on an approximation that the theorem itself does not supply. The Hohenberg-Kohn theorems guarantee a universal functional $F[n]$ and a variational principle for it, but they are existence theorems; all the many-body difficulty of the preceding chapters survives, relocated inside $F[n]$.

The Kohn-Sham construction is what makes the theory workable. By reintroducing orbitals for the kinetic energy – the one contribution that a local approximation handles badly, as Thomas-Fermi showed – it keeps shell structure exactly and leaves only a small remainder to be approximated. What is left is a self-consistent one-body problem of exactly the same form as Hartree-Fock, Eq. (6.26), differing in a single term: a local multiplicative v_{xc} in place of the non-local exchange operator.

The local density approximation takes that remainder from the electron gas of Chapter 6, and its success rests less on the density being slowly varying – it never is – than on the exchange-correlation hole sum rule, which fixes the quantity the energy is most sensitive to. Its two characteristic failures are visible in a system with four particles: exchange treated locally is wrong by a few per cent, and the self-interaction that Hartree-Fock cancels identically survives.

It is worth being clear about where this method sits relative to the others in this book. Configuration interaction, coupled cluster and the random-phase approximation are systematically improvable: there is a hierarchy, and one can in principle climb it towards the exact answer. Density functional theory is not of that kind. There is no sequence of functionals converging to $F[n]$, and no internal estimate of the error. What it offers instead is a cost that scales gently enough to treat thousands of atoms, and an accuracy that – for reasons that are now partly understood – is far better than the crudeness of the approximations would suggest.

The program `dft.py` performs every calculation of this chapter, and the notebook `dft.ipynb` runs it cell by cell.

8.9 Exercises

Exercise 1: densities and density matrices

a) Show that Eq. (8.4) and Eq. (8.5) define the same quantity, and that $\int n(\boldsymbol{r}) d^3\boldsymbol{r} = N$.
b) Derive Eq. (8.6) for a single Slater determinant, and show that $\rho^{(1)}$ is then the projector onto the occupied orbitals.
c) Derive the factorisation (8.13) from Wick's theorem, and verify that the diagonal vanishes at $x_1 = x_2$.
d) Show that for a correlated state the eigenvalues of $\rho^{(1)}$ are strictly between zero and one, and relate this to the occupation numbers of Section 1.31.

Answer to a).

Insert Eq. (8.4) into the expectation value and use the fact that Ψ is antisymmetric, so that every term of $\sum_i \delta(\boldsymbol{r} - \boldsymbol{r}_i)$ contributes equally:

$$\langle \Psi | \hat{n}(\boldsymbol{r}) | \Psi \rangle = N \int d^3\boldsymbol{r}_2 \cdots d^3\boldsymbol{r}_N \, |\Psi(\boldsymbol{r}, \boldsymbol{r}_2, \dots)|^2 , \tag{8.44}$$

which is Eq. (8.5). Integrating over $\boldsymbol{r}$ restores the full normalisation integral, giving $N \cdot 1 = N$.

Answer to b).

For a determinant $\rho^{(1)}(x';x) = \sum_{i=1}^{N} \psi_i(x')\psi_i^*(x)$, which follows from Eq. (8.9) by expanding the determinant and noting that only terms pairing the same orbital in bra and ket survive the integration over the remaining $N-1$ coordinates. Setting $x' = x$ gives Eq. (8.6). Regarded as an operator, $\rho^{(1)} = \sum_i |\psi_i\rangle\langle\psi_i|$, so $(\rho^{(1)})^2 = \rho^{(1)}$ by orthonormality and $\mathrm{tr}\,\rho^{(1)} = N$: a projector of rank N, which is Exercise 6.15d) in coordinate language.

Answer to c).

Write the pair density as $\langle\Phi_0|\hat{\psi}^\dagger(y_1)\hat{\psi}^\dagger(y_2)\hat{\psi}(x_2)\hat{\psi}(x_1)|\Phi_0\rangle$ and apply Wick's theorem. Only contractions pairing a creation with an annihilation operator survive, and there are exactly two ways to do it: $(y_1x_1)(y_2x_2)$ and $(y_1x_2)(y_2x_1)$, the second differing from the first by one interchange and hence carrying a minus sign. Each contraction is a factor $\rho^{(1)}$, giving Eq. (8.13). On the diagonal with $x_1 = x_2$ the two terms become $n(x_1)^2$ and $|\rho^{(1)}(x_1;x_1)|^2 = n(x_1)^2$, which cancel.

Answer to d).

$\rho^{(1)}$ is Hermitian and positive semi-definite, so its eigenvalues n_k are real and non-negative; they sum to N. For fermions no eigenvalue can exceed one, since $n_k = \langle\Psi|\hat{a}_k^\dagger\hat{a}_k|\Psi\rangle \leq 1$ by the Pauli principle. Equality for exactly N of them forces $\rho^{(1)}$ to be idempotent, hence forces Ψ to be a single determinant. Any correlated state therefore has all n_k strictly inside $[0,1]$, and the departure of the largest from unity is the diagnostic used in Table 1.8.

Exercise 2: the Hohenberg-Kohn theorems

a) Reproduce the proof of the first theorem, stating exactly where non-degeneracy is used.
b) Show that the theorem determines v_{ext} only up to an additive constant, and explain why that is harmless.
c) Prove the second theorem from the first, and state what "universal" means.
d) Explain the v-representability problem and how the Levy-Lieb constrained search, Eq. (8.21), removes it.
e) The theorems say the density determines everything. Why, then, can we not obtain excitation energies from $E[n]$?

Answer to a).

See Eqs. (8.17)–(8.19). Non-degeneracy enters in making the variational inequality *strict*: if the ground state were degenerate, $\Psi^{(2)}$ might be another ground state of $\hat{H}^{(1)}$ and the inequality would become an equality, so that adding the two relations would give $E^{(1)} + E^{(2)} \leq E^{(1)} + E^{(2)}$ – true, and no contradiction. This is exactly the loophole that the Levy-Lieb formulation closes.

Answer to b).

Adding a constant c to v_{ext} adds Nc to every eigenvalue and changes no eigenstate, hence no density. The map from densities to potentials therefore cannot possibly resolve the constant, and does not need to: energy differences, densities and all observables are unaffected. It is the same ambiguity as the zero of a potential in elementary mechanics.

Answer to c).

By the first theorem n_0 determines v_{ext}, hence $\hat{H}$, hence its ground state $\Psi[n]$. So $F_{\text{HK}}[n] = \langle\Psi[n]|\hat{T} + \hat{V}|\Psi[n]\rangle$ is a functional of n alone. "Universal" means that F contains no reference to v_{ext}: the very same functional applies to a hydrogen atom, a copper crystal and a stretch of DNA, since $\hat{T}$ and $\hat{V}$ are properties of electrons. For any trial n, the Rayleigh-Ritz principle gives $E[n] = F[n] + \int v_{\text{ext}} n \geq E_0$, with equality only at $n = n_0$.

Answer to d).

The construction of F_{HK} requires that n be the ground-state density of *some* potential in the admissible family – that it be v-representable. Densities are known that are not, and no useful characterisation of the v-representable ones exists. Levy and Lieb sidestep the issue by defining F through a minimisation over all antisymmetric wave functions yielding the given density, Eq. (8.21). This requires only N-representability, which is easy to characterise for a one-body density – positivity, correct normalisation and finite $\int |\nabla n^{1/2}|^2$ suffice – and it handles degenerate ground states.

Answer to e).

The theorems concern the ground state. The variational principle $E[n] \geq E_0$ singles out the minimum and says nothing about stationary points corresponding to excited states, and the functional $F[n]$ was constructed from ground-state wave functions only. Excitations require extra structure: time-dependent density functional theory obtains them from the linear response of the density, which is formally the density-functional analogue of the random-phase approximation of Chapter 7, with the exchange-correlation kernel $\delta v_{\text{xc}}/\delta n$ playing the part of the residual interaction in A and B.

Exercise 3: from the electron gas to a functional

a) Derive the Thomas-Fermi kinetic density (8.23) from the kinetic energy of Chapter 6, and confirm it numerically.
b) Derive the local exchange energy (8.38) and show that it reproduces the $-0.916/r_s$ of Section 6.13.6.
c) Show that $v_x = \frac{4}{3}\varepsilon_x$ for the local exchange functional, and explain why the factor is not one.
d) Derive the one-dimensional analogue (8.42) and the kinetic density $t[n] = (\pi^2/6)n^3$.
e) Verify the exchange-hole sum rule (8.40) analytically.

Answer to a).

Equation (6.91) gives the kinetic energy per particle as $\frac{3}{5}\hbar^2 k_F^2/2m$; multiplying by n and using $k_F = (3\pi^2 n)^{1/3}$ gives the energy per unit volume

$$t_0[n] = \frac{3}{10}\frac{\hbar^2}{m}\left(3\pi^2\right)^{2/3} n^{5/3}, \tag{8.45}$$

which is Eq. (8.23). The program compares the two expressions at three densities and finds them identical to eight digits.

Answer to b).

The exchange energy is $E_x = -\frac{1}{2}\iint v(|\boldsymbol{r}-\boldsymbol{r}'|)|\rho^{(1)}(\boldsymbol{r};\boldsymbol{r}')|^2$, the second term of Eq. (8.14) integrated against the interaction. For the uniform gas $\rho^{(1)}$ depends only on $s = |\boldsymbol{r}-\boldsymbol{r}'|$ and equals $3n[\sin y - y\cos y]/y^3$ with $y = k_F s$; doing the integral with $v = e^2/s$ gives $E_x/V = -\frac{3}{4}(3/\pi)^{1/3}n^{4/3}$, hence Eq. (8.38). Evaluating at $n = 3/(4\pi r_0^3)$,

$$\varepsilon_x = -\frac{3}{4}\left(\frac{3}{\pi}\right)^{1/3}\left(\frac{3}{4\pi}\right)^{1/3}\frac{1}{r_s} = -\frac{0.4582}{r_s}\ \text{Ha} = -\frac{0.9163}{r_s}\ \text{Ry}, \tag{8.46}$$

which is the exchange term of Eq. (6.108).

Answer to c).

$E_x = \int n\varepsilon_x(n)$ with $\varepsilon_x \propto n^{1/3}$, so the integrand is $\propto n^{4/3}$ and

$$v_x = \frac{d}{dn}[n\varepsilon_x(n)] = \frac{4}{3}\varepsilon_x(n). \tag{8.47}$$

The factor is not one because v_x is a functional derivative, not an energy per particle: changing the density at a point changes both the number of electrons there and the energy each of them carries. The distinction is the same as that between $\sum_i \varepsilon_i$ and the total energy in Eq. (8.33), and forgetting it is a standard first error.

Answer to d).

For a spinless one-dimensional gas, $n = \int_{-k_F}^{k_F} dk/2\pi = k_F/\pi$, so $k_F = \pi n$, and $\rho^{(1)}(s) = \int_{-k_F}^{k_F}(dk/2\pi)e^{iks} = \sin(k_F s)/\pi s$. Inserting into the exchange integral and using the symmetry of the two half-lines to cancel the factor $\frac{1}{2}$ gives Eq. (8.42). For the kinetic energy, $T/L = \int_{-k_F}^{k_F}(dk/2\pi)(k^2/2) = k_F^3/6\pi = (\pi^2/6)n^3$.

Answer to e).

Substituting Eq. (8.39),

$$\frac{4}{3\pi}\int_0^\infty dy\, y^2\,[g(y)-1] = -\frac{4}{3\pi}\cdot\frac{9}{2}\int_0^\infty dy\,\frac{(\sin y - y\cos y)^2}{y^4} = -\frac{6}{\pi}\cdot\frac{\pi}{6} = -1, \tag{8.48}$$

using the standard integral $\int_0^\infty(\sin y - y\cos y)^2 y^{-4}dy = \pi/6$. The program obtains -0.9976 by direct quadrature, the residue being the slowly decaying oscillatory tail.

Exercise 4: the Kohn-Sham equations

a) Derive Eq. (8.30) by varying Eq. (8.29) with respect to the orbitals subject to orthonormality.
b) Derive the total-energy expression (8.33) and explain each correction term.
c) Compare Eqs. (8.34) and (8.35) term by term. Which is variational, and with respect to what?
d) Why is $T_s[n]$, and not $T[n]$, computed from the orbitals? What is the difference, and where does it go?
e) Show that for one electron the exact E_{xc} must satisfy $E_{\text{xc}}[n] = -\mathscr{U}_{\text{H}}[n]$.

Answer to a).

Write T_s and n in terms of the orbitals and impose $\int \psi_i^* \psi_j = \delta_{ij}$ with multipliers ε_{ij}. Varying with respect to ψ_i^*,

$$-\frac{\hbar^2}{2m}\nabla^2\psi_i + \left[v_{\text{ext}} + \frac{\delta \mathscr{U}_{\text{H}}}{\delta n} + \frac{\delta E_{\text{xc}}}{\delta n}\right] \frac{\delta n}{\delta|\psi_i|^2}\psi_i = \sum_j \varepsilon_{ij}\psi_j, \tag{8.49}$$

and since $\delta n/\delta|\psi_i|^2 = 1$ the bracket is exactly v_{KS}. A unitary rotation among the occupied orbitals diagonalises ε_{ij} without changing the density – the same freedom as in Section 6.3 – giving Eq. (8.30).

Answer to b).

Multiplying Eq. (8.30) by ψ_i^*, integrating and summing over the occupied orbitals gives $\sum_i \varepsilon_i = T_s + \int v_{\text{KS}} n$. Substituting v_{KS} from Eq. (8.31) and comparing with Eq. (8.29),

$$E = \sum_i \varepsilon_i - \underbrace{\int v_{\text{H}} n}_{=2\mathscr{U}_{\text{H}}} - \int v_{\text{xc}} n + \mathscr{U}_{\text{H}} + E_{\text{xc}}, \tag{8.50}$$

which is Eq. (8.33). The band sum counts the Hartree energy twice, once for each partner, so one copy is removed; and it contains $\int v_{\text{xc}} n$, which is not E_{xc}, so the former is subtracted and the latter added. This is the same double-counting correction as Exercise 6.15b).

Answer to c).

The kinetic, external and Hartree terms are identical. The difference is the last term: Hartree-Fock has a non-local exchange *operator*, acting differently on each orbital, exact for a determinant and containing no correlation; Kohn-Sham has a local multiplicative *potential*, the same for all orbitals, approximate, but containing correlation as well. Hartree-Fock is variational with respect to the true energy – it is a genuine upper bound on E_0. Kohn-Sham is variational only with respect to the energy of the model functional it is given; with an approximate E_{xc} the result may lie above or below the exact energy, as Table 8.1 shows.

Answer to d).

$T[n]$, the true kinetic energy of the interacting system, is not accessible from the density; that was the problem in the first place. $T_s[n]$, the kinetic energy of the non-interacting system with the same density, is computed exactly from its orbitals. The difference $T[n] - T_s[n]$ is small – it is a correlation effect – and it is folded into E_{xc} by the definition (8.28). The whole construction is the observation that this is a far better place to put the difficulty than the kinetic energy itself.

Answer to e).

For $N = 1$ the exact energy contains no electron-electron interaction at all. But Eq. (8.29) contains $\mathscr{U}_{\text{H}}[n]$, which is non-zero for any density. The exact functional must therefore cancel it exactly, $E_{\text{xc}}[n] = -\mathscr{U}_{\text{H}}[n]$ for any one-electron density. The local density approximation violates this: the program finds a residue 0.0469 for the harmonic trap, Eq. (8.43). Imposing the one-electron condition exactly is the content of the Perdew-Zunger self-interaction correction.

Exercise 5: numerical explorations

a) Using `dft.py`, reproduce Table 8.1 and plot the Kohn-Sham, Hartree-Fock and non-interacting densities for $N = 4$. Where do they differ most?
b) Vary the interaction strength and follow the ratio $E_x^{\text{local}}/E_x^{\text{exact}}$. Does the local approximation improve or deteriorate as the interaction is made stronger?
c) Implement the Perdew-Zunger self-interaction correction for $N = 1$ – subtract $\mathscr{U}_{\text{H}}[n_i] + E_x[n_i]$ orbital by orbital – and check that the one-particle energy returns to 0.5.
d) Solve the Thomas-Fermi equation for the trap directly by minimising over $n(x)$ on a grid, and compare the resulting density with the Kohn-Sham one. Is the shell structure visible?
e) Add a crude gradient correction, $E_x \to E_x - \beta \int |\nabla n|^2/n^{4/3}$, and tune β to fit the exact exchange energies of Section 8.7.2. Does one value work for all N?

Answer.

Some pointers. In a) the densities differ most in the tails, where the Kohn-Sham density is too diffuse – the signature of self-interaction, since a spuriously self-repelled electron spreads out. In b) the ratio deteriorates as the interaction strengthens, because exchange grows relative to the kinetic energy and the errors in it become more visible; this is the same trend that made the mean field fail at strong coupling in Section 5.7. In c) the correction is exact by construction for one particle and the energy returns to 0.5 to machine precision; the interesting question is what it does for $N > 1$, where it is no longer exact and can over-correct. In d) the Thomas-Fermi density is a smooth dome with no oscillations, in contrast to the N visible maxima of the Kohn-Sham density – the clearest single illustration of what orbitals buy. In e) a single β does not fit all N; the failure of a one-parameter gradient correction is precisely why the construction of gradient functionals is done by imposing exact constraints instead of fitting.

Chapter 9
Many-body perturbation theory

Chapter 5 showed that the exact ground state is a superposition of all determinants, and that we cannot afford them. Chapter 6 found the best single determinant, and Chapter 7 described the fluctuations about it. Perturbation theory takes yet another route: it accepts the mean-field determinant as a starting point and generates corrections to it *order by order* in the interaction, each order being a finite, well-defined sum that can be written down and evaluated.

The appeal is obvious. There is no matrix to diagonalise, no iteration to converge, and no variational parameters – only sums over intermediate states, which for a two-body Hamiltonian are limited to particle-hole excitations of low rank. The price is equally clear and we shall be blunt about it: the method is not variational, so nothing guarantees that a higher order is a better answer; and the series need not converge at all.

This chapter develops the formalism algebraically. There is a beautiful diagrammatic language for organising these sums, and it is indispensable beyond third order, but it is a separate subject and we leave it out here in order to keep the algebra visible. We derive the two classical schemes – Brillouin-Wigner and Rayleigh-Schrödinger – write out the first, second and third-order terms, and then apply them to the four models that have run through the book: the pairing model of Section 4.2, the Lipkin model of Section 4.1, the Hubbard model of Section 4.3, and the pairing plus particle-hole model of Section 7.1. Every number is compared with the exact answer and with the methods of the preceding chapters.

9.1 An exact starting point

We are interested in the ground state, and we assume it is dominated by a single determinant $|\Phi_0\rangle$ that solves an unperturbed problem,

$$\hat{H}_0|\Phi_0\rangle = W_0|\Phi_0\rangle, \qquad \hat{H} = \hat{H}_0 + \hat{H}_I. \tag{9.1}$$

The exact state is expanded in the eigenstates of $\hat{H}_0$,

$$|\Psi_0\rangle = |\Phi_0\rangle + \sum_{m=1}^{\infty} C_m|\Phi_m\rangle, \tag{9.2}$$

with intermediate normalisation $\langle\Phi_0|\Psi_0\rangle = 1$, exactly as in Section 5.2. Note that $|\Psi_0\rangle$ is therefore not normalised to unity; this is a convenience, not an approximation.

Project the Schrödinger equation $\hat{H}|\Psi_0\rangle = E|\Psi_0\rangle$ onto the reference,

$$\langle\Phi_0|\hat{H}|\Psi_0\rangle = E\langle\Phi_0|\Psi_0\rangle = E, \tag{9.3}$$

and subtract the corresponding statement for $\hat{H}_0$,

$$\langle\Psi_0|\hat{H}_0|\Phi_0\rangle = W_0\langle\Psi_0|\Phi_0\rangle = W_0. \tag{9.4}$$

Both operators are Hermitian, so the two may be subtracted directly, giving

$$\boxed{\Delta E = E - W_0 = \langle\Phi_0|\hat{H}_I|\Psi_0\rangle.} \tag{9.5}$$

This is exact, and we call ΔE the correlation energy. Every perturbative derivation in this chapter starts from it.

As it stands, however, Eq. (9.5) is a rewriting of the Schrödinger equation and nothing more: the right-hand side contains the unknown exact state. To make progress we must expand $|\Psi_0\rangle$ in powers of $\hat{H}_I$.

9.2 Projection operators and the resolvent

Define the projector onto the reference and its complement,

$$\hat{P} = |\Phi_0\rangle\langle\Phi_0|, \qquad \hat{Q} = \sum_{m=1}^{\infty} |\Phi_m\rangle\langle\Phi_m| = 1 - \hat{P}, \tag{9.6}$$

so that our model space is one-dimensional. Both are idempotent, $\hat{P}^2 = \hat{P}$ and $\hat{Q}^2 = \hat{Q}$, both commute with $\hat{H}_0$ – they are built from its eigenstates – and $\hat{P}\hat{Q} = 0$. With intermediate normalisation,

$$|\Psi_0\rangle = \left(\hat{P} + \hat{Q}\right)|\Psi_0\rangle = |\Phi_0\rangle + \hat{Q}|\Psi_0\rangle. \tag{9.7}$$

Now introduce an energy parameter ω, to be fixed later, and rewrite the Schrödinger equation by adding and subtracting $\omega|\Psi_0\rangle$,

$$\left(\omega - \hat{H}_0\right)|\Psi_0\rangle = \left(\omega - E + \hat{H}_I\right)|\Psi_0\rangle. \tag{9.8}$$

Provided the operator on the left has an inverse – the *resolvent*, or unperturbed Green's function,

$$\left(\omega - \hat{H}_0\right)^{-1} = \frac{1}{\omega - \hat{H}_0}, \tag{9.9}$$

we may divide by it. Multiplying from the left by $\hat{Q}$ and using

$$\hat{Q}\frac{1}{\omega - \hat{H}_0}\hat{Q} = \hat{Q}\frac{1}{\omega - \hat{H}_0} = \frac{\hat{Q}}{\omega - \hat{H}_0}, \tag{9.10}$$

which holds because $\hat{Q}$ commutes with $\hat{H}_0$ and is idempotent, we obtain

$$|\Psi_0\rangle = |\Phi_0\rangle + \frac{\hat{Q}}{\omega - \hat{H}_0}\left(\omega - E + \hat{H}_I\right)|\Psi_0\rangle. \tag{9.11}$$

Equation (9.11) is still exact and still useless: it is a nonlinear equation for two unknowns, E and $|\Psi_0\rangle$. But it invites iteration. Substituting the whole right-hand side into itself repeatedly, starting from the natural guess $|\Psi_0\rangle \to |\Phi_0\rangle$, gives

$$|\Psi_0\rangle = \sum_{i=0}^{\infty}\left\{\frac{\hat{Q}}{\omega - \hat{H}_0}\left(\omega - E + \hat{H}_I\right)\right\}^i|\Phi_0\rangle, \tag{9.12}$$

and, inserted into Eq. (9.5),

$$\Delta E = \sum_{i=0}^{\infty}\langle\Phi_0|\hat{H}_I\left\{\frac{\hat{Q}}{\omega - \hat{H}_0}\left(\omega - E + \hat{H}_I\right)\right\}^i|\Phi_0\rangle. \tag{9.13}$$

This is a genuine expansion in powers of $\hat{H}_I$. Two things still spoil it: the undetermined parameter ω, and the appearance of the exact energy E. The two classical schemes are the two natural ways of disposing of them.

9.3 Brillouin-Wigner perturbation theory

The first choice is $\omega = E$, which annihilates the awkward factor $(\omega - E)$ and leaves

$$\Delta E = \langle \Phi_0 | \hat{H}_I + \hat{H}_I \frac{\hat{Q}}{E - \hat{H}_0} \hat{H}_I + \hat{H}_I \frac{\hat{Q}}{E - \hat{H}_0} \hat{H}_I \frac{\hat{Q}}{E - \hat{H}_0} \hat{H}_I + \cdots | \Phi_0 \rangle. \tag{9.14}$$

Every term is explicit and the structure is simple: alternating interactions and resolvents. The price is that E appears in every denominator, so the equation must be solved self-consistently – guess E, evaluate the right-hand side, update, repeat.

There is more to be said. Write $\hat{e} = E - \hat{H}_0$ and recall that $\hat{Q}$ commutes with $\hat{H}_0$ and is idempotent. The geometric series

$$\hat{Q} \frac{1}{\hat{e} - \hat{Q}\hat{H}_I\hat{Q}} = \hat{Q} \left[\frac{1}{\hat{e}} + \frac{1}{\hat{e}} \hat{Q}\hat{H}_I\hat{Q} \frac{1}{\hat{e}} + \frac{1}{\hat{e}} \hat{Q}\hat{H}_I\hat{Q} \frac{1}{\hat{e}} \hat{Q}\hat{H}_I\hat{Q} \frac{1}{\hat{e}} + \cdots \right] \hat{Q} \tag{9.15}$$

resums Eq. (9.14) completely, giving the closed form

$$\boxed{\Delta E = \langle \Phi_0 | \hat{H}_I + \hat{H}_I \hat{Q} \frac{1}{E - \hat{H}_0 - \hat{Q}\hat{H}_I\hat{Q}} \hat{Q}\hat{H}_I | \Phi_0 \rangle.} \tag{9.16}$$

This is not an approximation. Solved self-consistently it returns the exact ground-state energy, and the program `mbpt.py` verifies that it does, for the pairing model, to eight digits at every coupling:

g	BW(2)	BW(3)	BW resummed	FCI
0.25	1.73328927	1.73091635	1.73045985	1.73045985
0.50	1.43866884	1.42314335	1.41677428	1.41677428
1.00	0.79140813	0.70800963	0.63554847	0.63554847
1.50	0.09433209	−0.10095126	−0.34819051	−0.34819051
2.00	−0.63318105	−0.96772735	−1.48965216	−1.48965216

Equation (9.16) is, in other words, one more disguise of the Schrödinger equation. Its truncations are simple to write down and, for a single small system, converge quickly. They also have a fatal defect, which Section 9.8 exhibits.

9.4 Rayleigh-Schrödinger perturbation theory

The second choice is $\omega = W_0$, the *unperturbed* reference energy, which removes the exact energy from the denominators at the cost of leaving ΔE in the numerators,

$$\Delta E = \langle \Phi_0 | \hat{H}_I + \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \left(\hat{H}_I - \Delta E \right) + \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \left(\hat{H}_I - \Delta E \right) \frac{\hat{Q}}{W_0 - \hat{H}_0} \left(\hat{H}_I - \Delta E \right) + \cdots | \Phi_0 \rangle. \tag{9.17}$$

One simplification is immediate. Since ΔE is a number and $\hat{Q}$ annihilates the reference,

$$\hat{Q}\, \Delta E | \Phi_0 \rangle = \Delta E\, \hat{Q} | \Phi_0 \rangle = 0, \tag{9.18}$$

so the rightmost ΔE in each term drops out and

$$\Delta E = \langle \Phi_0 | \hat{H}_I + \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \hat{H}_I + \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \left(\hat{H}_I - \Delta E \right) \frac{\hat{Q}}{W_0 - \hat{H}_0} \hat{H}_I + \cdots | \Phi_0 \rangle. \tag{9.19}$$

Equation (9.19) is still implicit in ΔE, and the resolution is to expand it too. Writing

$$\Delta E = \sum_{i=1}^{\infty} \Delta E^{(i)}, \tag{9.20}$$

and collecting powers of $\hat{H}_I$, we obtain the first three orders:

$$\Delta E^{(1)} = \langle \Phi_0 | \hat{H}_I | \Phi_0 \rangle, \tag{9.21}$$

$$\Delta E^{(2)} = \langle \Phi_0 | \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \hat{H}_I | \Phi_0 \rangle, \tag{9.22}$$

$$\Delta E^{(3)} = \langle \Phi_0 | \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \hat{H}_I \frac{\hat{Q}}{W_0 - \hat{H}_0} \hat{H}_I | \Phi_0 \rangle - \langle \Phi_0 | \hat{H}_I | \Phi_0 \rangle \langle \Phi_0 | \hat{H}_I \frac{\hat{Q}}{\left(W_0 - \hat{H}_0 \right)^2} \hat{H}_I | \Phi_0 \rangle. \tag{9.23}$$

In a basis of unperturbed determinants, writing V_{mn} for the matrix elements of $\hat{H}_I$ and $D_m = W_0 - W_m$ for the energy denominators, these read

$$\Delta E^{(1)} = V_{00}, \tag{9.24}$$

$$\Delta E^{(2)} = \sum_{m \neq 0} \frac{V_{0m} V_{m0}}{D_m}, \tag{9.25}$$

$$\Delta E^{(3)} = \sum_{m,n \neq 0} \frac{V_{0m} V_{mn} V_{n0}}{D_m D_n} - V_{00} \sum_{m \neq 0} \frac{V_{0m} V_{m0}}{D_m^2}. \tag{9.26}$$

The second term of Eq. (9.26) is the one to pay attention to. It is the price of having removed E from the denominators, and it appears for the first time at third order. It is often called the *renormalisation* term, and it is not small: for the pairing model at $g = 0.5$ the program finds

$$\underbrace{-0.02300347}_{\text{principal}} - \underbrace{(-0.01258681)}_{\text{renormalisation}} = \underbrace{-0.01041667}_{\Delta E^{(3)}}, \tag{9.27}$$

so the correction is larger than the answer. Its function, as Section 9.8 shows, is to cancel exactly those contributions that would otherwise make the energy scale wrongly with the size of the system. This is the algebraic shadow of the linked-diagram theorem.

Generating the series to any order.

Rather than extend Eqs. (9.21)–(9.23) by hand, one uses the standard recursion. With $\hat{R} = \hat{Q}/(W_0 - \hat{H}_0)$,

$$|\Psi^{(n)}\rangle = \hat{R} \left[\hat{H}_I |\Psi^{(n-1)}\rangle - \sum_{k=1}^{n-1} \Delta E^{(k)} |\Psi^{(n-k)}\rangle \right], \qquad \Delta E^{(n+1)} = \langle \Phi_0 | \hat{H}_I | \Psi^{(n)} \rangle, \tag{9.28}$$

starting from $|\Psi^{(0)}\rangle = |\Phi_0\rangle$. Unwinding two steps reproduces Eqs. (9.22) and (9.23) exactly, renormalisation term included, and the program checks the agreement numerically for all four models. We use the recursion in Section 9.7 to look at what the series does at high order.

9.5 The wave operator, and the link to configuration interaction

Before turning to numbers it is worth seeing what perturbation theory is doing to the wave function, because it connects directly to Chapter 5.

Recall the correlation operator of Eq. (5.6),

$$|\Psi_0\rangle = \left(1+\hat{C}\right)|\Phi_0\rangle, \qquad \hat{C} = \sum_{ai} C_i^a \hat{a}_a^\dagger \hat{a}_i + \sum_{abij} C_{ij}^{ab} \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i + \cdots \tag{9.29}$$

In a configuration-interaction calculation the coefficients come out of a diagonalisation. Perturbation theory produces them instead as an expansion, and it is instructive to see the two agree at lowest order.

Take a Hartree-Fock basis, so that $\langle\Phi_0|\hat{H}_I|\Phi_i^a\rangle = 0$ by Brillouin's theorem, Eq. (6.31). A two-body interaction can connect the reference only to $2p$-$2h$ states, so Eq. (9.22) collapses to the single well-known expression

$$\Delta E^{(2)} = \frac{1}{4}\sum_{abij} \frac{\langle ij|\hat{v}|ab\rangle\langle ab|\hat{v}|ij\rangle}{\varepsilon_i+\varepsilon_j-\varepsilon_a-\varepsilon_b}. \tag{9.30}$$

Compare this with the exact correlation energy in the same basis, Eq. (5.30), which for a Hartree-Fock reference reduces to

$$\Delta E = \sum_{abij} \langle ij|\hat{v}|ab\rangle C_{ij}^{ab}. \tag{9.31}$$

The two agree provided

$$C_{ij}^{ab} = \frac{1}{4}\frac{\langle ab|\hat{v}|ij\rangle}{\varepsilon_i+\varepsilon_j-\varepsilon_a-\varepsilon_b}, \tag{9.32}$$

which is precisely the first-order perturbative estimate of the doubles amplitudes. Second-order perturbation theory is thus configuration interaction with the doubles amplitudes taken to first order instead of being solved for.

The difference matters. Full configuration interaction determines C_{ij}^{ab} to *all* orders in the interaction, whereas Eq. (9.32) is the leading term of an expansion. Going to higher order in perturbation theory brings in successively more complicated correlations – and, at fourth order and beyond, genuinely new excitation classes. This is the sense in which the two hierarchies are complementary: configuration interaction truncates the *excitation rank* and keeps all orders in the interaction; perturbation theory truncates the *order in the interaction* and keeps all excitation ranks that can contribute. Coupled-cluster theory, which we come to later, is designed to keep the best of both.

Choosing the partition. Nothing in the derivation fixed $\hat{H}_0$; only that it be soluble and that $|\Phi_0\rangle$ be one of its eigenstates. Two choices are standard. Taking $\hat{H}_0$ to be the one-body part of $\hat{H}$ – in a Hartree-Fock basis, the Fock operator – is the Møller-Plesset partition, and it is what we use below. Taking $\hat{H}_0$ to be the entire *diagonal* of $\hat{H}$ in the determinant basis is the Epstein-Nesbet partition; then $\Delta E^{(1)} = 0$ by construction and the denominators contain the diagonal interaction as well. The two sum to the same exact answer and disagree at every finite order. Section 9.12 compares them, and finds that neither is uniformly better – which is why a perturbative result quoted without its partition means nothing.

9.6 The pairing model

We now apply all of this. Take the pairing model of Section 4.2 with four doubly degenerate levels and four particles, $\xi = 1$, and let $\hat{H}_0$ be the one-body part. The reference is the determinant filling the two lowest levels, with $W_0 = 2$, and the first-order energy is

$$\Delta E^{(1)} = \langle\Phi_0|\hat{H}_I|\Phi_0\rangle = -g, \tag{9.33}$$

one factor $-g/2$ for each of the two occupied pairs. The energy through first order is therefore $2-g$, which is exactly the Hartree-Fock energy of Section 6.12: for this interaction the mean field does nothing, and first-order perturbation theory reproduces it. Every subsequent order is genuine correlation energy.

g	E_{ref}	MBPT2	MBPT3	FCI	CID	HF + RPA
0.25	1.75000	1.73177083	1.73046875	1.73045985	1.73050700	1.71634735
0.50	1.50000	1.42708333	1.41666667	1.41677428	1.41759645	1.37450235
1.00	1.00000	0.70833333	0.62500000	0.63554847	0.64890655	0.55458933
1.50	0.50000	−0.15625000	−0.43750000	−0.34819051	−0.29113175	−0.40624787
2.00	0.00000	−1.16666667	−1.83333333	−1.48965216	−1.35208670	−1.47622319

Table 9.1 Rayleigh-Schrödinger perturbation theory for the pairing model with four levels and four particles, compared with full configuration interaction, with doubles configuration interaction (Section 5.8) and with the particle-hole random-phase approximation (Section 7.9).

At $g = 0.25$ the series is superb: second order is accurate to 1.3×10^{-3} and third order to 9×10^{-6}. By $g = 1$ third order has crossed the exact answer, and at $g = 2$ it lies 0.34 below it while second order lies 0.32 above. Nothing prevents this. Perturbation theory is not variational, and neither the sign nor the magnitude of the next correction is under any control.

Contrast the behaviour of the two other approximations in the table. Doubles configuration interaction is variational, so it stays above the exact answer at every coupling and degrades smoothly. The random-phase approximation sums a restricted class of contributions to all orders; it is the worst of the three at $g = 0.25$ and the best at $g = 2$.

9.7 How far can the series be trusted?

The recursion (9.28) lets us stop guessing and look. The following table gives the size of the n-th term for the same model:

g	$\lvert E^{(2)}\rvert$	$\lvert E^{(5)}\rvert$	$\lvert E^{(10)}\rvert$	$\lvert E^{(20)}\rvert$	$\lvert E^{(40)}\rvert$	through 20	exact
0.50	7.3×10^{-2}	3.6×10^{-4}	5.7×10^{-8}	2.6×10^{-12}	1.9×10^{-22}	1.4167743	1.4167743
1.00	2.9×10^{-1}	1.2×10^{-2}	5.9×10^{-5}	2.8×10^{-6}	2.1×10^{-10}	0.6355478	0.6355485
1.50	6.6×10^{-1}	8.8×10^{-2}	3.4×10^{-3}	9.2×10^{-3}	2.3×10^{-3}	−0.3532205	−0.3481905
2.00	1.2×10^{0}	3.7×10^{-1}	6.0×10^{-2}	2.9×10^{0}	2.3×10^{2}	−3.7973859	−1.4896522
2.50	1.8×10^{0}	1.1×10^{0}	5.6×10^{-1}	2.5×10^{2}	1.7×10^{6}	−240.36	−2.7371418

Table 9.2 The magnitude of successive Rayleigh-Schrödinger terms for the pairing model. At $g = 0.5$ and $g = 1$ the series converges to the exact energy; at $g = 2$ and beyond the terms grow without limit.

At $g = 0.5$ and $g = 1$ the terms fall away steadily and the partial sums reproduce the exact energy to the digits shown. At $g = 2$ and $g = 2.5$ the terms grow by ten and by twelve orders of magnitude between the tenth and the fortieth, and the partial sums are worthless. The radius of convergence lies somewhere between $g = 1$ and $g = 2$; $g = 1.5$ sits close to it, and no finite number of terms settles the question there.

This deserves to be taken seriously rather than noted in passing, because of how it looks from the inside. At $g = 1.5$ the first three orders are

$$E_{\text{ref}} = 0.50000, \qquad \text{through second} = -0.15625, \qquad \text{through third} = -0.43750, \tag{9.34}$$

against an exact -0.34819. Those numbers look entirely reasonable. Nothing in them warns that the series behind them may not converge, and in a realistic calculation one rarely has the exact answer to compare against. Truncating a perturbation series at low order is an act of faith, and the faith is sometimes misplaced.

The underlying reason is the same one we have met twice already. The expansion is about the reference determinant, and once the true ground state has stopped resembling it, the expansion has nothing left to say. In Chapter 6 the same coupling made the mean field unstable; in Chapter 5 it made truncated configuration interaction deteriorate. Here it makes the series diverge.

9.8 Size extensivity, and why Brillouin-Wigner is abandoned

Section 5.9 showed that truncated configuration interaction fails a basic test: for two non-interacting subsystems it does not give twice the energy of one. We now apply the same test to the two perturbation schemes, and the outcome is the sharpest distinction between them.

Take M identical, non-interacting copies of a small pairing system – three levels holding one pair, with the interaction acting only within a subsystem. Any acceptable method must give exactly M times the one-subsystem energy. The program finds, for $M = 2$:

order	$2\times$ one copy	two copies	error
$g = 0.5$			
first	-0.500000000	-0.500000000	0
second	-0.093750000	-0.093750000	0
third	-0.007812500	-0.007812500	0
fourth	0.001098633	0.001098633	0
BW(2)	-0.583666572	-0.575845121	7.8×10^{-3}
$g = 1.0$			
first	-1.000000000	-1.000000000	0
second	-0.375000000	-0.375000000	0
third	-0.062500000	-0.062500000	0
fourth	0.017578125	0.017578125	0
BW(2)	-1.296377095	-1.249140538	4.7×10^{-2}

Table 9.3 Size extensivity for two non-interacting copies of a three-level pairing system. Rayleigh-Schrödinger is exact at every order, to machine precision; Brillouin-Wigner truncated at second order is not, and the error grows with the coupling.

Rayleigh-Schrödinger is exact *at every order separately*, to machine precision. This is the linked-diagram theorem in its most concrete form. The unlinked contributions – terms in which the intermediate state contains excitations in both subsystems with no interaction connecting them – would individually scale as M^2, and they are cancelled exactly by the renormalisation term of Eq. (9.26) and by its analogues at higher order. That is what the term is for.

Brillouin-Wigner has no such term. Its denominators contain the exact energy, which is itself extensive, so the denominators of a two-subsystem calculation differ from those of a one-subsystem calculation and the two can never match. The error is 7.8×10^{-3} at $g = 0.5$ and grows to 4.7×10^{-2} at $g = 1$, and it would grow linearly with the number of subsystems. For a calculation on bulk matter this is fatal, and it is the reason Brillouin-Wigner theory, for all its formal elegance and rapid convergence on small systems, is not used in many-body physics.

9.9 The Lipkin model

The Lipkin model of Section 4.1, with $W = 0$, is instructive because almost everything can be done in closed form. The interaction $(V/2)(\hat{J}_+^2 + \hat{J}_-^2)$ changes J_z by exactly two units, so

$$\Delta E^{(1)} = \langle \Phi_0 | \hat{H}_I | \Phi_0 \rangle = 0 \tag{9.35}$$

identically, and the reference energy is the unperturbed $-N\varepsilon/2$. The reference couples to exactly one state, $|J, -J+2\rangle$, at an excitation energy 2ε, with matrix element $\frac{V}{2}\sqrt{2N(N-1)}$, so

$$\Delta E^{(2)} = -\frac{N(N-1)V^2}{4\varepsilon}, \tag{9.36}$$

which the program confirms to ten digits.

The third order vanishes identically, and for a reason worth stating. The principal term of Eq. (9.26) requires a matrix element V_{mn} between two states both reachable from the reference; but every such state has $J_z = -J+2$, and $\hat{H}_I$ changes J_z by two units, so $V_{mn} = 0$. The renormalisation term carries a factor $V_{00} = 0$. Hence $\Delta E^{(3)} = 0$ exactly.

χ	E_{ref}	through $\Delta E^{(2)}$	through $\Delta E^{(3)}$	exact	HF + RPA
0.25	−2.00000	−2.02083333	−2.02083333	−2.02072594	−2.02109449
0.50	−2.00000	−2.08333333	−2.08333333	−2.08166600	−2.08796735
0.75	−2.00000	−2.18750000	−2.18750000	−2.17944947	−2.21691233
0.90	−2.00000	−2.27000000	−2.27000000	−2.25388553	−2.35114625
1.00	−2.00000	−2.33333333	−2.33333333	−2.30940108	−2.58578643

Here perturbation theory and the random-phase approximation may be compared directly, and the comparison is revealing. Both are close to the exact answer at small χ, with perturbation theory slightly the better of the two. As χ grows the two part company. The RPA collective frequency $\omega = \varepsilon\sqrt{1-\chi^2}$ of Section 7.7 goes soft as $\chi \to 1$, and since the correlation energy is $\frac{1}{2}(\sum_\nu \omega_\nu - \mathrm{Tr}A)$ the softening drives it steadily more negative: at $\chi = 1$ RPA gives -2.586 against an exact -2.309, an overshoot of 0.28. Beyond $\chi = 1$ the frequencies turn imaginary and the method stops altogether.

Perturbation theory does much better numerically – -2.333 at $\chi = 1$ – but it does so silently. It produces a perfectly ordinary number at the coupling where the mean field becomes unstable, and would go on producing ordinary numbers beyond it. RPA is less accurate here and more informative: its collapse is a diagnosis. Whether one prefers a method that fails loudly or one that fails quietly depends on whether an exact answer is available for comparison, and in a real calculation it is not.

9.10 The Hubbard model

For the Hubbard model of Section 4.3 we work in the momentum basis, where the plane-wave determinant is the Hartree-Fock solution (Section 5.7) and the first-order energy is the Hartree shift $UN_\uparrow N_\downarrow/L$. Six sites at half filling give $\Delta E^{(1)} = 3U/2$.

U/t	E_{ref}	MBPT2	MBPT3	MBPT4	FCI	CID
1	−6.50000	−6.60069444	−6.60069444	−6.60117251	−6.60115829	−6.59980159
2	−5.00000	−5.40277778	−5.40277778	−5.41042685	−5.40945685	−5.38877703
4	−2.00000	−3.61111111	−3.61111111	−3.73349623	−3.66870618	−3.41192866
8	4.00000	−2.44444444	−2.44444444	−4.40260631	−2.04813089	−0.36779217

Two features stand out. The third-order energy vanishes identically at this filling – the two middle columns agree to every digit, for every U – so we have included fourth order as well. This is special to the half-filled ring with a momentum-independent interaction; the pairing model of Section 9.6 has a perfectly good third-order term, and one should not expect the cancellation to survive a change of filling or of interaction.

The second feature is the familiar one. At $U/t = 1$ second order is already within 5×10^{-4} of the exact energy. At $U/t = 8$ fourth order overshoots it by more than a factor of two. This is the strong-correlation regime of Section 5.7 seen through the perturbation series: the reference determinant has ceased to be a good starting point, and the expansion about it fails.

9.11 The pairing plus particle-hole model

The model of Chapter 7 is the most interesting of the four, because its particle-hole term breaks pairs and therefore couples the reference to $1p$-$1h$ states. The second-order sum splits accordingly, and we may look at the two pieces separately:

g	f	E_{ref}	$E^{(2)}_{1p1h}$	$E^{(2)}_{2p2h}$	MBPT2	MBPT3	exact
0.50	0.025	1.45000	−0.00072917	−0.08856250	1.36070833	1.34684681	1.34713265
0.70	0.050	1.20000	−0.00291667	−0.18800000	1.00908333	0.96643444	0.96975878
1.00	0.050	0.90000	−0.00291667	−0.35425000	0.54283333	0.43194111	0.45058234
1.00	0.500	0.00000	−0.29166667	−1.30000000	−1.59166667	−2.25388889	−1.69173670

Table 9.4 Perturbation theory for the pairing plus particle-hole model, with the second-order energy split by the excitation rank of the intermediate state.

The singles contribution is small but non-zero, and it is entirely an artefact of our choice of basis: in a Hartree-Fock basis it would vanish identically by Brillouin's theorem. This is one more reason – beyond the variational one of Chapter 6 – for doing perturbation theory on a Hartree-Fock reference.

For the three weakly coupled rows the series behaves well, third order sitting within a few parts in a thousand of the exact answer. The last row is the interesting one. With $f = 0.5g$ the particle-hole term is strong, and third order overshoots by 0.56. Chapter 7 found RPA overshooting the same row in the same direction, -1.73461 against -1.69174, but by only 0.04. Summing a restricted class of contributions to all orders is, here, an order of magnitude safer than summing all of them to third order.

9.12 Does the partition matter?

Nothing forced our choice of $\hat{H}_0$, and it is worth checking how much the answer depends on it. Taking the whole diagonal of $\hat{H}$ instead of its one-body part gives the Epstein-Nesbet partition.

For three of our four models the two partitions give *identical* results, and the reason is worth seeing. The diagonal of $\hat{H}_I$ is the same on every determinant the reference can reach, so the two choices of $\hat{H}_0$ differ by a constant, which cancels in every energy denominator:

model	spread of $\text{diag}(\hat{H}_I)$
pairing	1.000000
Lipkin	0.000000
Hubbard	0.000000
pairing + particle-hole	2.000000

For the pairing model the spread is non-zero over the whole space, but the interaction conserves seniority, so every determinant the reference reaches has the same number of complete pairs and hence the same diagonal. Only the pairing plus particle-hole model, whose particle-hole term breaks pairs and contributes $-f$ per complete pair, has a genuinely varying diagonal within the active space. There the two partitions do disagree:

g	f	one-body $+\Delta E^{(2)}$	one-body $+\Delta E^{(3)}$	Epstein-Nesbet $+\Delta E^{(2)}$	Epstein-Nesbet $+\Delta E^{(3)}$	exact
0.50	0.25	0.60208333	0.51930556	0.62435065	0.51885823	0.55119157
1.00	0.50	−1.59166667	−2.25388889	−1.44791667	−2.25694444	−1.69173670
1.50	0.75	−4.58125000	−6.81625000	−4.17225275	−6.82273352	−4.34336785

Both partitions sum to the same exact answer, but at any finite order they disagree, and neither is uniformly better: at second order Epstein-Nesbet wins the last row and loses the first two, and at third order the two are almost indistinguishable and both are very poor. There is no theorem saying one partition is better than another. The choice is part of the approximation, and a perturbative number quoted without its partition means nothing.

9.13 Everything against everything

The pairing model is the one system every chapter has now treated, so we collect the results.

g	HF	MBPT2	MBPT3	CID	HF + RPA	FCI
0.25	1.75000	1.731771	1.730469	1.730507	1.716347	1.730460
0.50	1.50000	1.427083	1.416667	1.417596	1.374502	1.416774
1.00	1.00000	0.708333	0.625000	0.648907	0.554589	0.635548
1.50	0.50000	−0.156250	−0.437500	−0.291132	−0.406248	−0.348191
2.00	0.00000	−1.166667	−1.833333	−1.352087	−1.476223	−1.489652
errors						
0.25		1.3×10^{-3}	8.9×10^{-6}	4.7×10^{-5}	-1.4×10^{-2}	
0.50		1.0×10^{-2}	-1.1×10^{-4}	8.2×10^{-4}	-4.2×10^{-2}	
1.00		7.3×10^{-2}	-1.1×10^{-2}	1.3×10^{-2}	-8.1×10^{-2}	
1.50		1.9×10^{-1}	-8.9×10^{-2}	5.7×10^{-2}	-5.8×10^{-2}	
2.00		3.2×10^{-1}	-3.4×10^{-1}	1.4×10^{-1}	1.3×10^{-2}	

Table 9.5 The pairing model treated by every method of the book so far. Hartree-Fock from Chapter 6, doubles configuration interaction from Chapter 5, the random-phase approximation from Chapter 7, and full configuration interaction as the exact reference.

The errors are where the interest lies. Hartree-Fock does nothing at all, since the pairing interaction cannot couple the reference to a $1p$-$1h$ state. Beyond that, the ordering changes with the coupling and no method wins everywhere. At $g = 0.25$ third-order perturbation theory is best by three orders of magnitude, and the random-phase approximation is the worst of the four. At $g = 2$ the ordering has reversed completely: RPA is best, and the perturbation series – which by then is diverging – is the worst. Doubles configuration interaction is the only one of the four that is variational, so it alone is guaranteed to stay above the exact answer, and it alone degrades gracefully.

None of this is an argument for one method over another. It is an argument for knowing which regime one is in, and for having more than one method available.

9.14 Summary

Perturbation theory begins from the exact identity $\Delta E = \langle \Phi_0 | \hat{H}_I | \Psi_0 \rangle$ and turns it into a series by iterating the resolvent equation (9.11). The free parameter ω chooses the scheme. Setting $\omega = E$ gives Brillouin-Wigner, whose denominators contain the exact energy, which must therefore be found self-consistently, and whose full resummation is exact. Setting $\omega = W_0$ gives Rayleigh-Schrödinger, whose denominators are known in advance but whose higher orders carry renormalisation terms.

Those renormalisation terms are the whole point. They cancel the unlinked contributions exactly, which makes Rayleigh-Schrödinger size extensive at every order – verified here to machine precision – while Brillouin-Wigner is not. That single property decides the matter for many-body physics.

Applied to four models, the series behaves as its construction demands: very accurate when the reference determinant is good, and worthless when it is not. It is not variational, so it may approach the exact answer from either side and cross it; and it need not converge at all, as the pairing model demonstrates beyond $g \approx 1.5$, with no warning visible in the first three orders.

What perturbation theory offers in exchange is a systematic, order-by-order recipe requiring no diagonalisation, applicable where configuration interaction is far out of reach, and – crucially – correct in its scaling with system size. The next step is to keep that property while summing selected contributions to all orders, which is what coupled-cluster theory does.

The program `mbpt.py` performs every calculation of this chapter. The accompanying notebook runs it cell by cell.

9.15 Exercises

Exercise 1: the exact starting point

a) Derive Eq. (9.5), stating where intermediate normalisation and the hermiticity of $\hat{H}$ and $\hat{H}_0$ are used.
b) Show that $\hat{P}$ and $\hat{Q}$ of Eq. (9.6) are idempotent, orthogonal and commute with $\hat{H}_0$, and prove Eq. (9.10).
c) Obtain Eq. (9.11) from Eq. (9.8), and explain why it is exact but useless as it stands.
d) Why does the iteration (9.12) start from $|\Phi_0\rangle$ rather than from any other determinant?

Answer to a).

Projecting $\hat{H}|\Psi_0\rangle = E|\Psi_0\rangle$ onto $\langle\Phi_0|$ gives $\langle\Phi_0|\hat{H}|\Psi_0\rangle = E\langle\Phi_0|\Psi_0\rangle$, and intermediate normalisation sets the overlap to one, so the right-hand side is E. Similarly $\langle\Psi_0|\hat{H}_0|\Phi_0\rangle = W_0$. Since both operators are Hermitian we may read the second as $\langle\Phi_0|\hat{H}_0|\Psi_0\rangle = W_0$ and subtract: $\langle\Phi_0|\hat{H} - \hat{H}_0|\Psi_0\rangle = E - W_0$, which is Eq. (9.5). Without hermiticity the two matrix elements could not be identified and the derivation fails – which is a real restriction for non-Hermitian formulations such as coupled-cluster theory.

Answer to b).

$\hat{P}^2 = |\Phi_0\rangle\langle\Phi_0||\Phi_0\rangle\langle\Phi_0| = \hat{P}$ by normalisation, and $\hat{Q}^2 = \hat{Q}$ by orthonormality of the Φ_m; $\hat{P}\hat{Q} = 0$ since the reference is excluded from the sum. Both are built from eigenstates of $\hat{H}_0$, so $\hat{H}_0\hat{Q} = \sum_m W_m|\Phi_m\rangle\langle\Phi_m| = \hat{Q}\hat{H}_0$. Any function of $\hat{H}_0$ is therefore also diagonal in the same basis, and $\hat{Q}f(\hat{H}_0)\hat{Q} = \hat{Q}^2 f(\hat{H}_0) = \hat{Q}f(\hat{H}_0)$, which is Eq. (9.10).

Answer to c).

Divide Eq. (9.8) by $(\omega - \hat{H}_0)$, multiply from the left by $\hat{Q}$, use Eq. (9.10), and add $|\Phi_0\rangle = \hat{P}|\Psi_0\rangle$ from Eq. (9.7). It is exact because no term has been dropped. It is useless because it is one equation in two unknowns, $|\Psi_0\rangle$ and E, with the unknown state appearing on both sides.

Answer to d).

Any starting vector would generate a formally correct series, but the iteration converges usefully only if the starting vector already has a large overlap with the target. $|\Phi_0\rangle$ is chosen precisely because it is assumed to dominate $|\Psi_0\rangle$ – and when that assumption fails, as at strong coupling in Section 9.7, the series fails with it.

Exercise 2: the two schemes

a) Set $\omega = E$ in Eq. (9.13) and obtain the Brillouin-Wigner series (9.14).
b) Derive the resummation (9.16) using the geometric series (9.15), and explain why it is exact.
c) Set $\omega = W_0$ and obtain Eq. (9.17); then show that Eq. (9.18) reduces it to Eq. (9.19).
d) Expand Eq. (9.19) order by order and derive Eqs. (9.21)–(9.23). At which order does the renormalisation term first appear, and why not earlier?
e) Verify that the recursion (9.28) reproduces $\Delta E^{(2)}$ and $\Delta E^{(3)}$.

Answer to a).

With $\omega = E$ the factor $(\omega - E + \hat{H}_I)$ collapses to $\hat{H}_I$, and the i-th term of Eq. (9.13) becomes i factors of $\hat{H}_I\hat{Q}/(E - \hat{H}_0)$ acting on $\hat{H}_I|\Phi_0\rangle$, which is Eq. (9.14).

Answer to b).

Identify the bracketed sum in Eq. (9.15) as the expansion of $(\hat{e} - \hat{Q}\hat{H}_I\hat{Q})^{-1}$ in powers of $\hat{Q}\hat{H}_I\hat{Q}$, with $\hat{e} = E - \hat{H}_0$. Substituting it into Eq. (9.14) collapses the whole series into Eq. (9.16). It is exact because the geometric series has been resummed, not truncated: no term of Eq. (9.14) has been discarded. Equivalently, Eq. (9.16) is what one gets by solving the Schrödinger equation exactly in the $\hat{Q}$ space and substituting the result back, which is a Feshbach projection and involves no approximation at all.

Answer to c).

With $\omega = W_0$ the factor becomes $(W_0 - E + \hat{H}_I) = (\hat{H}_I - \Delta E)$ since $\Delta E = E - W_0$, giving Eq. (9.17). The rightmost factor always acts directly on $|\Phi_0\rangle$; there ΔE is a number and $\hat{Q}|\Phi_0\rangle = 0$, so that piece vanishes and each term loses its final ΔE, leaving Eq. (9.19).

Answer to d).

Count powers of $\hat{H}_I$, remembering that ΔE is itself of first order and higher. At order one only $\langle\Phi_0|\hat{H}_I|\Phi_0\rangle$ survives. At order two only the second term of Eq. (9.19) contributes, since the ΔE in the third term would make it third order. At order three the third term contributes both with $\hat{H}_I$ in the middle – the principal term – and with $-\Delta E^{(1)}$ in the middle, which gives $-V_{00}\sum_m V_{0m}V_{m0}/D_m^2$. The renormalisation term cannot appear earlier because it needs at least two resolvents, hence at least three interactions.

Answer to e).

With $|\Psi^{(0)}\rangle = |\Phi_0\rangle$ the recursion gives $|\Psi^{(1)}\rangle = \hat{R}\hat{H}_I|\Phi_0\rangle$, whence $\Delta E^{(2)} = \langle\Phi_0|\hat{H}_I\hat{R}\hat{H}_I|\Phi_0\rangle$, which is Eq. (9.22). At the next step

$$|\Psi^{(2)}\rangle = \hat{R}\left[\hat{H}_I\hat{R}\hat{H}_I - \Delta E^{(1)}\hat{R}\hat{H}_I\right]|\Phi_0\rangle, \tag{9.37}$$

and projecting with $\langle\Phi_0|\hat{H}_I$ gives exactly the two terms of Eq. (9.23), the second carrying $\hat{R}^2$ as required. The program checks the agreement numerically for all four models.

Exercise 3: the models to third order

a) For the pairing model show that $\Delta E^{(1)} = -g$ and that the energy through first order is the Hartree-Fock energy of Section 6.12.
b) For the pairing model with four levels and two pairs, evaluate $\Delta E^{(2)}$ by hand and compare with $-\frac{7}{24}g^2$, obtained in Exercise 5.13d).
c) For the Lipkin model derive Eq. (9.36) and prove that $\Delta E^{(3)} = 0$ identically.
d) Explain why the Hubbard model at half filling has $\Delta E^{(3)} = 0$ but $\Delta E^{(4)} \neq 0$, and why this should not be expected in general.
e) For the pairing plus particle-hole model, show that the $1p$-$1h$ part of $\Delta E^{(2)}$ vanishes in a Hartree-Fock basis.

Answer to a).

The pairing interaction $-\frac{g}{2}\sum_{pq}\hat{a}^\dagger_{p+}\hat{a}^\dagger_{p-}\hat{a}_{q-}\hat{a}_{q+}$ has a diagonal contribution only from $p = q$, giving $-\frac{g}{2}$ for each occupied pair. The reference has two, so $\Delta E^{(1)} = -g$ and $W_0 + \Delta E^{(1)} = 2 - g$. Section 6.12 found the Hartree-Fock energy to be exactly the reference energy, because $\langle\Phi_0|\hat{H}|\Phi_i^a\rangle = 0$ by seniority conservation. The two agree, as they must: first-order perturbation theory *is* the mean-field energy when the reference is the Hartree-Fock determinant.

Answer to b).

Only the four seniority-zero determinants reachable by moving one pair contribute. Their unperturbed excitation energies relative to the reference are $2, 4, 4, 6$ and each matrix element is $-g/2$, so

$$\Delta E^{(2)} = \frac{g^2}{4}\left(\frac{1}{-2}+\frac{1}{-4}+\frac{1}{-4}+\frac{1}{-6}\right) = -\frac{7}{24}g^2, \tag{9.38}$$

which is Eq. (5.36). At $g = 0.5$ this gives -0.0729, and the program's numerical value is -0.07291667.

Answer to c).

The reference is $|J,-J\rangle$ with $J = N/2$. The operator $(V/2)\hat{J}_+^2$ takes it to $|J,-J+2\rangle$ with matrix element

$$\frac{V}{2}\sqrt{J(J+1)-(-J)(-J+1)}\,\sqrt{J(J+1)-(-J+1)(-J+2)} = \frac{V}{2}\sqrt{2N(N-1)}, \tag{9.39}$$

and the unperturbed excitation energy is 2ε. Hence

$$\Delta E^{(2)} = -\frac{V^2\, 2N(N-1)/4}{2\varepsilon} = -\frac{N(N-1)V^2}{4\varepsilon}. \tag{9.40}$$

For the third order, every intermediate state reachable from the reference has $J_z = -J+2$; but $\hat{H}_I$ changes J_z by exactly two units, so $V_{mn} = 0$ between any two of them and the principal term vanishes. The renormalisation term carries $V_{00} = 0$. Hence $\Delta E^{(3)} = 0$ exactly, and the vanishing is a consequence of the selection rule, not an accident of the parameters.

Answer to d).

The interaction is momentum-independent and couples only opposite spins, and at half filling the ring is particle-hole symmetric. Under these conditions the third-order contributions cancel among themselves, and the program finds $\Delta E^{(3)}$ to be zero to machine precision for every U. The cancellation is a property of this filling and this interaction: change either – or take the pairing model, where the third order is a perfectly ordinary non-zero number – and it disappears. The fourth order is not protected by the same argument and is not zero.

Answer to e).

The $1p$-$1h$ part of $\Delta E^{(2)}$ is $\sum_{ai} |\langle \Phi_0 | \hat{H}_I | \Phi_i^a \rangle|^2 / D$, and by Brillouin's theorem, Eq. (6.31), every numerator vanishes in the Hartree-Fock basis. In the unperturbed oscillator basis used in Table 9.4 it does not, and the residue is a measure of how far the reference is from the Hartree-Fock solution. This is the practical argument for building perturbation theory on a Hartree-Fock reference: it removes an entire class of contributions at every order.

Exercise 4: size extensivity

a) For two non-interacting subsystems, show that the second-order energy is exactly additive.
b) Show that the principal term of $\Delta E^{(3)}$ alone is *not* additive, and that adding the renormalisation term makes it so.
c) Explain why Brillouin-Wigner theory fails this test at every truncation order.
d) Reproduce Table 9.3 and extend it to three and four subsystems. How does the Brillouin-Wigner error scale?
e) Relate all of this to the size-consistency failure of truncated configuration interaction in Section 5.9.

Answer to a).

With $\hat{H} = \hat{H}_A + \hat{H}_B$ and no coupling, the reference is a product and any singly excited intermediate state lives entirely in A or entirely in B. The sum in Eq. (9.25) therefore splits into two disjoint sums, and the denominators of the A states involve only A excitation energies. Hence $\Delta E^{(2)}(AB) = \Delta E^{(2)}(A) + \Delta E^{(2)}(B)$.

Answer to b).

At third order the intermediate states m and n may lie in different subsystems: m an excitation of A and n an excitation of B. Such a term is *unlinked* – no interaction connects the two – and it contributes V_{00} times a product of A and B second-order-like factors, which scales as the square of the system size. Written out, these are exactly the terms $V_{00}\sum_m V_{0m}V_{m0}/D_m^2$, and the renormalisation term subtracts them identically. What survives is linked and additive. The program verifies additivity to machine precision through fourth order.

Answer to c).

Brillouin-Wigner denominators contain the exact energy $E = W_0 + \Delta E$, and ΔE is extensive. Doubling the system therefore changes every denominator, so the two-subsystem sum cannot reduce to twice the one-subsystem sum, at any truncation order. There is no renormalisation term available to repair this because the scheme has none: that was the whole point of setting $\omega = E$.

Answer to d).

The extension is a one-line change to the demonstration. The Brillouin-Wigner error grows roughly linearly with the number of subsystems for weak coupling, so the *energy per subsystem* drifts – which is the practical statement of the failure. For a calculation on bulk matter, where the number of subsystems is effectively infinite, no useful answer survives.

Answer to e).

They are the same defect wearing two hats. Truncated configuration interaction fails because a product of two double excitations is a quadruple, which the truncation discards; Brillouin-Wigner fails because its denominators do not factorise. In both cases the symptom is an energy that does not scale linearly with the system size, and in both cases the cure is to make the ansatz multiplicative – $e^{\hat{T}}$ instead of $1+\hat{C}$, Eq. (5.29) – which is coupled-cluster theory. Rayleigh-Schrödinger achieves the same scaling order by order through the cancellation of unlinked terms, which is why it, and not Brillouin-Wigner, is the perturbative starting point for many-body theory.

Exercise 5: numerical explorations

a) Using `mbpt.py`, reproduce Table 9.1 and extend it to sixth order. Does every additional order help?
b) Reproduce Table 9.2 and locate the radius of convergence of the pairing series as accurately as you can. Why is this harder than it looks?
c) Compare the two partitions of Section 9.12 for the pairing plus particle-hole model over a range of f/g. Is there a regime where one is clearly preferable?
d) For the Hubbard model, verify that $\Delta E^{(3)} = 0$ and check whether it survives a change of filling or the addition of a next-nearest-neighbour hopping.
e) Implement a Padé resummation of the pairing series and see whether it extends the useful range beyond the radius of convergence.

Answer.

A few pointers. In a) the answer is no: at $g = 1$ the fourth-order term overshoots and the fifth brings it back, the partial sums oscillating about the exact value – convergence is not monotone and no individual order is guaranteed to improve matters. In b) the difficulty is that near the radius of convergence the terms decrease so slowly that any finite number of them is consistent with either behaviour; the ratio test needs terms of very high order, and those are contaminated by round-off once they have fallen many orders of magnitude below the leading ones. In c) neither partition wins uniformly, but Epstein-Nesbet is the more stable at second order when the diagonal interaction is large, which is the regime it was designed for. In d) the cancellation is fragile: it does not survive a filling that breaks particle-hole symmetry. In e) Padé approximants do extend the useful range, sometimes dramatically, because they can represent the singularity that limits the Taylor series – which is a reminder that a divergent series is not necessarily a useless one.

Chapter 10
Coupled cluster theory

Three defects have followed us through the last five chapters. Truncated configuration interaction, Section 5.9, is variational but not size consistent, so it becomes useless for large systems. Perturbation theory, Chapter 9, is size extensive but not variational, and its series need not converge. The random-phase approximation, Chapter 7, sums one class of contributions to all orders but collapses when the mean field goes unstable.

Coupled-cluster theory repairs the first two at once. It is size extensive by construction, it sums infinite classes of contributions, it is systematically improvable by raising the excitation rank, and its cost grows as a modest power of the basis size rather than exponentially. It is not variational – that is the price – but in practice this matters far less than one might fear. It is, by a wide margin, the most successful many-body method in use: nuclei up to lead have been computed with it, and in quantum chemistry it is the standard against which everything else is measured.

The idea is a single change of ansatz. Where configuration interaction writes $|\Psi\rangle = (1+\hat{C})|\Phi_0\rangle$, coupled cluster writes $|\Psi\rangle = e^{\hat{T}}|\Phi_0\rangle$. Everything follows from that exponential, and this chapter follows it through: the similarity-transformed Hamiltonian, the reason its expansion terminates exactly, the projected equations, and then full derivations of the doubles and the singles-and-doubles amplitude equations. We work algebraically throughout, as in Chapter 9; the diagrammatic language is a separate subject. The pairing model of Section 4.2 serves as the worked example, and we compare against every method the book has developed.

10.1 The exponential ansatz

We start, as always, from a reference determinant

$$|\Phi_0\rangle = \prod_{i\leq F} \hat{a}_i^\dagger |0\rangle, \tag{10.1}$$

usually but not necessarily the Hartree-Fock state of Chapter 6. The exact state is written

$$\boxed{|\Psi\rangle = e^{\hat{T}}|\Phi_0\rangle,} \tag{10.2}$$

with the *cluster operator*

$$\hat{T} = \hat{T}_1 + \hat{T}_2 + \cdots + \hat{T}_N, \tag{10.3}$$

$$\hat{T}_1 = \sum_{ia} t_i^a \, \hat{a}_a^\dagger \hat{a}_i, \tag{10.4}$$

$$\hat{T}_2 = \frac{1}{4} \sum_{ijab} t_{ij}^{ab} \, \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_j \hat{a}_i, \tag{10.5}$$

$$\hat{T}_3 = \frac{1}{36} \sum_{ijkabc} t_{ijk}^{abc} \, \hat{a}_a^\dagger \hat{a}_b^\dagger \hat{a}_c^\dagger \hat{a}_k \hat{a}_j \hat{a}_i, \tag{10.6}$$

and so on. The factors $1/4$ and $1/36$ compensate the repeated counting of antisymmetric index pairs, $(2!)^2$ and $(3!)^2$. The amplitudes are antisymmetric under interchange of the occupied or of the virtual indices.

Two features of $\hat{T}$ decide everything that follows.

The cluster operators commute.

Every term of $\hat{T}$ consists of particle creation and hole annihilation operators only – $\hat{a}_a^\dagger$ with $a > F$ and $\hat{a}_i$ with $i \leq F$ – and never the reverse. The same argument as in Eq. (6.35) therefore gives

$$\left[\hat{T}_m, \hat{T}_n\right] = 0 \tag{10.7}$$

for all m, n. This is what makes the exponential tractable.

Excitations cannot be undone.

Because $\hat{T}$ only excites, $\hat{T}^n$ produces at least n particle-hole pairs, and acting on a state with N particles it must vanish for large enough n. The exponential is therefore a finite sum in any given matrix element.

Why exponentiate at all?

Expanding Eq. (10.2),

$$e^{\hat{T}} = 1 + \hat{T}_1 + \left(\hat{T}_2 + \tfrac{1}{2}\hat{T}_1^2\right) + \left(\hat{T}_3 + \hat{T}_1\hat{T}_2 + \tfrac{1}{6}\hat{T}_1^3\right) + \cdots, \tag{10.8}$$

we see that the coefficient of a $2p$-$2h$ determinant is $t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a$, and so on. Comparing with the configuration-interaction expansion $|\Psi\rangle = (1 + \hat{C}_1 + \hat{C}_2 + \cdots)|\Phi_0\rangle$ of Eq. (5.5), the two are related by

$$\hat{C}_1 = \hat{T}_1, \qquad \hat{C}_2 = \hat{T}_2 + \tfrac{1}{2}\hat{T}_1^2, \qquad \hat{C}_3 = \hat{T}_3 + \hat{T}_1\hat{T}_2 + \tfrac{1}{6}\hat{T}_1^3, \tag{10.9}$$

and so forth. The exponential does not add anything the linear expansion cannot represent; the two are equivalent if both are complete. What it does is *reorganise*: the high-rank coefficients are built from products of low-rank ones rather than treated as independent. When $\hat{T}$ is truncated, that reorganisation is exactly what survives.

This is the point. Truncating $\hat{C}$ at doubles discards the quadruple excitations entirely, and Section 5.9 showed that this destroys size consistency because a product of two doubles on separate subsystems *is* a quadruple. Truncating $\hat{T}$ at doubles keeps that quadruple, as $\frac{1}{2}\hat{T}_2^2$, with its amplitude fixed by the doubles rather than free. Nothing else needs to be said: the disease of Chapter 5 is cured by construction.

10.2 The similarity-transformed Hamiltonian

Insert Eq. (10.2) into the Schrödinger equation $\hat{H}|\Psi\rangle = E|\Psi\rangle$ and multiply from the left by $e^{-\hat{T}}$:

$$\boxed{\overline{H}|\Phi_0\rangle = E|\Phi_0\rangle, \qquad \overline{H} \equiv e^{-\hat{T}}\hat{H}e^{\hat{T}}.} \tag{10.10}$$

This is worth pausing on. A similarity transformation does not change the spectrum – if $\hat{H}|E\rangle = E|E\rangle$ then $\overline{H}\,e^{-\hat{T}}|E\rangle = E\,e^{-\hat{T}}|E\rangle$ – so $\overline{H}$ has exactly the same eigenvalues as $\hat{H}$. What the transformation does is move the correlation out of the wave function and into the operator. Coupled cluster is the search for a transformation that makes the *uncorrelated* reference an exact eigenstate.

Because $e^{\hat{T}}$ is not unitary – $\hat{T}$ is not anti-Hermitian – $\overline{H}$ is not Hermitian. That is a genuine cost: left and right eigenvectors differ, and expectation values require both. It also brings a large benefit, as we are about to see.

Normal-ordering with respect to the reference, Section 3.5, splits the Hamiltonian as $\hat{H} = E_{\text{ref}} + \hat{H}_N$ with

$$\hat{H}_N = \sum_{pq} f_{pq}\left\{\hat{a}_p^\dagger \hat{a}_q\right\} + \frac{1}{4}\sum_{pqrs}\langle pq\|rs\rangle\left\{\hat{a}_p^\dagger \hat{a}_q^\dagger \hat{a}_s \hat{a}_r\right\}, \tag{10.11}$$

where f_{pq} is the Fock matrix of Eq. (6.25) and $\langle pq\|rs\rangle$ the antisymmetrised two-body element. Since E_{ref} is a number it commutes with $\hat{T}$, so

$$\overline{H} = E_{\text{ref}} + \overline{H}_N, \qquad \overline{H}_N = e^{-\hat{T}}\hat{H}_N e^{\hat{T}}, \tag{10.12}$$

and Eq. (10.10) becomes $\overline{H}_N|\Phi_0\rangle = \Delta E|\Phi_0\rangle$ with $\Delta E = E - E_{\text{ref}}$ the correlation energy.

10.3 Why the expansion terminates

The Baker-Hausdorff lemma of Exercise 6.15d) gives

$$\overline{H}_N = \hat{H}_N + \left[\hat{H}_N, \hat{T}\right] + \frac{1}{2!}\left[\left[\hat{H}_N, \hat{T}\right], \hat{T}\right] + \frac{1}{3!}\left[\left[\left[\hat{H}_N, \hat{T}\right], \hat{T}\right], \hat{T}\right] + \cdots \tag{10.13}$$

For a general operator this is an infinite series. Here it is not, and the reason is Eq. (10.7).

Because the cluster operators commute with one another, no commutator in Eq. (10.13) can be non-zero by a $\hat{T}$ acting on another $\hat{T}$: every $\hat{T}$ must contract with $\hat{H}_N$ directly. In the language of Section 3.6, each $\hat{T}$ must be *connected* to $\hat{H}_N$. But $\hat{H}_N$ contains at most four operators that can be contracted – two annihilation operators and two creation operators in its two-body part – so at most four cluster operators can attach to it. Hence

$$\boxed{\overline{H}_N = \hat{H}_N + \left[\hat{H}_N, \hat{T}\right] + \frac{1}{2!}\left[\left[\hat{H}_N, \hat{T}\right], \hat{T}\right] + \frac{1}{3!}[\cdots] + \frac{1}{4!}[\cdots],} \tag{10.14}$$

and the series stops *exactly* at the fourfold commutator. Every term is connected, which is why one writes $\overline{H}_N = (\hat{H}_N e^{\hat{T}})_c$ with the subscript denoting connected terms only.

This is the technical heart of the method, and it is worth appreciating what has been bought. The non-Hermiticity of $\overline{H}_N$, which looked like a defect, is precisely what makes the expansion finite: a unitary transformation with anti-Hermitian $\hat{T}$ would contain de-excitation operators, those would not commute among themselves, and the series would never terminate. Unitary coupled-cluster theory, which we meet again in the quantum-computing chapters, pays exactly that price.

10.4 The coupled-cluster equations

Equation (10.10) says $|\Phi_0\rangle$ is an eigenstate of $\overline{H}$. Projecting it onto the reference and onto each excited determinant turns that statement into a computable set of conditions:

$$\Delta E = \langle \Phi_0 | \overline{H}_N | \Phi_0 \rangle, \tag{10.15}$$

$$0 = \langle \Phi_i^a | \overline{H}_N | \Phi_0 \rangle, \tag{10.16}$$

$$0 = \langle \Phi_{ij}^{ab} | \overline{H}_N | \Phi_0 \rangle, \tag{10.17}$$

and so on up the excitation ladder. The first line gives the energy once the amplitudes are known; the rest determine them.

Read physically, Eqs. (10.16) and (10.17) say that the similarity-transformed Hamiltonian has *no* matrix elements connecting the reference to $1p$-$1h$ or $2p$-$2h$ states. The reference has been decoupled from the simple excitations. It has not been decoupled from everything: with $\hat{T} = \hat{T}_1 + \hat{T}_2$, the operator $\overline{H}_N$ still connects $|\Phi_0\rangle$ to $3p$-$3h$ and $4p$-$4h$ states, and that residual coupling is the entire error of CCSD.

Why is CCD not exact? A similarity transformation preserves all eigenvalues, so one might expect the transformed problem to be exactly soluble. The resolution is that we are not solving the transformed problem exactly either. With $\hat{T} = \hat{T}_2$ the full $\overline{H}_N$ contains up to six-body terms – $\hat{H}_N$ has four contractible operators, and each $\hat{T}_2$ adds two more – and it is that full operator which would reproduce the exact energy. What we impose is only that the $2p$-$2h$ block of $\overline{H}_N$ vanishes. Coupled cluster at any finite truncation is therefore a similarity transformation *plus* a projection, and the projection is where the approximation lives.

10.5 The correlation energy

Evaluating Eq. (10.15) is the easiest part of the calculation. Only terms of $\overline{H}_N$ that de-excite by exactly the amount $\hat{T}$ excites can contribute, so at most two cluster operators can attach, and

$$\Delta E = \langle \Phi_0 | \hat{H}_N \left(1 + \hat{T}_1 + \hat{T}_2 + \tfrac{1}{2}\hat{T}_1^2 \right) | \Phi_0 \rangle_c. \tag{10.18}$$

The one-body part of $\hat{H}_N$ contracts with $\hat{T}_1$, the two-body part with $\hat{T}_2$ and with $\frac{1}{2}\hat{T}_1^2$, and nothing else survives:

$$\boxed{\Delta E = \sum_{ia} f_{ia} t_i^a + \frac{1}{4} \sum_{ijab} \langle ij \| ab \rangle t_{ij}^{ab} + \frac{1}{2} \sum_{ijab} \langle ij \| ab \rangle t_i^a t_j^b.} \tag{10.19}$$

In a Hartree-Fock basis $f_{ia} = 0$ by Brillouin's theorem, Eq. (6.31), and the first term disappears. The singles then enter the energy only through the last term, quadratically – which is why CCSD energies are often close to CCD energies even when the singles amplitudes are not small.

Notice also that Eq. (10.19) has exactly the form of the configuration-interaction correlation energy, Eq. (9.31), with C_{ij}^{ab} replaced by $t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a$. That combination will appear so often that it is given a name,

$$\tau_{ij}^{ab} = t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a, \tag{10.20}$$

and its half-weighted cousin

$$\tilde{\tau}_{ij}^{ab} = t_{ij}^{ab} + \tfrac{1}{2} \left(t_i^a t_j^b - t_i^b t_j^a \right). \tag{10.21}$$

10.6 Deriving the CCD equations

We derive the doubles equation first, with $\hat{T} = \hat{T}_2$ alone. This is the *coupled-cluster doubles* approximation, and it is exact for any system whose interaction cannot produce a $1p$-$1h$ excitation – the pairing model, nuclear matter on a momentum grid – and the leading approximation in a Hartree-Fock basis for everything else.

The equation to satisfy is $\langle \Phi_{ij}^{ab} | \overline{H}_N | \Phi_0 \rangle = 0$ with

$$\overline{H}_N = \left(\hat{H}_N e^{\hat{T}_2}\right)_c = \left(\hat{H}_N \left[1+\hat{T}_2+\tfrac{1}{2}\hat{T}_2^2\right]\right)_c. \tag{10.22}$$

The series stops at $\hat{T}_2^2$: a third $\hat{T}_2$ would need two more contractible operators in $\hat{H}_N$ than it has, once the two free particle and two free hole lines of the $2p$-$2h$ projection are accounted for.

We take the three terms in turn. Throughout we use the permutation operators

$$P(ij)X_{ij} = X_{ij} - X_{ji}, \qquad P(ab)X^{ab} = X^{ab} - X^{ba}, \tag{10.23}$$

and their product $P(ij)P(ab)X_{ij}^{ab} = X_{ij}^{ab} - X_{ji}^{ab} - X_{ij}^{ba} + X_{ji}^{ba}$, which enforce the antisymmetry of the amplitudes under interchange of the two occupied or the two virtual labels. Repeated indices are summed.

No cluster operator.

$\langle \Phi_{ij}^{ab}|\hat{H}_N|\Phi_0\rangle$ is the Condon-Slater matrix element between the reference and a double excitation, Section 3.13, which is simply

$$\langle ab\|ij\rangle. \tag{10.24}$$

This is the driving term: without it all amplitudes would vanish.

One cluster operator.

$\langle \Phi_{ij}^{ab}|\left(\hat{H}_N\hat{T}_2\right)_c|\Phi_0\rangle$ requires all four external lines of the projection to be accounted for, with $\hat{T}_2$ supplying two particle and two hole lines and $\hat{H}_N$ connecting to the rest. Enumerating the ways to contract, the one-body part of $\hat{H}_N$ can attach to a particle line or to a hole line,

$$P(ab)\sum_c f_{bc}\, t_{ij}^{ac} - P(ij)\sum_k f_{kj}\, t_{ik}^{ab}, \tag{10.25}$$

and the two-body part in three distinct ways – both particle lines, both hole lines, or one of each:

$$\frac{1}{2}\sum_{cd}\langle ab\|cd\rangle\, t_{ij}^{cd} + \frac{1}{2}\sum_{kl}\langle kl\|ij\rangle\, t_{kl}^{ab} + P(ij)P(ab)\sum_{kc}\langle kb\|cj\rangle\, t_{ik}^{ac}. \tag{10.26}$$

These three are the *particle-particle ladder*, the *hole-hole ladder* and the *ring* term. The factors $\frac{1}{2}$ compensate the double counting of the two equivalent contractions in each ladder; the ring has no such factor because the two lines are inequivalent, and instead carries the full permutation $P(ij)P(ab)$.

Two cluster operators.

$\frac{1}{2}\langle \Phi_{ij}^{ab}|\left(\hat{H}_N\hat{T}_2^2\right)_c|\Phi_0\rangle$ needs both $\hat{T}_2$ connected to $\hat{H}_N$, which forces the two-body part and uses all four of its contractible operators. Only $\langle kl\|cd\rangle$ with two hole and two particle indices can do this. There are four distinct topologies, according to how the four contractions are distributed between the two amplitudes:

$$
\begin{aligned}
&+\frac{1}{4}\sum_{klcd}\langle kl\|cd\rangle\, t_{ij}^{cd}\, t_{kl}^{ab} \\
&+P(ij)P(ab)\,\frac{1}{2}\sum_{klcd}\langle kl\|cd\rangle\, t_{ik}^{ac}\, t_{jl}^{bd} \\
&-P(ij)\,\frac{1}{2}\sum_{klcd}\langle kl\|cd\rangle\, t_{ik}^{cd}\, t_{lj}^{ab} \\
&-P(ab)\,\frac{1}{2}\sum_{klcd}\langle kl\|cd\rangle\, t_{kl}^{ac}\, t_{ij}^{db}. \qquad (10.27)
\end{aligned}
$$

The first is the product of a hole-hole and a particle-particle ladder; the second is a ring on each side; the third and fourth attach both indices of one amplitude to the interaction and leave the other amplitude as a spectator.

The result.

Collecting Eqs. (10.24)–(10.27) and writing the diagonal Fock elements on the left, the CCD amplitude equation is

$$
\begin{aligned}
D_{ij}^{ab}\, t_{ij}^{ab} = {}& \langle ab\|ij\rangle + P(ab)\sum_{c} f'_{bc}\, t_{ij}^{ac} - P(ij)\sum_{k} f'_{kj}\, t_{ik}^{ab} \\
&+\frac{1}{2}\sum_{cd}\langle ab\|cd\rangle\, t_{ij}^{cd} + \frac{1}{2}\sum_{kl}\langle kl\|ij\rangle\, t_{kl}^{ab} + P(ij)P(ab)\sum_{kc}\langle kb\|cj\rangle\, t_{ik}^{ac} \\
&+\frac{1}{4}\sum_{klcd}\langle kl\|cd\rangle\, t_{ij}^{cd} t_{kl}^{ab} + \frac{1}{2}P(ij)P(ab)\sum_{klcd}\langle kl\|cd\rangle\, t_{ik}^{ac} t_{jl}^{bd} \\
&-\frac{1}{2}P(ij)\sum_{klcd}\langle kl\|cd\rangle\, t_{ik}^{cd} t_{lj}^{ab} - \frac{1}{2}P(ab)\sum_{klcd}\langle kl\|cd\rangle\, t_{kl}^{ac} t_{ij}^{db}, \qquad (10.28)
\end{aligned}
$$

with the energy denominator

$$
D_{ij}^{ab} = f_{ii} + f_{jj} - f_{aa} - f_{bb} = \varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b \qquad (10.29)
$$

and f' the off-diagonal part of the Fock matrix, which vanishes in a canonical Hartree-Fock basis.

Equation (10.28) is a set of coupled quadratic equations, one for each amplitude, and it is solved by iteration: start from $t_{ij}^{ab} = \langle ab\|ij\rangle / D_{ij}^{ab}$, which is the second-order perturbation amplitude of Eq. (9.32), evaluate the right-hand side, divide by D, and repeat.

Intermediates.

Evaluated naively, the quadratic terms cost $\mathscr{O}(n^8)$. Factoring them through intermediates that are built once per iteration reduces this to $\mathscr{O}(n^6)$, which is what makes the method usable. Defining

$$
\begin{aligned}
\chi_{kj} &= \frac{1}{2}\sum_{lcd}\langle kl\|cd\rangle\, t_{jl}^{cd}, & \chi_{cb} &= -\frac{1}{2}\sum_{kld}\langle kl\|cd\rangle\, t_{kl}^{bd}, \\
\chi_{klij} &= \frac{1}{2}\sum_{cd}\langle kl\|cd\rangle\, t_{ij}^{cd}, & \chi_{bkcj} &= \frac{1}{2}\sum_{ld}\langle kl\|cd\rangle\, t_{lj}^{db}, \qquad (10.30)
\end{aligned}
$$

the last four lines of Eq. (10.28) become

$$
-P(ij)\sum_{k} t_{ik}^{ab}\chi_{kj} + P(ab)\sum_{c} t_{ij}^{ac}\chi_{cb} + \frac{1}{2}\sum_{kl} t_{kl}^{ab}\chi_{klij} + P(ij)P(ab)\sum_{kc} t_{ik}^{ac}\chi_{bkcj}, \qquad (10.31)
$$

each a single matrix multiplication. This is how the program implements it, and how every production code does.

10.7 Deriving the CCSD equations

With $\hat{T} = \hat{T}_1 + \hat{T}_2$ the derivation is the same in principle and far longer in practice, because $\hat{T}_1$ can appear up to four times. We give the result in the standard factored form, which is both the clearest statement and the one that is implemented.

The organising trick is to collect the recurring combinations into intermediates. Three carry two indices,

$$F_{ae} = f'_{ae} - \frac{1}{2}\sum_m f_{me} t_m^a + \sum_{mf} t_m^f \langle ma\|fe\rangle - \frac{1}{2}\sum_{mnf} \tilde{\tau}_{mn}^{af} \langle mn\|ef\rangle, \tag{10.32}$$

$$F_{mi} = f'_{mi} + \frac{1}{2}\sum_e t_i^e f_{me} + \sum_{en} t_n^e \langle mn\|ie\rangle + \frac{1}{2}\sum_{nef} \tilde{\tau}_{in}^{ef} \langle mn\|ef\rangle, \tag{10.33}$$

$$F_{me} = f_{me} + \sum_{nf} t_n^f \langle mn\|ef\rangle, \tag{10.34}$$

and three carry four,

$$W_{mnij} = \langle mn\|ij\rangle + P(ij)\sum_e t_j^e \langle mn\|ie\rangle + \frac{1}{4}\sum_{ef} \tau_{ij}^{ef} \langle mn\|ef\rangle, \tag{10.35}$$

$$W_{abef} = \langle ab\|ef\rangle - P(ab)\sum_m t_m^b \langle am\|ef\rangle + \frac{1}{4}\sum_{mn} \tau_{mn}^{ab} \langle mn\|ef\rangle, \tag{10.36}$$

$$\begin{aligned} W_{mbej} = {} & \langle mb\|ej\rangle + \sum_f t_j^f \langle mb\|ef\rangle - \sum_n t_n^b \langle mn\|ej\rangle \\ & - \sum_{nf} \left(\tfrac{1}{2} t_{jn}^{fb} + t_j^f t_n^b\right) \langle mn\|ef\rangle. \end{aligned} \tag{10.37}$$

These are the intermediates of Stanton, Gauss, Watts and Bartlett, and their structure repays a moment's attention: each is a bare integral, plus corrections in which the amplitudes dress it. F_{ae} and F_{mi} are the Fock matrix dressed by correlation; W_{mnij} and W_{abef} are the hole-hole and particle-particle ladders dressed likewise; and W_{mbej} is the dressed ring.

With them, the amplitude equations take a compact form. The singles:

$$\begin{aligned} D_i^a t_i^a = {} & f_{ia} + \sum_e t_i^e F_{ae} - \sum_m t_m^a F_{mi} + \sum_{me} t_{im}^{ae} F_{me} \\ & - \sum_{nf} t_n^f \langle na\|if\rangle - \frac{1}{2}\sum_{mef} t_{im}^{ef} \langle ma\|ef\rangle - \frac{1}{2}\sum_{men} t_{mn}^{ae} \langle nm\|ei\rangle, \end{aligned} \tag{10.38}$$

with $D_i^a = \varepsilon_i - \varepsilon_a$. The doubles:

$$
\begin{aligned}
D_{ij}^{ab} t_{ij}^{ab} = \langle ij \| ab \rangle &+ P(ab) \sum_e t_{ij}^{ae} \left(F_{be} - \tfrac{1}{2} \sum_m t_m^b F_{me} \right) \\
&- P(ij) \sum_m t_{im}^{ab} \left(F_{mj} + \tfrac{1}{2} \sum_e t_j^e F_{me} \right) \\
&+ \frac{1}{2} \sum_{mn} \tau_{mn}^{ab} W_{mnij} + \frac{1}{2} \sum_{ef} \tau_{ij}^{ef} W_{abef} \\
&+ P(ij) P(ab) \sum_{me} \left(t_{im}^{ae} W_{mbej} - t_i^e t_m^a \langle mb \| ej \rangle \right) \\
&+ P(ij) \sum_e t_i^e \langle ab \| ej \rangle - P(ab) \sum_m t_m^a \langle mb \| ij \rangle . \qquad (10.39)
\end{aligned}
$$

Reading the equations.

Several structural features are worth noting, and each connects to an earlier chapter.

Setting all amplitudes on the right to zero leaves $D_{ij}^{ab} t_{ij}^{ab} = \langle ij \| ab \rangle$, which is the first-order perturbation amplitude, and $D_i^a t_i^a = f_{ia}$, which vanishes in a Hartree-Fock basis. The first iteration of a coupled-cluster calculation is therefore an MP2 calculation, exactly as Eq. (9.32) said.

Setting $\hat{T}_1 = 0$ makes $\tau = \tilde{\tau} = t_2$, kills the last two lines of Eq. (10.39) and reduces the intermediates to those of Eq. (10.30); Eq. (10.28) is recovered.

The singles equation is not driven in a Hartree-Fock basis: with $f_{ia} = 0$ the leading surviving term is $-\frac{1}{2} \sum_{mef} t_{im}^{ef} \langle ma \| ef \rangle$, which is first order in t_2 and hence second order in the interaction. Singles therefore start one order later than doubles, which is Brillouin's theorem showing up again.

The equations are nonlinear in both t_1 and t_2, and it is the nonlinearity that performs the infinite resummations. The Section 10.8 makes that statement quantitative.

Cost. The most expensive term in Eq. (10.39) is the particle-particle ladder, $\frac{1}{2} \tau_{ij}^{ef} W_{abef}$, which costs $\mathscr{O}(n_o^2 n_v^4)$. With n_o occupied and n_v virtual orbitals the total scales as n^6, against the exponential scaling of full configuration interaction, Section 5.5. This single fact is why coupled cluster reaches systems that configuration interaction cannot approach: for a hundred orbitals, $n^6 = 10^{12}$ operations is a large but finite calculation, while $\binom{100}{50} \approx 10^{29}$ determinants is not. Adding triples, CCSDT, costs n^8; the perturbative CCSD(T) approximation costs n^7 and is the standard compromise.

10.8 Coupled cluster as a resummation of perturbation theory

Chapter 9 built the correlation energy as a series in the interaction and found that it may converge slowly or not at all. Coupled cluster contains the same series, but does not sum it term by term. The relation can be made exact and, for the pairing model, checked numerically to every digit.

The CCD amplitude equation (10.28) is *quadratic* in the amplitudes – nothing of higher degree appears – so it may be written

$$
Dt = C + L[t] + Q[t,t], \qquad (10.40)
$$

with C the driving integral, L the linear terms and Q the bilinear ones. Expanding $t = \sum_k t^{(k)}$ with $t^{(k)}$ of order k in the interaction and collecting powers,

$$
Dt^{(1)} = C, \qquad Dt^{(k)} = L\left[t^{(k-1)}\right] + \sum_{i+j=k-1} Q\left[t^{(i)}, t^{(j)}\right], \qquad (10.41)
$$

since L and Q each carry one power of the interaction. The energy contribution of $t^{(k)}$ is of order $k+1$.

This is a perturbation expansion, and it is the Møller-Plesset one: the first amplitude is $\langle ab\|ij\rangle/D_{ij}^{ab}$ and the first energy is MP2. The program checks the identification against the Rayleigh-Schrödinger machinery of Chapter 9, computed independently in the determinant basis:

g	order	from CCD	from MBPT
0.50	2	−0.0623931624	−0.0623931624
	3	−0.0165183724	−0.0165183724
	4	−0.0038210766	−0.0038210766
1.00	2	−0.2190476190	−0.2190476190
	3	−0.1004988662	−0.1004988662
	4	−0.0403096858	−0.0403096858

They agree to every digit through fourth order. That CCD reproduces MP2 and MP3 exactly is general – those orders contain only doubles. That it also gets MP4 right here is special to the pairing model, where the interaction produces neither singles nor triples and the quadruple excitations enter only through the disconnected $\frac{1}{2}\hat{T}_2^2$ that the exponential already supplies. For a general interaction, MP4 contains connected triples that CCD does not have, and the agreement stops at third order.

A warning about partitions.

The numbers above are *not* those of Chapter 9, and the difference is instructive. There we took $\hat{H}_0$ to be the bare one-body operator; here the denominators contain the Fock matrix, whose occupied elements are shifted down by $g/2$. For the pairing model the second-order energy in the two partitions is

$$\Delta E^{(2)}_{\text{one-body}} = -\frac{7}{24}g^2, \qquad \Delta E^{(2)}_{\text{Møller-Plesset}} = -\frac{g^2}{4}\left[\frac{1}{2+g} + \frac{2}{4+g} + \frac{1}{6+g}\right], \tag{10.42}$$

which differ already at order g^3: at $g = 1$ they give -0.2917 and -0.2190. Section 9.12 warned that a perturbative number without its partition is meaningless, and here is the warning in force. Coupled cluster uses the Møller-Plesset partition because the Fock operator is what makes the singles vanish at first order.

Summing the series.

The point of the exponential is that the expansion need not be summed term by term. For the pairing model:

order	$g = 0.5$	$g = 1.0$	$g = 1.5$
2	-6.24×10^{-2}	-2.19×10^{-1}	-4.40×10^{-1}
3	-1.65×10^{-2}	-1.00×10^{-1}	-2.68×10^{-1}
4	-3.82×10^{-3}	-4.03×10^{-2}	-1.43×10^{-1}
5	-6.58×10^{-4}	-1.23×10^{-2}	-5.86×10^{-2}
6	-2.98×10^{-5}	-1.31×10^{-3}	-9.80×10^{-3}
7	$+3.49 \times 10^{-5}$	$+1.62 \times 10^{-3}$	$+1.16 \times 10^{-2}$
sum of 16	−0.0833623	−0.3695742	−0.8857526
CCD	−0.0833623	−0.3695572	−0.8841653

Table 10.1 The order-by-order content of coupled-cluster doubles for the pairing model, and its resummation. The expansion converges back onto the CCD energy; solving the nonlinear equation reaches the same answer directly and without the intermediate oscillation.

The series does converge here – but slowly, and non-monotonically, changing sign at seventh order. Solving Eq. (10.28) by iteration reaches the same number in twenty-odd steps, with no reference to the order structure at all. That is the practical content of "infinite resummation": not that new physics appears, but that the physics already present in the series is obtained without having to sum it.

The classes summed are those the truncation can represent. The particle-particle ladder term of Eq. (10.28), iterated, sums the ladder series to all orders; the ring term sums the ring series, which is what the random-phase approximation of Chapter 7 does in its own channel; and the quadratic terms couple them. CCD is neither a ladder theory nor a ring theory but a self-consistent combination of both.

10.9 The pairing model

We now do the calculation. The pairing Hamiltonian of Section 4.2,

$$\hat{H} = \xi \sum_{p\sigma} (p-1)\, \hat{a}^\dagger_{p\sigma} \hat{a}_{p\sigma} - \frac{g}{2} \sum_{pq} \hat{a}^\dagger_{p+} \hat{a}^\dagger_{p-} \hat{a}_{q-} \hat{a}_{q+}, \tag{10.43}$$

has $\langle p{+}\, p{-} \| q{+}\, q{-} \rangle = -g/2$ and nothing else. Two consequences follow immediately, and both simplify the calculation enormously.

First, the interaction moves complete pairs, so it has no matrix element between the reference and a $1p$-$1h$ state. Hence $t_i^a = 0$ identically and CCSD reduces to CCD. The program verifies this: the singles amplitudes come out as exactly zero, and the two energies agree to every digit.

Second, only three of the interaction channels are non-zero – $\langle ab\|cd\rangle$, $\langle ab\|ij\rangle$ and $\langle ij\|kl\rangle$. There is no $\langle ab\|id\rangle$ or $\langle ai\|jb\rangle$, because those would require breaking a pair. The ring term of Eq. (10.28) therefore vanishes, and only the two ladders and the quadratic terms survive.

The Fock matrix is diagonal, with

$$\varepsilon_i = \xi(p_i - 1) - \frac{g}{2}, \qquad \varepsilon_a = \xi(p_a - 1), \tag{10.44}$$

the occupied levels shifted down by $g/2$ and the empty ones untouched – which is the Hartree-Fock solution found in Section 6.12. Taking four levels and four particles, the results are

g	E_{ref}	MP2	CCD	FCI	CCD error	iterations
0.25	1.75000	1.73320261	1.73045621	1.73045985	-3.6×10^{-6}	13
0.50	1.50000	1.43760684	1.41663766	1.41677428	-1.4×10^{-4}	18
1.00	1.00000	0.78095238	0.63044275	0.63554847	-5.1×10^{-3}	26
1.50	0.50000	0.05974026	−0.38416526	−0.34819051	-3.6×10^{-2}	30
2.00	0.00000	−0.70833333	−1.60959440	−1.48965216	-1.2×10^{-1}	31

Table 10.2 Coupled-cluster doubles for the pairing model with four doubly degenerate levels and four particles, $\xi = 1$. The reference energy is the Hartree-Fock energy $2 - g$ of Chapter 6, and the exact column is the full configuration interaction of Chapter 5.

At weak coupling CCD is very nearly exact – four parts in a million at $g = 0.25$, where MP2 is already wrong in the third decimal. It over-correlates, and the error grows with the coupling, reaching 0.12 at $g = 2$. What is missing is the *connected* quadruple excitations: the exponential supplies the $4p$-$4h$ amplitude as $\frac{1}{2}\hat{T}_2^2$, the disconnected part only, and for four particles in four levels the connected part is not negligible once the interaction is strong. Including $\hat{T}_4$ would make the calculation exact, since the exact state has no components beyond $4p$-$4h$.

10.10 When singles matter

To see CCSD do something CCD cannot, we need an interaction that breaks pairs. The pairing plus particle-hole model of Section 7.1 is exactly that: its particle-hole term

$$\hat{V}_{\mathrm{ph}} = -\frac{f}{2}\sum_{pqr}\left(\hat{a}^{\dagger}_{p+}\hat{a}^{\dagger}_{p-}\hat{a}_{q-}\hat{a}_{r+} + \mathrm{h.c.}\right) \tag{10.45}$$

connects the reference to $1p$-$1h$ states and the singles amplitudes are no longer zero. We run the calculation twice, once on the bare oscillator reference and once on the Hartree-Fock reference of Chapter 6, where Brillouin's theorem holds.

g	f	CCD	CCSD	CCSD (HF)	FCI	$\max\lvert t_1\rvert$	$\max\lvert t_1\rvert$ (HF)
0.50	0.025	1.3476038	1.3469075	1.3469088	1.3471327	1.1×10^{-2}	5.4×10^{-4}
0.70	0.050	0.9707634	0.9681476	0.9681644	0.9697588	1.9×10^{-2}	1.5×10^{-3}
1.00	0.050	0.4447429	0.4423659	0.4424119	0.4505823	1.7×10^{-2}	1.8×10^{-3}
0.50	0.250	0.6033126	0.5467210	0.5472745	0.5511916	8.4×10^{-2}	7.3×10^{-3}
1.00	0.500	−1.6764657	−1.7957374	−1.7805416	−1.6917367	1.0×10^{-1}	1.3×10^{-2}

Table 10.3 CCD and CCSD for the pairing plus particle-hole model of Chapter 7, on the bare oscillator reference and on the Hartree-Fock reference.

Three things to read off.

The singles matter. At $g = 0.5$, $f = 0.25$ they move the energy from 0.6033 to 0.5467 against an exact 0.5512, turning a nine per cent error into one of under one per cent.

Brillouin's theorem is visible. The singles amplitudes are an order of magnitude smaller on the Hartree-Fock reference, because there they start at second order in the interaction rather than first, Eq. (10.38). This is the practical reason for working in a Hartree-Fock basis: not that the singles vanish, but that they are small.

Coupled cluster is not invariant under a change of reference. The two CCSD columns are close but not identical. A truncated cluster operator depends on the determinant it is built from, and only the untruncated theory is reference-independent. In a Hartree-Fock basis, where $\hat{T}_1$ has little to do, the dependence is weak – which is the usual situation and the usual justification.

Note also the last row. At $g = 1$, $f = 0.5$ the particle-hole term is strong, CCSD overshoots the exact energy by 0.11, and CCD – which cannot see the singles at all – happens to be closer. This is not a reason to prefer CCD. It is a reminder that an uncontrolled error can have either sign, and that agreement at a single point proves nothing.

10.11 Size extensivity

The property that motivated the whole construction can now be verified. For two non-interacting subsystems A and B, the cluster operator separates, $\hat{T} = \hat{T}_A + \hat{T}_B$, the two pieces commute, and therefore

$$e^{\hat{T}}|\Phi_0\rangle = e^{\hat{T}_A}e^{\hat{T}_B}|\Phi_A\rangle|\Phi_B\rangle = \left(e^{\hat{T}_A}|\Phi_A\rangle\right)\left(e^{\hat{T}_B}|\Phi_B\rangle\right). \tag{10.46}$$

The wave function factorises exactly, and the energy is therefore additive – at any truncation level, since $\hat{T}_A$ and $\hat{T}_B$ are truncated independently. This is the whole argument; it is one line, and it is the reason coupled cluster exists.

The program confirms it for identical copies of a three-level pairing system:

g	$E(1)$	$E(2)-2E(1)$	$E(3)-3E(1)$
0.25	−0.012167069342	0	0
0.50	−0.050059379847	-2.1×10^{-13}	-3.2×10^{-13}
1.00	−0.205303400091	5.6×10^{-17}	-3.0×10^{-13}

Exact to machine precision. Compare Section 5.9, where doubles configuration interaction – the *linear* theory with the same excitation rank – had errors growing from 6×10^{-5} to 1.1×10^{-2} over the same range of couplings. The two methods keep the same excitations; only the ansatz differs; and that difference decides whether the method can be used on a solid.

10.12 Everything against everything

The pairing model has now been treated by every method in this book.

g	HF	MBPT2	MBPT3	CID	HF + RPA	CCD	FCI
0.25	1.75000	1.73177	1.73047	1.73051	1.71635	1.7304562	1.7304598
0.50	1.50000	1.42708	1.41667	1.41760	1.37450	1.4166377	1.4167743
1.00	1.00000	0.70833	0.62500	0.64891	0.55459	0.6304428	0.6355485
1.50	0.50000	−0.15625	−0.43750	−0.29113	−0.40625	−0.3841653	−0.3481905
2.00	0.00000	−1.16667	−1.83333	−1.35209	−1.47622	−1.6095944	−1.4896522
errors							
0.25		1.3×10^{-3}	8.9×10^{-6}	4.7×10^{-5}	-1.4×10^{-2}	-3.6×10^{-6}	
0.50		1.0×10^{-2}	-1.1×10^{-4}	8.2×10^{-4}	-4.2×10^{-2}	-1.4×10^{-4}	
1.00		7.3×10^{-2}	-1.1×10^{-2}	1.3×10^{-2}	-8.1×10^{-2}	-5.1×10^{-3}	
1.50		1.9×10^{-1}	-8.9×10^{-2}	5.7×10^{-2}	-5.8×10^{-2}	-3.6×10^{-2}	
2.00		3.2×10^{-1}	-3.4×10^{-1}	1.4×10^{-1}	1.3×10^{-2}	-1.2×10^{-1}	

Table 10.4 The pairing model treated by every method of this book: Hartree-Fock (Chapter 6), second- and third-order perturbation theory and doubles configuration interaction (Chapters 9 and 5), the random-phase approximation (Chapter 7), coupled-cluster doubles, and full configuration interaction as the exact reference.

At weak and moderate coupling coupled cluster is the best of the five, by two to three orders of magnitude at $g=0.25$ and by a factor of two over third-order perturbation theory at $g=1$. Unlike the perturbation series it does not fall apart as the coupling grows: at $g=2$, where MBPT3 is wrong by 0.34, CCD is wrong by 0.12.

It does not win everywhere. At $g=2$ the random-phase approximation happens to be closer, which is a reminder that a method summing a different class of contributions can prevail in the regime it was designed for. And CCD is not variational – the errors change sign across the table – so a single number carries no guarantee.

What distinguishes it is that no other entry in the table is simultaneously size extensive, systematically improvable and accurate across the whole range. Configuration interaction is variational but not extensive; perturbation theory is extensive but divergent; the random-phase approximation is neither systematically improvable nor stable. Coupled cluster is extensive by construction, improvable by raising the rank of $\hat{T}$, and polynomial in cost. That combination is why it, and not the others, became the standard.

10.13 Beyond CCSD

The hierarchy continues. Including $\hat{T}_3$ gives CCSDT, with a third amplitude equation

$$0 = \langle \Phi_{ijk}^{abc} | \overline{H}_N | \Phi_0 \rangle, \tag{10.47}$$

whose structure is the same – a driving term, linear terms, and nonlinear couplings – and whose length is formidable. The cost rises to n^8.

Empirically, CCSD recovers around 90% of the correlation energy of a well-behaved system and CCSDT around 99%. Since the triples are expensive and their effect modest, the standard compromise is to compute them perturbatively rather than iteratively. Approximating

$$t_{ijk}^{abc} \approx \frac{N_{ijk}^{abc}(t_1, t_2; \langle pq \| rs \rangle)}{D_{ijk}^{abc}}, \qquad D_{ijk}^{abc} = \varepsilon_i + \varepsilon_j + \varepsilon_k - \varepsilon_a - \varepsilon_b - \varepsilon_c, \tag{10.48}$$

with a numerator built from the converged singles and doubles, gives CCSD(T) at a cost of n^7 and with no iteration. This method is accurate enough, and cheap enough, to have earned the name “the gold standard of quantum chemistry”.

Two further directions deserve mention. Because $\overline{H}_N$ is not Hermitian, its left and right eigenvectors differ, and computing properties other than the energy requires both – this is the price of the exponential noted in Section 10.2. And diagonalising $\overline{H}_N$ in the space of excited determinants, rather than merely demanding that its $1p$-$1h$ and $2p$-$2h$ blocks vanish, gives excitation energies: this is equation-of-motion coupled cluster, and it stands to the present chapter as the random-phase approximation of Chapter 7 stands to Hartree-Fock.

10.14 Summary

Coupled-cluster theory replaces the linear ansatz $(1+\hat{C})|\Phi_0\rangle$ by the exponential $e^{\hat{T}}|\Phi_0\rangle$. The two are equivalent when complete, and they differ decisively when truncated: the exponential builds high-rank excitations as products of low-rank ones, so truncating $\hat{T}$ at doubles retains the disconnected quadruples that truncating $\hat{C}$ discards. That single reorganisation makes the method size extensive at every truncation level, which configuration interaction is not.

Requiring $e^{-\hat{T}}\hat{H}e^{\hat{T}}$ to have the reference as an eigenstate turns the Schrödinger equation into the projected equations (10.15)–(10.17). Because the cluster operators commute among themselves, every $\hat{T}$ in the Baker-Campbell-Hausdorff series must connect to $\hat{H}_N$, which has only four contractible operators; the expansion therefore terminates exactly at the fourfold commutator. The non-Hermiticity that this exploits is the same non-Hermiticity that complicates properties – one cannot have the one without the other.

The resulting equations, (10.28) for doubles and (10.38)–(10.39) for singles and doubles, are nonlinear and are solved by iteration. Their first iteration is MP2; their order-by-order expansion is Møller-Plesset perturbation theory, verified here through fourth order; and solving them to convergence sums that expansion within the retained excitation classes. For the pairing model CCD is accurate to four parts in a million at weak coupling and still the best method available at strong coupling, while remaining exactly size extensive.

The program `coupledcluster.py` performs every calculation of this chapter.

10.15 Exercises

Exercise 1: the exponential ansatz

a) Prove Eq. (10.7), that the cluster operators commute, and identify exactly where the particle-hole structure is used.
b) Derive the relations (10.9) between the configuration-interaction and cluster amplitudes, and extend them to $\hat{C}_4$.
c) Show that for two non-interacting subsystems the exponential ansatz factorises, Eq. (10.46), and that the linear ansatz truncated at doubles does not.
d) Why does the exponential series terminate when acting on an N-particle determinant?

Answer to a).

Take two terms of $\hat{T}$, each a string of particle creation and hole annihilation operators. Since particle and hole indices are disjoint, every annihilation operator in one string anticommutes with every creation operator in the other, with no delta function generated. Moving one string past the other therefore produces a sign $(-1)^{n_1 n_2}$ from the interchange, and the same sign appears in the reverse ordering, so the two products are equal. This is Eq. (6.35) generalised, and it is exactly the disjointness of the index sets that makes it work.

Answer to b).

Expand $e^{\hat{T}}$ and collect terms by excitation rank. Rank one gives $\hat{C}_1 = \hat{T}_1$; rank two, $\hat{C}_2 = \hat{T}_2 + \frac{1}{2}\hat{T}_1^2$; rank three, $\hat{C}_3 = \hat{T}_3 + \hat{T}_1\hat{T}_2 + \frac{1}{6}\hat{T}_1^3$; and rank four,

$$\hat{C}_4 = \hat{T}_4 + \hat{T}_1\hat{T}_3 + \tfrac{1}{2}\hat{T}_2^2 + \tfrac{1}{2}\hat{T}_1^2\hat{T}_2 + \tfrac{1}{24}\hat{T}_1^4. \tag{10.49}$$

The term $\frac{1}{2}\hat{T}_2^2$ is the one that matters: it is present in CCD, where $\hat{T}_3 = \hat{T}_4 = \hat{T}_1 = 0$, and absent from CID.

Answer to c).

With no interaction between the subsystems, no amplitude can carry indices from both, so $\hat{T} = \hat{T}_A + \hat{T}_B$ with $[\hat{T}_A, \hat{T}_B] = 0$. Hence $e^{\hat{T}} = e^{\hat{T}_A} e^{\hat{T}_B}$ and, since the operators act on disjoint orbitals, the state factorises. The energy of a product state under $\hat{H}_A + \hat{H}_B$ is the sum of the two energies. For the linear ansatz, $(1 + \hat{C}_2^A + \hat{C}_2^B)$ is *not* the product $(1 + \hat{C}_2^A)(1 + \hat{C}_2^B)$: the cross term $\hat{C}_2^A\hat{C}_2^B$ is a quadruple excitation, which CID has thrown away. This is Section 5.9 in one line.

Answer to d).

Each application of $\hat{T}$ annihilates at least one hole, and there are only N holes. Hence $\hat{T}^n|\Phi_0\rangle = 0$ for $n > N$, and more sharply for a truncated $\hat{T}$: with $\hat{T} = \hat{T}_2$ the series stops at $\hat{T}_2^{N/2}$.

Exercise 2: the similarity transformation

a) Show that $\overline{H} = e^{-\hat{T}}\hat{H}e^{\hat{T}}$ has the same eigenvalues as $\hat{H}$ for any $\hat{T}$.

b) What condition on $\hat{T}$ would make $\overline{H}$ Hermitian? Why does coupled cluster not impose it?
c) Prove that the Baker-Campbell-Hausdorff series terminates at the fourfold commutator, and say precisely which property of $\hat{H}_N$ is used.
d) If a similarity transformation preserves all eigenvalues, why is the CCD energy not exact?

Answer to a).

If $\hat{H}|E\rangle = E|E\rangle$ then

$$\overline{H}\left(e^{-\hat{T}}|E\rangle\right) = e^{-\hat{T}}\hat{H}e^{\hat{T}}e^{-\hat{T}}|E\rangle = e^{-\hat{T}}\hat{H}|E\rangle = E\,e^{-\hat{T}}|E\rangle, \tag{10.50}$$

so $e^{-\hat{T}}|E\rangle$ is an eigenstate of $\overline{H}$ with the same eigenvalue. The map $|E\rangle \mapsto e^{-\hat{T}}|E\rangle$ is invertible, so the spectra coincide. Note that eigen*values* are preserved but eigen*vectors* are not, and neither are norms: $\overline{H}$ is not Hermitian and its left and right eigenvectors differ, satisfying $\langle L_\alpha|R_\beta\rangle = \delta_{\alpha\beta}$.

Answer to b).

$\overline{H}$ is Hermitian if $e^{\hat{T}}$ is unitary, that is if $\hat{T}^\dagger = -\hat{T}$. Coupled cluster does not impose it because an anti-Hermitian $\hat{T}$ contains de-excitation operators $\hat{a}_i^\dagger\hat{a}_a$ as well as excitations, those do not commute with the excitations, and the Baker-Campbell-Hausdorff series then never terminates. One would gain Hermiticity and lose the finite expansion. Unitary coupled cluster makes precisely that trade, and is used where the exponential can be realised directly rather than expanded – on a quantum computer, for instance.

Answer to c).

By Eq. (10.7) no commutator can be made non-zero by one $\hat{T}$ acting on another, so in $[[\cdots[\hat{H}_N,\hat{T}],\ldots],\hat{T}]$ every $\hat{T}$ must be contracted with $\hat{H}_N$ itself. Each such contraction consumes one of $\hat{H}_N$'s operators that can be contracted to the right – and its two-body part has exactly four, $\hat{a}_p^\dagger\hat{a}_q^\dagger$ and $\hat{a}_s\hat{a}_r$ in normal-ordered form. A fifth $\hat{T}$ has nothing left to attach to, so the term vanishes. The property used is that $\hat{H}_N$ is *at most two-body*; a three-body force would extend the series to the sixfold commutator.

Answer to d).

Because we do not solve the transformed problem exactly either. With $\hat{T} = \hat{T}_2$ the full $\overline{H}_N$ contains up to six-body terms, and it is that operator whose spectrum equals the exact one. What CCD imposes is only that the $2p$-$2h$ block of $\overline{H}_N$ vanish – the reference is decoupled from doubles but not from triples and quadruples. Coupled cluster at finite truncation is a similarity transformation *plus* a projection, and the error lives in the projection.

Exercise 3: deriving the CCD equations

a) Show that the expansion (10.22) stops at $\hat{T}_2^2$ when projected onto a $2p$-$2h$ state.
b) Derive the linear terms (10.25) and (10.26) and explain the factors $\frac{1}{2}$ and the permutation operators.
c) Derive the four quadratic terms (10.27).
d) Verify that setting all amplitudes on the right-hand side to zero gives the MP2 amplitude.
e) Show that the intermediates (10.30) reduce the cost of the quadratic terms from $\mathscr{O}(n^8)$ to $\mathscr{O}(n^6)$.

Answer to a).

A $2p$-$2h$ projection supplies two hole and two particle external lines. Each $\hat{T}_2$ must connect to $\hat{H}_N$, consuming two of its contractible operators, and $\hat{H}_N$ has four. Two cluster operators therefore exhaust it, and a third has nothing to attach to.

Answer to b).

Use the generalised Wick theorem of Section 3.12 on $\langle \Phi_{ij}^{ab} | \hat{H}_N \hat{T}_2 | \Phi_0 \rangle$. The one-body part can contract with either a particle or a hole line of the amplitude, giving the two terms of Eq. (10.25); the permutation operators appear because the external labels a, b and i, j must be antisymmetrised. For the two-body part, contracting both of its creation operators with the amplitude's particle lines gives the particle-particle ladder, both annihilation operators with the hole lines gives the hole-hole ladder, and one of each gives the ring. The factor $\frac{1}{2}$ in each ladder compensates the two equivalent ways of assigning the summed pair of indices; the ring has inequivalent lines and instead carries the full $P(ij)P(ab)$.

Answer to c).

Now both $\hat{T}_2$ connect to $\hat{H}_N$, which forces the two-body part and uses all four of its operators, so only $\langle kl \| cd \rangle$ contributes. The four topologies are fixed by how the four contractions are shared: two with each amplitude in the "ladder times ladder" pattern (first term), one particle and one hole with each (second term, the ring-ring), or both particle and both hole indices with one amplitude leaving the other a spectator (third and fourth). The combinatorial factors follow by counting equivalent assignments as in b).

Answer to d).

Setting $t = 0$ on the right leaves $D_{ij}^{ab} t_{ij}^{ab} = \langle ab \| ij \rangle$, so

$$t_{ij}^{ab} = \frac{\langle ab \| ij \rangle}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}, \tag{10.51}$$

which is Eq. (9.32). Substituting into Eq. (10.19) gives the MP2 energy. The program checks this against the closed form $-\frac{g^2}{4}[1/(2+g) + 2/(4+g) + 1/(6+g)]$ for the pairing model.

Answer to e).

Written out, a term such as $\sum_{klcd} \langle kl \| cd \rangle t_{ik}^{ac} t_{jl}^{bd}$ has four free external indices and four summed ones, hence $\mathscr{O}(n^8)$. Building $\chi_{bkcj} = \frac{1}{2} \sum_{ld} \langle kl \| cd \rangle t_{lj}^{db}$ first costs $\mathscr{O}(n^6)$ – four free, two summed – and contracting it with the remaining amplitude costs $\mathscr{O}(n^6)$ again. The saving is the standard one of never forming an object with more indices than necessary, and it is what makes the n^6 scaling of CCSD achievable.

Exercise 4: CCSD and its perturbative content

a) Derive the energy expression (10.19) and explain why no term with three or more cluster operators appears.
b) Show that Eqs. (10.38) and (10.39) reduce to the CCD equation when $\hat{T}_1 = 0$.

c) Show that in a Hartree-Fock basis the singles amplitudes are of second order in the interaction while the doubles are of first order.
d) Establish the order-by-order recursion (10.41) and show that the first two energies are MP2 and MP3.
e) Why does CCD reproduce MP4 for the pairing model but not in general?

Answer to a).

$\langle \Phi_0|\overline{H}_N|\Phi_0\rangle$ requires that the excitation produced by the cluster operators be exactly undone by $\hat{H}_N$, which can de-excite by at most two ranks. Hence at most $\hat{T}_2$ or $\frac{1}{2}\hat{T}_1^2$ can appear, plus $\hat{T}_1$ against the one-body part. Three cluster operators would produce at least a $3p$-$3h$ excitation, which $\hat{H}_N$ cannot annihilate.

Answer to b).

With $\hat{T}_1 = 0$ we have $\tau = \tilde{\tau} = t_2$; the last two lines of Eq. (10.39) vanish, as does the second half of the ring term; and the intermediates collapse to $F_{ae} \to f'_{ae} - \frac{1}{2}t_2 \cdot v$, $W_{mnij} \to \langle mn\|ij\rangle + \frac{1}{4}t_2 \cdot v$ and so on, which are exactly the combinations of Eq. (10.30). What remains is Eq. (10.28).

Answer to c).

In a Hartree-Fock basis $f_{ia} = 0$, so the driving term of Eq. (10.38) vanishes. The leading surviving term is $-\frac{1}{2}\sum_{mef} t_{im}^{ef}\langle ma\|ef\rangle$, first order in t_2; since t_2 is itself first order in the interaction, t_1 is second order. The doubles, by contrast, are driven directly by $\langle ij\|ab\rangle$ and are first order. The program shows this concretely: the singles amplitudes of the pairing plus particle-hole model drop by an order of magnitude when the reference is rotated to the Hartree-Fock one.

Answer to d).

The CCD equation is quadratic, so $Dt = C + L[t] + Q[t,t]$ exactly. Substituting $t = \sum_k t^{(k)}$ and noting that C, L and Q each carry one power of the interaction, the terms of order k are $L[t^{(k-1)}]$ and $Q[t^{(i)}, t^{(j)}]$ with $i + j = k - 1$, which is Eq. (10.41). The first amplitude is the MP2 one by d) of the previous exercise, and its energy is MP2. The second satisfies $Dt^{(2)} = L[t^{(1)}]$, whose energy contribution is the sum of the particle-particle ladder, hole-hole ladder and ring contractions of two MP2 amplitudes – which is precisely the third-order energy.

Answer to e).

At fourth order the exact energy receives contributions from singles, doubles, triples and quadruples. CCD has the doubles, and it has the quadruples in their disconnected form $\frac{1}{2}\hat{T}_2^2$, which is the whole of the fourth-order quadruple contribution. It lacks the singles and the connected triples. For the pairing model those are absent anyway – the interaction cannot produce a $1p$-$1h$ or a $3p$-$3h$ excitation – so CCD is complete at fourth order and the program finds exact agreement. For a general interaction the connected triples are present and CCD misses them, so agreement stops at third order. This is exactly why CCSD(T) exists.

Exercise 5: numerical explorations

a) Using `coupledcluster.py`, reproduce Table 10.2 and extend it to six and eight levels. Does the CCD error grow or shrink with the size of the space?
b) Verify the size-extensivity result for four and five subsystems, and compare with doubles configuration interaction over the same range.
c) Add $\hat{T}_4$ to the pairing calculation, or equivalently solve the seniority-zero problem exactly, and confirm that the remaining CCD error is entirely due to connected quadruples.
d) Study the convergence of the CCD iteration as a function of g. At what coupling does it stop converging, and how does that compare with the radius of convergence of the perturbation series in Section 9.7?
e) Implement the CCSD(T) correction of Eq. (10.48) for the pairing plus particle-hole model and see how much of the remaining error it removes.

Answer.

Some pointers. In a) the relative error shrinks as the space grows, because the neglected connected quadruples become a smaller fraction of a larger correlation energy – the opposite of the trend for truncated configuration interaction, whose size-consistency error grows. In b) the coupled-cluster error stays at machine precision for any number of subsystems, while the CID error grows roughly linearly. In c) the pairing model with four levels has no $6p$-$6h$ states, so CCD plus connected quadruples is exact, and the difference from CCD is precisely the number in the error column of Table 10.2. In d) the CCD iteration continues to converge well past the point where the perturbation series diverges, which is the practical meaning of resummation; it eventually fails too, and the failure is associated with the same physics that made the mean field unstable in Chapter 6. In e) the perturbative triples remove a large part of the residual error at weak coupling and much less of it at strong coupling, where the expansion underlying them is itself unreliable.

Part III
Stochastic methods

Part IV
Machine Learning based approaches

Part V
Quantum Computing

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